

The Energy Balance Project

What an Economy Must Keep Alive

**The physical theory and design of an Energy
Balance Unit**

A continuous explanatory edition

Expanded theoretical integration

Konrad Grzyb

Author and project originator

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Preface

Every generation inherits a world that makes its activity possible. Money helps people exchange, coordinate and plan within that world. Yet a completed sale cannot, by itself, tell us whether the water line will still work, whether a field will still produce food or whether a community can absorb its next disturbance. Those are physical questions. This book asks what an economy would look like if its account of action answered them as well.

We begin with the reasons money arose and the incentives it creates. From there the argument moves through light, energy, thermodynamics and the living body's continuous work of homeostasis. These comparisons help us distinguish local action from the larger conditions that make it useful. The recurring clinic and water examples keep that question close to ordinary experience.

The Energy Balance Unit enters once the problem is clear. We declare a physical state, a complete change and a potential expressing a functional comparison. Finite EBU is the decrease of that potential between the endpoints. The rest of the book develops the consequences of taking this definition seriously. Its central question stays the same: how can society account coherently for the state-changing consequences of action, including shared effects and delayed response, while keeping physical description and institutional choice clear?

A path integral explains why the marginal changes during a finite action. Telescoping explains how receipts compose into a history. Coalition tables and Möbius inversion explain how joint work differs from isolated proposals. Discrete Taylor structure, continuous curvature and composite response connect the same comparison at several mathematical levels.

Dynamics then gives the account a time dimension. Water already in transit becomes part of the present state. Feedback can create memory and oscillation, yet a common linear response retains an exact pair-only interaction structure under a quadratic potential. Nonlinear response explains where that simplification ends. Canonical probability and the specified entropy relation provide further physical interpretations under their own assumptions.

At each stage, the law has a practical meaning. It can prevent duplicate claims, reduce unnecessary calculation, make a shared bottleneck visible or explain why another delivery would be premature. These features matter because an economy should make useful maintenance and reliable service easier to support, while preserving access for the people who need them.

Book 1 contains the theoretical foundation needed for that argument. The later books develop test designs, simulations, applications and results. Established mathematics is presented as such, with the assumptions that give it its exact scope. The institutional chapters then explain how the information can be used while keeping rights, attribution and access explicit.

We can begin where people already make their choices: with money, the work it does for us and the physical conditions it cannot describe by itself.

The reading route

Chapters 1–13 establish motivation, physical state, reference, potential and local field. Chapters 14–24 connect finite actions to paths, permission, locality, history and joint value. Chapters 25–30 build the interaction hierarchy from coalition tables through exact discrete Taylor structure and degree bounds.

Chapters 31–35 connect that hierarchy to motion, generators, feedback and nonlinear response. Chapters 36–42 develop dependency structure, canonical probability, entropy, changing fields, uncertainty and the central continuity theorem. Chapters 43–49 assemble attribution, routes, institutional rules and a complete theoretical system. Five appendices contain the longer proofs and worked comparisons.

Each chapter opens with its purpose and returns the mathematics to an EBU capability. The detailed contents can be used as a reading map; the symbol guide provides a compact reference. Readers can follow the main argument before returning to the appendix proofs.

This is an author-review manuscript integrating the independently cleared R-stage, S-MG and feedback sources. A separate independent book audit remains required. The edition note records provenance and scientific status in one place.

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What does an economy keep alive?

An economy makes promises about the future. A wage promises purchasing power; a maintenance plan promises a working bridge; a water service promises something usable when a household opens its tap. These promises depend on different records and different kinds of evidence. This book begins by asking how they can remain connected, then develops a precise account of the physical changes on which continuing service depends.

1.1 Why people made money

No household can produce everything it needs, and an exchange is difficult when each person must want exactly what the other can offer at that moment. Communities have found ways to record obligations, compare contributions and carry claims across time. Historical records include accounts on clay tablets, measured commodities, weighed metal and coins. Money developed through several forms and institutions; its history is not one universal passage from simple barter to coinage.[1]

A shared monetary unit makes a remarkable form of cooperation possible. A carpenter can receive payment without taking whatever food the customer happens to have. A clinic can order supplies from a distant manufacturer. An obligation can be recorded today and discharged later. Trust, enforceable agreements and functioning institutions support these arrangements. Money's familiar roles as a unit of account, means of exchange and store of value help cooperation extend beyond the immediate encounter.[2]

The achievement is real. It also has a defined scope. Suppose the clinic pays its electricity bill and orders medicine, but a failed pipe leaves its treatment room without water. The payments and orders may all be correctly recorded. They do not make the missing service available. That service depends on reservoirs, treatment, pipes, electricity, stocks already in transit and people able to maintain the chain.

The reverse situation is possible too. Someone repairs the pipe without making a sale, and usable service returns. The repair has changed the town even if no revenue records its benefit. A financial account and a physical account can therefore describe the same day accurately while reporting different things.

1.2 A measure can be excellent at the wrong question

A monetary sum is useful partly because it can travel across very different transactions. It need not describe every property of the object purchased. Yet two products with the same sale price can use different materials, leave different wastes and have different consequences for future service. Equal receipts do not imply equal physical changes.

This is not a claim that profitable activity is necessarily damaging. Maintenance can be profitable, and a loss-making project can waste resources. The more precise point is that no accounting identity makes private financial return equal the improvement of a declared physical condition. Connecting the two requires knowledge and a rule.

Existing societies already make such connections through contracts, public services, standards, taxes, regulation and shared obligations. A water-quality standard identifies a condition a supplier must respect; a maintenance contract makes a future service part of a present agreement. Their effectiveness depends on specifying the condition, observing it and responding when the promise fails. The financial number does not perform those tasks by itself.

The EBU proposal begins at this measurement problem. It asks what physical state an action starts from, what complete change it produces, and how that change compares with a declared potential. The potential is a mathematical description of a specified concern. It is not a claim that one number can contain every purpose of human life. The rest of the book will make both the power and the boundary of that account explicit.

1.3 Want, need and the power to pay

A person's ability to make an offer is not the same as the urgency of their need. A buyer may be able to pay for a second convenience while another person cannot afford a first safe water supply. Both can be represented in a market, but their offers do not automatically rank the consequences of going without.

The difficulty becomes sharper when those affected cannot participate in the transaction. A future child cannot bid today for an intact aquifer. A downstream household may have no contract with the activity that changes its water. Institutions can represent those interests, but doing so is a deliberate achievement. It does not follow from the existence of a price.

A physical account could make some neglected consequences more inspectable. For example, a maintenance record could show the source stock used, the usable receiver stock restored and the material sent to a sink. Such a record

would give a community evidence for deciding how to support essential repair. It would not decide the repairer's pay, the patient's entitlement or the community's priorities.

This distinction is also a source of efficiency. When a problem is described accurately, people can spend less effort disputing the wrong question. They can distinguish an empty reservoir from a broken route, a delivery still pending from one that never occurred, and a repaired service from a paid invoice. The possible social gain is more dependable provision. It depends on credible observations and institutions willing to use them.

The question of this book

What does an action take from the represented world, what does it leave there, and which continuing function depends on the change? A useful physical account makes that comparison clear enough to inform a human decision without pretending to replace it.

1.4 A world inherited rather than purchased

Every generation receives a world it did not manufacture. Sunlight arrives, water circulates, organisms reproduce, and people build and maintain the networks through which these processes support shelter, food and care. Keeping that inheritance usable does not mean freezing every stock at its present value. A living system continually changes; continuing function depends on the right changes occurring in the right places and at the right times.

A repaired wetland, a maintained water line or unpaid care can preserve possibilities that a sales total describes poorly. Conversely, an activity can produce revenue while reducing tomorrow's options. The challenge is to make those consequences visible without claiming that every worthwhile human contribution can already be measured on one universal scale.

We shall build the argument in that order. First identify states and complete physical transitions. Then learn how sequential changes fit together and why simultaneous actions may have a joint effect that differs from their separate effects. Only after valuation is clear will we ask how a response law, memory or feedback produces an endpoint, and under which additional conditions probability and entropy describe the same comparison.

The immediate next step is closer to ordinary life: a workshop must choose what to do with a day's labour. Following that choice will show how an accurate physical account might inform an incentive while remaining distinct from the incentive itself.

What money rewards, and what it cannot see

A useful account must enter a decision somewhere. This chapter follows a workshop choosing between a paid order and an essential repair. The example separates three things that will remain distinct throughout the book: what can physically happen, how its consequences are evaluated, and which rule makes someone willing or able to act.

2.1 The decision at the workshop

The workshop can repair the town's water line today or complete an order from a customer who will pay immediately. Its workers need wages, and its tools and electricity cost money. The paid order is a serious claim on its time. So is the pipe: households and a clinic depend on it.

A simplified private calculation is

$$\text{expected private return} = \text{expected receipts} - \text{private expenditure.} \quad (2.1)$$

This expression leaves out many reasons people work, including care, duty, curiosity and legal obligation. It nevertheless matters to an organisation that must meet its bills. A repair may protect many people while leaving the workshop unable to cover its own expenditure. Another activity may pay well while shifting physical costs onto people outside the agreement.

Economics describes an effect on others that a market price does not include as an externality. Pollution is a familiar example. The IMF's explanations of externalities and energy subsidies discuss circumstances in which environmental costs are not reflected in the price paid by a user.[4, 5] The omitted consequence remains a physical consequence whether or not an invoice names it.

For EBU, the practical starting point is to declare the boundary before evaluating the change. A repair that restores a receiver while depleting an unrecorded source has an incomplete physical description. A fuller account can expose the omitted effect. Turning that information into a contract, payment or obligation is a subsequent institutional decision.

2.2 A dollar is a claim, not a physical forecast

Money makes claims transferable. A deposit can pay a supplier who then pays someone else, without each payment carrying a complete description of the earlier activity. Modern bank lending can create deposits, subject to the constraints described by the Bank of England.[3] Those monetary mechanisms do not themselves forecast the stock of safe water or working equipment available when the money is spent.

Imagine two orders bringing the same revenue to the workshop. One uses a scarce spare part in a short-lived installation; the other repairs a pump needed for continuing service. The financial inflow is equal, but the future physical possibilities differ. A good contract or public rule may take that difference into account. It needs a description of the difference before it can do so reliably.

Nor does observing a favourable condition prove how it was produced. Water may arrive because of today's repair, yesterday's dispatch or an external change upstream. Later chapters will distinguish the physical outcome from a response model and from causal attribution. That distinction begins here: a receipt, a forecast and an observation are different kinds of record.

A central EBU feature is therefore traceability. A claim about improvement should point to the state, boundary and completed transition on which it rests. This can make duplicated claims and missing consequences easier to detect. It may reduce the cost of verification, but it does not remove the need for trusted measurements or accountability when a record is wrong.

2.3 Demand, desire and indispensable service

Offers in a market depend on purchasing power as well as preference. An additional convenience for one person can attract a larger offer than an indispensable service for another. This does not establish what producers feel about either person. It identifies a limitation of using offers alone to rank needs.

Public provision, insurance, subsidies, rights and regulation can alter that allocation. Their existence shows that an economy already contains many mechanisms for connecting payment to broader purposes. The question is how those mechanisms learn whether the promised service was actually delivered and whether maintaining it has displaced an equally important need elsewhere.

Consider a clinic whose water is restored for an hour and then fails again. A final stock reading may look satisfactory at the chosen observation time, while

the interruption has still mattered to patients. The eventual EBU endpoint account and a measure of service continuity would answer different questions. A responsible institution may need both. Naming the distinction early prevents a single physical number from acquiring a role its definition does not support.

2.4 Where an incentive points

Suppose a supplier earns ten monetary units by exporting water and six by supplying the clinic. If the clinic's consequences do not enter the supplier's contract, export has the larger private return. A service obligation can change that choice. A drought can change which withdrawals are physically possible. These are three separate changes: in a payment comparison, an institutional rule and a physical constraint.

The proposed account can inform such decisions by describing a declared physical transition consistently. It cannot manufacture water, appoint an owner or decide whose need takes priority. Those boundaries make the account more useful: the same physical result can be inspected even when people disagree about the appropriate rule.

Two ledgers for one decision

A repair costs eight monetary units and earns five today, giving private return -3 . If observations also show that it prevents a specified water loss, that fact does not change the arithmetic of the financial account. It gives the community evidence for designing a payment, duty or public service that makes the repair possible.

The efficiency opportunity is to connect incentives to verifiable outcomes rather than merely to activity labels. A rule that rewards reported repairs needs a way to distinguish completed restoration, repeated reporting of the same event and damage followed by repair. The later laws of finite valuation and sequential closure will provide part of that checking structure. The institution can then make its maintenance and access rules explicit, using the physical history as their common reference.

2.5 The next difficulty: who can reach what exists?

Even an accurate account of quantity and price does not establish access. A resource can be present but unusable, unreachable, already committed or withheld under existing rights. The next chapter separates these failures so that the response addresses the actual obstacle.

Scarcity is more than an empty store

The clinic's missing water could have several causes. Treating them as one shortage would make the response wasteful: more dispatch will not repair a broken pipe, and a new pipe will not settle a right of access. This chapter explains why the represented state must distinguish resource, route, timing and entitlement before any number can describe an intervention usefully.

3.1 Water behind a locked gate

A reservoir contains water, yet the clinic receives none. Perhaps the pipe has failed. Perhaps the water is unsafe. Perhaps a delivery is still pending, or the owner has refused access. Each explanation is consistent with a positive stock total. Each calls for a different action.

Three questions help separate them. What usable resource exists within the declared boundary? Can it reach the required function in time? Who is permitted and able to receive it? Supply and demand, infrastructure and access can all contribute to water scarcity; the FAO's account likewise treats scarcity as more than a single stock figure.[7]

Some constraints are physical. The district cannot distribute more usable water than it can obtain, treat or safely reuse over the relevant interval. Other constraints arise from a route, a schedule or a rule. A physically full store can coexist with exclusion. Neither observation proves that all scarcity is natural or all scarcity is artificial. It tells us to identify the constraint before recommending a remedy.

For an EBU system, this implies a practical requirement: the physical description must include the coordinates that distinguish relevant alternatives. If pending water is omitted, a second order can look like a first response. If unusable water is pooled with safe water, a favourable total can conceal failed service. A smaller model is computationally cheaper only when it still preserves the distinctions needed for the decision.

3.2 Competition is a choice of allocation rule

When several people seek a constrained resource, a monetary bid is one possible way to allocate it. A queue, service priority, entitlement or negotiated share is another. An account of the physical change can inform these rules without selecting one of them.

The same distinction applies to cooperation. Shared need does not guarantee agreement, and a common resource does not guarantee that independently chosen actions will fit together. Two deliveries may draw on the same source; two repairs may support the same service. Adding their separate descriptions can miss what happens when both occur. The later coalition mathematics will make this joint comparison exact for a declared table.

There is no need to condemn every act of maximisation to see the difficulty. The decisive questions are what is being improved and which constraints remain binding. Improving private return, reducing a represented burden and protecting essential access are different objectives. An institution can choose how to combine them, but it should make that choice visible.

Better visibility can improve efficiency by directing effort toward the actual bottleneck. A verified permission problem belongs with governance; a source problem belongs with supply; a timing problem may require coordination. Calling all three a low EBU value would obscure useful information rather than organise it.

3.3 Inequality changes exposure as well as income

The World Bank's *Poverty, Prosperity, and Planet Report 2024* estimated that about 700 million people lived in extreme poverty under its then-used \$2.15-a-day line. It described stalled progress and the combined challenge of poverty, unequal opportunity and climate risk. The figure is a dated estimate under a named monetary definition, not a count of every kind of deprivation.[8]

The WHO's 2025 report on social determinants of health equity examines how conditions of birth, work, housing and access to power, money and resources shape health. Its comparisons show why a national average cannot represent everyone's circumstances.[11] A physical model has an analogous responsibility: aggregation must not erase the differences relevant to the claim being made.

A district can have sufficient water in total while one clinic has none. A road may work for people with vehicles but provide no practical route for others. A medicine can exist while being unavailable at the time a patient needs it. Representing these differences does not solve distribution. It allows

a distributional claim to be checked against something more informative than an average.

The possible benefit is a more accountable discussion of service. People could challenge a reassuring total with evidence about an omitted place, time or condition. Such an account would need privacy, accessible explanation and a way to correct errors; measurement alone does not guarantee that those affected will be heard.

3.4 Resources, grievance and conflict

Resource control and exclusion can be part of a conflict, but they do not provide a universal explanation of violence. The World Bank identifies economic exclusion among contributors to fragility alongside governance and other causes.[9] The IPCC finds conditional links between climate variability, extremes and conflict risk, while giving greater weight to major social and political drivers.[10]

An EBU account should respect that complexity. Showing that two towns share one river does not settle their rights or motives. It can help prevent a negotiation from proceeding as though each town had the full flow independently available. A common physical description supports the discussion; it does not decide what a just agreement is.

The later distinction between a joint outcome and an allocation of responsibility has the same purpose. Mathematical decomposition can identify a dependence in a declared comparison. It cannot infer consent, ownership or blame from that dependence alone.

Find the failure before choosing the remedy

Too little usable resource, no workable route in time, and no effective right to receive it are different failures. A useful account keeps them distinguishable so that an intervention addresses the problem actually present.

3.5 From a district to a planet

Humanity cannot procure a replacement biosphere. The next chapter follows planetary resources through the same questions of quantity, boundary, timing and consequence.

The planet inside the account

A physical account becomes useful when its numbers can be traced to observations and interpreted within their scope. Planetary assessments offer a demanding example: their quantities differ, their consequences unfold over different timescales, and their projections depend on future choices. The same discipline will be needed when a small EBU model moves from an exact equation to a claim about the world.

4.1 A changing inheritance

The IPCC's 2023 synthesis concluded that human activities had unequivocally caused global warming, with global surface temperature in 2011–2020 about 1.1 °C above 1850–1900. It assessed widespread changes and unequal contributions and exposure.[12] This is a comparison between specified periods. It should not be presented as today's instantaneous reading.

The IPBES global assessment estimated that roughly one million animal and plant species were threatened with extinction. Threatened does not mean already extinct, and the estimate is not a census of every species.[13] The International Resource Panel's 2024 outlook described a conditional path under which global extraction could rise by almost sixty percent from 2020 to 2060 without urgent concerted action.[14] The projection names a possible course under assumptions, not an inevitable future.

Temperature, biodiversity and extraction are different objects. Their headline numbers cannot be added to obtain one planetary score. They nevertheless show why an economic action can affect conditions that support people long after the transaction closes.

An EBU account must begin with the same respect for distinctions. It needs a declared physical state and potential, appropriate units and a boundary within which the comparison makes sense. A proof about that representation is valuable, but it cannot establish that the representation includes every important harm.

4.2 The short return and the long consequence

A company can receive revenue from a water withdrawal this month while consequences for groundwater, soils or downstream ecology develop over years. Legal duties and public scrutiny may already address some of those effects. Others may remain uncertain or outside the immediate decision.

Several clocks are involved. A transaction closes at one time. Material moves through a system at another. Regeneration, degradation and ecological response may have still different timescales. A complete receipt for the first event is not evidence that the later processes have ended.

This gives the future EBU system two related jobs. It should state what transition has actually been evaluated, and it should distinguish that record from predictions about later response. A model of pending effects or feedback may help with the prediction. It must supply its own assumptions and evidence; a potential difference does not secretly contain a law of motion.

The benefit could be better-timed maintenance and intervention. A clear distinction between completed improvement, expected recovery and continuing exposure can reduce premature declarations of success. For society, that can matter when a service must remain dependable throughout a recovery, rather than merely look better at its final inspection.

4.3 A projection is a conditional warning

The IPCC compares outcomes under different emissions pathways. A pathway states assumptions about future activity; its appearance in a table does not itself assign it a probability.[12] Comparing alternatives is useful because a decision can still affect which conditions develop.

Our later coalition tables serve a narrower mathematical purpose: they compare specified combinations of actions under one protocol. Those alternatives must not be mistaken for observed histories or probabilities merely because the table is complete. Nor does a model-generated endpoint become a measurement by passing through exact algebra.

This separation supports a practical workflow. Keep observations, response assumptions and valuation rules identifiable. Then a corrected observation can be distinguished from a changed forecast or a newly chosen field. Decisions can improve without silently rewriting what an older record meant.

Read the year and the condition

The UNEP/IRP projection of almost sixty percent more extraction by 2060 uses 2020 as its comparison and assumes a continuation without urgent concerted action. It describes neither an already observed 2060 nor an equal increase in every resource.

4.4 What is missing from the dashboard?

A district can report rising income while its aquifer falls, its clinic loses reliable service and its waste accumulates. Each report may be accurate. The error would be to expect one to substitute for all the others.

The constructive question is what information a particular decision requires. Which stocks or qualities can the action change? Which source, receiver and sink belong in its physical boundary? Which effects are already present, which remain pending, and which require a separate forecast? These questions can produce a tractable local account even when a complete model of the planet is unavailable.

Later we shall prove that a local calculation can equal a declared global potential change when unchanged factors cancel. That is an efficiency result with exact assumptions, not permission to ignore a remote consequence that genuinely changes. It offers a way to make a physical account usable without demanding global knowledge from every person opening a valve.

Before developing it, we look at familiar physical examples of local relations with consistent consequences. They will introduce the difference between a descriptive principle, a quantity being compared and a law that actually moves a system.

How does the world know which way to go?

How can many local decisions form a coherent account? We begin with light because it offers a familiar example of a local relation with a precise global consequence. The comparison also establishes an essential distinction for EBU: a quantity accumulated along a route and a change of a state function answer different questions.

5.1 No central calculator

When light crosses from air into water, its route bends. The light does not pause at the boundary while an invisible planner surveys every possible trajectory. When warmth spreads from a hot object into a cooler one, no administrator compares all the molecules. The behaviour follows local physical relations, and those relations can have consequences that are described by compact global statements.

This is the feature we are looking for, not a claim that societies should behave like photons. No person could calculate the complete state of the planet each time a tap is opened or a delivery truck moves. If a proposed physical account required such an all-knowing observer, it would fail at the first ordinary decision. We need to understand how a local change can be assessed without pretending that the entire world is locally known.

Physics provides two kinds of discipline. Conservation tells us that a carrier has not appeared from nowhere inside a closed account. A relation of motion or change tells us which transitions occur under specified conditions. Neither is the same as a moral rule or a social allocation policy. Their value for our purpose is structural: they show how to make claims precise enough that hidden creation, omitted boundaries and impossible action can be detected.

5.2 The bent path of light

Imagine a point above a flat water surface and another below it. In a single uniform medium, the straight segment is the shortest geometrical distance. Across the boundary, light can instead follow two straight segments joined at an angle. The route is longer in space than the direct line, yet the light

travels at different speeds in the two media. Distance alone is not the relevant quantity.

In elementary geometrical optics the travel time of a candidate path γ is

$$T[\gamma] = \frac{1}{c} \int_{\gamma} n(\mathbf{r}) ds, \quad (5.1)$$

where c is the speed of light in vacuum, $n(\mathbf{r})$ is the refractive index and ds is a small piece of distance. The integral adds the time cost of all those pieces. Fermat's principle identifies the observed ray, in its domain of application, by stationary optical time: a small permitted displacement of the path produces no first-order change in T . The familiar phrase *least time* is useful for the elementary two-medium case, but stationarity is the more general statement.[16]

We can see the rule in a drawing with no mystery. Let the light enter the surface at a horizontal coordinate u . If its starting point is a horizontal distance a from the chosen origin and height h_1 above the surface, while its destination lies at horizontal coordinate b and depth h_2 , then the candidate time is

$$T(u) = \frac{n_1}{c} \sqrt{(u-a)^2 + h_1^2} + \frac{n_2}{c} \sqrt{(b-u)^2 + h_2^2}. \quad (5.2)$$

Differentiate with respect to the crossing point u . At a stationary path,

$$\frac{dT}{du} = 0 \implies n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad (5.3)$$

which is Snell's law. The angles are measured from the normal to the interface. If $n_2 > n_1$, the ray bends toward that normal. A single rule accounts for the bending because the cost of another small distance is different on the two sides.

Why the straight line can lose

Take $n_1 = 1$ and $n_2 = 1.5$. If $\sin \theta_1 = 0.6$, stationarity requires $\sin \theta_2 = 0.4$. The path changes direction at the interface. It is not the shortest geometrical connection, because distance in the slower medium contributes more time.

5.3 What the analogy gives us, and what it does not

The optical integral teaches us to name the quantity being accumulated and the path over which it is accumulated. It also teaches caution. Light is not awarded a payment for finding a route; Fermat's principle is not a doctrine that people should choose the fastest path or that an economy should minimise travel time. It is a physical result for light under specified optical conditions.

Later in this book another path integral will appear. It has a different mathematical character. If a smooth scalar potential V is declared, integrating its gradient along a path between two states gives a difference of endpoint values. Such a gradient integral is path-independent while the optical time in Eq. (5.1) generally depends on which route is proposed. The similarity is the discipline of summing local contributions; the distinction is essential to the mathematics.

Nor should the expression *the world knows* be taken literally. Local laws can generate orderly patterns without foresight. A photon does not inspect an economic objective, and molecules do not vote on the future. For human systems we must declare the actions, observations and decision rules ourselves. For EBU this means recording three separate objects: the field that values a state, the response rule that changes it, and the institution that uses the resulting account. Keeping these objects explicit makes a design easier to improve. A community can repair a slow delivery rule without secretly changing yesterday's unit of account; it can revise an unsuitable reference without claiming that the pipes acquired a new physical law.

5.4 From path to balance

A water movement between a reservoir and a treatment site also has a route. It may require a pump, electricity, maintenance and a pipe. Describing only the final delivery could hide the resource used along the way; counting the same transformation twice could invent an extra cost. The first task is therefore to keep the physical account closed: what left one place, what entered another, and what crossed the boundary? Two routes can have the same complete endpoints and therefore the same finite EBU, yet differ in travel time and in the service supplied during the journey. If one route consumes more electricity, its complete energy endpoint differs and that consequence belongs in the state comparison. If it merely delivers later, waiting and interruption belong in the time description. This separation lets a planner compare efficiency without charging twice for one physical effect.

But conservation is only the beginning. Even when an amount is conserved, useful energy can become dispersed; a system may remain energetically possible while losing its ability to perform a function. The next chapter examines the first and second laws of thermodynamics, then asks how a human body maintains function in a world where every activity consumes and transforms resources.

Energy, life and functional balance

A useful economy keeps functions alive through continuing flows. Energy conservation explains why these flows require sources; thermodynamics explains why maintenance cannot be made free by recycling an accounting entry. Homeostasis then supplies the central design question: how can a system continue to serve people while its conditions change?

6.1 The rule that prevents a free lunch

A pump lifts water because something supplies energy. Electricity reaches its motor through a network; that electricity was generated from an identified source. The pump cannot deliver indefinitely from an empty energy supply, however persuasive a contract or a price might be. This is the ordinary force of the first law of thermodynamics.

For a closed thermodynamic system under a familiar sign convention,

$$\Delta U = Q - W, \quad (6.1)$$

where U is internal energy, Q is heat entering and W is work done by the system. An open system also carries energy with matter crossing its boundary. The equation does not say that every useful form of energy remains available for reuse; it says that a claimed change requires a corresponding transfer or transformation. It is the scientific reason a perpetual-motion machine of the first kind cannot be built.[17]

The bookkeeping lesson is immediate. If a pump moves water while the account contains no energy source, either the physical description is incomplete or the pump has been granted impossible power. Recording the electricity as a flow does not decide whether pumping is wise. It prevents us from calling the action self-sustaining when it depends on something outside the picture.

6.2 Why conserved energy is not enough

The second law addresses a different question: which transformations are possible and what happens to the ability of energy to do useful work. For an isolated system, total entropy does not decrease. A real heat engine cannot

convert every unit of heat from one reservoir into work while leaving no other change. Friction and mixing can preserve energy while dispersing it in forms less useful for the task at hand.[17]

The Earth is not an isolated box. It receives energy from the Sun and radiates energy to space. A living organism can build and maintain local order by taking in energy and matter and exporting heat and waste. The second law does not forbid life; it explains why maintenance has a physical cost and why an account that shows only the organised result is incomplete.

At the clinic, a refrigerator preserves medicine by moving heat from its interior to the surrounding room while consuming electricity. Saying that the medicine remained cold without recording the power and the rejected heat would mistake a continuing service for a free property of the refrigerator. A durable economy likewise has to ask what flows sustain its useful arrangements.

6.3 The body does not maximise its temperature

Now consider a human body. Its temperature, blood pressure and concentrations of important substances vary while the body remains alive. Regulation does not mean setting every variable to zero, holding every amount perfectly fixed or making one quantity as large as possible. Too little and too much can both be dangerous. Physiology describes homeostasis as the maintenance of conditions compatible with function through interacting processes of sensing and response. Negative feedback is a central pattern: a departure triggers a response that tends to oppose it.[15, 18]

The point is clearer in a simple tank. Water enters and leaves, yet its level can remain within a useful region. If outflow increases, a sensor may detect a fall and a valve may admit more water. If the inlet fails, the level continues to fall despite the sensor. If the response is delayed, it may overshoot. The water already on its way matters too. A controller that sees only the current shortage can keep ordering after enough water has entered the pipe. When those orders arrive together, the tank overshoots. Representing pending delivery turns this apparent surprise into a calculable state. Later we will derive exactly how a pending stock creates memory, when a linear controller has decaying oscillations, and why that temporal behaviour need not create higher-order interaction between actions.

We sometimes speak as if growth or accumulation were an adequate objective for every system. The body shows why that cannot be a general physical rule. More heat is not always better; more glucose is not always better; more pressure is not always better. The same warning applies to a society that

treats increased throughput or income as sufficient evidence of continuing health. A gain has to be assessed in relation to the conditions that make the system function.

6.4 A balance that remains active

Suppose the clinic's water tank contains ten units at the beginning and end of an hour. During the hour it receives four and supplies four. Another tank also begins and ends at ten but has both valves closed. Their equal snapshots conceal different service histories. This is why functional balance is not simply equality of two numbers: it can be a maintained process.

The stock account for the first tank is

$$x_{t+1} = x_t + q_{\text{in}} - q_{\text{out}}, \quad (6.2)$$

with each q measured in the same physical unit over the same interval. A constant stock means $q_{\text{in}} = q_{\text{out}}$ for that interval; it does not tell us whether the water was safe or whether a patient received it. Those are additional facts. At a larger scale, regeneration, consumption and disturbance have to be represented if we want a claim about continuing function.

Three parts of a functioning system

Conservation closes the physical account. Valuation compares represented states. Dynamics describes how those states evolve. EBU connects all three while keeping their roles visible.

This separation supports a practical feature: a service can be maintained without inventing a reward for every fluctuation of its controller. Water supplied, energy consumed and the final change of potential can each be recorded in their proper place. A town can then recognise necessary maintenance even when it produces little net change in a stock, and avoid mistaking a large volume of corrective activity for an equally large improvement in public service.

6.5 Where should the system return?

Negative feedback responds to departure from a condition. That condition can differ between places and functions: two towns need not hold equal reserves, and a body is not healthy because every variable is equal.

The next chapter separates a *reference* used for comparison, an *equilibrium* of a stated law and *homeostasis* through continuing change. These distinctions explain restoration before we introduce its unit of account.

Equilibrium is not a maximum

A reference, a steady operating point and a stable response are different objects. This chapter makes those differences concrete before we define a unit of improvement. They explain why an economy should care about maintained function, and why a controller can be stable while keeping a system away from the minimum of its declared burden.

7.1 What is being kept in balance?

The body does not seek the highest possible temperature. The clinic does not become more secure merely by filling one tank without regard to the pressure in its pipes or the water left elsewhere. A living arrangement has several conditions of continued function. Some resources can be too scarce; some can be present in unusable places; some processes can run too quickly or consume the very reserve on which tomorrow depends.

This observation gives us a different kind of question from the one posed by a price or a sale. Instead of asking how large one return can become, we ask which physical conditions must be maintained and how a proposed action changes them. The answer cannot be a single universal target imposed on every place. A patient, a reservoir and an electrical network are not the same object. Their relevant variables and acceptable conditions must be declared for the case at hand.

Physiologists use homeostasis for the maintenance of conditions compatible with life through processes of regulation. The word does not mean motionless perfection. Breathing, circulation and temperature regulation continue while a person is well. In an economy, too, a stable stock might conceal a continuous flow of production, use, repair and replacement. A useful physical account must be able to distinguish a functioning flow from an identical-looking snapshot.

7.2 Reference, equilibrium and stability

To measure departure we first need a reference. Suppose two water stores, A and B, contain twenty physical units between them. A reference of ten in

each is

$$x^* = (10, 10). \quad (7.1)$$

The star marks the declared comparison state. A reference may encode useful reserves, safe ranges or a functional operating target. It does not by itself describe what the system will do.

An equilibrium belongs to a rule of change. If that rule is $\dot{x} = b(x)$, an equilibrium $\bar{x}$ satisfies $b(\bar{x}) = 0$. If the rule is discrete, $x_{t+1} = T(x_t)$, the condition is $T(\bar{x}) = \bar{x}$. A rule that never acts has every state as an equilibrium, including a state with an empty clinic tank. Stationarity alone cannot define adequate service.

Stability describes the response to a displacement. An asymptotically stable equilibrium attracts nearby states under the stated law. The approach can be monotone or can include decaying oscillations. A minimum of a potential has this dynamical significance only when the response law connects motion to that potential in a suitable way.

A continuously used tank gives a particularly useful example. Let X be available water, Q water pending arrival, $h > 0$ consumption, $\tau > 0$ a residence time and $\nu > 0$ a replenishment coefficient. A shortage-driven dispatch rule and first-order arrival law give

$$\dot{X} = Q/\tau - h, \quad \dot{Q} = \nu(L - X) - Q/\tau. \quad (7.2)$$

Its steady state is $Q^* = \tau h$ and $X^* = L - h/\nu$. Consumption requires continuing dispatch. The stock therefore remains below the controller's target L , even though both derivatives vanish. With positive coefficients this unconstrained affine system is asymptotically stable; Chapter 33 derives the roots and identifies the admissible physical branch.

A stable service can have a nonzero burden

A replenishment system can maintain a constant supply below its target because a continuing shortage signal is what sustains dispatch. Stability of this arrangement and reduction of its shortage burden are distinct mathematical statements.

For EBU, this distinction prevents a useful service from disappearing into a misleading zero: zero stock derivative does not mean zero throughput, zero resource use or a zero value of the potential. It also identifies a concrete design opportunity. Recording pending delivery gives a regulator information that the available-stock snapshot omits. Better coordination can reduce needless corrective deliveries and leave more physical capacity available for essential service.

7.3 A physically possible reference

Conservation places a useful limit on the discussion. The two stores together hold twenty units. Lossless transfers between them preserve that total. The reference $(10, 10)$ is compatible with the total, whereas $(12, 12)$ requires twenty-four units and cannot be reached by internal transfers alone. No better incentive or smarter calculation can supply those missing four units without an explicitly represented input.

Compatibility is not enough. A route may be absent or blocked. A source may lack enough available stock for a proposed transfer. The action may use electricity or lose water between the two measurements. We must therefore separate the question of where we would like a system to be from the question of what its physical network can actually do.

Now compare $(18, 2)$ with $(14, 6)$. Both contain twenty units. The latter is closer to $(10, 10)$ in an ordinary geometric sense. But what does closer mean when two resources have different units, or when a four-unit shortage at B matters differently from a four-unit surplus at A? We need a declared measure of deviation before we can assign a number to improvement. Chapter 11 will give each coordinate a reference and a scale; Chapter 12 will show why a Gaussian-shaped construction supplies a transparent, smooth measure.

7.4 Why the direction must depend on the state

The same transfer can help in one state and harm in another. Moving four units from A to B changes $(18, 2)$ into $(14, 6)$, bringing both stores toward the symmetric reference. Starting instead from $(10, 10)$, the identical movement produces $(6, 14)$, moving both away. An action name such as water delivery cannot carry a permanent positive sign independently of its starting state.

This is precisely why maximising every local transaction is not a satisfactory physical rule. A payment may reward the same delivery at both starting points; a field concerned with the represented balance must distinguish them. It must also account for the whole finite movement, because an action that initially helps can overshoot if carried too far. A local slope tells us a direction near the start, but the value of a completed action must include what happens as the state changes.

There is a caution here that strengthens rather than weakens the proposal. A mathematical measure of departure is not itself a law that opens valves or allocates water. It can inform incentives and expose consequences, but actions still require physical permission and an institution that decides what to do with the information. The account therefore supplies a useful comparison to a controller without becoming the controller itself. This allows EBU to describe

the cost of maintaining a service, to reveal oversupply, and to preserve the distinction between a successful regulator and a desirable operating condition.

7.5 The question that leads to a new account

Imagine that a community has two possible transfers. Both are feasible. One reduces an immediate shortage but consumes a scarce reserve; another produces less immediate delivery while leaving the stores better prepared for the next disturbance. A monetary invoice records payment, but it does not itself calculate the represented change in physical balance. An endpoint-only stock description can miss the energy carriers, losses and intermediate states needed to describe how either transfer happened.

We want an account whose sign and amount respond to what the physical action actually changes, under a stated reference and boundary. The account should not pretend that a desirable endpoint creates its own water, that a smooth mathematical minimum guarantees recovery, or that one stock coordinate describes every need. But it can make a crucial question answerable: did this accepted finite action reduce the declared burden of the represented system, and by how much?

What an Energy Balance Unit is

We can now define the account precisely. EBU measures the finite change of a declared physical potential. The definition is simple enough to calculate for one transfer, and strong enough to support the rest of the book: path accounting, complete transaction histories, joint action, feedback and equilibrium probability all connect back to this same finite comparison.

8.1 A physical consequence needs an account

The previous chapters have brought us to a practical question. The clinic cannot operate without payments, but a successful payment is not evidence that the water system remains capable of serving it. Conversely, water can be physically available while the people who need it cannot gain access. We do not have to abolish one description to improve the other. We need a second account that makes physical consequences visible at the point of action.

An Energy Balance Unit, or EBU, is the signed unit of that account. Given a declared physical potential V , an accepted finite action has the decrease of V between its complete before-state and after-state. If the represented state moves toward the declared reference, the EBU can be positive; if it moves away, it can be negative. Zero means that this particular potential did not change. It does not mean that no water moved, no energy was used, or no human work occurred.

This definition is deliberately connected to a physical event. A promise to deliver water is not the delivery. A payment for it is not the delivery. The account must identify the state before the action, the change that actually occurred, and the physical boundary within which that change is assessed. It is equally important to name what the account is measuring: a field built from specified carriers and functions, not a moral score for the person who acted.

8.2 The finite definition comes first

Let x^b be the complete represented state before an action and let $x^b + \delta$ be its complete successor under one fixed potential. Define

$$E = V(x^b) - V(x^b + \delta). \quad (8.1)$$

The minus sign makes a decrease in represented burden positive. No gradient approximation, motion law or actor payment is needed to define this finite comparison. The field, physical boundary and complete endpoints must be declared before the number has meaning.

8.3 Following the change along a path

Recall the bent ray from Chapter 5. Fermat's optical travel-time integral can depend on the spatial route. A fixed-potential EBU integral has a different structure: it integrates the differential of a supplied state function. Its role is to explain how changing local marginals add to the finite endpoint value already defined.

Choose an admissible piecewise C^1 path γ from x^b to $x^b + \delta$, and suppose V is C^1 on a neighbourhood of that path. The chain rule and fundamental theorem of calculus give

$$E = - \int_0^1 \nabla V(\gamma(s))^\top \gamma'(s) ds. \quad (8.2)$$

Chapter 14 proves this identity carefully. A straight evaluation path need not be an observed physical trajectory; that interpretation requires a physical response model. The total is path independent under these assumptions because it is the difference of one single-valued potential.

One definition, a conditional path representation

Finite EBU is the before-minus-after potential difference. Under the regularity assumptions, a changing-field integral gives the same number. Neither expression grants physical permission, chooses a motion law or assigns ownership.

8.4 Where the rest of the physical process belongs

A water transfer may require pumping, filtration or another physical transformation. Those consequences do not become invisible merely because our

first example follows water alone. If the pump consumes electricity, a fuller physical model records the energy carrier and its transformation. If water is lost, the lost amount enters a declared sink coordinate or crosses an explicit boundary. A potential defined on that fuller state can then evaluate all the consequences it is meant to value.

This is different from subtracting an unexplained “process cost” from the EBU equation. The general finite definition in this book remains

$$E = V(x^b) - V(x^b + \delta). \quad (8.3)$$

An engineering or institutional account may keep additional, separately named quantities. Such a quantity is not physical EBU by a change of notation, and the same effect must not be counted twice. Our first water transfer is an intentionally lossless teaching world. Its simplicity lets us learn the equation; it does not assert that operating a real pump is free.

8.5 A four-unit transfer

A holds 18X, B holds 2X, and the declared reference is (10,10). Here X denotes one unit of the represented physical stock. Each location has scale 2X. The potential we will construct in Chapter 12 is

$$V(x) = \frac{1}{2} \left(\frac{x_1 - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{x_2 - 10}{2} \right)^2. \quad (8.4)$$

Its value at (18, 2) is 16. If four units are transferred from A to B, the state becomes (14, 6) and the potential becomes 4. The exact EBU is therefore 12EBU. Four physical units arrived at B; twelve EBU express the decrease of this particular potential. They are different units describing different properties of the same event.

If eight units are transferred, the endpoint is the reference and the EBU is 16EBU. Transferring sixteen units instead reaches (2, 18), with potential 16 once again, so its EBU is zero. The last movement was not empty: it simply overshot far enough to cancel the initial gain in this account. At seventeen units the final potential is 20.25 and the EBU is -4.25 EBU. These calculations show why a fixed reward for each delivered stock unit cannot substitute for valuation along the full finite action.

8.6 What would an actor receive EBU for?

Imagine two feasible actions from the same measured state. One restores a depleted supply without pushing another represented location into trouble. The other earns the same sales revenue while worsening the declared



Total stock: 20 before and after

Potential: 16 before, 4 after; field value: +12

Figure 8.1: A four-unit lossless transfer preserves twenty physical units while the declared potential falls from sixteen to four.

balance. A conventional invoice can look alike in both cases. The EBU account distinguishes their represented physical consequences. That distinction is the proposed incentive: useful restoration becomes visible and potentially rewardable, while depletion cannot be made to look like restoration merely by earning more money.

There is an important institutional step between a positive EBU calculation and a spendable personal balance. An institution would have to specify who may claim the action, how multiple contributors are treated, when a measured outcome becomes a receipt, and which future actions such a receipt could enable. A capacity design might allow demonstrated restorative value to support later physically permissible activity. The finite definition supplies a verifiable total for such a design. Attribution and the rules for using a receipt supply the remaining institutional structure. Later chapters keep a group result, an individual attribution and an institutional rule distinct.

The incentive also cannot be reduced to making every EBU number as large as possible. A well-functioning system may need continuing service and repair even when the represented state is already near its reference. An action may be essential despite a negative result in one field, and access to a clinic must not be made contingent on a person's ability to manufacture credits. The physical account can show a cost or benefit honestly; institutions must decide how essential provision, rights and obligations are protected. The resulting design can recognise maintenance, protect essential access and account honestly for resource use at the same time. These functions are compatible because the physical valuation is not made to carry every institutional decision.

8.7 Permission and value are different questions

Before a transfer is accepted, the source must have enough usable stock, the route must exist and the declared physical constraints must hold. A positive hypothetical quote for an impossible transfer creates no water. A negative quote for an urgent, permitted service does not by itself forbid the service. Permission answers whether the physical change is allowed; EBU answers how the declared potential changes.

Likewise, neither the source's loss nor the destination's gain alone determines the group result when several actions occur together. The potential is evaluated for the actual combined state change. Its exact group total can be known without a justified division into personal claims. We will keep that distinction as the mathematics develops.

Several useful properties follow from this finite definition. Successive receipts under one field telescope to the net endpoint change. Joint actions can be compared against all their subgroups to identify exactly what coordination contributes. A response model can supply the endpoint at a stated horizon, including pending flows and feedback. These are complementary capabilities of one account: reliable history, visible joint dependence and a clear connection to physical dynamics. We have named the unit and seen its consequence in one small world. The next chapter draws the world as cells and routes, so that a sentence such as four units reached B becomes an action we can test for physical possibility before calculating its value.

Cells, nodes, and edges

A physical account begins with a map of what can change. We build that map from cells, routes and quantities, then distinguish it from two other maps: the dependencies of the potential and the interactions of actions. This distinction allows EBU to compute locally while retaining the shared resources and pending processes that make coordination necessary.

9.1 Why the map comes before the actor

A model of economic activity can begin by giving an actor a goal. The actor seeks a resource, maximizes a reward, or chooses a trading partner. Physical accounting must first establish a world in which these actions have a definite meaning. Where is the resource stored? Which transformations are available? Which places can exchange it? Otherwise a clever decision rule can deliver stock that was never withdrawn or use a route that the represented world does not contain.

The simplest useful map is a directed graph. A graph is a collection of places together with connections between them. A directed connection has an orientation. Here a place is called a cell or node, and a directed connection is called an edge. The graph describes available physical transformations; it does not prescribe which one an actor selects.

A cell need not be a square on a geographical grid. It may be one well-mixed tank, a warehouse, or a region for which a particular stock can be measured. Good boundaries keep important exchanges visible. Combining a reservoir and treatment plant into one cell may conceal a transformation between them. Dividing one well-mixed tank into a thousand cells may add notation without adding physical information.

9.2 Vocabulary before notation

Word	Meaning in the present model
Cell or node	One represented place carrying a scalar stock.
Index	A stable label used to identify a cell or an edge.
Edge	One declared directed transfer possibility.
Source	The cell from which an action withdraws stock.
Destination	The cell that receives stock.
Path	An ordered sequence of adjoining edges.
Topology	The pattern of available connections.
Support	The coordinates changed by a specified action.

An index is a label, not a size or a ranking. Writing $i = 1$ for A and $i = 2$ for B does not declare A more important. It fixes the order in which their numbers appear. That order must remain stable throughout a calculation and its record.

We write x_i for the stock at cell i . Thus $x_1 = 18$ and $x_2 = 2$ are two scalar statements. A scalar is one number with its declared unit. A vector collects several numbers in a fixed order.

9.3 A vector is an ordered physical record

The complete state of N cells is written

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \in \mathbb{R}^N. \quad (9.1)$$

The notation $\mathbb{R}^N$ means the set of ordered lists of N real numbers. Our physical domain is narrower: every stock is nonnegative, and their sum is the fixed total M . Writing the larger mathematical space is convenient for derivatives; it does not permit negative water in an executable state.

For A and B the state is $x = (18, 2)^T$. The superscript T means transpose. It turns a row into a column or a column into a row. It is neither a power nor a time label. A row multiplied by a matching column gives the dot product:

$$a^T b = \sum_{i=1}^N a_i b_i. \quad (9.2)$$

For $a = (2, 3)^T$ and $b = (5, 7)^T$, the result is $2 \cdot 5 + 3 \cdot 7 = 31$. Later, a marginal vector multiplied by a state change will add the local contributions

to a change of potential. The operation remains this ordinary multiply-and-add procedure.

9.4 An edge is a directed possibility

Write an edge as $e = (i \rightarrow j)$. One unit sent on this lossless edge removes one stock unit from source i and adds one to destination j . The existence of $i \rightarrow j$ does not imply that $j \rightarrow i$ exists. A pipe may operate only in one direction. A permitted road route may differ from its reverse. If both directions are available, represent them as separate edges.

The graph also does not say how much will move. A road's existence does not fill a truck. The action contains a finite nonnegative quantity q_e , expressed in stock units. Direction is carried by the edge; we do not silently use a negative quantity to create a reverse process that the graph has not declared.

Nor is an edge a universal relation of distance. Two distant tanks may have a direct pipe, while neighbouring tanks may have no connection. Distance can matter to delay, equipment, or an extended physical model. It does not enter this field through an assumed inverse-square law. The graph and the potential have explicitly defined jobs.

9.5 One column for one transfer

For edge $e = (i \rightarrow j)$, define the column S_e by

$$(S_e)_k = \begin{cases} -1, & k = i, \\ +1, & k = j, \\ 0, & \text{otherwise.} \end{cases} \quad (9.3)$$

Read the subscript outside the parentheses as “entry k of the column.” It does not multiply two symbols. The minus sign marks withdrawal, the plus sign marks delivery, and zero marks an unchanged coordinate.

The action's state change is

$$\delta_e = S_e q_e. \quad (9.4)$$

For A to B, $S_e = (-1, 1)^T$. Sending four units gives $\delta_e = (-4, 4)^T$, so the successor is $(18, 2)^T + (-4, 4)^T = (14, 6)^T$. This is the calculation already performed in words. The column has made its signs reusable.

Read the column aloud

One lossless edge column says: withdraw one unit here, deliver one unit there, change nothing else directly. Multiplying by a finite quantity gives the physical increment. The column records where the change occurs; it does not choose the quantity or value the action.

9.6 From columns to a matrix

Place all edge columns side by side to form the incidence matrix,

$$S = [S_1 \ S_2 \ \cdots \ S_m]. \quad (9.5)$$

There is one row per cell and one column per edge. The letter m here counts graph edges; the symbol E is reserved elsewhere for an EBU value. If $q = (q_1, \dots, q_m)^T$ collects finite quantities, then Sq is the sum of their state increments.

Consider a separately declared three-cell illustration: A is cell 1, a store is cell 2, and the clinic tank is cell 3. Edges run $1 \rightarrow 2$ and $2 \rightarrow 3$. Both are lossless. Then

$$S = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}. \quad (9.6)$$

For quantities $q = (4, 2)^T$, multiplication gives

$$Sq = \begin{bmatrix} -4 \\ 4 - 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 2 \end{bmatrix}. \quad (9.7)$$

The store receives four and sends two, for a net gain of two. The net entries add to zero. This matrix product is physical bookkeeping, not a proof that the proposed actions are jointly feasible. If the store begins empty and the event rule forbids use of simultaneous incoming stock, its outgoing proposal still fails. Event timing is not determined by matrix notation.

9.7 A shared source has one stock

Now let A supply three destinations: the clinic, a food store, and a fire tank. In this new four-cell example,

$$S = \begin{bmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (9.8)$$

With quantities $(2, 1, 0.5)^T$, the state increment is $(-3.5, 2, 1, 0.5)^T$. Every source entry appears in the same row. The total withdrawal is 3.5, not three independent uses of the same stock.

This geometry matters twice. Physical feasibility must consider aggregate use of the source. Valuation must consider the combined change of any shared potential coordinate. Independently asking whether each action fits can miss a joint conflict. Independently valuing each action from the full initial state can also count the same improvement more than once. Those are distinct problems, and later chapters address them separately.

9.8 A path is not a single edge

A route from A through a depot to B is a sequence of edges. Its intermediate stock may matter. Water inside a pipe is neither still available at the source nor already available to the clinic. Represent it as a pending coordinate Q whenever its future arrival affects the response. Dispatch then moves stock from source to Q ; arrival moves it from Q to the usable destination. This makes the physical history intelligible without allowing both endpoints to claim the same water. So may timing, physical availability, and the order in which each action becomes possible. Combining the route into one arrow can be appropriate only if the model explicitly justifies the coarser transformation and retains every relevant boundary.

Suppose the depot begins with two units. A simultaneous event requests four units into the depot and three out. The endpoint has three units, so it is nonnegative. But the initial depot does not contain the three outgoing units. A model may allow synchronized passage through a real continuous pipe; another may require receipt before onward dispatch. Both can be specified. Neither follows from the positive endpoint alone.

The same caution applies more strongly to fuel, electricity, and delivered water. They are different carriers, and an energy conversion is not a lossless relocation of a homogeneous scalar. Their actions should form a linked physical chain with appropriate units and conversion accounts. The simple graph notation can teach how to record support, but it cannot erase the engineering of the chain.

9.9 Three maps that must stay distinct

The physical graph tells us which transformations exist. The potential's dependency structure tells us which coordinates each potential term reads. The action-interaction structure tells us which actions overlap in a way relevant to joint valuation or feasibility. These maps need not coincide. A fourth

distinction will become useful when motion is introduced: a response law can transmit a change through time, creating endpoint dependence beyond the coordinates touched directly at dispatch. Its state must include the variables needed to determine that response. If pending stock is omitted, the remaining available-stock coordinate can acquire memory; the underlying physical graph has not thereby become a new potential.

In our separable potential, each local term reads one cell, so the potential has no cross-coordinate terms. Nevertheless two actions withdrawing from A both change A's local quadratic. They interact in the joint finite value. A diagonal potential therefore does not imply independent actions.

Conversely, two geographically close places may have no common action support. Their values can be independent within this model if no potential factor couples them. A richer model could add such a factor, but then the affected neighbourhood used in a local quote must include it. The general principle is simple: locate dependence from the actual equations, not from a picture or a metaphor.

9.10 Building a graph from a practical description

Begin with the carrier and its unit. Next choose cells whose stocks can be measured consistently. Name each transfer process and its direction. Assign stable row and column indices. For each lossless edge put -1 at its source, $+1$ at its destination, and zeros elsewhere. Keep a separate record of physical quantity limits, timing, and permitted information.

Do not place EBU values or reference stocks inside S . The incidence matrix does not depend on whether A currently has excess or deficit relative to its reference. It describes the change per unit of the named transfer. Stocks belong in x , references in x^* , scales in σ , and finite quantities in q . Separating these objects makes later changes auditable.

An edge record might state its identifier, endpoints, carrier, quantity unit, and physical validity conditions. A version identifier can freeze that declaration for an event. This helps prevent a route or parameter from changing halfway through an account. This stable declaration makes local records composable. A clinic and a depot can describe their part of a delivery using the same carrier and event identifiers, allowing shared-source use to be reconciled once. The efficiency gain is concrete: a consistent state map removes duplicate availability claims and directs coordination to the resources actually shared.

9.11 Notation laboratory

For a four-cell transfer from cell 2 to cell 4, the column is $S_e = (0, -1, 0, 1)^T$. Its second entry is -1 , its fourth is $+1$, and its dot product with the all-ones vector is zero. A quantity $q_e = 3$ gives $(0, -3, 0, 3)^T$.

If there are N cells and m edges, the matrix has shape $N \times m$. The quantity vector has shape $m \times 1$, so the result has shape $N \times 1$. The inner dimensions match. This simple size check catches many transcription mistakes before any arithmetic is attempted.

For any cell i , the component formula is

$$(Sq)_i = \sum_{e: \text{dest}(e)=i} q_e - \sum_{e: \text{src}(e)=i} q_e. \quad (9.9)$$

It says to add all incoming quantities and subtract all outgoing quantities. A reader who finds the full matrix abstract can always return to this one-cell account. Matrix form is especially useful for proofs; component form is especially useful for tracing a real event.

9.12 Guided problem: build a returning route

A depot sends resource to a clinic and to a water plant. The clinic can return unused resource to the depot. In this illustration all transfers involve the same homogeneous carrier and are lossless. Index depot 1, clinic 2, plant 3, and edges $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 1$. The matrix is

$$S = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}. \quad (9.10)$$

For $q = (2, 1, 0.5)^T$, the change is $(-2.5, 1.5, 1)^T$. Its entries add to zero. The reverse edge is its own event component. It is not a negative number concealed in the first quantity.

Now suppose the starting state is $(5, 1, 0)^T$. The endpoint is $(2.5, 2.5, 1)^T$, with total six units before and after. The depot's outgoing total is three, and the clinic's outgoing quantity is one-half, each available from its initial stock. Under a source-availability rule excluding unreceived inflow, those particular source checks pass. Other physical constraints would still need their own declared checks.

9.13 Disconnected regions and unreachable references

The graph can impose conservation statements stronger than the global total. Suppose there are two disconnected regions and every edge begins and ends

inside one region. Summing the incidence entries over either region gives zero for every edge. Each region therefore retains its own stock under the declared internal actions.

For a concrete four-cell illustration, let cells 1 and 2 form one connected pair and cells 3 and 4 another, with no edge between the pairs. Starting stocks are $(8, 2, 3, 7)$. The two regional totals are both ten. A reference $(5, 5, 5, 5)$ is compatible with these regional totals. A reference $(6, 6, 4, 4)$ has the same global total 20 but demands twelve units in the first region and eight in the second. It cannot be reached using only the declared internal edges.

This is an exact obstruction supplied by topology. No improvement in quote arithmetic removes it. A new cross-region route would change the physical graph and could remove the obstruction, but its construction is another physical and institutional question. The field should expose the remaining deviation without pretending that the actor has an unavailable action.

More generally, a cut is a partition of the cells into two sets. The only internal transfers that can change the total on one side are edges crossing that cut. Writing the sum of the component ledger on that side cancels every wholly internal edge. This simple cancellation is useful when auditing a region, even when its individual cells are numerous.

9.14 Renaming cells should not change the physics

If A and B exchange their numerical indices in a data file, every associated object must be reordered together: state, references, scales, and incidence rows. The physical result should be unchanged. A consistent permutation changes notation, not the model.

For example, the main state $(18, 2)$ and column $(-1, 1)$ become $(2, 18)$ and $(1, -1)$ when the two row labels are exchanged. The same four-unit transfer produces the reordered endpoint $(6, 14)$. This is the earlier endpoint $(14, 6)$ written in the new order. It must not be confused with physically reversing the action while keeping the original labels.

This distinction is elementary but important in large records. A matrix with correct numbers and the wrong identifier-to-row map can conserve total stock while moving it between the wrong places. Physical identity belongs in the record alongside arithmetic consistency. Stable identifiers make an account reproducible across software ordering choices.

9.15 Common questions and practice

Does S calculate the best route? No. It records effects of declared edge quantities. A route selector is another object.

Are all entries of a physical state allowed real numbers? The state is represented by real coordinates, but admissibility restricts them to the declared nonnegative fixed-mass domain and any further physical conditions.

Does a zero entry mean a place cannot be affected indirectly? It means this action column does not directly change that coordinate. Later actions or a richer potential can create other dependencies, which must be represented separately.

For practice, draw A connected to two destinations and add a return edge from one destination. Write all columns and compute their sum for quantities $(3, 2, 1)$. Your source increment should be -4 , the returning destination's net increment $+2$, and the other destination's increment $+2$. Explain how the result depends on the declared ordering of coordinates but not on the order in which the three columns are added.

Next remove the return edge. A proposal to return one unit is now unavailable even if it would improve the potential. This illustrates a central boundary: an algebraically attractive direction is not automatically an edge in the physical world.

What to carry forward

A cell is a state location. An edge is a directed physical possibility. Its incidence column records source withdrawal and destination delivery. The product Sq adds finite increments. Feasibility, timing, value, and action selection require their own declarations.

With the map established, we can audit quantities, units, and boundaries. The next chapter follows one physical change without letting a mathematical shorthand create or destroy stock.

Quantities, boundaries, and one physical change

Every useful valuation rests on a complete physical change. Here we follow quantities across a boundary, include losses and pending stocks, and distinguish gross activity from net change. The resulting conservation identities make it possible to combine many local records without creating resources in the act of accounting.

10.1 The ledger underneath every valuation

Before asking whether an action improves the represented world, ask whether its physical account closes. A transfer cannot deliver more of the homogeneous resource than the declared source withdraws in a lossless model. A missing quantity cannot disappear because a potential decreases. An extra quantity cannot enter because an account has enough capacity to pay for it.

The main model has a finite number of nonnegative stocks and a fixed total M . It represents internal relocation of one homogeneous scalar. The physical state space is

$$\mathcal{X}_M = \left\{ x \in \mathbb{R}^N : x_i \geq 0, \quad \sum_i x_i = M \right\}. \quad (10.1)$$

This set is called a simplex. The unfamiliar name adds no new assumption: every coordinate is nonnegative and all coordinates share one total. In two cells with total 20, specifying A's stock determines B's stock as $20 - x_1$. The two numbers cannot vary independently while the total remains fixed.

10.2 Quantity is not rate

We use q for a finite source-side quantity. It has the same stock unit as x_i . If a flow rate j is constant over a duration Δt , the quantity is

$$q = \Delta t j. \quad (10.2)$$

The rate has units X/time; multiplying by duration gives X. Once the finite quantity has been formed, the action increment is $S_e q$. Multiplying it by Δt again would count time twice.

For example, a rate of eight units per hour acting for half an hour moves four units. The same quantity is moved by a rate of two units per hour for two hours, if the declared process is constant and no other event intervenes. The endpoint stock change is the same in this ideal example. Other physical consequences, such as equipment heating or delay, may differ in a richer model and would need representation there.

If the rate varies, quantity is the integrated rate, $q = \int j(t) dt$. One cannot replace this integral by initial rate times duration without an assumption that makes that replacement valid. The finite account begins once the event's quantity has been established; it need not pretend every physical flow is constant.

Object	Unit	Example
Stock x_i	X	18 units in A
Rate j_e	X/hour	8 units per hour
Duration Δt	hour	one-half hour
Quantity q_e	X	4 units sent
Increment $S_e q_e$	X per coordinate	$(-4, 4)^T$

10.3 The exact internal-transfer identity

Let $\mathbf{1}$ be the column vector whose entries are all one. Its transpose sums a state: $\mathbf{1}^T x = \sum_i x_i$. Every lossless incidence column has one -1 and one $+1$, so

$$\mathbf{1}^T S_e = 0. \tag{10.3}$$

For a group of additive transfer increments $\delta_G = \sum_{a \in G} S_a q_a$, linearity gives $\mathbf{1}^T \delta_G = 0$. Hence

$$\mathbf{1}^T (x^b + \delta_G) = \mathbf{1}^T x^b = M. \tag{10.4}$$

This is an exact algebraic conservation identity under the declared transfer model. It does not require that the action lowers potential or that an actor acts wisely. A damaging relocation conserves the same total as a helpful one.

The identity also does not by itself ensure nonnegative stocks. A proposal to send 25 units from A's initial 18 has changes $(-25, 25)$, whose sum is zero, but its endpoint is $(-7, 27)$. Conservation and feasibility are separate conditions. The negative coordinate rejects the proposal under our physical domain.

10.4 Worked ledger for A and B

The starting state is $x^b = (18, 2)^T$. For $q = 4$, the source loses four and the destination gains four. The physical residual is

$$r_M = (14 + 6) - (18 + 2) = 0. \quad (10.5)$$

At $q = 8$, the state is $(10, 10)^T$, with the same zero residual. At $q = 17$, it is $(1, 19)^T$, again with zero residual. Yet their finite field values are different: 12, 16, and -4.25 , respectively, under the main potential.

This comparison is worth reading slowly. The physical total cannot determine the distributional potential. The potential cannot replace the stock ledger. They answer complementary questions about the same action. A programme that reports only one of them has left the other question unanswered.

10.5 The event boundary determines what is credited

The symbol x^b denotes the shared pre-action baseline. In an undisturbed illustrative event, it is simply the current state. In a later forcing-first model, an external event changes an earlier state x to x^b ; the actor then acts from x^b . Both sides of the actor's finite comparison use that same baseline.

Suppose an external conservative relocation moves the system from $(10, 10)$ to $(18, 2)$. Its total stock is still 20, but it has increased potential from zero to 16. An actor then transfers four back toward the reference, producing $(14, 6)$ and potential 4. The actor's field reduction is 12. The external increase of 16 belongs in the external account, not as an actor debit or credit created by relabelling time.

The three states expose the role of the baseline: each comparison has a definite physical owner and interval. If an account instead compared the final potential 4 directly with the earlier reference potential zero and called the result the actor's effect, it would combine the external event and the action. The resulting -4 would answer another question.

10.6 No hidden creation by clipping

Suppose a proposed update produces $(-1, 21)$. Replacing -1 by zero may seem like harmless housekeeping. It changes total stock from 20 to 21. If the positive coordinate is also adjusted, a new redistribution has occurred. Either way, the operation is a physical or mathematical correction requiring an explicit account.

For the main model, infeasible actions are excluded before execution. We do not repair their successors silently. A broader model may include spill,

saturation, external replenishment, or emergency corrections, but those processes must carry named increments or boundary entries. A state projection can be a useful numerical procedure; it is not automatically a law of physical conservation.

This principle is more general than water. An inventory programme that sets a negative medicine count to zero has not produced medicine, but it has changed its reported record. An account should expose the disagreement so that its cause can be investigated. Concealing it creates an apparently consistent state while destroying the information needed to explain the event.

10.7 What changes when a route loses usable stock

Consider a transfer with efficiency $\eta = 0.8$. A source sends five units, a destination receives four, and one enters a declared loss compartment W . The extended increment is

$$(\Delta x_i, \Delta x_j, \Delta W) = (-5, 4, 1). \quad (10.6)$$

The usable-stock total falls by one; the complete carrier account remains conserved. An explicitly open account can instead record the unit crossing its boundary. In either description, the amount has a destination in the accounting.

A route with pending stock makes the same discipline more informative. In the coordinate order (S, Q, X, W) , dispatch and arrival have directions

$$s_D = (-1, 1, 0, 0)^T, \quad s_A = (0, -1, \eta, 1 - \eta)^T. \quad (10.7)$$

Both conserve the complete carrier. Dispatch removes available source stock immediately. Arrival releases pending stock later, splitting it between useful delivery and loss. The route is thus a sequence of complete changes, not two descriptions of the same instantaneous event.

The potential may value all four coordinates or an explicitly reduced set. If $\mu = \nabla V$, the arrival's marginal improvement is

$$f_A = -\mu^T s_A = \mu_Q - \eta\mu_X - (1 - \eta)\mu_W. \quad (10.8)$$

A valued sink contributes its own term. The shorter expression $\mu_Q - \eta\mu_X$ is valid when the potential omits W or its contribution vanishes. This is the same chain rule as for a lossless edge, applied to a complete direction.

Completeness makes losses visible once

A complete carrier ledger identifies where the lost amount went. A potential over that ledger determines how the change is valued. Keeping the two declarations together prevents both an omitted consequence and a second charge for a consequence already included.

The numbers $(-5, 3, 2)$ also sum to zero, but violate the declared efficiency. Conservation of a total and correctness of a transformation are separate invariants. Their combination supports a useful EBU feature: a route record can explain both the amount consumed and the function delivered. Communities can compare improvements in delivery efficiency using a consistent boundary, with less room for shifting a burden into an unnamed compartment.

10.8 A homogeneous carrier is a strong simplification

A conserved scalar is not a claim that all useful things can be added in one physical unit. One litre of water and one kilowatt-hour of electricity cannot become two units of a single conserved stock merely because both support a clinic. Their linked use requires separate carriers and a transformation relation with consistent dimensions.

The same applies to physical degradation. Energy can remain conserved while its usefulness changes. A stock of usable doses may fall when medicines expire although their material constituents remain. The carrier named by a coordinate must therefore be explicit: usable resource under a declared quality definition, elemental mass, energy, or another measured quantity. Its conservation statement applies to that declaration and boundary.

Our ideal transfers preserve the homogeneous scalar exactly. They are a clean setting for learning local finite accounting. Extending them to heterogeneous resource chains is scientific work, not a change of labels. It must preserve what is conserved, represent relevant changes of quality, and avoid an unlicensed conversion between different EBU accounts.

10.9 Net change and gross movement

Two opposite transfers can leave stocks unchanged while moving a substantial gross quantity. Suppose the two lossless directions between A and B are both declared, and an event sends three units each way. The increments $(-3, 3)$ and $(3, -3)$ add to zero. The endpoint is the initial state.

The stock ledger correctly reports zero net change. A separate action record correctly reports six units of gross source-side movement, three on each route.

Neither description should erase the other. If the complete physical model includes process energy or equipment wear, those can change even when the represented stock increment cancels. The ideal cost-free scalar example records zero field value for the net event because its endpoints coincide; it does not make a claim about an actual pump's energy use.

This distinction prevents another kind of hidden correction. An auditor should not delete both action lines merely because their increments cancel. The lines identify what was attempted or executed, while the endpoint describes the resulting stock distribution. Physical feasibility may depend on gross use of a route or shared machine, so it too can differ between an empty event and a nonempty cancelling group.

10.10 Conservation under a change of accounting boundary

Imagine observing only B. A four-unit delivery raises its stock by four. Within this smaller boundary the delivery is an inflow. Now enlarge the boundary to include A. The same event is an internal transfer with zero total change. The physical event did not change; the boundary did.

Write this explicitly before comparing two ledgers. A local total can rise while the global total is conserved, because the local boundary has an incoming edge. A statement such as "stock increased" is incomplete until its region and carrier are named. EBU's local potential difference can still be calculated, but a source contribution must not disappear just because the reader looks only at the destination's tank.

For any region, add the stock changes of its cells. Every transfer with both endpoints inside contributes minus and plus the same quantity, cancelling internally. Edges entering the region contribute positively; edges leaving contribute negatively. This cancellation is the regional version of the all-ones-vector identity. It explains why enlarging a boundary changes which terms are called external without changing the complete physical account.

In a real study, changing the boundary can also change which consequences are represented and hence which potential terms are needed. It must therefore be recorded as a model revision, not as a way to make an inconvenient burden vanish from a report. A narrower account can be useful if its scope is stated honestly.

10.11 Measurement is not a new flow

Real gauges have finite resolution, calibration limitations, and timing uncertainty. Removing numerical error-controller material from the introductory

model does not make these practical issues disappear. They belong in the interpretation of observed records and in later domain validation.

If a gauge reports 18 units, the underlying quantity may not be known with infinite precision. The algebraic example treats the input as exact so that its mathematical implications are visible. An empirical application must state how the input was obtained and what uncertainty accompanies it. It must not insert measurement uncertainty into the stock equation as though uncertainty were additional water.

A measured residual can reflect an omitted flow, incompatible timestamps, sensor bias, or computational rounding. A declaration of acceptable measurement agreement must precede judging a real record. Choosing a convenient allowance after observing an inconvenient mismatch would not establish conservation. The equations here specify the physical target against which such a measurement protocol would be assessed.

10.12 Guided problem: reconstruct a branching event

Consider a new three-cell state $(9, 3, 0)^T$, with lossless edges from cell 1 to cells 2 and 3. The finite quantities are two and four. The increments are $(-2, 2, 0)^T$ and $(-4, 0, 4)^T$. Their sum is $(-6, 2, 4)^T$, and the endpoint is $(3, 5, 4)^T$.

Check the ledger three ways. Cell by cell, source withdrawal is six and total delivery is six. By totals, the starting 12 equals the ending 12. By matrix structure, each column sums to zero, so every linear combination does too. The agreement is not three independent empirical tests; it is three views of one declared algebraic identity.

Now alter only the second quantity to eight. Aggregate withdrawal becomes ten, exceeding the source's nine. The formal endpoint is $(-1, 5, 8)^T$. The total remains 12, but the state is infeasible. Do not clip its first entry or report the eight-unit delivery as executed service. The example demonstrates why a joint source check belongs before an accepted event.

Finally convert rates correctly. If the valid quantities two and four were delivered over one-quarter hour at constant rates, the rates were eight and sixteen units per hour. If those same rates acted for half an hour, the quantities would double. A duration change can therefore alter physical feasibility even when the rates appear unchanged.

10.13 Practice with answers

A lossless edge carries a constant rate of three units per hour for two-thirds of an hour. The finite quantity is two units. Starting from $(7, 1)$, the endpoint

is $(5, 3)$, and total stock remains eight. Explain why writing $x' = x + \Delta t S q$ with this already integrated $q = 2$ would be wrong.

In a separate extended loss ledger, a source sends ten units at $\eta = 0.9$. Destination delivery is nine and the loss-compartment increment is one. A record showing destination eight and loss two has zero total residual but violates the stated efficiency. Identify the missing component check.

For the main model, a no-action event has $q = 0$, so $\delta = 0$ and state is unchanged. If a real observer nevertheless sees a change, at least one premise of the no-action, no-external-change illustration does not describe the observation. The response is to locate the process or measurement issue, not to declare that zero caused a nonzero transition.

A complete quantity ledger now tells us what changed and what remained conserved. It still cannot say whether that change improved the declared condition. The next chapter supplies the reference and scale that make such a comparison meaningful.

Declaring a reference and a scale

A useful reference says what physical condition the account compares; a scale says how strongly a departure counts. These declarations make unlike places comparable without pretending their needs are identical. We also distinguish the dimensionless EBU normalization from a native burden unit used in feedback equations, so that a controller's speed cannot silently redefine the accounting unit.

11.1 A comparison needs a declared centre

A tank reading of ten units is a physical fact within a measurement protocol. Whether it represents shortage, adequate provision, or excess cannot be read from the number alone. The same quantity can support a small clinic and be inadequate for a large hospital. A model that measures deviation must therefore declare what each stock is being compared with.

Write the reference for cell i as x_i^* . The star identifies a reference value; it is not multiplication and does not mean that the value has been proved optimal. The vector x^* collects all references in the same cell order as the state. In our main example it is $(10, 10)^T$.

The reference is a model declaration whose practical meaning requires domain justification. It might be informed by engineering function, an agreed service profile, or a research design. Algebra cannot decide which interpretation is appropriate. The important first step is to record that interpretation explicitly and keep it fixed during a comparison.

11.2 Reference, observation, and equilibrium

Three ideas are easily conflated. An observed mean summarizes data. A reference states the centre used by a model. A dynamic equilibrium is a state that remains unchanged under a particular evolution rule. These objects may coincide in a carefully justified application, but their symbols do not make them identical.

If a community's water distribution is persistently unequal, its historical mean distribution may reflect that inequality. Choosing a reference equal to that

mean is a substantive decision. Conversely, choosing a more balanced reference does not prove that available routes or institutions can reach it. A reference makes the question precise; it does not answer every practical question associated with it.

For the Gaussian potential taught here, x^* is the unique zero of the full quadratic when all scales are positive. That mathematical minimum is a statement about the chosen function. It does not prove that an actor process converges to the point, that a society should adopt it, or that all living functions are satisfied there.

Three sentences with different meanings

“The mean stock was ten” describes a dataset. “The reference is ten” defines a comparison. “Ten is an equilibrium” describes a specified dynamic law. Use the sentence supported by the available definition or evidence.

11.3 Compatibility with a conserved total

The first model preserves total stock M . We therefore choose references satisfying

$$x_i^* \geq 0, \quad \sum_i x_i^* = M. \quad (11.1)$$

This ensures that the reference lies in the same nonnegative fixed-mass domain as the physical states. In the main example, $10 + 10 = 20$. A reference of $(12, 12)$ would require 24 units, which a closed 20-unit world cannot supply.

The compatibility condition is simple but scientifically important. Without it, a positive minimum deviation may reflect an impossible reference rather than poor action choices. A model might intentionally study an incompatible target in another project, but it would need to explain that problem. We do not introduce it silently here.

Compatibility is weaker than reachability. If the only physical edge runs from B to A, then starting from $(18, 2)$ the direct transfer toward $(10, 10)$ is unavailable. If the graph is disconnected, each component preserves its own total under internal transfers. The reference must then also be compatible with those component totals to be reachable by those actions. Declaring a global total alone does not construct a route.

11.4 The scale tells us how to compare deviations

Let $\sigma_i > 0$ be a declared physical scale in the same units as x_i . The standardized deviation is

$$y_i = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.2)$$

The numerator says how many stock units the cell differs from its reference. Dividing by the scale expresses that difference in units of the declared width. The result is dimensionless.

With reference ten and scale two, stock fourteen has standardized deviation two. Stock six has standardized deviation minus two. The signs distinguish above from below. Both are the same standardized distance from the reference. A quadratic potential will give them equal local burden because that symmetry is part of its definition.

Here σ_i controls the geometry of the potential. A measurement error describes uncertainty in an observation; a tolerance specifies an acceptable range; a statistical standard deviation belongs to a probability distribution. Those other quantities can inform a choice of scale, but they retain their own definitions. Declaring the scale explicitly allows EBU to recognise different functional needs while keeping the comparison reproducible.

11.5 A scale is not a boundary

At $x_i = x_i^* + \sigma_i$, the standardized deviation is one. Nothing discontinuous happens there. There is no switch from safe to unsafe and no automatic prohibition. The potential and its derivative change smoothly through this point.

This distinguishes the Gaussian construction from a model that uses a flat acceptable interval with penalties outside it. The present field has one declared centre and positive curvature everywhere. Every nonzero local deviation contributes to its potential. Physical feasibility still includes nonnegative stock and any separately declared domain constraints; those are not manufactured by treating σ_i as a threshold.

If an application requires a hard storage capacity or a protected ecological condition, it must declare that physical or institutional rule separately. Calling a scale a safety certificate would give it a job its definition does not perform. Conversely, the lack of an introductory threshold term does not claim that real systems lack critical boundaries.

11.6 Worked standardization table

For one cell with $x^* = 10$ and $\sigma = 2$, the relevant values are

Stock x	Deviation $x - 10$	Standardized y	$\frac{1}{2}y^2$
2	-8	-4	8
6	-4	-2	2
8	-2	-1	0.5
10	0	0	0
12	2	1	0.5
14	4	2	2
18	8	4	8

Read the third column before the fourth. Standardization is a physical-unit conversion within the model; squaring and halving is the declared valuation shape. Separating these operations helps a reader locate where an assumption enters.

A deviation of four stock units is two scales, and its quadratic contribution is two. Doubling that physical deviation to eight makes the contribution eight, four times as large. This is a transparent consequence of the square. It is not an empirical theorem that doubling every real shortage multiplies its harm by four.

11.7 Different cells may have different scales

Consider a new illustrative reference (10,10) with scales (2,4). The main stocks remain (18,2). A is four scales above reference and B is two scales below it. Their contributions are eight and two, for total ten.

The field now assigns a smaller contribution to B’s eight-unit deviation than before. No physical stock changed; the declared geometry changed. This example is a sensitivity calculation across two different models. It must not be inserted into one action record as though the actor caused the difference between potential 16 and potential 10.

Unequal scales can be useful when a stock deviation has different functional significance in different places. They can also conceal unjustified weights. A narrow scale steepens the field and makes changes at that cell count more strongly. Its provenance belongs in the model definition, not in an after-the-fact explanation of why a preferred action won.

11.8 Changing physical units must not change the answer

Suppose one stock unit represents one cubic metre, and we switch to litres. Every stock, reference, scale, and finite quantity is multiplied by 1000. The standardized deviation remains unchanged:

$$\frac{1000x_i - 1000x_i^*}{1000\sigma_i} = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.3)$$

The potential and its finite difference are therefore unchanged. The marginal's numerical value changes because it is now per litre rather than per cubic metre. Multiplying it by a correspondingly converted quantity restores the same dimensionless product.

If an implementation changes stocks to litres but leaves σ numerically unchanged, it has changed the model, not just the unit. Its potential becomes much steeper. This is a useful audit test because every piece of the declaration is visible. A physical unit conversion must transform all quantities with that unit consistently.

11.9 Why a reference must remain fixed during an action

An action comparison uses one potential before and after. If the reference or scale changes between the two evaluations, the result includes a model change. A zero physical increment can then appear to create value.

For example, take a cell with stock 14, reference 10, and scale 2. Its contribution is two. If an editor moves the reference to 14 without moving stock, the contribution becomes zero. The number fell by two, but no physical action supplied that reduction. The field definition changed.

References can be revised in a research programme or calibrated between declared epochs. Such revisions need an explicit version boundary and an account of the redefinition. They belong in a field-change account, separate from the physical action receipt. The same principle applies to a scale revision or to adding and removing potential terms. This lets an institution improve its model without rewriting the physical history of contributors. A clearer reference can guide future decisions while past receipts remain interpretable under the rule that produced them.

A further normalization matters when we study feedback. A native burden V_B can carry units B , whereas our standardized V is dimensionless. If an application declares a positive burden scale B_0 and a fixed offset c_B , it may define $V = (V_B - c_B)/B_0$. Then $E = (V_B^b - V_B^a)/B_0$. The constant offset cancels; the scale does not. In the quadratic feedback model $V_B = w(L - X)^2$, the response coefficient contains the product $2wM$ of burden weight

and mobility. Observing that response rate cannot identify the weight and mobility separately, or supply B_0 . This preserves a stable meaning for the unit while allowing controller response to be designed independently.

11.10 Changing cell granularity changes the declaration

Suppose one well-mixed tank has stock x , reference x^* , and scale σ . An editor divides it into two bookkeeping cells with equal half-stocks and half-references. It is tempting to copy the old scale into both cells and assume the sum of their potentials still means the same thing. It does not.

With copied scales, each deviation is half the original, and the two squared contributions sum to half the original potential. If each new scale is instead set to $\sigma/2$, each standardized deviation equals the old one, and their total potential doubles. Neither result is a physical consequence of moving water; it comes from changing the accounting representation.

If the halves are constrained always to remain equal and the aim is to preserve this particular aggregate quadratic exactly, choosing each scale as $\sigma/\sqrt{2}$ makes the two contributions sum to the original. That is a conditional algebraic mapping, not a universal rule for spatial refinement. Once the halves can vary independently, the finer model can represent an internal imbalance that the original one-cell state could not see.

The example shows why cell boundaries, references, and scales should be calibrated together. Merely increasing the number of cells is not a neutral way to gain apparent precision. A change of granularity may reveal useful new physical structure, but it also changes the field's meaning and must be declared.

11.11 The reference is not a quota for each transaction

A ten-unit reference at B does not mean that every delivery should contain ten units. The reference is a state coordinate, whereas an action quantity is a change. If B already contains six, adding four reaches ten; if it contains nine, adding four reaches thirteen. The same delivery has different consequences at the two baselines.

Nor does a cell's reference assign ownership of that stock to a particular actor. The physical index identifies the represented place. Ownership and permission require their own maps and rules. Keeping those objects separate allows a potential to be studied without pretending that an equation has decided all institutional arrangements.

Finally, a reference at one time does not guarantee adequacy at every future time. A continuing-life application may have changing demands and carriers. Such extensions require a time-varying model or additional physical state. They cannot be simulated simply by shifting the reference after an outcome and reporting the resulting lower potential as earned improvement.

11.12 A reference document that a reader can audit

A useful declaration identifies the carrier, cell boundary, stock unit, reference value, scale, and provenance for each coordinate. It states the total-mass compatibility condition and whether the action graph can connect the relevant places. It also states the interpretation intended for the potential: a standardized model deviation rather than a complete measure of welfare.

For teaching, the provenance is openly illustrative. We choose (10,10) and (2,2) to make arithmetic readable. These choices define the teaching model. An application records the origin of its own references and scales in the same declaration, preserving the same precision in the account.

The document should also distinguish fixed parameters from state variables. If carrying capacity, demand, or another functional characteristic is allowed to change physically, it may need its own represented coordinate. Quietly editing a reference whenever conditions change can make a poor process appear successful by moving its target after the fact.

11.13 Guided problem: asymmetric needs

For a separate two-cell exercise, declare reference (12,8), scales (2,2), total stock 20, and current state (18,2). Both coordinates are six units from reference, or three scales. Potential is $\frac{1}{2}(3)^2 + \frac{1}{2}(-3)^2 = 9$.

A lossless six-unit transfer reaches (12,8), the declared reference. An eight-unit transfer reaches (10,10), leaving deviations (-2,2), each one scale, and potential one. Equal raw stocks are therefore not the minimum of this changed model. The reference embodies different declared functional stocks at the two cells.

Do not conclude that either reference is ethically correct from the calculation. The example establishes the mathematical dependence of value on the declaration. An author who changes from equal to asymmetric references must explain the domain reason, rather than claim that the equations discovered it.

11.14 Guided problem: incompatible mass

Suppose total stock is 20 and someone proposes reference (12, 12). The reference lies outside the fixed-mass domain. With equal scales, the best balanced feasible point is (10, 10), but it still has a positive deviation from the proposed reference. In fact each coordinate differs by minus two and contributes one-half at scale two, for total one.

This is not a result showing that an actor failed to recover to a reachable reference. The reference itself demands more stock than the closed model contains. The correct first-model response is to require a compatible reference or explicitly change the physical boundary and problem. The example explains why compatibility is checked before controller performance is interpreted.

11.15 Misconceptions and practice

“Sigma is the noise level.” Only if a separate declaration connects it to measurement uncertainty. In this model it is a physical reference scale.

“One scale from the reference is a safe boundary.” No such boundary is implied. It is a location on a smooth quadratic.

“Gaussian means actual stocks follow a normal distribution.” The Gaussian shape motivates the potential geometry. An actual distribution requires a stochastic process and evidence or a mathematical derivation for that process.

“A compatible reference must be reachable.” Conservation compatibility removes one obstruction; graph structure and allowed actions may create others.

For practice, take reference 15, scale 3, and stocks 9, 15, and 21. Their standardized deviations are minus two, zero, and two; their local potentials are two, zero, and two. Now convert every physical quantity to a unit one-tenth as large. The numerical stocks and scale become ten times larger, while all three potentials remain unchanged.

For a three-cell reference (4, 6, 10) and total stock 20, check compatibility. Then imagine no edge connects the first two cells to the third, and the third begins with eight. The reference is globally compatible but unreachable by internal transfers within the two disconnected components. Explain the missing condition in words before attempting any controller design.

What to carry forward

The reference names the centre of comparison; the positive scale measures deviations in physical units. Neither is automatically an empirical mean, uncertainty bound, or equilibrium. Keep them fixed within an action account and compatible with the declared physical domain.

We can now assemble these declarations into a potential and see why a Gaussian landscape gives especially transparent finite mathematics.

Homeostasis as a Gaussian landscape

The Gaussian construction turns reference and scale into a transparent state potential. Its quadratic form gives more than a convenient drawing: exact finite quotes, explicit overlap between actions and a rigorous upper limit on interaction order whenever the endpoint response is affine. Those properties make the simplest model a useful foundation for a much broader theory.

12.1 From comparison to a function

A function is a repeatable rule that maps an input to an output. Our input is a physical state. The output will be a nonnegative number measuring standardized deviation from the declared reference. Calling this number a potential gives it a mathematical role: its differences and derivatives can be evaluated consistently across actions.

For cell i , define

$$V_i(x_i) = \frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2, \quad \sigma_i > 0. \quad (12.1)$$

The global potential is

$$V(x) = \sum_{i=1}^N V_i(x_i). \quad (12.2)$$

The word global describes the argument of this mathematical function. It does not require every actor to read the whole state. A later locality identity shows exactly which terms a local action needs.

The factor one-half is a convenient normalization. It makes the derivative especially simple. Omitting it would create a consistently rescaled model if every dependent equation were changed, but mixing the two conventions would change numerical quotes. We retain the one-half throughout this edition.

12.2 Why the landscape is called Gaussian

Consider the unnormalized shape

$$g_i(x_i) = \exp \left[-\frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2 \right]. \quad (12.3)$$

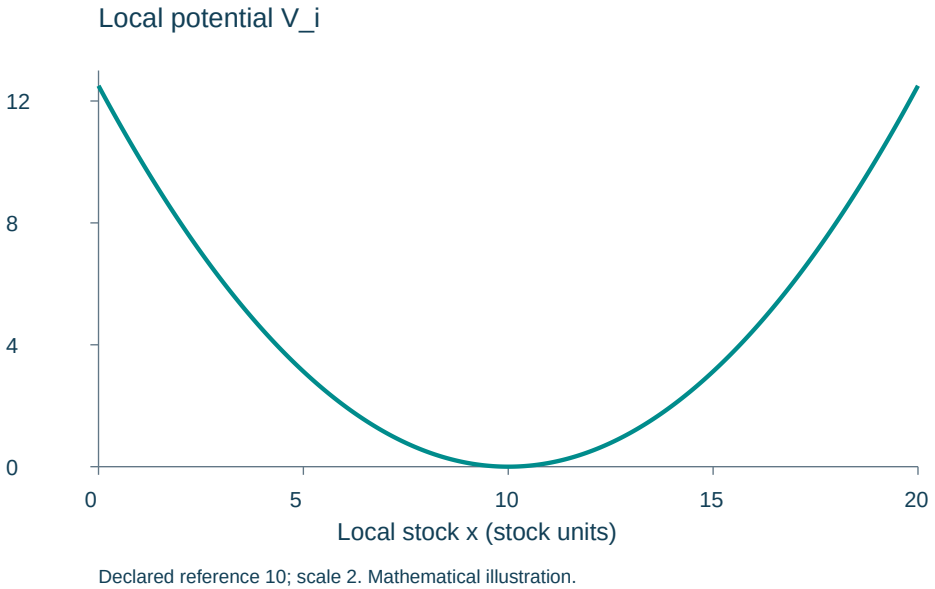


Figure 12.1: The one-coordinate quadratic penalty with reference ten and scale two. Its height is a declared deviation measure, not an amount of physical stock.

It has the familiar Gaussian bell shape, with its highest value at the reference. Taking the negative logarithm gives $-\log g_i(x_i) = V_i(x_i)$. The potential is therefore the relative negative-log geometry of that shape, with its minimum set to zero.

The construction fixes the relative negative-log geometry. A probability interpretation requires a domain, a reference measure and a normalization, which we supply in Chapter 37. For now the shape gives a smooth and explicit way to compare states. Its value, gradient and curvature can be calculated exactly without first specifying a random mechanism.

On the fixed-mass nonnegative domain, coordinates are also constrained together. Even though the quadratic is a sum of one-coordinate terms, the actual stocks cannot vary independently: their sum is fixed. A product of Gaussian shapes in an unconstrained mathematical space is not automatically a product probability law for the physical system.

12.3 The bowl and its centre

Each local potential is a smooth quadratic bowl. It is zero at x_i^* , positive elsewhere, and symmetric about that centre. Moving one scale above or below

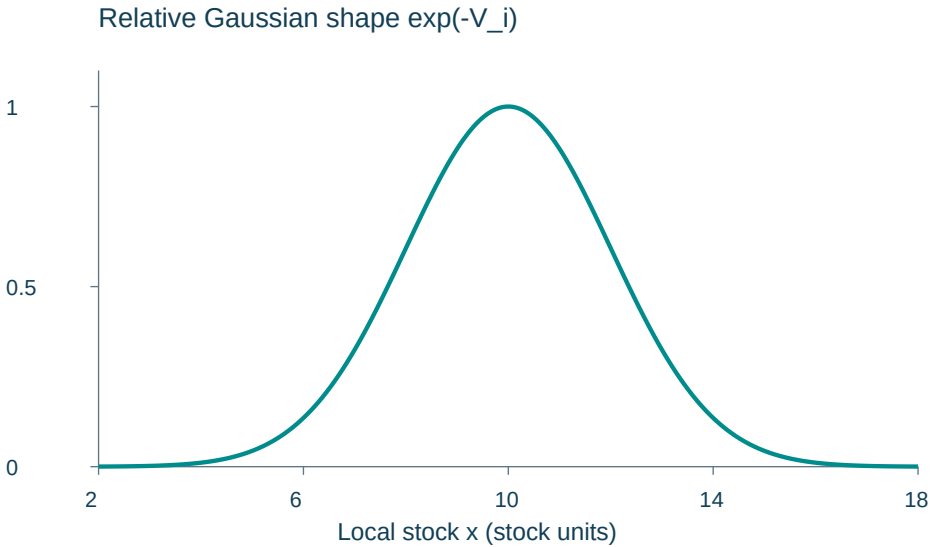


Figure 12.2: The exponential shape associated with the quadratic. This relative shape does not itself supply a probability model or prove that observations are normally distributed.

the reference gives $V_i = 1/2$. Moving two scales gives $V_i = 2$. Moving four scales gives $V_i = 8$.

The symmetry is an explicit modelling choice. It does not assert that real deficit and excess have equal biological consequences. A clinic with too little water may face different problems from one with too much. The first Gaussian model isolates a transparent geometry for studying an accounting and action process. A later asymmetric or multi-coordinate field would need its own declaration and review.

Definition and immediate consequence

With fixed positive scales, $V(x) = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2 \geq 0$. Equality holds exactly when every represented coordinate equals its reference. This follows from a sum of nonnegative squares. It is not a statement that every unrepresented human or ecological need is satisfied.

12.4 Equal total, unequal potential

Return to the main world. At (18, 2), standardized deviations are (4, -4) and potential is 16. At (14, 6), they are (2, -2) and potential is 4. At (10, 10), both are zero. Each state contains the same total stock, 20.

The landscape distinguishes configurations of a conserved quantity. It does not create or destroy the quantity it evaluates. This is why a conserved physical ledger can coexist with a changing potential. The total-mass invariant concerns a sum of stocks; the potential concerns squared departures from a reference.

The reflected state (2, 18) again has potential 16. It is different from (18, 2), although the scalar potential is equal. A single potential value does not reconstruct the state. If an account records only V , it loses the information about which place is above and which below its reference.

12.5 Reading the quadratic along the physical domain

For two cells with total 20, every physical state can be written $(r, 20 - r)$, with $0 \leq r \leq 20$. Under equal references ten and scales two,

$$V(r, 20 - r) = \frac{1}{2} \left(\frac{r - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{10 - r}{2} \right)^2 \quad (12.4)$$

$$= \frac{(r - 10)^2}{4}. \quad (12.5)$$

This one-dimensional curve is the potential restricted to the physically allowed line. Its centre is $r = 10$. The two endpoints $r = 0$ and $r = 20$ each have potential 25. Every state in this particular closed world therefore has a finite potential between zero and 25.

The boundedness follows from the domain before an EBU-guided process is tested. A comparison between two action rules cannot claim success merely because both remain bounded in this finite simplex. Useful questions concern how often they occupy large-deviation states, whether they recover after forcing, whether they cycle, or whether affordability makes them refuse useful opportunities.

12.6 The marginal: a local slope

The local marginal is the derivative of the potential with respect to its stock:

$$\mu_i(x_i) = \frac{dV_i}{dx_i} = \frac{x_i - x_i^*}{\sigma_i^2}. \quad (12.6)$$

Below reference, the marginal is negative: adding a small amount lowers local potential. Above reference, it is positive: adding more raises local potential. At the reference it is zero.

Derive the formula rather than memorize it. Write $V_i = (x_i - x_i^*)^2 / (2\sigma_i^2)$. The reference and scale are fixed. Differentiating the square gives $2(x_i - x_i^*)$; the two cancels the one-half. The remaining denominator is σ_i^2 .

The potential is dimensionless in this model. The marginal therefore has units X^{-1} , or potential units per stock unit. It is neither a stock nor a complete action value. Multiplying by a small stock increment gives the first-order potential change. A finite increment also encounters curvature, which we derive in Chapter 15.

12.7 A gentle derivative laboratory

For one cell with reference ten and scale two, $V(x) = (x - 10)^2 / 8$. Starting at 14, add a small increment h . Direct expansion gives

$$V(14 + h) - V(14) = h + \frac{h^2}{8}. \quad (12.7)$$

Divide by $h \neq 0$: the ratio is $1 + h/8$. As h approaches zero, the ratio approaches one. That limit is the marginal at 14. The formula also shows what the derivative leaves out for a finite change: the quadratic term $h^2/8$.

At stock six, the same calculation gives marginal minus one. Adding a small positive amount lowers potential. At ten, the marginal vanishes, but a finite nonzero displacement raises the potential by its square divided by eight. Zero initial slope is therefore not permission to move a finite distance without effect.

This is the bridge between an intuitive landscape and calculus. A slope is a nearby rate of change. It describes the tangent at one point. The finite EBU account compares two actual endpoints within the declared model.

12.8 The gradient and the Hessian

Collect the marginals into the gradient vector,

$$\mu(x) = \nabla V(x) = \left(\frac{x_1 - x_1^*}{\sigma_1^2}, \dots, \frac{x_N - x_N^*}{\sigma_N^2} \right)^T. \quad (12.8)$$

The symbol ∇ , read “nabla,” denotes the vector of partial derivatives. A partial derivative varies one coordinate while holding the others fixed in the mathematical function. Physically allowed directions can then be built from combinations that respect conservation.

The Hessian records derivatives of those slopes:

$$H = \text{diag} \left(\frac{1}{\sigma_1^2}, \dots, \frac{1}{\sigma_N^2} \right). \quad (12.9)$$

“Diagonal” means the off-diagonal entries are zero. Each local slope changes at the constant rate $1/\sigma_i^2$ with its own stock. The Hessian is positive definite because all scales are positive. For any nonzero vector d , $d^T H d = \sum_i d_i^2 / \sigma_i^2 > 0$.

The matrix form of the potential is

$$V(x) = \frac{1}{2}(x - x^*)^T H(x - x^*). \quad (12.10)$$

It compresses the sum of standardized squares. It does not replace their physical interpretation. The row-vector, matrix, and column-vector multiplication expands back into the same local contributions.

12.9 Ellipses, distance, and unequal scales

In an unconstrained two-coordinate drawing, points with a fixed potential satisfy

$$\frac{(x_1 - x_1^*)^2}{\sigma_1^2} + \frac{(x_2 - x_2^*)^2}{\sigma_2^2} = 2V. \quad (12.11)$$

These curves are ellipses. With equal scales they are circles in the chosen physical coordinate units. A larger scale stretches the contour along its coordinate: a larger physical deviation is needed to produce the same contribution. This is the geometric meaning of scale weighting.

The physical fixed-mass states occupy only a slice of this drawing. For two cells they lie on a line of constant sum. The field evaluated along that line is the restriction of the same quadratic, not a second potential. A picture of a circular bowl should therefore not suggest that every direction in the plane is a permitted physical motion.

The quantity $\sqrt{2V}$ is the Euclidean length of the standardized deviation vector. It can be described as a distance in these standardized coordinates. It is not geographical distance, and it is not itself the signed action value. Taking its square root changes finite differences and derivatives. We retain V as the declared potential even when the distance picture helps intuition.

12.10 Diagonal geometry does not mean independent outcomes

The diagonal Hessian says that each local marginal depends on its own coordinate. Conservation and shared action support still connect changes. If two

actions withdraw from A , both alter the same squared deviation there. The joint value therefore contains a pair contribution even though the potential has no cross-coordinate terms.

For a quadratic potential with Hessian H , and additive increments h_i, h_j , that contribution is exactly $-h_i^T H h_j$. The weighted overlap determines its sign. No third derivative of V is present, and every interaction involving three or more distinct actions is zero when the endpoint is affine in their membership indicators. Chapter 30 proves this degree bound.

The same conclusion applies when a common linear feedback law generates the increments over a finite horizon. Pending stock and decaying oscillations do not change it: a linear system superposes its responses, and a quadratic reads that superposition with degree at most two. Nonlinear response changes the conclusion, because the composed endpoint value can then have higher degree even when V remains quadratic.

This yields an efficient EBU feature. A system whose assumptions satisfy the degree bound can represent a joint value by its single-action and pair terms exactly. It need not invent or calculate higher-order corrections. The saved effort can instead support a larger, more complete physical boundary. A richer nonseparable quadratic changes which pairs interact, but by itself still does not create a third-order term under affine response.

12.11 A bounded domain gives a bound before any experiment

Because nonnegative stocks sum to M , every coordinate lies between zero and M . Its squared departure is bounded by the larger endpoint departure. Thus

$$V(x) \leq \frac{1}{2} \sum_i \frac{\max\{(x_i^*)^2, (M - x_i^*)^2\}}{\sigma_i^2}. \quad (12.12)$$

For finitely many cells and fixed positive scales this is finite. It is a convenient bound, although not generally the tight maximum because all coordinates cannot reach their separate extremes simultaneously.

The bound is an established property of the domain and potential. It applies to every admissible policy, including a poor one. Its practical use is to delimit the accounting range and to separate boundedness from better questions about function: closeness to the reference, persistence of a shortage, recovery and the resources required to maintain service. An EBU design can use the bound in its theoretical specification without treating it as a special achievement of its controller.

12.12 Convexity without mystery

A quadratic bowl with positive curvature is convex. Geometrically, the straight chord joining two points on its graph lies on or above the graph. Algebraically, the tangent at a starting point underestimates the potential at a finite displaced point by a nonnegative quadratic amount.

This will matter for valuation. If the initial slope predicts an improvement, multiplying that slope by a large action quantity overstates the finite field improvement. The potential curves upward relative to its tangent. Exact endpoint comparison automatically includes that curvature.

Convexity is a property of the potential, not a promise about action choices. A random actor can choose a direction that raises a convex potential. A capacity rule can permit cycles. A graph can block a path toward the minimum. A simple bowl is mathematically helpful precisely because these separate mechanisms remain visible; the shape does not decide them by itself.

12.13 Why the potential is not a controller

Drawing the landscape tells us the value at every state. Moving through it requires a law of response. A gradient rule, a constrained valve controller and a rule for selecting human actions can all use the same potential while generating different paths.

This separation is productive. The field gives a common way to compare their endpoints; the dynamics explains timing, pending stock and recovery. In particular, a replenishment controller can be stable while the native shortage burden rises over part of its return toward a driven operating point. We will derive a storage function for that stability calculation without replacing the EBU potential.

EBU can therefore support two useful views of the same service. Its receipts describe the declared change of physical burden. Its response model describes whether delivery settles, overshoots or stays within a permitted region. The combination makes coordination more informative than either a bare transaction count or a single stability label.

12.14 Do not replace the function while keeping its name

The bounded function $1 - \exp(-V)$ can be formed from the quadratic. It is not the same potential. Its derivative multiplies the quadratic gradient by $\exp(-V)$, and its finite differences give different values. Large deviations become compressed toward one.

Such a transformation might be studied under another declared model, but it cannot be substituted merely because both expressions use a Gaussian exponential. The active definition is the quadratic negative-log potential itself. Field, marginal, curvature, and finite account must all be derived from that same definition.

Likewise, dropping a coordinate, adding a weight, or changing a scale changes the functional. Precise notation prevents an appealing diagram from concealing a substantive amendment. Every quantitative claim in the main example traces back to Equations 12.1 and 12.2.

12.15 Guided problem: three cells

In a new three-cell illustration, choose reference $(4, 6, 10)$, scales $(2, 2, 4)$, and state $(2, 8, 10)$. Total stock and reference total are both 20. Standardized deviations are $(-1, 1, 0)$, so $V = 1$.

The marginals are $(-1/2, 1/2, 0)$. Moving a small lossless quantity from cell 2 to cell 1 has direction $(1, -1, 0)^T$. Its initial potential derivative is $(-1/2)(1) + (1/2)(-1) = -1$, so the direction initially lowers potential. A two-unit transfer reaches the reference, giving exact field reduction one. Initial slope times quantity would predict two, twice the available reduction.

This calculation uses a new reference and scale declaration, identified openly. It reinforces the same mathematical structure without quietly changing the parameters of A and B. The reader can check each coordinate independently before adding the total.

12.16 One potential value, several physical descriptions

The scalar potential deliberately compresses a state vector. That compression is useful for finite identities, but it loses information. In the symmetric two-cell fixture, the same value four occurs at $(14, 6)$ and $(6, 14)$. These states place the shortage at different locations. A clinic at B experiences them differently even though the total standardized squared deviation is equal.

In a larger graph many more states can share one value. They can differ in which cell is far from reference, whether an available edge points toward relief, and how much stock a prospective actor can access. Equal potential does not imply equal feasible menus, equal service, or equal future behaviour.

For this reason, the account retains the state and its distribution across cells alongside V . The finite account uses the scalar because its changes have a precise meaning. Interpretation should not discard the coordinates that gave

it that meaning. A contour of equal potential is a mathematical level set, not a class of socially or physically interchangeable worlds.

12.17 Practice and reflection

With reference ten and scale two, calculate local potential and marginal at stocks 4, 10, and 16. The potentials are 4.5, 0, 4.5; the marginals are $-1.5, 0, 1.5$. Explain why a nonnegative potential can have a negative derivative.

For the main two-cell state, verify $\mu = (2, -2)^T$ and $H = \text{diag}(1/4, 1/4)$. Then explain why equal total stock does not imply equal potential, and why equal potential does not imply equal state. Use $(18, 2)$, $(10, 10)$, and $(2, 18)$ to support the answers.

Finally, write one statement that this potential establishes and one statement it leaves open. A valid pair is: “Its value is zero exactly at the declared reference” and “The proposed actor process may or may not spend much time near that reference.” The first follows from the definition; the second needs a specified process and further analysis.

The potential gives the whole state a declared value; its derivatives tell us how that value changes nearby. We next restrict those derivatives to the physical directions supplied by the network, so that a local field refers to an actual possible transfer.

From a marginal to an edge field

The gradient turns the landscape into local information: how would a small permitted change alter the potential? We derive that directional field from complete physical directions, then distinguish the field from mobility, dispatch policy and finite value. This makes local advice useful without confusing it with a command to the physical system.

13.1 What the word field means here

A map of local potential values is not yet an action account. To understand a proposed transfer, we need to know how the potential changes in the direction of that transfer. The marginal at each cell tells us how its local contribution responds to a small addition. The edge field combines the source and destination responses with the physical signs of the action.

In this book, the edge field is a directional derivative of a declared potential. It is sometimes called a force by analogy with potential-based physics. Its unit here is potential per stock, not newtons. We derive its formula from the action column and the chain rule; no gravitational or electromagnetic law is assumed.

This careful use of the word field matters. It gives us a useful mathematical structure without claiming that the potential is a newly discovered physical interaction. A stock distribution and a declared Gaussian reference determine the values. An actor or physical law still has to specify what happens next.

13.2 The direction of one lossless transfer

For an edge $e = (i \rightarrow j)$, the state after a finite quantity q is $x^b + S_e q$. Its source coordinate is $x_i^b - q$; its destination coordinate is $x_j^b + q$. To find the initial response, imagine changing q by a very small amount while holding the initial baseline fixed.

The source changes at minus one stock unit per unit sent, and the destination changes at plus one. The chain rule gives

$$\left. \frac{d}{dq} V(x^b + S_e q) \right|_{q=0} = -\mu_i(x_i^b) + \mu_j(x_j^b). \quad (13.1)$$

Define the edge field with the opposite sign, so a burden-reducing direction has a positive field:

$$f_e(x^b) = \mu_i(x_i^b) - \mu_j(x_j^b) = -\mu(x^b)^T S_e. \quad (13.2)$$

This sign convention is the bridge to signed EBU. The potential change is negative when a small action improves the field. The corresponding field improvement is positive.

13.3 Deriving the equation one term at a time

The source contribution to the derivative is $\mu_i dx_i/dq$. Because $dx_i/dq = -1$, this is $-\mu_i$. The destination contribution is $\mu_j dx_j/dq = +\mu_j$. Every unaffected coordinate has derivative zero and contributes nothing. Adding these terms gives the potential derivative; negating it gives f_e .

At the main state (18, 2), the marginals are 2 and -2 . Removing a small amount from A lowers its local potential, because A is above reference. Adding that amount to B also lowers its local potential, because B is below reference. Thus

$$f_e = 2 - (-2) = 4. \quad (13.3)$$

The two beneficial endpoint effects add. An infinitesimal sent quantity dq changes potential to first order by $-4dq$.

The field does not mean that every finite quantity earns $4q$. It is the starting slope. As A loses stock and B gains it, both marginals move toward zero. The exact finite calculation must include that change.

A small quantity and its changing slope

Starting at (18, 2), sending $q = 0.1$ gives exact field reduction $4(0.1) - (0.1)^2/4 = 0.3975$. The initial-slope estimate is 0.4. Their difference, 0.0025, comes from curvature. It is a known algebraic term in this model, not an unexplained numerical discrepancy.

13.4 Four endpoint patterns

The signs of the endpoint marginals help interpret the direction, but the difference determines the result. An above-reference source has positive marginal. A below-reference destination has negative marginal. Together they give positive field. If both are below reference, removing source stock is locally harmful while adding destination stock is locally helpful; their magnitudes compete.

If both are above reference, withdrawal helps the source while delivery worsens the destination. Again, the field compares the two effects. A below-reference source sending to an above-reference destination has negative field because both endpoint contributions raise burden.

Source	Destination	Initial effect
Above reference	Below reference	Both changes lower local burden.
Below reference	Below reference	Source harm competes with destination relief.
Above reference	Above reference	Source relief competes with destination harm.
Below reference	Above reference	Both changes raise local burden.

These statements concern the declared potential. They do not decide whether the transfer serves an urgent need or violates another constraint. An application can require additional measurements of service and feasibility.

13.5 Raw stock is not the field

Two cells can contain the same stock and have different marginals because their references or scales differ. In a separate two-cell illustration, let both stocks be eight, references be ten and six, and scales both two. The marginals are $-1/2$ and $+1/2$. Sending from the second cell to the first initially lowers potential, although the raw stocks are equal.

The reference totals and stock totals in this example are both 16, so the comparison is compatible with the closed domain. The field expresses deviation relative to function as declared by the model. It is not simply a pressure to equalize every raw number.

Unequal scales provide another distinction. Keep the main stocks and references but change scales to (2,4) for this sensitivity illustration. Marginals become 2 and $-1/2$, so the field is 2.5, rather than four. The physical action column is unchanged. The model’s valuation geometry has changed, and the record must identify that new declaration.

13.6 Field along a finite path

For the Gaussian potential, the marginal changes linearly with stock. Along $x^b + S_e q$,

$$\mu_i(q) = \frac{x_i^b - x_i^* - q}{\sigma_i^2}, \quad (13.4)$$

$$\mu_j(q) = \frac{x_j^b - x_j^* + q}{\sigma_j^2}. \quad (13.5)$$

Their difference is

$$f_e(q) = f_e(0) - q \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (13.6)$$

The slope of the field along the action is negative because both positive curvature contributions add. In the main example this becomes $f_e(q) = 4 - q/2$.

At $q = 0$, the field is four. At $q = 4$, it is two. At $q = 8$, it is zero. Beyond eight, it is negative. These are field values at points along the same declared path, not different schedules chosen after the result. The whole function follows from the initial state, reference, scales, and action direction.

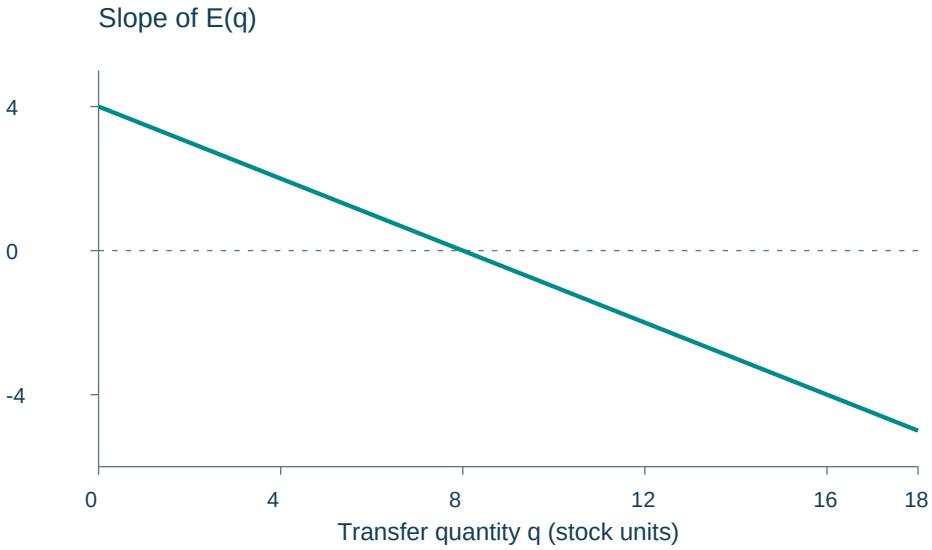
The quantity eight reaches the reference for both cells in the symmetric fixture. That coincidence is special to the compatible two-cell setup. In a larger system, equality of marginals on one edge does not imply that every coordinate equals its reference.

13.7 Zero field deserves careful reading

A zero edge field says the initial directional derivative is zero. It does not say the direction has zero finite cost. In the Gaussian model the curvature along every nonzero transfer direction is positive. Therefore a finite move from a point with $f_e = 0$ increases the potential, unless the net physical increment is zero.

At the reference (10,10), both marginals vanish. Sending two units gives (8,12). Each endpoint is one scale from reference and contributes one-half, so total potential rises to one. The initial field was zero; exact finite EBU is minus one. The difference is fully explained by the quadratic term.

A zero field can also arise from equal nonzero endpoint marginals in a larger or differently constrained world. It means there is no first-order improvement along that edge. It does not certify that the complete represented world is at its minimum or that its real needs are satisfied.



The slope crosses zero at $q=8$; it does not select an action.

Figure 13.1: The instantaneous field along the running transfer falls as the quantity increases. Integrating it, rather than multiplying its starting value by every finite quantity, gives the exact finite difference.

13.8 The graph form of the field

Collect the edge fields into a vector f . With the incidence matrix S , the compact expression is

$$f = -S^T \mu. \quad (13.7)$$

The transpose turns each edge column into a row that selects the source and destination marginals. If there are N cells and m edges, S^T has shape $m \times N$, so the result has one field value per edge.

For a separately declared three-cell chain with marginals $(2, 0, -1)^T$ and edges $1 \rightarrow 2$, $2 \rightarrow 3$, the fields are two and one. The field vector reports the two directional slopes. It does not choose the first edge, allocate stock, or decide whether both actions can occur together.

This is an example of local computation with global notation. The formula is compact for a proof, but each edge needs only the relevant endpoint marginals under the separable model. The whole vector can support an aggregate diagnostic, while a local action uses the inputs appropriate to its own decision rule.

13.9 Reverse directions and cycles

If the reverse lossless edge exists, its column is $-S_e$ and its field at the same state is $-f_e$. At the main baseline, A-to-B has field four and B-to-A has field minus four. At a later state their values must be reevaluated. A reverse action performed after a forward action does not use the old reverse marginal.

Around a directed cycle of lossless edges evaluated at one common state, marginal differences cancel. For a three-cell cycle,

$$(\mu_1 - \mu_2) + (\mu_2 - \mu_3) + (\mu_3 - \mu_1) = 0. \quad (13.8)$$

This is an identity for initial fields at that state. It does not claim that arbitrary unequal finite quantities around the cycle have zero value, nor that a real cycle uses no energy. The complete physical state, including any affected energy carrier or sink, determines the finite account.

If equal simultaneous quantities circulate around a lossless closed cycle, the net stock increment can be zero. The field component is then zero because the endpoint equals the baseline. Actual circulation may use energy, time, or equipment in a fuller model. Those carriers must be represented before claims about their burden are made.

13.10 The field around a shared source

Consider a separate three-cell fixture with state $(14, 3, 3)$, reference $(8, 6, 6)$, and equal scales two. Both state and reference sum to twenty. The marginals are $(1.5, -0.75, -0.75)$. The source-to-each-destination field is therefore 2.25.

The equality of the two starting fields says the destinations are symmetric under this declaration. It does not allocate a source quantity to either actor. If one actor sends two units, the source marginal falls by one-half and its destination marginal rises by one-half. Recomputing the other edge at the resulting state gives a smaller source contribution even though its own destination has not changed.

This is why several actors cannot each treat a shared initial field as an unlimited personal resource. Their actions use one physical coordinate. The correct joint finite value includes that shared curvature. A simultaneous attribution convention can divide a joint value under declared path semantics, but it cannot create several independent copies of the source's starting state.

An experiment should therefore distinguish equal local fields, equal finite quotes, equal physical permissions, and equal selection probabilities. They coincide only under additional symmetry conditions. For example, unequal quantity menus can produce unequal group frequencies even when endpoint

parameters are identical. These are protocol properties, not new forces in the potential.

13.11 A field can be useful even when it is not followed

A thermometer remains informative when nobody turns the heater on. Similarly, the derivative of a potential can diagnose the initial effect of a candidate action even when the actor chooses differently. The value's explanatory role does not require an automatic law of obedience.

This makes negative outcomes interpretable. If a permitted and affordable random action moves uphill, the field account can record the increase without treating it as a malfunction of the equation. The question for later research is how the complete policy changes the distribution of such events over time. That question needs the actor process, forcing, and account rules in addition to the local derivative.

13.12 An explicitly separate loss-aware boundary illustration

For a lossy transfer, the complete direction includes both useful delivery and the sink. In the pending route of Chapter 10, arrival has direction $s_A = (0, -1, \eta, 1 - \eta)^T$. Consequently

$$f_A = \mu_Q - \eta\mu_X - (1 - \eta)\mu_W. \quad (13.9)$$

Dispatch has a different direction, $s_D = (-1, 1, 0, 0)^T$, and field $f_D = \mu_S - \mu_Q$. A controller that uses the eventual receiver shortage to request dispatch is using a completed-delivery signal. That signal is not generally the derivative of the instantaneous dispatch event.

For example, if a reduced potential values only receiver stock, immediate dispatch leaves that potential unchanged while arrival changes it. The controller can still sensibly order water in anticipation of the future. EBU accounts for the event boundary it actually declares; the controller describes the response across time.

This precision has a direct efficiency benefit. A route can avoid awarding the same completed improvement once at dispatch and again at arrival. Pending stock remains visible, and the final service consequence can be associated with the complete declared action or with its correctly separated stages.

13.13 Where the Onsager connection comes from

Lars Onsager's two 1931 papers studied coupled irreversible processes, including heat conduction, electrical transport and diffusion. Their reciprocal

relations connect response coefficients under microscopic-reversibility assumptions. The theory concerns thermodynamic forces and fluxes in a linear-response setting; it is not a general statement that every system follows a gradient of any function one chooses [19, 20].

The useful idea for EBU is to distinguish a driving difference from the strength of response. Think of two otherwise similar routes with the same endpoint marginal difference. One route can carry a large flow; the other responds slowly. The field difference and the route's responsiveness describe different things. This is why the original EBU architecture introduced an Onsager-type force–mobility–flux structure rather than treating the marginal itself as a transported quantity.

13.14 Force, mobility and flux in the EBU model

For a lossless edge, the already derived field force is $f_e = \mu_i - \mu_j$. The earlier EBU controller used

$$J_e = M_e[f_e - \theta_e]_+, \quad M_e > 0, \quad \theta_e \geq 0. \quad (13.10)$$

Here J_e is a flow rate and M_e is mobility: the declared response per unit force. The positive-part operation $[a]_+ = \max(a, 0)$ permits flow only in the declared edge direction and only above the activation threshold θ_e . This edge activation threshold is not an Allee threshold or a lower/upper homeostatic stock band. It belongs to this optional controller, not to the Gaussian potential.

In the dimensionless Gaussian normalization, force has units X^{-1} , mobility has units X^2/time , and their product has units of stock per time. A finite duration gives quantity $q = \Delta t J_e$. Physical permission still has to check that quantity; high mobility cannot create stock that the source does not hold.

At the main baseline the force is four. As a separate controller illustration, choose mobility one-half and activation threshold zero. The requested rate is two stock units per time unit. Over half a time unit it requests one stock unit. The exact finite field value of that quantity is $15/4$, not the initial-force estimate four. Mobility sets the response rate; the finite equation values the complete accepted change. The example separates response speed from the value of the accepted quantity.

13.15 Why borrow this structure, and where does it stop?

The force–mobility structure gives a precise descent result. If $\dot{x} = SJ$ has no external drive, the chain rule gives $\dot{V} = -\sum_e f_e J_e$. With the threshold-mobility response, every active product is nonnegative, so this particular con-

tinuous law cannot increase V . Chapter 31 derives the result and the extra restriction for finite steps.

This is useful when a designer wants a controller with a known direction of change. It also explains why feedback must be examined on its own terms. Once delivery is pending, consumption continues or the dispatch signal differs from the instantaneous gradient, the simple descent proof need not apply. Another storage function can establish stability without turning V into a thermodynamic free energy or making its decrease an entropy-production law.

The advantage for EBU is a modular design: a local field can inform action, mobility can regulate speed, physical constraints can restrict permission, and the finite account can value the accepted consequence. Each part can be improved without silently changing the others.

13.16 Why the actor law remains an explicit choice

It would be possible to propose a flow proportional to positive field, or to choose whichever edge has the largest field. Either choice would add a controller. It might be useful for a different research question, but it would not follow from Equation 13.2 alone.

A declared actor model keeps valuation distinct from choice. Feasible actions can have positive, zero or negative values. A capacity policy can restrict affordability, and a selector can prioritise service, rank values or use another stated rule. These choices produce different dynamics while sharing the same finite evaluator.

This distinction also prevents an easy but circular success claim. If a plant is given autonomous restoring flow, observing movement toward the reference does not by itself demonstrate an economic accounting mechanism's contribution. The field may inform analysis without being a hidden force that physically compels the actors.

13.17 Guided problem: equal field, unequal finite behaviour

Two separate hypothetical edges have the same initial field four. Edge A connects cells with scales two and two, so its curvature along quantity is $1/4 + 1/4 = 1/2$. Edge B connects cells with scales one and one, so its curvature is two. Their initial tangents match, but their fields fall at different rates as quantity increases.

For a quantity of two, the exact field reductions, derived in Chapter 15, are

$$F_A(2) = 4(2) - \frac{1}{2}(1/2)(2)^2 = 7, \quad (13.11)$$

$$F_B(2) = 4(2) - \frac{1}{2}(2)(2)^2 = 4. \quad (13.12)$$

Both calculations assume compatible declared states that give initial field four and permit the quantity. The example isolates curvature; it is not a recommendation between two real services. It shows why the starting field cannot replace a finite schedule.

Now ask what would happen at quantity four. Edge A gives $16 - 4 = 12$, while B gives $16 - 16 = 0$. The same initial field is consistent with very different finite values. A complete quote requires the potential along the action direction, not merely one slope sample.

13.18 A sign routine for every new example

When a formula feels abstract, first state the physical change: the source loses and the destination gains. Next ask how each change affects its local potential. Translate those answers into derivative signs. Only then combine the terms.

For an above-reference source, $\mu_i > 0$, so its negative stock increment contributes a negative potential change. For a below-reference destination, $\mu_j < 0$, so its positive stock increment also contributes a negative change. The negative of their sum is positive field. This routine is slower than memorizing a sign, but it is much harder to misuse when the carrier or action direction changes.

The routine also exposes ambiguous action descriptions. If one cannot say which coordinate decreases, it is too early to calculate the field. A transformation involving several inputs and outputs needs a correspondingly complete column before the dot product has physical meaning.

13.19 Practice with solutions

After four units move in the running example, the state is $(14, 6)$, marginals are $(1, -1)$ and the field is two. After eight units it is zero; after twelve it is minus two. The direction changes because the stock changes.

For state $(9, 7)$, reference $(10, 6)$ and scales two, the marginals are $(-1/4, 1/4)$. Sending from the second cell to the first initially improves the field despite the source's smaller raw stock. Reference and scale, not raw quantity alone, determine the derivative.

A field of $4X^{-1}$ is not four stock units. Integrating its changing value over quantity gives finite EBU. The next chapter proves that connection.

Why the whole path has one endpoint value

Local marginals change while an action unfolds. The path identity shows why adding those changes gives exactly the finite EBU already defined. It also separates improvement between endpoints from service, exposure and resource use accumulated through time, giving each a clear place in an EBU account.

14.1 Many small changes, one complete comparison

A pipe can transfer water steadily or in pulses. During the movement the local marginal changes: the source becomes less full and the destination less empty. If we keep using the first marginal, we will count later units as though the earlier ones never arrived. The integral solves precisely this problem. It follows the changing marginal along the entire represented path.

There are two objects here. The finite definition compares states, $E = V(x_0) - V(x_1)$. The path theorem explains when a continuous calculation gives that same number. The theorem does not define a new kind of EBU or add a charge for traveling between the endpoints. It connects the finite account to local differential information.

Think of measuring elevation along a walk. Adding all upward and downward elevation changes gives the difference between the final and initial heights. A long detour changes the sequence of readings, but not that height difference. Distance walked and effort expended are different quantities. In an EBU application, pumping energy and wear likewise need their own physical representation; they cannot be inferred from the endpoint change of an inventory potential.

14.2 The assumptions that make the connection exact

Theorem 14.1 (Fixed-field path identity). Let V be a single-valued C^1 function on an open neighbourhood of an admissible piecewise C^1 path $\gamma : [0, 1] \rightarrow \mathcal{X}$, with $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Hold the potential and its

parameters fixed throughout the comparison. Then

$$E = V(x_0) - V(x_1) = - \int_0^1 \nabla V(\gamma(s))^\top \gamma'(s) ds. \quad (14.1)$$

The same statement holds intrinsically on a smooth constrained state manifold, using its differential dV along a piecewise smooth path.

Proof. On each smooth segment the chain rule gives $\frac{d}{ds} V(\gamma(s)) = \nabla V(\gamma(s))^\top \gamma'(s)$. Integrating gives the potential at that segment's end minus the potential at its start. Summing over segments cancels the intermediate endpoints. Negating the result gives the stated sign convention.

□

The proof is short because the important work is in the declaration. We have supplied a state function, the path lies in its domain, and the derivative is regular enough for the fundamental theorem of calculus. We have not started with an arbitrary vector field and tried to prove that a potential exists. That distinction removes a frequently misplaced topological condition: *no simply connected domain is required when the single-valued potential is already supplied*. A hole in the domain does not invalidate the chain rule for a path on which V is defined.

If instead someone gives only a vector field and asks whether it is a gradient, global compatibility becomes a separate problem. Vanishing local curl alone can fail on a domain with holes. That is not the problem solved by defining EBU through an already declared V .

14.3 What the local differential says

Write

$$dE = -dV = -\nabla V(x)^\top dx. \quad (14.2)$$

This is an infinitesimal relation along the fixed field. It says how a tiny represented displacement contributes to the full comparison. It does not say that a finite displacement can be priced by one derivative sample. For an edge increment $dx = S_e dq$, it becomes $dE = f_e(x) dq$, where $f_e = -\nabla V^\top S_e$.

A useful analogy is a meter whose reading changes while a tank fills. Multiplying the first rate by the full duration works only when the relevant rate stays constant. Integrating changing readings is not an optional correction. It is the calculation needed to describe the whole event.

The path parameter s is not necessarily clock time. Replacing it by a smooth increasing parameter changes both the velocity and the integration element, leaving their product and the integral unchanged. Thus the mathematical

account need not depend on how quickly an interpolation is traversed. A physical consequence that does depend on speed, such as heating or spoilage, requires a state and action model that actually represents it.

14.4 The same endpoints do not mean the same operation

Two pumps can transfer the same water while using different electricity. A potential over tank stocks assigns equal values to their matching water endpoints. A potential over the fuller carrier state can distinguish their energy consequences. The identity remains exact in either case; the boundary determines which comparison it makes.

Time introduces another distinction. Along a physical trajectory $x(t)$ on $[0, T_f]$,

$$-\int_0^{T_f} \nabla V(x(t))^T \dot{x}(t) dt = V(x(0)) - V(x(T_f)), \quad \mathcal{A} = \int_0^{T_f} V(x(t)) dt. \quad (14.3)$$

The first expression integrates the *rate of change* of the potential. The second integrates its *level*. They are different functions of a trajectory and have different units: $\mathcal{A}$ includes an extra unit of time. Two deliveries reaching the same endpoint can have equal EBU while one leaves the clinic short of water for longer.

Feedback makes this distinction especially valuable. Overshoot, settling time, peak shortage and cumulative exposure describe different aspects of service. A controller's oscillation can affect all of them without producing an extra transaction or changing the endpoint definition. A delivery policy can therefore be compared on both restoration and continuity of service, with neither measure hidden inside the other.

For EBU, path independence gives a concrete protection: changing the book-keeping route between fixed complete endpoints cannot manufacture additional total value. The separate time description gives a complementary benefit: a system can value prompt, reliable service without mislabelling duration as another endpoint EBU. These two capabilities support clearer comparisons of maintenance and route choices.

14.5 An integral checked against the endpoints

For the familiar two-cell transfer, $V(q) = 16 - 4q + q^2/4$ and $f(q) = 4 - q/2$. A four-unit movement therefore has

$$E = \int_0^4 (4 - q/2) dq = 16 - 4 = 12. \quad (14.4)$$

Direct evaluation gives the same answer from $V(0) = 16$ and $V(4) = 4$. A two-stage physical execution gives seven for the first two units and five for the next two. The changing contributions sum to twelve because their baselines join.

The agreement is more informative than a numerical coincidence. If a software quote returns sixteen, it probably kept the first field value for the whole movement. If it returns fourteen by adding two same-baseline two-unit quotes, it probably treated two counterfactual beginnings as one physical history. The endpoint identity gives an independent way to recognize both mistakes.

14.6 A path that is declared is not necessarily observed

A straight segment between the endpoints is often convenient. On a convex stock domain it stays within that domain. But it is a physical trajectory only if the declared mechanism supplies that timing and simultaneous flow. Endpoint measurements alone cannot prove that every intermediate point occurred.

For the total, every admissible path satisfying the theorem gives the same result. For a decomposition among children or actors, the chosen path can matter. We will therefore calculate the joint outcome and its interaction structure before treating a path split as an attribution rule. A convenient interpolation must remain visible as a convention.

14.7 Regularity is part of the theorem

The C^1 assumptions above are sufficient and easy to check. A weaker formulation can use absolute continuity of $V \circ \gamma$ together with its almost-everywhere chain rule and an integrable derivative. Merely having a derivative almost everywhere is not enough: singular functions can change while their derivative is zero almost everywhere. Appendix A explains the corresponding issue for repeated finite differences.

A jump retains its endpoint EBU whenever both potentials exist, even when its actual mechanism has no smooth-path description. The finite account therefore survives the boundary of the differential representation. The next chapter evaluates it exactly for a quadratic, including the complete curvature correction.

Finite changes and curvature

The finite account contains the whole movement, including the point at which further delivery ceases to help. For a quadratic potential this correction is exact and simple. We use it to understand oversupply, shared-source interaction and why splitting one event cannot remove its curvature.

15.1 The whole movement matters

A slope describes the landscape at one point. A finite action travels a distance through that landscape. If the slope changes along the way, its starting value multiplied by the distance cannot give the exact endpoint difference. The Gaussian model makes this distinction unusually transparent because the omitted term can be written exactly. That exactness is useful operationally: a quote can display how its value changes with quantity, allowing a permitted delivery to be assessed before its size is fixed.

Return to the two-cell state (18, 2). The starting edge field is four. A four-unit transfer therefore has initial linear estimate $4 \cdot 4 = 16$. Its exact field reduction is twelve. An eight-unit transfer has linear estimate 32 but exact reduction sixteen. At sixteen units the linear estimate is 64, while the exact reduction is zero. No measurement noise or numerical integration is needed to explain these differences. They follow from the curvature of the declared potential.

This chapter retains the original book's lesson about overshoot while deriving it directly for finite quantities. We are not choosing a timestep for a restoring controller. We are evaluating a declared physical increment exactly.

15.2 Expand one square first

For one coordinate, write its initial deviation as $a = x_i - x_i^*$ and its physical increment as d . Then

$$(a + d)^2 = a^2 + 2ad + d^2. \quad (15.1)$$

Dividing by $2\sigma_i^2$ gives

$$V_i(x_i + d) - V_i(x_i) = \frac{x_i - x_i^*}{\sigma_i^2} d + \frac{d^2}{2\sigma_i^2}. \quad (15.2)$$

The first term is marginal times increment. The second is the curvature contribution. It is nonnegative and vanishes only when $d = 0$, given a finite positive scale.

This is exact expansion, not an approximation truncated after a convenient term. A quadratic has no cubic or higher terms to omit. That fact makes the Gaussian domain a clear introductory setting: the finite correction can be derived with the algebra of a square before using vector calculus.

15.3 Add the coordinates

Let δ be any declared finite state increment. Sum the one-coordinate expansions:

$$V(x^b + \delta) = V(x^b) + \mu(x^b)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (15.3)$$

The dot product $\mu^T \delta$ is the initial linear contribution. The quadratic form $\delta^T H \delta$ equals $\sum_i \delta_i^2 / \sigma_i^2$. The word “form” simply describes this matrix expression producing a scalar.

Define the finite EBU as $E = V(x^b) - V(x^b + \delta)$. In the calculations of this chapter we also write $F = E$ when emphasizing the field integral. Then

$$F = -\mu(x^b)^T \delta - \frac{1}{2} \delta^T H \delta. \quad (15.4)$$

The sign reverses because improvement means lower potential. A beneficial initial direction makes the first term positive. Curvature subtracts from that initial linear estimate.

Exact quadratic finite-change identity

For fixed references, fixed positive scales, and the declared additive Gaussian potential, every finite increment satisfies Equation 15.3. Therefore exact field reduction equals the initial linear improvement minus one-half the squared standardized increment. Physical feasibility is a separate requirement on the increment and event.

15.4 Specialize to one lossless edge

For $i \rightarrow j$, the increment is $\delta_i = -q$, $\delta_j = q$, with every other entry zero. Substituting into the exact identity gives

$$F_e(q) = q(\mu_i(x_i^b) - \mu_j(x_j^b)) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (15.5)$$

Name the positive directional curvature

$$k_e = \frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} > 0. \tag{15.6}$$

Then the schedule is the simple parabola $F_e(q) = f_e(0)q - k_e q^2/2$. This notation makes the roles visible: $f_e(0)$ is the initial benefit per stock unit, while k_e tells how quickly that benefit falls as more is sent.

For the main fixture, $f_e(0) = 4$ and $k_e = 1/2$. Thus

$$F(q) = 4q - \frac{q^2}{4}. \tag{15.7}$$

The physical source limit for this isolated lossless transfer is $0 \leq q \leq 18$. The algebraic parabola can be evaluated outside that interval, but such values are not physically executable points of the declared event.

15.5 Read the complete schedule

<i>q</i>	A	B	After-potential	Field reduction
0	18	2	16	0
2	16	4	9	7
4	14	6	4	12
6	12	8	1	15
8	10	10	0	16
12	6	14	4	12
16	2	18	16	0
17	1	19	20.25	-4.25
18	0	20	25	-9

Every row is an independent counterfactual from the same baseline, not the next step of a trajectory. That distinction is essential. The $q = 6$ row means send six units from (18,2), not send six after the $q = 4$ row has already happened.

Potential initially falls as the action moves toward the reference. It reaches its minimum at eight. Further movement crosses the reference and climbs the other side. The total field reduction remains positive until sixteen because the final potential is still below the original 16. Positive total value therefore does not imply that every part of the path is downhill.

15.6 The maximum of a schedule is not a selector

Differentiating the cost-free field schedule gives

$$F'_e(q) = f_e(0) - k_e q. \quad (15.8)$$

If $f_e(0) > 0$, its unconstrained vertex is at $q_v = f_e(0)/k_e$. In the main fixture this is eight. The positive root other than zero is $q_0 = 2f_e(0)/k_e = 16$.

These quantities describe the function. They do not instruct the actor to choose q_v . A physical cap may make the vertex unavailable. A service obligation may require another quantity. Naming the maximum allows us to understand the geometry without quietly installing optimization as a physical law.

If the initial field is nonpositive, every positive quantity has negative field reduction in this quadratic lossless model. At zero initial field, curvature alone gives $-k_e q^2/2$. At negative initial field, both terms are negative. This is a conditional mathematical statement about one fixed direction, not a claim that the action must be morally prohibited.

15.7 An integral gives the same finite difference

The field along the path is $f_e(s) = f_e(0) - k_e s$. Integrating from zero to the finite quantity gives

$$\int_0^q f_e(s) ds = f_e(0)q - \frac{1}{2}k_e q^2 = F_e(q). \quad (15.9)$$

This is the fundamental theorem of calculus in a particularly simple form. The area under the changing field equals the endpoint potential reduction. It is not a different competing definition of single-action value.

For $q = 4$, the field falls from four to two. Its average is three, so the integral is $3 \cdot 4 = 12$. For $q = 8$, the average is two and the integral is sixteen. For $q = 16$, the field falls from four to minus four; positive and negative signed areas cancel.

The integral uses a mathematical path through state space. For an isolated linear transfer whose physical model realizes constant-rate motion, that path can also describe the modeled physical progression. If only endpoints have been declared, the path is an interpolation used to derive the identity. A mathematical identity alone does not establish which physical timing occurred.

15.8 Average, initial, and final slopes

For $q > 0$, the average field is

$$\bar{f}(q) = \frac{F_e(q)}{q} = f_e(0) - \frac{1}{2}k_e q. \quad (15.10)$$

It lies halfway between the initial and final fields because the field is linear along this quadratic path. The exact total is quantity times this average, not quantity times the initial value.

At the main $q = 12$ point, initial field is four, final field is minus two, average field is one, and total reduction is twelve. The final field is already negative even though the total remains positive. This distinction helps interpret an action that has passed the potential minimum: the action as a whole still improves on its starting state, but its last increments undo some earlier improvement.

An intensity diagnostic such as $F(q)/q$ can be informative, but it is not the total finite value. Ranking by intensity and ranking by total answer different questions. Each ranking becomes a selection rule only when the policy explicitly uses it.

15.9 At the reference every nonzero net change costs field value

At $x^b = x^*$, the gradient is zero. Equation 15.4 becomes

$$F = -\frac{1}{2}\delta^T H \delta. \quad (15.11)$$

It is negative for every nonzero net increment. This follows from positive definiteness of H . The reference is a strict minimum of the potential.

The phrase “net increment” matters for groups. Several nonzero actions can cancel so that their sum is zero. In the declared state the group’s endpoint then equals its baseline and its EBU is zero. It would be false to infer negative EBU merely from the presence of nonzero child actions. If their operation changes another represented physical carrier, that carrier belongs in the complete state and the potential must be evaluated there.

The negative sign does not itself forbid departure. A clinic may need to draw from a well-placed reserve to treat a patient. Physical permission and service are separate questions from the potential difference. An institution that makes a negative EBU unaffordable adds a rule of its own; the geometry neither requires nor justifies that rule.

15.10 Two coordinates with unequal scales

Use a separate sensitivity example with the main stocks and references but scales (2, 4). Initial potential is ten, field is 2.5, and directional curvature is $1/4 + 1/16 = 5/16$. Therefore

$$F(q) = \frac{5}{2}q - \frac{5}{32}q^2. \quad (15.12)$$

The vertex remains eight in this two-cell compatible setup, because a transfer of eight reaches both references simultaneously. The maximum is ten, the entire starting potential. At four units, the reduction is $10 - 2.5 = 7.5$, and the after-potential is 2.5.

The unchanged vertex should not be generalized to all networks. Here there is only one mass-conserving degree of freedom, and both deviations are opposite amounts. In a multi-cell world an action can improve one pair while leaving other deviations untouched. Different scales then affect tradeoffs among available directions and finite quantities.

15.11 Several actions introduce cross terms

For two additive increments δ_a and δ_b , the net increment is their sum. Expanding its quadratic term gives

$$(\delta_a + \delta_b)^T H (\delta_a + \delta_b) = \delta_a^T H \delta_a + 2\delta_a^T H \delta_b + \delta_b^T H \delta_b. \quad (15.13)$$

Consequently the joint field reduction differs from the sum of standalone reductions evaluated at one baseline by

$$F_{ab} - F_a - F_b = -\delta_a^T H \delta_b. \quad (15.14)$$

This previews the group chapters. The cross term is present even though the potential is separable. The overlap occurs because both actions change a shared coordinate.

If both remove stock from the same source and change disjoint destinations, their source components have the same sign, giving a positive inner product and a negative correction. If one adds to a coordinate while the other removes, that shared contribution can have the opposite sign. No universal assertion of positive synergy follows from the existence of a group.

This is a comparison of fixed increments at the same baseline. If each hypothetical subset reruns a resolver and gets new quantities, it defines another set of comparisons. The simple fixed-increment expansion must not be used to describe it without checking those new dependencies.

15.12 Worked shared-source correction

Use a new three-cell declaration: state (14, 3, 3), reference (8, 6, 6), scales two at every cell, and lossless edges from the source to both destinations. Initial potential is

$$\frac{1}{2} \left(3^2 + (-1.5)^2 + (-1.5)^2 \right) = 6.75. \quad (15.15)$$

Each actor proposes quantity two. Alone, either action has initial field 2.25 and directional curvature $1/2$, so its standalone reduction is $2.25(2) - (2)^2/4 = 3.5$. The two standalone values sum to seven.

Together the increment is $(-4, 2, 2)$, giving endpoint (10, 5, 5). Standardized deviations there are $(1, -0.5, -0.5)$, so potential is 0.75. Exact group reduction is $6.75 - 0.75 = 6$, not seven.

The cross-term calculation gives the same correction. The action increments are $\delta_a = (-2, 2, 0)$ and $\delta_b = (-2, 0, 2)$. With $H = I/4$, their weighted inner product is one, entirely from the common source. Thus $F_{ab} = 3.5 + 3.5 - 1 = 6$. No potential coupling between different cells was needed.

This group is physically possible under an ordinary frozen-source availability rule because aggregate withdrawal four does not exceed fourteen. Physical feasibility therefore does not solve the separate overcounting problem. Each check has its own job.

15.13 Worked cancellation correction

At the main baseline, suppose both directed lossless edges exist. Let action A send four from A to B, and action B send four in the reverse direction. The reverse standalone action is not feasible from B's two units under a frozen-source rule, so its same-baseline value must not be presented as an admissible physical comparator there.

To make the comparison legitimate, declare a separate baseline (10, 10), with the same reference and scales. Both four-unit standalone transfers are now physically feasible. Each alone raises potential from zero to four, so each has field value minus four. The joint cancelling group has net increment zero and field value zero.

The same-baseline correction is therefore $0 - (-4 - 4) = 8$. It is positive because the hypothetical standalone endpoints each depart from the reference while the group endpoint does not. This does not identify an extra eight units to issue in addition to the group value. It is a diagnostic difference between two different accounting comparisons.

The explicit baseline change in this example is essential. Without it, an appealing algebraic illustration would depend on a standalone action that

cannot be performed in the declared physical world. Mathematical formulas must retain the admissibility assumptions of the comparisons they claim to represent.

15.14 A fuller physical account changes the state

A complete increment can include source stock, pending stock, usable delivery, an energy carrier and a loss compartment. The same quadratic identity applies to the resulting vector whenever the declared potential is quadratic. It is the actual increment, not the familiar name of the action, that enters the calculation.

Suppose dispatch initially changes only source and pending coordinates. Valuing it as though all pending water had already arrived would use a different endpoint. One may instead define a complete delivery action evaluated at an agreed horizon; its response model then determines how much has arrived by that time. The finite difference is unchanged in form, but its inputs now describe the promised physical comparison accurately.

This is also where the later generator theory joins the account. A common linear system with additive action forcing produces endpoint increments that superpose. A quadratic potential then retains exactly its singleton and pair terms, even if those increments were generated by an oscillating response. Nonlinear dispatch, saturation or a coalition-dependent controller can change the endpoint map and create higher orders. The distinction identifies where a more complex calculation is necessary, instead of treating every delayed service as an arbitrary nonlinear surcharge.

15.15 Why dividing the action does not erase curvature

Take the main four-unit transfer and split it into two consecutive two-unit actions under one fixed potential and no intervening external change. The first action goes from potential 16 to nine, giving seven. The second begins at the new state (16,4) and goes from nine to four, giving five. Their sum is twelve, the unsplit four-unit field reduction.

If both pieces were instead valued independently from the original state, each would claim seven, for a total of fourteen. That is a different calculation and overstates the actual four-unit endpoint reduction. The second piece has been allowed to reuse the first piece's initial field.

This example distinguishes sequential telescoping from independent same-baseline claims. The correction is not an arbitrary penalty for using two names. It is the mathematical consequence of evaluating the second physical

change where it actually begins. Simultaneous groups require a joint value and a separately declared attribution convention; those are developed later.

15.16 Guided problem: derive an entire schedule

Use a separate lossless two-cell example with total 16, state (12, 4), reference (8, 8), and equal scales two. Initial marginals are one and minus one, so the field is two. Directional curvature is one-half. The exact schedule is

$$F(q) = 2q - \frac{q^2}{4}, \quad 0 \leq q \leq 12. \quad (15.16)$$

At $q = 2$, field reduction is three; the endpoint (10, 6) has potential one, compared with initial four. At $q = 4$, the reference is reached and reduction is four. At $q = 8$, the endpoint is (4, 12) and reduction is zero. At $q = 10$, reduction is $20 - 25 = -5$.

Now perform the same exercise by endpoint evaluation alone. This independent form of arithmetic verifies signs and normalization. Finally integrate the field $2 - q/2$ from zero to each quantity. The three methods agree because they are three expressions of the same quadratic identity, not three separate scientific findings.

15.17 Practice and reflection

For the main schedule, solve $F(q) = 12$. The two roots are four and twelve. Both provide the same total field reduction but leave different endpoint distributions, (14, 6) and (6, 14). Explain why a value alone cannot identify what physically happened.

Calculate the difference between the linear estimate and exact field reduction at $q = 3$. It is $q^2/4 = 2.25$; the linear estimate is twelve and the exact reduction is 9.75. This difference is a curvature contribution, not an optional audit adjustment.

At the reference, let $\delta = (-1, 1)^T$ with scales two. The quadratic term is $\frac{1}{2}(1/4 + 1/4) = 1/4$, so the field value is minus one-quarter. Explain how a zero gradient and a negative finite value coexist without contradiction.

We now have an exact finite schedule, including its oversupply correction. Before using its favourable region, we must check whether the proposed quantity and its complete route are physically possible. That is the role of permission in the next chapter.

Permission before value

A useful number must refer to a possible action. Feasibility supplies the physical domain on which EBU comparisons and interaction theorems apply. We distinguish individual permission, shared resource limits and conditions along a trajectory, so that a mathematically attractive endpoint cannot conceal an impossible route.

16.1 A favourable number cannot create stock

The potential tells us what a specified state change would mean in the declared account. It does not make that change possible. A source holding three units cannot supply an isolated four-unit lossless transfer; a road that is closed cannot deliver medicine because the delivery would be useful. Before an EBU quote can describe an executable action, a separate physical question must be answered: *can this complete transformation occur now?*

For one isolated lossless edge $i \rightarrow j$, with nonnegative stock and no other bound, the answer requires $0 \leq q \leq x_i^b$. This simple interval already separates value from permission. From the state (18, 2), a transfer of seventeen units is possible under that physical rule even though it raises the potential and has negative EBU. A request for nineteen units can be entered into the quadratic formula, but it lies outside the event's physical domain. A mathematical value for an impossible endpoint is a counterfactual, not a permit.

16.2 A group can fail even when each child passes

Suppose one source holds eight units and two actions each request five. Either action can pass an isolated source check. Their simultaneous group cannot pass a frozen-source rule because its aggregate request is ten. For that rule the necessary condition is

$$\sum_{a: \text{src}(a)=i} q_a \leq x_i^b. \quad (16.1)$$

The same principle applies to shared pump capacity, receiving storage and any other physical limit that a group uses together. Conservation of the final total does not settle these limits.

Now imagine a chain in which the middle node begins empty, receives four units and sends four out during one event. Its final stock is zero, not negative. Yet a warehouse that must receive and inspect stock before dispatching it may refuse this event, while a continuously operating pipe could allow a declared flow-through process. The endpoint is identical; the physical timing rule is not. We cannot settle the difference by choosing the more attractive EBU calculation.

Three questions, not one

Physical feasibility asks what can occur. EBU asks how the complete accepted change alters the declared potential. Institutional authority asks who may decide and on what terms. A mathematical answer to one question does not silently answer the other two.

16.3 The domain of a consistent comparison

A physical model may admit a continuous interval of quantities while an institution offers a finite menu. The menu restricts available choices; it does not change the carrier's divisibility or the potential. Choosing the largest EBU, prioritising urgent service and selecting randomly among permitted plans are separate decision rules over that menu.

Coalition analysis adds a further domain requirement. To calculate an interaction for a set of actions, every subset appearing in its finite difference must have a defined response under the same background conditions. If a subset is infeasible, one cannot silently fill its missing value with zero. One must restrict the comparison or explicitly define the resolver that determines each subset's accepted outcome.

For a response through time, nonnegative endpoints alone may be insufficient. Stock, capacity or branch constraints may have to hold at every intermediate time. A linear differential equation can have a perfectly well-defined mathematical solution outside the physical domain. Its use as an EBU response therefore includes a proof or declaration of admissibility on the chosen interval.

These conditions enable a useful system feature: comparable proposals can be evaluated within one explicit feasible set. Shared stock is reserved once, pending delivery is not spent early, and a favourable score cannot override a physical limit. This makes coordination more efficient while protecting the continuity of essential services.

16.4 What a permission record must say

For an accepted action, record the predecessor state and its epoch, route or other physical action identifier, requested and accepted quantities, shared resources, governing constraints and the resulting complete increment. If a request was reduced, the rule that reduced it belongs in the record. If only endpoints were checked, the record must not imply that transient path constraints were checked as well.

This discipline leaves us with a physically meaningful candidate for valuation. The next chapter can now calculate its exact EBU without asking the potential to perform the job of a route, a stock certificate or a human decision-maker.

The exact local signed EBU quote

We now combine boundary, feasibility and finite valuation into an exact quote. Its most useful property is that unchanged parts of the potential cancel, so a complete local computation can equal the global difference. This supplies a practical foundation for an account that many sites can use without recomputing the entire network for every delivery.

17.1 The action finally has a number

We have taken care to identify a physical state, a reference, a potential and an action that can actually occur. Only now is it safe to attach a number to the action. The number is not the quantity delivered, the price paid or the initial slope of the potential. It is the complete change in the declared potential between the accepted predecessor and successor.

Let x^b be the state immediately before an action, let q be its finite source-side quantity, and let S_e be the declared increment per unit of that quantity. Under one fixed field V ,

$$E_e(q) = V(x^b) - V(x^b + S_e q). \quad (17.1)$$

Positive E_e means the represented potential fell. Negative E_e means it rose. Zero means the two endpoints have equal potential; it does not prove that the action was empty. The complete increment and the version of V belong beside the number on any trustworthy quote.

The finite definition

For a fixed, declared potential, exact finite EBU is before-potential minus after-potential for the complete accepted physical change. It is not defined by the local gradient approximation, a monetary price or an extra charge invented after the event.

17.2 Why a local computation can give the whole answer

Suppose the potential is a sum of factors, $V = \sum_{\alpha} \phi_{\alpha}$, each with a declared set of input coordinates. An action changes only some of those coordinates. In

$V(x^b) - V(x^b + S_e q)$ every factor whose inputs are unchanged cancels exactly. Thus a local evaluator needs the changed factors and their complete inputs, not a new calculation over every untouched place in the system. This is the precise mathematical answer to the opening question of how a local action can be assessed without a global recalculation each time.

Local evaluation does not license incomplete observation. If a factor couples the source to a downstream or environmental coordinate, that entire factor must be reevaluated. If an action changes an energy carrier as well as water, the complete physical increment must say so. Computational locality follows from a declared factorization; it is not a promise that the physical world has no long-range dependencies.

17.3 The Gaussian expression is a theorem below the definition

For the quadratic potential of Chapter 12, write $H = \text{diag}(1/\sigma_i^2)$ and $\mu(x^b) = H(x^b - x^*)$. Exact expansion of the squares gives

$$E_e(q) = -q \mu(x^b)^T S_e - \frac{q^2}{2} S_e^T H S_e. \quad (17.2)$$

The first term uses the initial slope. The second tells us how the slope changes over the finite action. No small- q approximation has been made. For a lossless transfer from i to j , $S_e = -e_i + e_j$, and the expression becomes

$$E_e(q) = q(\mu_i - \mu_j) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (17.3)$$

This edge polynomial is a corollary for this physical increment and this quadratic potential. It is not the general definition of EBU. A lossy route, a valued sink or a coupled potential changes the specialized formula while leaving Equation 17.1 intact.

17.4 One calculation with three independent readings

Return to the lossless two-cell state (18, 2), reference (10, 10) and scales (2, 2). A four-unit transfer reaches (14, 6). Direct evaluation gives $V_{\text{pre}} = 16$ and $V_{\text{post}} = 4$, hence $E = 12$.

The Gaussian expansion gives the same result: the initial edge field is 4 per stock unit and the directional curvature is $1/2$, so $E = 4(4) - (1/2)(4^2)/2 = 12$. Along the linear evaluation path the field falls from 4 to 2; its average is 3, and integrating over four units again gives 12. These are three derivations of one number, not three independent claims of value.

The three readings give complementary ways to inspect a quote: endpoints establish the finite total, curvature explains the correction, and the integral explains its accumulation. Alongside the quote, permission describes availability, the event record describes occurrence, and a service condition describes what the clinic received. A well-designed account presents these facts together, allowing a reader to understand both the number and its practical consequence.

17.5 The cost of an omitted carrier

If a pump consumes electricity and that consumption matters to the claim, one may extend the physical state with an energy coordinate and declare the pump's full transition. The potential over that extended state then evaluates its chosen consequences. If some loss crosses the model boundary, the boundary crossing must be explicit. By contrast, adding a number called "process burden" to the right of Equation 17.1 would create a different account. It cannot silently redefine the physical EBU.

There is room for engineering cost ledgers and institutional charges. Their units, provenance and decision role should be written alongside EBU, not confused with it. That separation lets a reader ask which question each number answers and prevents the same physical loss from being counted twice.

17.6 From a quote to a record

A quote specifies the predecessor, fixed potential and complete successor. An immediate transfer gives $x^b + S_e q$; a dynamic service gives the endpoint $T_{T_f}(S, x^b)$ at its declared horizon. The finite definition values both.

Water arriving after that horizon remains pending. If the background moves, a no-action comparison can identify the incremental effect; Chapter 25 specifies that baseline for each subgroup.

The receipt then identifies what actually occurred. Changed quantities or conditions can change the value, while the original quote remains available for comparison. Joint valuation fixes the total before attribution. The next chapter follows this passage from a calculable proposal to an identified outcome.

From a pre-action schedule to a measured record

A quote describes a possible event; a receipt describes an identified outcome. This chapter connects the two without losing the original promise, the physical boundary or the time at which the consequence is evaluated. The same discipline also prevents the continuing response of a controller from being counted repeatedly as fresh actor activity.

18.1 A promise is not yet a physical event

An operator proposes to send four stock units from cell 1 to cell 2. Before anything moves, we can calculate what that action would mean under the frozen model. Afterward, a meter may show that only three units moved. The proposed quantity, permitted quantity and observed quantity answer different questions. Keeping all three is part of a useful account.

The preceding chapter gave an exact value for a finite action. Here we ask what information makes that equation reproducible across time. The object of interest is still a physical change. A message requesting a transfer does not move the water. A signed agreement does not prove delivery. A correct calculation of a planned successor does not prove that the successor occurred.

The progression from quote to receipt connects a proposed physical outcome to the event that occurred. A later settlement rule can use that record to change an institutional balance. Keeping the stages distinct makes both the physical promise and its institutional consequence understandable.

Three quantities in one story

The request says what an actor asks to do. The accepted domain says what the declared rules permit. The measured record says what the evidence supports as having occurred. A scientifically useful record never silently substitutes one for another.

18.2 Freeze the whole function, not one attractive point

For the recurring two-cell example, let

$$x^b = (18, 2), \quad x^* = (10, 10), \quad \sigma = (2, 2).$$

A lossless transfer of quantity q changes the state to $(18 - q, 2 + q)$. Its exact schedule is

$$E(q) = 4q - \frac{q^2}{4}. \quad (18.1)$$

If the accepted domain is $0 \leq q \leq 4$, the quoted value at full execution is $E(4) = 12$. The equation also specifies every smaller accepted quantity. At $q = 3$ it gives $12 - 9/4 = 9.75$. That is the value of the three-unit event on this schedule.

Multiplying the full quote by the fraction executed would give $12(3/4) = 9$, which is different. The first three units and the fourth unit do not have the same marginal effect. We can see this without calculus: the three-unit successor is $(15, 5)$, its potential is $25/4 = 6.25$, and $16 - 6.25 = 9.75$. The direct endpoint calculation agrees with the function.

This is why a pre-action account should identify its whole domain and formula. A few selected values may illustrate the formula, but they do not replace it. A table can represent an exact schedule only if its interpolation rule is also justified. Connecting points with straight lines usually changes a curved function.

18.3 The information held fixed

To reproduce the schedule, the reader needs the complete baseline state, coordinate ordering, action map, reference vector, positive scales, physical boundary, units and accepted domain. A version label should identify the equation and those definitions. A timestamp alone is insufficient: two records at the same time can describe different fields.

The scope must also say whether q is quantity or rate. Here q is a quantity. If a meter reports a rate j over duration Δt , then $q = \Delta t j$ when the rate is constant. A variable rate requires its time integral. Replacing a reported average rate with an instantaneous endpoint reading can change the quantity, even if the numerical values look similar.

For a group, the record needs the accepted vector, the joint successor and the path convention if child attributions are displayed. A child's measured short-fall can change the group's field value and the attributions of other children. It cannot always be handled by independently scaling one child's old number. Chapter 24 explains the mathematical reason.

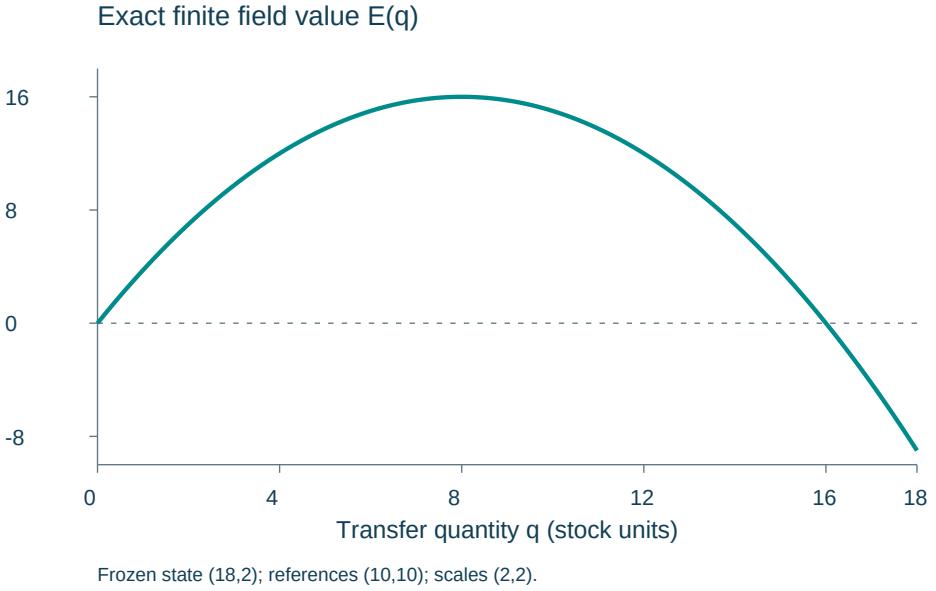


Figure 18.1: The full finite-value curve for the running two-cell example. A positive initial slope does not make every finite quantity beneficial.

18.4 The shared baseline removes an external change

Suppose an external, lossless redistribution changes the physical state from (16, 4) to (18, 2) before the actor's epoch. Under our fixed potential,

$$V(16, 4) = 9, \quad V(18, 2) = 16.$$

The external change increased represented deviation by seven. The actor then transfers four units, ending at (14, 6) with $V = 4$. The actor's field value is $16 - 4 = 12$.

Using the earlier state as the actor's baseline would give $9 - 4 = 5$. It would mix the external worsening with the actor's improvement. Neither number is arithmetically mysterious; they answer different questions. The external record is +7 in potential change, while the action record is +12 in avoided potential. Together they explain why the complete interval changed potential by $4 - 9 = -5$.

Write the general relationship with pre-forcing state x , frozen post-forcing baseline x^b and action successor y :

$$D_{\text{ext}} = V(x^b) - V(x), \tag{18.2}$$

$$E = V(x^b) - V(y), \tag{18.3}$$

$$V(x) - V(y) = E - D_{\text{ext}}. \tag{18.4}$$

The subtraction makes the attribution boundary explicit. Calling the external change a disturbance does not require it to increase burden. It can also reduce burden, in which case D_{ext} is negative.

No-action neutrality

At the shared baseline, $E(0) = V(x^b) - V(x^b) = 0$. An idle actor receives no field reduction from an external event merely because that event preceded the observation. This is an algebraic statement within the declared comparison.

18.5 An epoch has a beginning and an expiry

A quote is valid under a declared state and field version. Its epoch tells participants which observation and rule produced it; its expiry specifies when those conditions must be refreshed. A later change of stock, route availability or potential does not retroactively alter the old proposal. It changes the conditions for a new one.

A finite-horizon quote additionally fixes the start and end of its response interval. If a pump action is valued after T_f , the endpoint includes usable stock, pending delivery and every other represented consequence at that time. The account should not move the horizon after seeing which oscillation peak gives the largest number.

A controller may continue adjusting after the action is registered. That continuation is part of the declared response and its background ledger. It does not create new actor receipts merely because the potential rises and falls. If further actions are registered, they receive their own identities and comparisons. Otherwise settlement follows the original action's declared boundary.

This feature is essential for an efficient EBU system: one physical intervention supports one defined claim, even when its consequences unfold over many observations. Continuous monitoring can improve service without turning every sensor update into another charge or credit.

18.6 Shortfall, overexecution and missing evidence

For a measured quantity within the valid domain, the committed schedule can be evaluated at that quantity when its other assumptions remain satisfied. Our three-unit example is such a case. It does not require a new potential or a proportional approximation.

If the accepted domain ended at four but the meter reports five, the physical report must retain five. A rule limiting accepted quantity cannot erase an extra unit from the reservoir. At the same time, the earlier commitment did not authorize five. Describing the physical consequence and deciding the institutional response are separate tasks.

The mathematical function may be evaluable beyond its accepted domain, but mathematical evaluability is not permission. A retrospective diagnostic can explicitly calculate the consequence of overexecution under the frozen field. It must not be mislabeled as an authorized receipt or a precommitted positive credit.

If the sensor is missing or inconsistent, “not established” is a possible outcome. Zero is a measured or derived value, not a universal substitute for uncertainty. Crediting the full request would invent execution; charging an arbitrary negative amount would invent a physical burden. Any sanctions or compensation need their own declared basis.

18.7 Identity prevents counting the same change twice

One transfer can appear in the source meter, the destination meter, a pump controller and a transporter’s report. Those records can corroborate one another. They do not automatically describe four actions. A stable event identity connects them to one physical transformation.

Conversely, a chain can contain several physical transformations under one invoice. Fuel use, electricity generation, delivery and water pumping must retain their physical boundaries where they are in scope. A commercial reference number can link the chain without collapsing its quantities or converting its resource units into one scalar.

Repeated actions can be distinguished by their execution intervals and predecessor states. Splitting an actual continuous movement into bookkeeping pieces must not let each piece claim the same original improvement. If the pieces are sequential, each uses the preceding piece’s successor. If they are simultaneous, the group account must reconcile their joint effect.

18.8 A hash protects a record, not a sensor

A digital fingerprint can help establish that the equation and inputs shown after an event match the committed bytes. It does not establish that the water existed, that a meter was calibrated, or that the declared reference was appropriate. Integrity of a file and truth of a measurement are different properties.

For a physical account, the evidence chain can include device identity, calibration records, location, time interval, unit conversion, independent observations and documented corrections. The appropriate set depends on the domain. A teaching tank does not prescribe a surveillance system for a hospital.

Records should explain uncertainty without burying it in the potential's scale. In the Gaussian model, σ_i defines the reference geometry. A sensor's uncertainty concerns knowledge of x_i . Treating those as the same parameter would change both the meaning of the field and the meaning of confidence in the observation.

18.9 Corrections preserve the earlier account

Suppose a later calibration finds that a meter overstated delivery. A correction should identify the earlier record, the evidence for the correction and the authorized correction procedure. Deleting the earlier line conceals what participants actually knew and acted on at the time.

A physical reversal is different. Returning water from B to A is a new action with a new predecessor state. It may reverse the earlier endpoint change, but it is not a correction to the fact that the first action occurred. Its value must be computed from the complete physical transition under the applicable fixed field.

Versioning the potential is also different from correcting a measurement. If new evidence justifies a different reference, the same unchanged state can acquire a different potential. That is a model revision. It needs a separate reconciliation term when accounts span the change; it is not physical restoration by the actor.

18.10 A complete worked action record

Consider the three-unit shortfall again. A readable record contains the following facts:

Item	Declared or observed content
Physical object	Lossless A-to-B transfer of one homogeneous stock
Baseline	$(18, 2)X$, ordered A then B
Field	Reference $(10, 10)X$; scales $(2, 2)X$
Domain	Accepted quantity between zero and four stock units
Schedule	$E(q) = 4q - q^2/4$ under the fixed field
Observation	Three stock units transferred; endpoint $(15, 5)X$
Calculation	Before 16; after 6.25; signed field value 9.75
Status	Illustrative arithmetic; no real event or spendable credit claimed

The record is longer than the number 9.75 because it explains what the number means. Someone can check the arithmetic without knowing the actor's income, preferences or occupational status. Whether that person is entitled to compensation remains a separate question.

18.11 Guided problem: a changed observation

Keep the same frozen field and accepted domain. The source reports that $2.5X$ left A; the destination reports that $2.5X$ arrived. Calculate the successor, its potential and the exact value. Then compare the result with scaling the full four-unit quote by the delivered fraction.

The successor is $(15.5, 4.5)X$. Each endpoint differs from its reference by $5.5X$ in magnitude. Thus $V = 2(1/2)(5.5/2)^2 = 7.5625$, and the field reduction is $16 - 7.5625 = 8.4375$. Equivalently, the schedule gives $4(2.5) - (2.5)^2/4 = 8.4375$.

Proportional scaling would give $12(2.5/4) = 7.5$. It is smaller because it assigns the fourth unit's lower marginal contribution to earlier units as well. The endpoint equation avoids that substitution. The two meters agree on conservation in this illustration; agreement alone does not establish their calibration in a real application.

18.12 Practice and reflection

First, evaluate $E(1)$, $E(2)$ and $E(4)$. The answers are 3.75, 7 and 12. Explain why doubling quantity from one to two does not double value. Next, suppose an external event takes $(18, 2)$ to $(16, 4)$ before a two-unit action. The new baseline has $V = 9$; the action ends at $(14, 6)$ with $V = 4$, so the action value is 5. The external potential change was -7 . Neither should be replaced by the other.

Finally, write an example of a missing measurement, a physical reversal and a field-version change. For each, name the record that must be retained and the new information that must be added. The purpose is to recognize distinct events before assigning numbers to them.

What to carry forward

A schedule is a conditional account over a declared domain. Verified execution supplies the point to evaluate. External changes, expired conditions, missing evidence and later corrections remain visible. The equation determines the field difference; an institution must separately define the consequences of recording it.

The next chapter follows these signed values into the real-needs examples that motivated the book. Understanding a positive number is incomplete until we can also explain a zero or a negative number without confusing the action with the person it serves.

Positive, zero and negative EBU in real needs

A signed EBU value describes an action in a particular physical state. It is not a ranking of human needs. By working through positive, zero and negative cases, we can see how an account can support necessary service, reveal recurring burdens and guide redesign without making access depend on the ability to generate a positive receipt.

19.1 The sign belongs to an action in a state

The same stock transfer can arrive where it is urgently needed, where stock is already well placed, or where delivery displaces something else. The name of the product does not determine the sign of its EBU. The declared before-state, complete physical transformation and fixed potential determine the calculation.

This statement is sometimes uncomfortable because everyday language uses “valuable” for several different things. A treatment can be valuable to a patient. A delivery can have a high market price. A redistribution can lower a particular stock potential. These statements may all be true, but they do not express the same relationship.

The examples in this chapter keep the physical account visible alongside service and human need. All numbers are invented teaching inputs. They are not clinical guidance, a valuation of a life or a claim that medicine can be represented adequately by two stocks.

Read the sign with its noun

Positive EBU means the declared fixed potential fell across the complete represented transition. It does not mean a good person, an adequate wage, a complete cure or a socially optimal decision. Negative values are equally specific.

19.2 One delivery, three starting states

Use two storage locations with compatible references $(10, 10)X$ and scales $(2, 2)X$. A delivery sends $2X$ from the first to the second, with perfect quantity retention in this ideal illustration. Any energy use would require an additional represented carrier and is outside this two-stock example.

Start first from $(18, 2)X$. The potential is 16; the successor $(16, 4)X$ has potential 9. The delivery's field value is $+7$. The clinic still has less than its declared reference, so a positive result does not mean all relevant needs have been met.

Now start from $(11, 9)X$. The potential is $1/4$. The same delivery produces $(9, 11)X$, also with potential $1/4$. Its field value is zero. Two physical stocks changed, and medicine reached the clinic, but the chosen quadratic assigns equal deviation to the endpoints.

Finally start from $(10, 10)X$. Potential is zero. The same delivery produces $(8, 12)X$, whose potential is 1. The field value is -1 . This is the simplest example of a permitted physical change that moves the modeled system away from its reference.

Before	After	V before/after	E
$(18, 2)$	$(16, 4)$	$16 \rightarrow 9$	$+7$
$(11, 9)$	$(9, 11)$	$1/4 \rightarrow 1/4$	0
$(10, 10)$	$(8, 12)$	$0 \rightarrow 1$	-1

The common action description is not enough to reproduce any of the three answers. The starting state is indispensable. So is the fixed field: using different references would ask a different question.

19.3 A zero is not an empty event

In the middle example, the source and destination exchange their deviations. The endpoint states differ, although their potentials agree. A scalar function assigns one number to a state containing several numbers; equal potential generally does not imply equal state.

This matters for interpretation. One endpoint may have a different available route, future need or exposure to failure. Those differences can change later opportunities even when current V is equal. They cannot be recovered from the scalar alone. A complete action record keeps the state, not merely its potential.

Zero can also arise when the total increment of a group cancels. Two opposing ideal transfers may leave the net stocks unchanged. That does not establish that real pumps consumed no electricity. If pumping energy is in scope, its transformation must be represented through the appropriate carrier or boundary. A richer state can therefore give a different EBU from the ideal two-stock calculation without adding an unrepresented charge to the equation.

19.4 The full quantity can reverse the initial sign

Return to (18, 2)X. A small A-to-B transfer improves the Gaussian field, but the exact schedule is

$$E(q) = 4q - q^2/4.$$

At $q = 8$, the endpoint is the reference and $E = 16$. At $q = 16$, the endpoint is (2, 18) and $E = 0$. At $q = 17$, the endpoint is (1, 19), whose potential is $81/4 = 20.25$, giving $E = -4.25$.

The positive initial slope did not promise that every amount in that direction would be beneficial. The transfer passed through the reference and continued until the full action increased deviation. A finite calculation includes the entire movement. A slope is a nearby statement.

One could design a controller that chooses $q = 8$ to maximize this particular schedule. That is an additional selection rule, not part of the definition of EBU. A different actor might choose an amount for a deadline, a random menu or another objective. This book teaches the measurement without silently building a maximizer into the word “field.”

19.5 An essential action can have negative field value

Suppose a patient needs a dose at the second location, while the simplified inventory reference currently places equal stock at both locations. The example starting at (10, 10) says a two-unit transfer has field value -1 . It does not tell us whether the patient should receive treatment.

The potential may not represent clinical timing, patient condition, dose compatibility, expiry or access. A reader who turns -1 into automatic refusal has supplied a social rule and a claim of model completeness that the equation never contained. A reader who changes the result to positive because the patient needs help has replaced the declared physical comparison with a different judgment.

An honest account can state both: the represented inventory deviation increased by one, and the delivery met a separately described urgent need. If the scientific model is inadequate for the intended decision, its scope must

be revised through explicit evidence and governance. The old event remains described by the field actually used.

An action is not a person

Do not say that a patient “is a burden of one EBU.” Say which delivery action changed which represented state under which field. A human need and a person’s claim to care are not coordinates in this two-tank demonstration.

19.6 Feasibility, value and service are separate tests

A movement can have positive hypothetical value while being physically impossible. If A contains only one unit, an instruction to remove two cannot be made valid by a favourable clinic term. Conservation and source availability constrain the action before its potential difference is treated as an executable opportunity.

A feasible action can fail a service requirement. A delivery arriving after a deadline may preserve a stock balance while failing its intended use. Time can be part of the represented state, a constraint or a separately reported outcome, but it cannot be omitted and then assumed satisfied.

A negative action can be feasible and successful on a service dimension. The three dimensions therefore form a small report, not a single ranking:

Physical status	Field status	Service interpretation
Feasible	Positive	Can improve the field while leaving demand unmet
Feasible	Zero	Can move goods without changing the chosen potential
Feasible	Negative	Can meet a need while worsening represented deviation
Infeasible	Any hypothetical sign	No executed candidate follows from the sign

The report avoids calling every dimension “success.” A later scientific comparison must declare which outcomes matter and how conclusions are drawn. Choosing that interpretation after seeing results would change the question being tested.

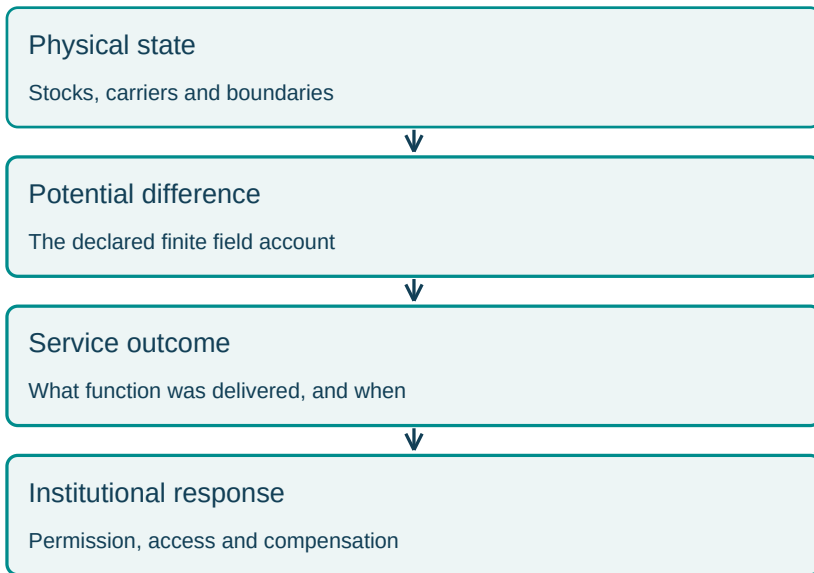


Figure 19.1: Physical stock, potential and an action account answer different questions. Their records must be linked without treating them as the same quantity.

19.7 A repair needs a represented physical consequence

A mechanic repairs a pump. A water-stock model that observes only the current litres may show no immediate change in x . Under that model, the stock field effect is zero. This is not evidence that repair is useless; the model lacks the equipment condition or other capacity coordinate needed to describe the benefit.

One option for future research is to represent a certified physical equipment state. That requires a definition, unit, measurement method and justified contribution to a potential. Merely writing “pump repaired” in a field does not supply these. Another option is to report the repair in a linked physical record while observing its later effects without issuing an invented current field reward.

The distinction also applies to diagnosis. Finding a leak provides useful information. If no physical coordinate in the model changes, a separate claim of positive physical field reduction needs an information or service model that actually defines such a change. Professional effort deserves consideration in compensation, but occupation and effort do not replace the state difference.

19.8 Five layers in an infrastructure decision

Consider building a shared cold store. First describe construction: materials, energy, land, transport and waste, with appropriate carriers and boundaries. Second describe what capability has been established and how it is measured. Third identify who has authority to approve the construction. Fourth determine who bears its institutional responsibilities. Fifth specify how workers and providers are compensated.

These five questions can refer to one project without becoming one equation. A high future service expectation does not erase present construction burden. Conversely, a negative present physical account does not show that investment should never occur. Long-term planning can consider future outcomes, but its forecast is not an observed action receipt.

A changed route or machine can also alter future action maps and efficiencies. That is a versioned physical and modeling change. The old route's receipts should remain interpretable under the old description; the new route's receipts should identify the new description. Rewriting old efficiencies would make improvement impossible to audit.

19.9 The rural clinic as a linked case

The clinic needs medicine, refrigeration, water and a working backup supply. In ordinary conversation these form one service. Physically they contain different carriers and transformations. The medicine transfer changes dose stocks; refrigeration transforms energy while maintaining temperature; water delivery changes water stocks; repair changes equipment condition when that state can be measured.

Linking these records preserves dependence: medicine can become unusable without refrigeration, and a clinic can cease functioning without water. Adding their numerical EBU values without a declared common calibration would hide differences of meaning. A small positive water account cannot automatically compensate for an unrepresented failure of medicine storage.

The teaching scalar is useful because it allows complete arithmetic. It is not useful if we forget why it is small. For a real clinic, the scientific task includes deciding which states and dependencies are necessary for the intended claim. The potential must be evaluated as a model of that task, not as a universal vocabulary of health.

19.10 Recurring burdens can inform redesign

An essential service can repeatedly consume a reserve or move a represented state away from its reference. Recording that burden honestly makes a design problem visible. It can point toward a more efficient device, a better route, a local reserve or a change in the timing of supply. Removing the negative line would conceal precisely the information needed to improve the service.

Feedback adds a useful distinction. A stable replenishment system can maintain a continuing shortage signal because that signal sustains dispatch against consumption. Its steady burden can be positive, and its burden can rise temporarily even while departures from the operating point decay. A stability calculation describes recovery under the controller; the finite EBU account describes the change of the declared potential. Chapter 33 gives an exact example in which these signs differ.

For a clinic, this suggests a constructive use of the account. Keep the water service available, record its complete resource consequences, and use the separation between delivery, pending stock and control response to locate avoidable burden. The efficiency improvement is then attached to a physical change—for example, less wasted pumping or fewer duplicate orders—while the institution preserves the patient’s access to care.

19.11 Guided problem: different needs at equal potential

Use the same field and compare $(14, 6)$ with $(6, 14)$. Both have potential 4. From each, consider a two-unit first-to-second transfer. The first ends at $(12, 8)$ with potential 1 and has value $+3$. The second ends at $(4, 16)$ with potential 9 and has value -5 .

The starting potentials were equal, but the local directions were different. This shows why recording only V loses decision-relevant state information. Now suppose the second location in both cases has a separate urgent deadline. That service fact does not change the arithmetic. It changes the complete interpretation and may reveal that the current reference model needs additional representation.

19.12 Practice and reflection

For $x^b = (12, 8)$, calculate a two-unit and a four-unit transfer. The initial potential is 1. The first transfer reaches $(10, 10)$ and gives $+1$; the second reaches $(8, 12)$ and gives zero. State a reason why the zero-value action might still have a separately measured service outcome.

Then describe a useful maintenance action whose immediate benefit is missing from a stock-only model. Identify the missing physical quantity, possible evidence for it, and the institutional question that would remain even after it was measured. Do not assign a number to the benefit without defining the new field.

Finally rewrite the sentence “This household caused five negative EBU” as an action-specific report. A good answer identifies the device or transport event, model version, physical cause, service delivered and measurement status. It avoids converting a receipt into a judgment of the household’s worth.

These signs make resource consequences legible without ranking the people who need the service. A larger system must compute them without reading every unchanged coordinate each time. The next chapter derives the exact locality that makes this possible.

Local actions and exact global identities

Local exactness is a cancellation theorem. It explains how a large EBU system can assess a small action efficiently, and it states precisely when more information must be included. The relevant neighbourhood follows the potential and the endpoint response, including hidden pending state, rather than geographical distance alone.

20.1 The apparent paradox

An action affecting only two cells can be valued using data from those cells. Yet we have described its field effect as a change in a potential summed across a whole network. How can a calculation that does not read the other locations give the complete result?

The answer is cancellation. Under the declared separable potential, every untouched location contributes exactly the same term before and after the action. When those terms are subtracted, they disappear. The local calculation is not a statistical estimate of the remaining world; it is the part of an exact difference that survives.

This distinction is central to the project. It explains how a local action account can be mathematically complete within a declared model. It also identifies the boundary of that claim: the model's potential must include the relevant effects, and the action's evaluation must include every term that actually changes.

20.2 Support means the coordinates changed

Let δ be an action's finite increment. Define its write support by

$$W = \{i : \delta_i \neq 0\}. \quad (20.1)$$

For a positive lossless transfer along $i \rightarrow j$, this set contains i and j . A group can change more coordinates. A proposed edge of zero quantity changes none, even though its name still refers to two endpoints.

Support is an algebraic description of the state change, not a prediction of every future consequence. Water arriving at B may enable a later transfer to

another place. That later event has its own increment and baseline. It does not automatically appear in the present endpoint difference.

For the Gaussian model,

$$V(x) = \sum_{i=1}^N V_i(x_i), \quad V_i(x_i) = \frac{(x_i - x_i^*)^2}{2\sigma_i^2}.$$

Each local term depends on one coordinate and fixed local parameters. This is what makes endpoint support sufficient for the present model.

20.3 The cancellation proof, one line at a time

Write the complete difference as a sum:

$$V(x^b) - V(x^b + \delta) = \sum_{i=1}^N [V_i(x_i^b) - V_i(x_i^b + \delta_i)]. \quad (20.2)$$

For an index outside W , the increment is zero. Its summand is therefore $V_i(x_i^b) - V_i(x_i^b) = 0$. Removing those zero summands gives

$$V(x^b) - V(x^b + \delta) = \sum_{i \in W} [V_i(x_i^b) - V_i(x_i^b + \delta_i)]. \quad (20.3)$$

Define V_{loc} as the sum over W . Equation (20.3) says that its before/after difference is the same as the global difference. The proof does not need the values of the unchanged terms.

Exact local-to-global cancellation

Under one fixed separable potential, with a correctly declared action increment, the sum of the changed local terms equals the complete potential difference. Every unchanged term cancels. This equality concerns the declared model; it does not prove the model contains every consequence that matters.

Notice what the proof did not use. It did not require equal references, equal scales or a Gaussian shape. The same cancellation works for any fixed sum of one-coordinate functions. Gaussian geometry provides convenient explicit formulas for the changed terms; additivity provides locality.

20.4 A five-cell numerical account

Take five cells with reference 10X and scale 2X each. Let

$$x^b = (18, 2, 12, 8, 10)X.$$

The corresponding potential terms are $(8, 8, 1/2, 1/2, 0)$, so the global potential is 17. Transfer 4X from cell 1 to cell 2. The successor is $(14, 6, 12, 8, 10)X$, with terms $(2, 2, 1/2, 1/2, 0)$ and total 5.

The global reduction is $17 - 5 = 12$. The local reduction is $(8+8) - (2+2) = 12$. Reading the last three stocks would add no information needed for this action value, provided the declared state transition really leaves them unchanged and the potential has the stated separable form.

The whole network still has potential 5 after the transfer. The action improved its own affected region; it did not certify that all cells reached their references. Local correctness and system-wide completion are separate propositions.

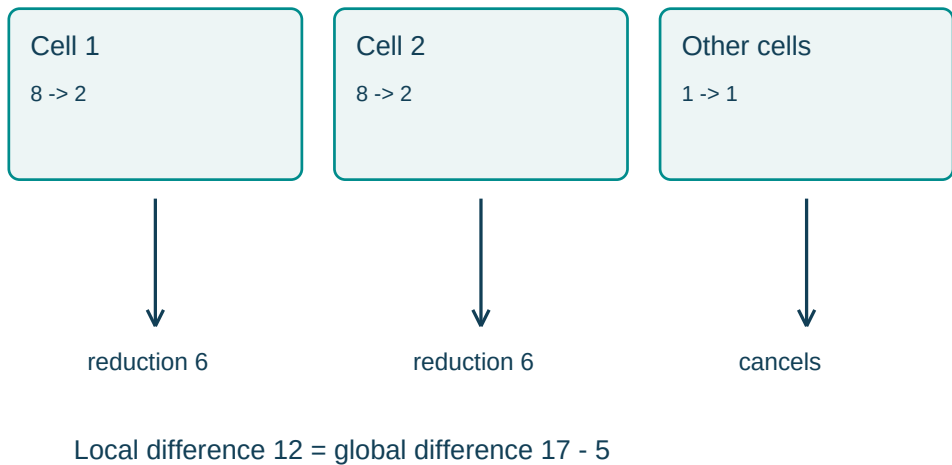


Figure 20.1: The unchanged cells jointly contribute one unit to both global totals and cancel. The local calculation includes both changed endpoints.

20.5 Global refers to the declared boundary

The adjective “global” can sound larger than the mathematics. Here it means all coordinates in the declared model. It does not mean all of Earth, all future time, every ecosystem, every financial consequence or every dimension of wellbeing.

Suppose the modeled network contains only five tanks. Equation (20.3) gives the exact change of that five-tank potential. If the pump also consumes electricity outside the represented boundary, the identity remains true for the tanks while the physical description is incomplete for a claim about the whole operation.

The proper response is to identify the missing carrier and intended scope. A linked electrical account can preserve the physical transformation. A calibrated extension might later place several effects in one declared functional. Neither extension follows merely from calling the five-tank sum global.

20.6 When a neighbour must also be read

A future potential might contain a factor depending on two coordinates, such as a declared interaction $w_{ij}(x_i, x_j)$. Changing x_i can then change that factor even if x_j stays fixed. Reading only the one-coordinate term at i would miss part of the difference.

To express the general rule, suppose

$$V(x) = \sum_{f \in \mathcal{F}} \phi_f(x_{D_f}), \quad (20.4)$$

where D_f is the set of coordinates used by factor f . The affected factors satisfy $D_f \cap W \neq \emptyset$. Every other factor cancels. Evaluation needs the union of coordinates read by the affected factors, which can be larger than the write support.

This is a conditional extension of the same cancellation argument. It does not adopt a new coupled potential for the present book's examples. It explains how to check locality if a later model needs such a potential.

20.7 A counterexample to endpoint-only reading

Consider an explicitly different teaching functional,

$$\tilde{V}(x) = \sum_{i=1}^3 V_i(x_i) + \frac{1}{2}(x_2 - x_3)^2,$$

with dimensionless numerical stock coordinates for this illustration, references $(5, 5, 3)$ and unit scales. Move one unit from cell 1 to cell 2, starting from $(6, 4, 3)$. The successor is $(5, 5, 3)$. Both the reference and physical state have total stock thirteen.

The separable part changes from $1/2 + 1/2 + 0 = 1$ to zero, a reduction of 1. The interaction changes from $1/2(4 - 3)^2 = 1/2$ to $1/2(5 - 3)^2 = 2$, an increase of $3/2$. The complete field reduction is therefore $1 - 3/2 = -1/2$.

An endpoint calculation that retained only V_1 and V_2 would report $+1$, including the wrong sign. A correct local calculation reads x_3 because the interaction factor involving x_2 depends on it. The direct write support is $\{1, 2\}$; the information neighbourhood is $\{1, 2, 3\}$.

The example also shows why the potential's dependency list is part of its scientific declaration. Locality cannot be imposed independently of the equation. Read too little and a changed term disappears. Read more than necessary and a selector may gain information that its design did not intend to expose.

20.8 Three different graphs

The physical graph connects locations by possible actions. The factor graph connects variables used together in the potential. The action-interaction graph connects action increments whose joint value contains a cross term. These graphs can differ.

Our main potential is separable, so its coordinate Hessian is diagonal. Nevertheless, two actions can both withdraw from A. Their increments overlap in the A coordinate, producing a quadratic cross term. A diagonal potential therefore does not imply independent action values.

Conversely, two physical edges can be far apart in a drawing but belong to the same modeled factor if the declared functional couples their variables. Geographic distance alone is not a reliable definition of a mathematical neighbourhood. The relevant question is dependency, followed by whether that dependency is physically justified.

Keeping the three graphs separate helps a reader avoid a common shortcut: “local” does not mean every kind of interaction is absent. It means the required information and support have been specified.

20.9 Global audit and actor information

A local evaluator reads the complete inputs of affected factors. An aggregate account can then reconcile the resulting events across the declared boundary. These are different information tasks. A participant need not receive every global stock merely because the total identity is global.

Dynamics determines which endpoint coordinates are affected. A dispatch action may initially change only source and pending stock but later change the receiver and a shared downstream resource. For a quote at that later horizon, cancellation is based on the resulting endpoint differences. Reading only the immediate action support would omit affected factors.

Hidden state also matters. Two tanks with the same available stock and different pending deliveries can have different future endpoints. If the pending coordinate is retained, the response can be a local state law; if it is eliminated, the remaining description may require a memory integral and an initial-history

term. Omitting both pending state and its memory is a change of model, not a computational saving supplied by the locality theorem.

This gives EBU a disciplined route to scale. It can avoid reading unaffected data while preserving the dependencies needed for a correct quote. The resulting economy of computation and information can make participation easier for small sites and reduce unnecessary disclosure, while shared dependencies still receive joint treatment.

20.10 Local computation does not choose governance

The theorem shows that some calculations can use small neighbourhoods. Different devices could compute them, and independent auditors could reproduce them. These possibilities do not settle who may certify references, own sensors, approve model changes or resolve a dispute.

A centrally governed system can calculate local differences. A distributed institution can perform occasional global audits. Neither arrangement is made inevitable by a sum of potential terms. Political authority must be described in its own language rather than inferred from a graph drawing.

Privacy raises a related issue. A valid physical calculation may need quantities and dependencies while requiring little personal information. Whether a public record must identify an individual is an institutional decision. Mathematical support tells us which variables the calculation needs; it does not demand unlimited disclosure.

20.11 Aggregation can erase the effect

Suppose someone replaces A and B by a single cell containing their total 20X. The internal four-unit transfer leaves that total unchanged. If the aggregate potential depends only on the total, it reports zero change. The fine account reports a reduction of 12.

There is no algebraic contradiction. The aggregate description discarded distribution between the two locations, precisely the information on which the fine potential depends. An aggregate account that is intended to preserve that effect needs additional state, such as internal deviation, or a justified summary of its potential.

This matters for hierarchical systems. A building can expose boundary flows to a district, but a district total alone does not reproduce every room's temperature or water distribution. A sound hierarchy must state what each level retains and which questions cannot be answered after aggregation.

The introductory locality theorem is not an automatic theorem of lossless compression. It gives a starting point for designing summaries and a test for checking them: does the summary preserve the specific field difference the application claims to compute?

20.12 Guided problem: determine the necessary reads

Take six independent Gaussian terms. An action changes coordinates 1 and 4. Which terms must be evaluated? Only V_1 and V_4 , using their before and after coordinates and fixed parameters. Every other term cancels.

Now add a factor $w_{45}(x_4, x_5)$ and another $w_{56}(x_5, x_6)$. The action changes x_4 but not x_5 . It changes w_{45} , so x_5 must be read. It does not directly change w_{56} , whose two inputs remain the same. The necessary coordinates are 1, 4 and 5 for this declared instantaneous action.

The answer would change if the transformation also updated x_5 or if another factor depended on x_4 . This is why one must inspect the actual action map rather than propagate “affected” indefinitely through a diagram. Future influence and present algebraic dependence are different things.

20.13 Practice and reflection

Replace the four-unit transfer in the five-cell example by two units. The affected terms change from (8, 8) to (4.5, 4.5); the others still sum to one. Both the local reduction and the global change $17 - 10$ equal seven.

Draw a four-location route and one transfer’s write support. Add a pair factor linking an endpoint to a neighbour. Identify the required reads and explain why predicting the neighbour’s later response needs more than this factor list.

Explain the aggregation counterexample: the total stayed fixed while distribution changed. Conservation and exact cancellation both held; the coarse model discarded information.

What to carry forward

Unchanged factors cancel exactly. The required information follows the potential and endpoint response, while selection, privacy and governance remain separate choices.

We can now make the factor structure explicit, including functions that depend on several resources together. That is the next chapter’s task.

Beyond isolated coordinates

Many useful functions depend on combinations of physical conditions. A factorized potential represents those combinations while retaining exact local cancellation. It also shows why shared valuation, shared equipment and causal influence are separate kinds of dependence, each with a different role in coordinating EBU actions.

21.1 What the simple field has taught us

The two-stock examples made one fact unusually clear: an action is valued by what it does to a declared physical state, not by the name of the action or the intention of its actor. In a real service network, however, an energy carrier can support a water transfer, a store can protect a medicine shipment, and a damaged route can change the meaning of available stock. We need a way to keep those relationships visible without calculating the condition of the entire planet after every action.

The Gaussian used so far has one term per coordinate. That separability is a useful starting point, not a requirement of exact EBU. The finite definition only asks for one sufficiently defined potential, the same field before and after, and a physical transition represented completely enough for the claim at hand. We can therefore enlarge the field while retaining the identity that made local accounting possible.

21.2 Factors, not necessarily separate cells

Write the potential as a sum of declared factors,

$$V(x) = \sum_{\alpha \in \mathcal{F}} V_{\alpha}(x_{D_{\alpha}}), \quad (21.1)$$

where D_{α} is the set of physical coordinates on which factor α depends. One factor might describe water availability at a reservoir, another a coupled relation between water and pumping energy, and another a temperature-dependent condition of medicine storage. Their formulas, units, references and data sources must be specified; the symbols alone do not confer physical meaning.

Suppose an action changes only coordinates in a write set W . A factor whose dependency set does not meet W has exactly the same value before and after. Consequently,

$$E = V(x) - V(x + \delta) = \sum_{\alpha: D_\alpha \cap W \neq \emptyset} [V_\alpha(x_{D_\alpha}) - V_\alpha((x + \delta)_{D_\alpha})]. \quad (21.2)$$

This is the precise sense in which the calculation can be local. We evaluate every factor affected by the action, including coupled factors that read unchanged neighbouring coordinates, while unchanged factors cancel. It is not a promise that a poorly chosen global factor can be evaluated from two nearby sensors. Locality is a property to establish for the declared factor graph.

To see the distinction, take three dimensionless coordinates and choose

$$V(x_1, x_2, x_3) = \frac{1}{2}(x_1 - x_2)^2 + \frac{1}{2}x_3^2. \quad (21.3)$$

An action adds q to x_2 and changes neither x_1 nor x_3 . Its exact EBU is

$$E(q) = q(x_1 - x_2) - \frac{1}{2}q^2. \quad (21.4)$$

The x_3 term cancels, but x_1 must be read even though it is not written. This is how a nearby dependency enters a local account. Replacing the first term by a function of x_2 alone would make the calculation easier only by changing the scientific model. The example also shows finite curvature again: the initial slope is $x_1 - x_2$, whereas the value of a whole action includes $-q^2/2$.

Finite locality

For a fixed factorized potential and a represented state increment, all factors untouched by that increment cancel exactly. The result is an algebraic theorem. Whether the chosen factors faithfully represent the physical consequence remains a scientific question.

21.3 What coupling changes

Near a reference x^* , a quadratic extension can be written

$$V(x) = \frac{1}{2}(x - x^*)^\top H(x - x^*), \quad (21.5)$$

with a symmetric positive-semidefinite matrix H on the declared state domain. Nonzero off-diagonal entries represent interactions in the chosen field. For a finite increment δ , direct expansion gives

$$E = -\nabla V(x)^\top \delta - \frac{1}{2}\delta^\top H\delta. \quad (21.6)$$

No small-action approximation appears here. If H depends on state, the same exact endpoint definition still works, but this particular quadratic expansion does not.

Coupling also changes the reading neighbourhood. An action may write one coordinate while its value depends on another coordinate because a factor reads both. The calculating node needs the relevant neighbouring state or a trustworthy authenticated summary. It does not need every unrelated coordinate merely because those coordinates exist in the world.

This distinction matters for governance. A mathematical proof that unaffected factors cancel is not a proof that every participant has access to the required measurements, that private data should be disclosed, or that a distributed network will reach agreement. The data protocol and the institution each require their own design.

21.4 Coupling prepares the joint-action question

Suppose a potential factor reads both a water reserve and the electrical reserve needed to deliver it. An action on either reserve can alter that factor. Two actions with disjoint direct coordinate supports can therefore interact through the same coupled valuation.

For a quadratic potential with symmetric Hessian H , the pair term is $-h_i^T H h_j$. Off-diagonal entries allow this product to be nonzero even when h_i and h_j change different coordinates. Conversely, a diagonal H still gives interaction when both actions touch the same coordinate. The formula identifies the actual dependency without requiring a metaphor about distance.

A shared pump can also couple *feasibility* without adding a cross term to the potential. A nonlinear controller can couple *response*, changing the endpoint produced by an action when another is present. The later composite theorem puts response dependence and valuation dependence into one exact calculation.

These distinctions make coordination more selective. EBU can group actions that share a physical limit, reevaluate factors whose inputs change, and leave independent components separate when the equations permit it. This directs collective effort to the places where it can change the outcome, instead of requiring every participant to negotiate with the entire system.

21.5 The limit of a single number

A physical system may carry water, energy, medicine and ecological function in unlike units. Declaring a scalar V requires a scientifically and institution-

ally justified relation among represented effects. Where that relation is not justified, a vector of separate accounts is more honest. One can still ask whether each physical balance closes and whether each service requirement is met. What one cannot do is add unlike consequences under an unstated exchange rate and call the result a law of nature.

The question of substitution is particularly important near essential limits. An extra unit of one abundant carrier may not replace a missing unit of another carrier that performs a specific function. A potential can encode such dependence through a coupled factor or a hard feasibility constraint, but it has to say which. A hard physical impossibility cannot be cured by a sufficiently favourable value elsewhere in the potential. Conversely, a scientific claim that two effects are exchangeable needs a measured basis; it cannot be read from the fact that a scalar sum is convenient.

There is a practical way to test the meaning of a proposed factor. Hold its input coordinates fixed while changing something it claims to represent. If the represented function changes but the factor cannot respond, the model is incomplete for that claim. If the factor responds to a coordinate that cannot be observed or verified at action time, the proposed local evaluator is incomplete. These are design tests, not failures of the algebra. The algebra is deliberately transparent enough to make them visible.

The factor view therefore extends the exact mathematics without pretending that model construction is finished. We now know how to calculate a consequence in a fixed field. The next question follows complete events through time: how do their signed records add when every successor becomes the next predecessor?

Action receipts and exact closure

Successive receipts form a physical history because each event starts where the preceding one ends. Their values therefore telescope. This exact closure is the foundation for reconciling local EBU records, detecting duplicate claims and preserving the contribution of necessary actions with either sign.

22.1 What a physical receipt records

A shop receipt records a commercial exchange. An EBU action receipt names a physical predecessor, an accepted transformation, its successor and the version of the potential used to compare them. The two kinds of receipt may coexist, but one does not validate the other. A payment cannot prove that a reservoir recovered, and a positive EBU does not establish who was paid.

The preceding factor chapters showed why a complete change can often be evaluated from affected factors alone. We now connect several correctly evaluated changes. The central fact is simple: the state after one action is the state before the next. Its potential appears once with each sign, and cancels.

22.2 Follow the actual chain

Consider a sequence of authoritative states under one fixed potential,

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_m. \quad (22.1)$$

For each complete transition define

$$E_r = V(x_{r-1}) - V(x_r). \quad (22.2)$$

The arrows are physical or modeled events, not an arbitrary sort order for files. If several actions occur together, one arrow can be their joint transition, retaining the identities of its children. A later allocation of that group's total is another question.

Summing the consecutive differences gives

$$\sum_{r=1}^m E_r = V(x_0) - V(x_1) + V(x_1) - V(x_2) + \cdots + V(x_{m-1}) - V(x_m) \quad (22.3)$$

$$= V(x_0) - V(x_m). \quad (22.4)$$

Equivalently,

$$\sum_r E_r + V(x_m) - V(x_0) = 0. \quad (22.5)$$

This is exact telescoping for any state function V . Gaussian geometry is not needed. Neither is an actor willing to improve the state. The intermediate states still matter: without them we cannot test whether each action was physical, whether its recorded value was correct or whether an event is missing.

What closure proves

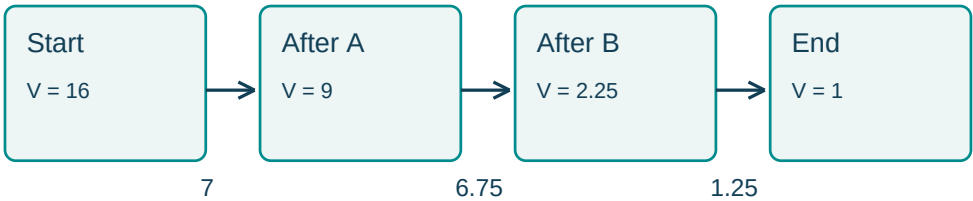
Correctly linked, complete transitions under one fixed potential have a total EBU equal to their endpoint potential reduction. The identity gives the account an exact reconciliation rule. Ownership, service and recovery retain their separately declared meanings.

22.3 One sequence, not three fresh beginnings

Use the two-cell potential with reference (10,10) and scales (2,2). Three successive lossless transfers carry the state through

$$(18, 2) \rightarrow (16, 4) \rightarrow (13, 7) \rightarrow (12, 8), \quad V : 16 \rightarrow 9 \rightarrow \frac{9}{4} \rightarrow 1.$$

The action values are 7, 27/4 and 5/4, which add to 15 = 16 - 1. The middle transfer starts from (16,4), not from (18,2). Reusing the initial slope for it would quote a different imagined action, not the second event in this sequence.



Action values: $7 + 6.75 + 1.25 = 15$

Endpoint difference: $16 - 1 = 15$

Figure 22.1: Three consecutive finite differences telescope to fifteen. Every action uses the preceding endpoint as its baseline.

Suppose a record assigned the first and third lines to one actor and the second to another. Their historical contributions would be $33/4$ and $27/4$, adding to the same 15. This partition uses an explicit owner rule; the potential did not choose the owners. Individual accounts can remain signed. There is no physical theorem requiring a negative contribution to be prohibited or prefunded.

22.4 Why the negative line must remain

Let one action take potential from 1 to 4 and another take it from 4 to 0. Their EBU values are -3 and $+4$. The total is $+1$, exactly the endpoint reduction. Keeping only the positive line would report four units of improvement where the complete history contains one.

An institution can support the first action when it supplies necessary service. Support belongs in its allocation rules; the sign belongs in the physical history. Maintaining both gives the institution a more useful record: it can see what restoration achieved and what service required.

The same principle covers autonomous feedback. If a registered action starts a response that later overshoots and settles, successive observations can explain the path, but they are not automatically new actor events. A no-action background can have nonzero physical change. Its line must remain distinguishable from the incremental action comparison.

Exact closure is therefore socially useful as well as algebraically neat. It makes selective reporting harder, allows small local records to be reconciled into a common total, and keeps necessary burdens visible so they can be reduced through better design rather than hidden through accounting.

22.5 External events need their own line

Now let an external event take x_t to x_t^b , followed by an actor action from x_t^b to x_{t+1} . Define the external potential change by

$$D_{\text{ext},t} = V(x_t^b) - V(x_t).$$

The actor's EBU is $E_t = V(x_t^b) - V(x_{t+1})$, so

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t}. \quad (22.6)$$

An external worsening can create deviation that a later actor reduces. An external improvement can remove deviation without actor contribution. Both signs belong in the record. For example, if an external redistribution moves

potential from 0 to 4 and an actor then reduces it to 1, the actor records 3, not the -1 obtained by comparing the wrong endpoints.

If the potential itself changes between events, the calculation needs an explicit field-change line too. Comparing values produced by two different rulers as though they were one fixed V breaks Equation 22.5. We return to that harder question after the fixed-field theory is complete.

22.6 Balance is not carrier conservation

In a lossless transfer, physical stock is conserved while V changes. States (18, 2) and (14, 6) each contain twenty stock units; their potentials are sixteen and four. A twelve-EBU reduction did not create or destroy twelve litres. Equation 22.5 is a reconciliation of values, not a conservation law for matter or energy.

An auditor therefore checks both ledgers. First follow the physical states, routes, carriers and boundaries. Then verify potential versions, event identities and signed EBU lines. A residual can expose an omitted event, duplicate receipt, stale baseline or arithmetic error; it cannot by itself reconstruct the missing physical cause. The next chapter uses this discipline to examine closed loops and action splitting.

Closed cycles, splitting and external change

A complete return loop has zero net EBU under a fixed potential. That law prevents damage and repair from appearing as net value creation, but it does not erase the resources used or the service interrupted along the way. We use cycles to make this distinction precise and to show why action histories remain more informative than their final scalar total.

23.1 Can damage and repair mint value?

Imagine moving water away from a useful distribution and then returning it. The return movement may have positive EBU. If an actor reports only that movement, it looks like a renewable source of credit. The first movement, however, raised the represented potential and had the opposite sign. A complete account follows both events from their actual baselines.

For an undriven sequence that begins and ends at the same complete state $x_0 = x_m$ under one fixed potential, Chapter 22 gives

$$\sum_{r=1}^m E_r = V(x_0) - V(x_m) = 0. \quad (23.1)$$

The theorem does not need a Gaussian. It needs the complete chain and the same state function at every comparison. If a transition is a simultaneous group, its exact group total is one line in the chain. The equation still does not dictate how that line is divided among people.

At reference (10, 10) with scales (2, 2), transfer two units one way. The new state (8, 12) has potential one, so the action is worth -1 . Transfer the same amount back and the return action is worth $+1$. The positive restoration is real within the account; the claim of positive *net* creation from the complete loop is false.

23.2 The represented state must really return

A battery can recover its displayed charge while losing health. A tank can recover its volume while losing quality. A vehicle can return to the depot after using fuel. If those changes belong to the physical claim, a loop in the

displayed coordinate is not a loop in the complete state. Their carriers and boundaries determine the finite comparison.

Even a genuine return of the complete represented state does not erase the service history. A clinic may experience an interruption before the tank is refilled. Endpoint EBU for the loop is zero, while cumulative shortage and interruption time can be positive. As Chapter 14 showed, an integral of the potential level is a different quantity from its endpoint change.

Equal potential is weaker still than equal state. (18, 2) and (2, 18) have equal Gaussian potential but different source availability and future options. Their direct EBU is zero although the physical state did not return.

The practical consequence is a richer and more reliable EBU record. The total identifies net restoration; the complete states identify resource consequences; the time description identifies continuity of service. Keeping those views together prevents a damage–repair loop from being praised for its positive half or dismissed as harmless merely because its net scalar value is zero.

23.3 Dividing an action cannot divide away its consequence

From (18, 2), an unsplit four-unit transfer ends at (14, 6) and reduces potential by twelve. Split it into two genuine sequential two-unit actions. The first moves potential $16 \rightarrow 9$ and records seven. The second begins from that new state, moves $9 \rightarrow 4$ and records five. Thus $7 + 5 = 12$. Calling both pieces seven would reuse the original state twice.

This is refinement invariance of a state-function difference:

$$[V(x) - V(y)] + [V(y) - V(x^b)] = V(x) - V(x^b).$$

It does not claim that physically stopping and restarting a pump always has the same complete effects as running it continuously. Different energy use, heat or wear would change the complete physical transition if those carriers were represented. The identity says that a consequence cannot be made to disappear by changing how its record is sliced.

23.4 An aggregate cycle need not restore every actor's history

Suppose a return loop contains an action attributed to one owner and its reverse attributed to another. The total EBU is zero, but the two historical accounts can change by opposite amounts. A three-cell example from the physical foundation begins at (4, 4, 4): $B \rightarrow C$ has EBU -1 assigned to B , and the reverse $C \rightarrow B$ has EBU $+1$ assigned to C . The state returns; the

owner vector changes by $(0, -1, +1)$. This is an existence example, not a claim that every possible attribution rule has such loops.

The distinction matters because exact group closure is sometimes mistaken for a fairness theorem. It is not. Aggregate EBU is a difference of state values, while a person's historical attribution can depend on the path of events and on the declared ownership rule.

23.5 External change can recreate work

If a disturbance moves a system away from its reference and an actor restores it, positive actor EBU may recur. That does not violate Equation 23.1: the interval contains an external event. In the notation of Chapter 22, the action total is

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t}.$$

If repeated disturbances increase potential by four and subsequent actions reduce it by four, the positive actor entries are balanced by real external changes. A request or desire alone is not a physical sink. To claim a recurring demand caused a stock change, a model must represent consumption, transformation or boundary exchange.

Nor may an actor-created disturbance be silently renamed an external one. The equation tells us what entries are needed; provenance and causal review tell us whether they were classified honestly. The mathematics cannot reconstruct an omitted event from an apparently convenient total.

23.6 Changing the ruler is a third kind of event

At the same state $(14, 6)$, scales $(2, 2)$ give potential four, while scales $(4, 4)$ give potential one. No water moved. The three-unit difference came from changing the field. If old and new values are joined without a field-change entry, intermediate potentials will not telescope as though one ruler had been used throughout.

This is not an argument against improving a model. It is a demand that learning about the world not be disguised as an action upon the world. Later we ask a harder question: whether EBU values earned under different physical fields can nevertheless share one stable physical interpretation.

The cycle law establishes a property of accounting, while stability belongs to a specified dynamics. Both are useful: one reconciles value, the other explains response. We will connect them after developing the missing comparison in between—several actions sharing one nonlinear field at the same time.

Several actors on one nonlinear field

Joint action must be evaluated as a joint physical change. Two individually correct quotes can add to the wrong group total when they share one nonlinear field. This chapter establishes the exact pair correction, then prepares a general language for identifying what each larger coalition contributes.

24.1 Known quantities do not make value linear

Two couriers each deliver two units from A to B. We know their identities, quantities and directions. Yet valuing each delivery as though the other did not exist gives a different sum from valuing the combined four-unit movement. The difficulty is not missing quantity data. It is the curvature of the potential.

Physical increments can add linearly while the value of their sum remains nonlinear. This distinction is already present in a diagonal Gaussian potential. It does not require a mysterious nonlocal force or a new physical interaction between the couriers.

The shared-source example gives a concrete reason for coordination. A system can identify when proposals rely on the same remaining improvement, compute the joint value once, and preserve each contributor's physical record. The group result is then available for an explicit attribution rule, rather than being inflated by adding incompatible singleton beginnings.

24.2 The two-courier counterexample

Use the repeated state $(18, 2)X$, references $(10, 10)X$ and scales $(2, 2)X$. Let the increments be

$$\delta_A = (-2, 2)X, \quad \delta_B = (-2, 2)X.$$

Either courier alone would produce $(16, 4)X$, reducing potential from 16 to 9. The two same-baseline singleton values therefore sum to $7 + 7 = 14$.

Their actual joint increment is $(-4, 4)X$, producing $(14, 6)X$ with potential 4. The joint field value is 12. The difference of two is not a rounding error or an unobserved physical loss. It is the discrepancy between a joint finite difference and a sum of different counterfactual finite differences.

A counterfactual sum is not the joint account

Each singleton value asks what would happen if that child acted alone at the frozen baseline. The joint value asks what happens when the accepted children act together. On a nonlinear field these are different comparisons, even when the child quantities are known exactly.

If the couriers actually act sequentially, the values are 7 and 5, which sum to 12. A physically meaningful sequence supplies different baselines. If execution is simultaneous, inventing that sequence afterward gives an allocation convention, not an observed intermediate history.

24.3 The exact group formula

Let the accepted increments be δ_a , and define $\delta_G = \sum_{a \in G} \delta_a$. Under fixed Gaussian geometry,

$$E_G = -\mu(x^b)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G \quad (24.1)$$

for the complete represented physical transition. Here $H = \text{diag}(1/\sigma_i^2)$ and $\mu(x^b) = H(x^b - x^*)$.

The same-baseline singleton sum uses δ_a separately in each quadratic term. Expanding the joint square gives

$$E_G - \sum_a E_a^{(0)} = - \sum_{a < b} \delta_a^T H \delta_b. \quad (24.2)$$

The superscript (0) marks the common frozen baseline. It does not denote an observed first step. The right-hand side is a signed cross-term diagnostic for these fixed increments and this fixed potential.

For the two couriers, $H = \text{diag}(1/4, 1/4)$ and $\delta_A^T H \delta_B = 2$, giving $12 - 14 = -2$. The shared coordinates and directions determine the sign.

24.4 The discrepancy can have either sign

Convexity does not make every group discrepancy negative. Consider opposing increments from the reference state (10, 10):

$$\delta_A = (-2, 2), \quad \delta_B = (2, -2).$$

Each action alone increases potential to 1, so each singleton value is -1 . Their joint increment is zero and the ideal group field value is zero. Thus $E_G - (E_A^{(0)} + E_B^{(0)}) = 2$.

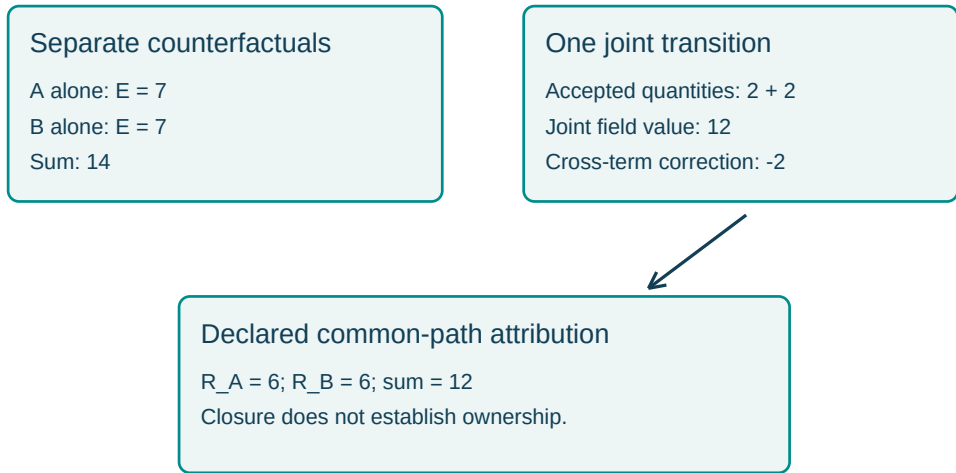


Figure 24.1: Two separately evaluated two-unit actions each have value seven at the frozen baseline. The joint four-unit action has value twelve, not fourteen. A later declared straight-path decomposition assigns six to each.

The cross product is -2 , agreeing with equation (24.2). Two same-direction increments gave a negative discrepancy; opposing increments give a positive one. The physical feasibility of the combined flows must still be declared. Endpoint cancellation does not automatically permit unlimited circulation or establish that the real process is free.

For disjoint coordinate supports under diagonal H , the cross product is zero. The field values then add. Shared physical constraints can nevertheless couple the permitted quantities. An arithmetic additivity result does not remove a common pump-capacity or source-availability constraint.

24.5 Three comparisons that must stay separate

The group-versus-singletons discrepancy compares simultaneous execution with two or more counterfactuals that each start alone. A parallel-versus-sequential comparison uses a complete sequential history. An institutional allocation compares ways of distributing a given joint number.

With fixed additive increments and the same endpoint, each physically feasible sequential order and the group have the same total field value. A feasible joint endpoint alone does not guarantee feasible intermediate states in every order. The singleton sum can still differ. Thus a nonzero singleton discrepancy does not establish that simultaneity created a physical efficiency gain.

If a resolver changes quantities when actions are replayed sequentially, or a delay changes the state, the sequential endpoint can differ. The comparison then has to identify the ordering, resolver, timing and endpoint. Reporting an “interaction benefit” without naming this comparator leaves its meaning incomplete.

24.6 Grouping follows dependencies

Shared coordinates and shared potential factors identify dependencies in valuation. Shared pump capacity, source stock and reservation intervals identify dependencies in permission. A response law can add further links: one action may change the delivery, timing or controller used by another.

A useful group therefore follows the declared dependencies of the complete event. Actions connected through a third action cannot automatically be separated just because their direct supports are disjoint. Conversely, common membership in an organisation does not itself create a mathematical interaction.

For dynamic groups, fix the initial state, background forcing, controller and horizon before comparing subsets. If a coalition changes the controller, that is part of its declared response map. It cannot be treated as though the original fixed increments still applied. The affine-response theorem later tells us exactly when increments can be superposed despite pending delivery and oscillation.

Every group retains its child identities, quantities and routes. Joint evaluation supplies one consistent total while preserving the information needed to allocate responsibility or improve the physical process. This is the practical advantage of the interaction framework: it allows shared infrastructure to be coordinated without making each participant claim the full improvement alone.

24.7 The next question is the complete interaction landscape

The two-action cross term is the first member of a larger hierarchy. Before dividing a joint value among actors, we will examine all admissible subsets, subtract lower-order contributions and identify what truly belongs to a higher order. Chapter 43 later returns to the common-path allocation and its limits.

The full group value already includes every interaction under its declared response. An interaction diagnostic must never be issued again as additional value on top of that total. The next chapter defines the common-baseline table that makes those comparisons precise.

A landscape of possible groups

To understand cooperation, we need a common set of alternatives. A coalition landscape records what each subset of actions produces under the same comparison conditions. It provides the complete input for the interaction laws and gives EBU a disciplined way to compare a coordinated project with its component proposals.

25.1 Before synergy, specify what is being compared

Suppose a clinic needs a delivery, refrigeration and a repaired access route. The three activities may be useful together in a way that their isolated descriptions miss. Yet the phrase “useful together” is incomplete until we specify the alternatives. Does the delivery have the same quantity when refrigeration is absent? Does a failed route reduce the delivered amount? Does the baseline change between attempts? These choices define the comparison.

A coalition table makes the alternatives explicit. It does not assume that actions are additive. Each permitted group is assigned its complete endpoint by a declared physical response rule. We can then ask how the values of the groups fit together. The endpoint rule might be a fixed sum of increments, an engineering resolver, or a well-posed flow. The arithmetic to come works at the level of the resulting table.

25.2 The complete Boolean cube

Definition 25.1 (Coalition landscape). Let $N = \{1, \dots, n\}$ index distinct actions. For every $S \subseteq N$, declare an endpoint x_S under the same initial condition x_0 , field V , boundary and comparison protocol. Set

$$E(S) = V(x_0) - V(x_S). \quad (25.1)$$

For the normalized fixed-baseline landscape require $x_\emptyset = x_0$, so $E(\emptyset) = 0$.

A group is a subset, so its label does not specify an order. With two actions there are four entries: neither, A alone, B alone, and both. With three there are eight. The mathematical structure is the Boolean lattice, ordered by set

inclusion. The word Boolean means that each action is absent or present in the table; it does not require the physical state to be binary.

All entries must refer to one field and comparator rule. Changing a reference halfway through the table changes the question. So does replacing A's fixed quantity in one row with a resolver-dependent quantity in another without saying so. A resolver-dependent table is legitimate when declared, but it is a different landscape from the fixed-increment table.

25.3 One baseline, several counterfactual endpoints

The table is a family of alternatives. One physical event visits one endpoint; the table also specifies what the other declared groups would produce. In a mathematical model these endpoints follow from the response rule. In an application their provenance accompanies the comparison.

A dynamic background may move the system even when no action is present. Fix an initial state, horizon T_f , background inputs and controller, and write $x_S(T_f)$ for each endpoint. There are then two useful tables:

$$E^{\text{raw}}(S) = V(x_0) - V(x_S(T_f)), \quad (25.2)$$

$$E^{\text{rel}}(S) = V(x_{\emptyset}(T_f)) - V(x_S(T_f)). \quad (25.3)$$

The relative table has $E^{\text{rel}}(\emptyset) = 0$ and removes the common no-action change. The two tables differ by the constant $V(x_0) - V(x_{\emptyset}(T_f))$. Every nonempty interaction coefficient is therefore identical in the two tables; only the empty coefficient changes.

This is especially useful for a maintained service with consumption and feedback. Natural depletion or autonomous correction does not become an actor's contribution merely because it occurred during the same interval. The account can retain the raw physical change and the incremental action comparison, each with its own meaning.

The common initial state includes pending stock. Holding only the available receiver stock fixed while changing water already in transit would compare different backgrounds. Shared source conditions, control gains and horizon likewise belong to the protocol. These declarations make a joint project comparable with its parts instead of allowing each row to start from the most favourable world.

25.4 A complete two-action table

Return to two couriers each moving two units from (18, 2) under the fixed reference (10, 10) and scales (2, 2). All four alternatives are physically permitted in this teaching domain:

Group	Endpoint	Potential	$E(S)$
$\emptyset$	(18, 2)	16	0
$\{A\}$	(16, 4)	9	7
$\{B\}$	(16, 4)	9	7
$\{A, B\}$	(14, 6)	4	12

The duplicate singleton endpoints are not an error: distinct couriers can have equal physical increments. Their identities are distinct, while their effects under this declaration coincide. The table says the joint value is twelve and the sum of isolated comparisons is fourteen. It does not say that twelve plus a further interaction reward may be issued. The joint value already includes the interaction.

25.5 Conditional addition is a change of baseline in the table

For disjoint groups A and B , define the conditional increment

$$E(B \mid A) = E(A \cup B) - E(A) = V(x_A) - V(x_{A \cup B}). \quad (25.4)$$

This is an exact difference between table entries. If the declared physical execution of B after A actually takes x_A to $x_{A \cup B}$, it also represents that live sequential action. Without this compatibility it is only a table contrast. A sequential resolver may instead reach another endpoint.

For two actions, the discrepancy between B after A and B alone is

$$E(B \mid A) - E(B) = E(AB) - E(A) - E(B), \quad (25.5)$$

using $E(\emptyset) = 0$ and writing AB for $\{A, B\}$. In the courier example, B after A gives five rather than seven, a difference of minus two. The interaction measures how the context changes the increment.

An actual sequential endpoint $y_{A \rightarrow B}$ gives total $V(x_0) - V(y_{A \rightarrow B})$. Comparing it with the simultaneous group yields

$$E(AB) - E_{A \rightarrow B} = V(y_{A \rightarrow B}) - V(x_{AB}). \quad (25.6)$$

This is zero when the complete endpoints match, even though the same-baseline interaction can be nonzero. The distinction prevents a curvature correction from being advertised as proof that simultaneous organization is more efficient.

25.6 When the full table cannot be physically constructed

Two actions can each be permitted alone but impossible together because they share a source or a machine. Conversely, an action may be possible only with another action supplying a necessary service. If a required row is physically inadmissible, the full Boolean interaction is not an observed physical quantity on that domain.

There are several honest responses: restrict the study to a fully admissible motif; declare a modeled extension and label its unobserved rows; or work on the actual feasible partially ordered set. These are different scientific objects. Filling an impossible row with zero does not recover the original comparison: zero is a value, not a marker for missing permission.

For a finite feasible poset P , inversion uses its own incidence function μ_P , determined by

$$\mu_P(a, a) = 1, \quad \sum_{a \leq c \leq b} \mu_P(a, c) = 0 \quad (a < b). \quad (25.7)$$

One must inspect the intervals rather than transplant Boolean signs. There is a useful nuance: if the feasible family is downward closed, every interval $[A, B]$ with $A \subseteq B$ in that family is still a full Boolean interval. Its signs remain $(-1)^{|B \setminus A|}$. There may nevertheless be no feasible top group on which to ask the intended high-order question.

25.7 What a system gains from declaring the alternatives

A coalition table states exactly whose presence varies and what remains fixed. It can distinguish shared-source competition from complementarity between a water repair and an energy repair. It can also show when an attractive bundle contains no contribution beyond its smaller components.

For EBU, this makes coordinated planning explainable. A planner can show which combination restores a function and which shared resource limits it. A community can then compare complete projects without paying several times for the same represented improvement. The following transform extracts these relationships without losing any part of the total.

Finding the part that only a group contributes

A group's value contains the contributions already visible in smaller groups. Möbius inversion removes those contributions in the right order and leaves the part specific to each coalition. Its practical strength is exact reconstruction: the pieces explain the whole without creating an additional pool of value.

26.1 Subtract what has already been explained

When two actions act together, their total can include what each would do alone and a further correction for being combined. With three actions, subtracting only the singleton values is insufficient. Some of the difference may already be explained by pairs. A genuine three-way coefficient is what remains after every lower-order contribution has been accounted for exactly once.

This idea is easier to understand as bookkeeping than as new terminology. If a sum contains individual terms and pair terms, those terms must be removed before naming the remainder “three-way.” Möbius inversion provides the general rule for that subtraction. The name refers to a well-established mathematical construction, not an additional physical force discovered by EBU.

26.2 The interaction coefficient

Definition 26.1 (Boolean Möbius coefficient). For a complete finite coalition table $E : 2^N \rightarrow \mathbb{R}$, define

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} E(S). \quad (26.1)$$

In particular $m_E(\emptyset) = E(\emptyset)$. For the normalized landscape the empty coefficient is zero.

The sign alternates as entries are added or removed. A singleton has $m_E(\{i\}) = E(i) - E(\emptyset)$. A pair has

$$m_E(ij) = E(ij) - E(i) - E(j) + E(\emptyset). \quad (26.2)$$

Thus the two-courier coefficient is $12 - 7 - 7 + 0 = -2$. It is a negative correction to the same-baseline singleton sum. It is neither a two-unit physical loss nor a two-unit charge added to a twelve-unit receipt.

For three actions the alternating sum is

$$\begin{aligned} m_E(123) = E(123) - E(12) - E(13) - E(23) \\ + E(1) + E(2) + E(3) - E(\emptyset). \end{aligned} \quad (26.3)$$

The singleton signs become positive because each singleton was subtracted twice inside the three pair terms. Inclusion–exclusion restores the correct count.

26.3 Why all the pieces reconstruct the whole

Theorem 26.2 (Boolean inversion). For every $S \subseteq N$,

$$E(S) = \sum_{T \subseteq S} m_E(T). \quad (26.4)$$

The coefficients are unique.

Proof. Insert the definition of $m_E(T)$ and collect the coefficient of one fixed $E(A)$ with $A \subseteq S$. That coefficient is $\sum_{A \subseteq T \subseteq S} (-1)^{|T|-|A|} = (1-1)^{|S \setminus A|}$. It is zero unless $A = S$, when it equals one. This proves reconstruction. Uniqueness also follows recursively: the empty value fixes the empty coefficient, and the value at each larger set fixes its coefficient after all proper-subset coefficients are subtracted. $\square$

No derivatives, dynamics or probability laws appear in this proof. Any finite table has the decomposition. The physical quality of an EBU interaction therefore comes from the quality of its declared coalition endpoints and potential, not from the sophistication of the transform applied afterward.

26.4 Three equal actions reveal what pairs already explain

Let three couriers each carry one unit on the same ideal edge from $(18, 2)$. All groups are admissible. A group of size r has value $4r - r^2/4$, so the singleton, pair and triple values are $15/4$, 7 and $39/4$.

Each pair coefficient is $7 - 15/4 - 15/4 = -1/2$. The triple coefficient is

$$\frac{39}{4} - 3(7) + 3\left(\frac{15}{4}\right) = 0. \quad (26.5)$$

Yet the triple value differs from the singleton sum by $39/4 - 45/4 = -3/2$. All of that discrepancy is the sum of the three pair corrections. Calling the entire $-3/2$ a three-way interaction would confuse the size of the group with the order of its unexplained interaction.

This distinction matters when discussing cooperation. A large project can contain many pair dependencies without containing a nonzero high-order coefficient. Conversely, a modest three-action motif can contain a truly three-way dependence that no pair comparison reveals. A system should report the order it measured, not the number of participants in its photograph.

26.5 Interaction is not necessarily beneficial

A positive interaction coefficient says that the joint contrast is more positive than the lower-order contributions explain. A negative one says it is less positive. Neither sign alone tells us whether the whole group has positive EBU, delivers an essential service or uses fewer physical resources.

The two opposing transfers at the reference illustrate this well. Their pair coefficient is positive because they cancel each other's negative standalone effects. The joint endpoint value is still zero. By contrast, two deliveries in the same direction can have a negative pair term while their joint EBU remains strongly positive. Interaction describes dependence relative to a specified comparison.

This distinction makes the coefficient more useful. It can reveal redundant proposals, shared-source competition or enabling combinations without replacing the full project evaluation. For society, the useful feature is transparency about coordination: a group can explain why acting together changes the account, while service and resource consequences remain visible in the complete endpoints.

26.6 The EBU sign comes from the potential difference

For nonempty T , the constant $V(x_0)$ cancels in the alternating sum. Therefore

$$m_E(T) = - \sum_{S \subseteq T} (-1)^{|T|-|S|} V(x_S). \quad (26.6)$$

The minus sign is essential. A positive potential interaction represents an extra burden in the composite endpoint landscape; its EBU coefficient has the opposite sign. Retaining this convention throughout will connect the same coefficient to derivatives, log probabilities and, under narrower assumptions, medium entropy.

An additive constant in V changes no EBU and no nonempty interaction. A positive rescaling of V rescales all values and coefficients, while changing references or coordinate weights can change their relative sizes and signs. Interactions are meaningful within their declared denomination and field version.

26.7 Established mathematics, a specific EBU application

Rota's incidence-algebra treatment gives the general inversion language. Cooperative-game theory calls the corresponding coefficients Harsanyi dividends; equivalent representations also occur in pseudo-Boolean functions, capacities and multilinear game extensions [31, 32, 33, 34, 35]. Epistasis studies use related decompositions to describe effects that depend on combinations of factors [36].

EBU's contribution here is a disciplined application: define one physically interpretable endpoint table, retain the sign of potential reduction, and connect that table to the other declared layers. It is not a claim to have invented inclusion–exclusion, cooperative-game dividends or high-order interaction. Nor should a baseline-specific dividend be confused with a distribution-averaged sensitivity index; the averaging measure would add another declaration.

What this law makes possible

The transform gives a lossless decomposition of a complete coalition table. It prevents pair effects from being renamed three-way effects and prevents interaction from being added twice to the group value. A complete EBU account can use these coefficients to explain a joint total, locate shared dependence and avoid charging for the same contribution twice.

We have a formula for every order. The next step is to understand why each higher order asks how a lower-order relationship changes in a new context.

When the relationship itself changes

A higher-order interaction asks how a lower-order relationship changes when another action is present. This recursive view turns a long alternating sum into an intelligible question about context. It is particularly helpful for services whose components enable one another, such as water delivery, refrigeration and electrical supply.

27.1 A third action can change a pair

A water delivery and a refrigeration repair may have a certain combined effect when a clinic already has electricity. If the electrical connection is absent, their relationship may be different. The question is not merely what the third action contributes alone. It is whether the third action changes what the first two contribute together.

This is the intuitive meaning of the third-order coefficient. Higher order continues the same pattern: one more action changes a lower-order interaction. The recursion gives a way to teach, calculate and interpret the hierarchy without treating every order as an unrelated formula.

27.2 A finite difference adds one action

For $i \notin S$, define

$$(\Delta_i E)(S) = E(S \cup \{i\}) - E(S). \quad (27.1)$$

The operator compares a row with and without i while holding its background S fixed. For distinct actions i, j , the difference operators commute wherever all required rows exist. Expanding either order gives the same four entries.

For disjoint T and S , write $\Delta_T = \prod_{i \in T} \Delta_i$. Then

$$(\Delta_T E)(S) = \sum_{A \subseteq T} (-1)^{|T|-|A|} E(S \cup A). \quad (27.2)$$

It is the interaction among the actions in T at background S . At the empty background it is exactly $m_E(T)$.

Theorem 27.1 (Discrete difference and recursion). On a complete Boolean table,

$$m_E(T) = (\Delta_T E)(\emptyset), \quad (27.3)$$

$$(\Delta_{T \cup \{k\}} E)(S) = (\Delta_T E)(S \cup \{k\}) - (\Delta_T E)(S), \quad (27.4)$$

$$(\Delta_T E)(S) = \sum_{R \subseteq S} m_E(T \cup R), \quad (27.5)$$

where $S, T, \{k\}$ are disjoint as needed.

Proof. The first identity is the alternating expansion of the product of differences. The second is the definition of one more difference. For the third, expand each $E(S \cup A)$ by Boolean inversion. A coefficient $m_E(B)$ survives the differences precisely when B contains every element of T and its remaining elements lie in S . Every such coefficient survives once. $\square$

This proof also explains why context matters. A pair difference at a populated background can include the empty-background pair coefficient and higher-order coefficients involving that background. Calling all of it “the pair coefficient” without naming the background hides those contributions.

27.3 The triple written as a comparison of pairs

Define the A–B interaction with C present by

$$I_{AB|C} = E(ABC) - E(AC) - E(BC) + E(C). \quad (27.6)$$

With C absent it is $I_{AB|\emptyset} = E(AB) - E(A) - E(B) + E(\emptyset)$. Their difference is

$$m_E(ABC) = I_{AB|C} - I_{AB|\emptyset}. \quad (27.7)$$

Thus a third-order finding means the pair relationship changed when the third action entered. It does not mean that three standalone benefits were simply added.

27.4 A nonlinear response makes the recursion visible

Consider dimensionless controls $z_i \in \{0, 1\}$ and the constructed response

$$x(z) = z_1 + z_2 + z_3 + z_1 z_2, \quad V(x) = \frac{1}{2} x^2. \quad (27.8)$$

Here $x_0 = 0$. The extra $z_1 z_2$ says that the first two controls jointly change the endpoint beyond their separate increments. It is a teaching response map, not a calibrated clinic model. Its complete table is

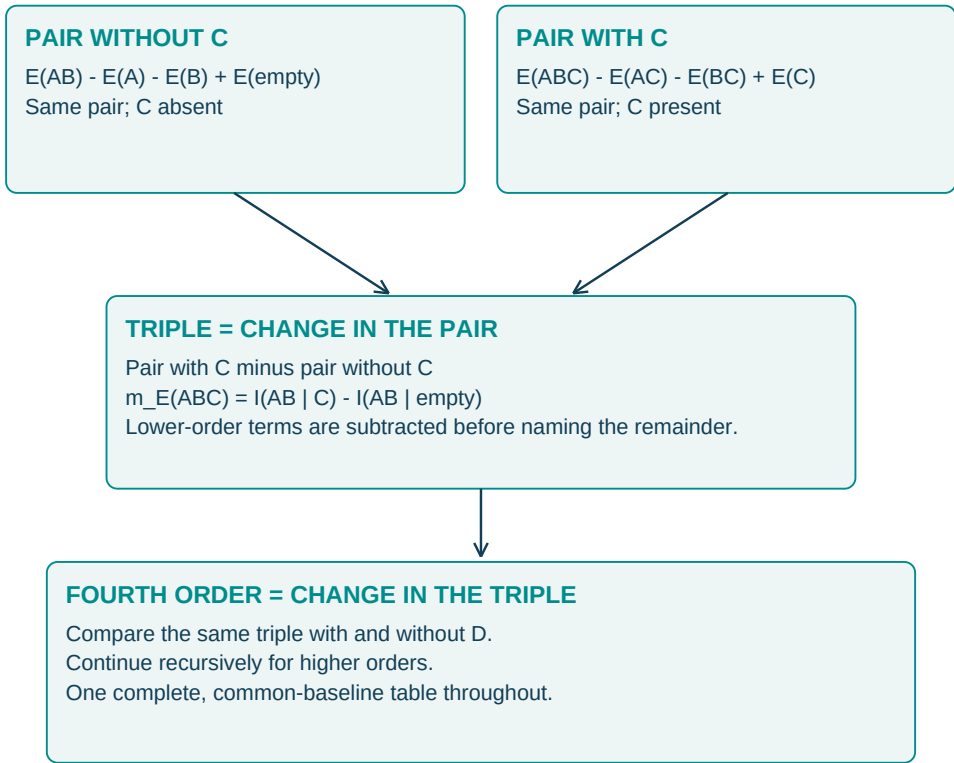


Figure 27.1: Interaction as a changing relationship. The third-order coefficient compares the same pair contrast with and without C. The fourth order asks how this difference changes when D is present. Every box uses one declared coalition table.

Group	x_S	$E(S)$
$\emptyset$	0	0
1, 2, 3 individually	1	$-1/2$ each
12	3	$-9/2$
13, 23	2	-2 each
123	4	-8

The A–B pair contrast without C is $-9/2 + 1/2 + 1/2 = -7/2$. With C it is $-8 + 2 + 2 - 1/2 = -9/2$. Their difference is -1 , the genuine triple coefficient.

All endpoint values here are negative because the example moves away from the minimum of V . This is useful: it stops a discussion of structure from quietly turning into a success story. The response illustrates higher-order dependence regardless of whether a particular institution would want the action.

27.5 A quadratic null also has an interpretation

For fixed additive increments and a quadratic potential, the A–B pair coefficient is independent of whether C is included. Therefore the triple difference is zero. The cancellation is a substantive prediction about this declared landscape. It is not a statement that C has no standalone effect, that all actors are independent, or that the physical system contains no shared constraints.

If a measured triple differs from zero beyond its uncertainty, one or more assumptions behind that null may fail. The response could be nonlinear, the potential nonquadratic, the baseline inconsistent, or the measurements biased. The result narrows a question; it does not uniquely identify the responsible mechanism. Later chapters make these alternatives explicit.

27.6 Fourth order does not require a new kind of arithmetic

Let $I_{ABC|D} = (\Delta_A \Delta_B \Delta_C E)(\{D\})$. Then

$$m_E(ABCD) = I_{ABC|D} - I_{ABC|\emptyset}. \quad (27.9)$$

The pattern continues at every order. This is an exact recurrence, not an approximation valid only for small actions. It works because the table has discrete absent/present coordinates and all needed rows are supplied.

In an EBU research programme the recursion supports small, interpretable studies. Begin by checking two-action interactions. Add a third action only when the scientific question concerns a change in that relationship. This can reduce the temptation to fit a large opaque model whose high-order coefficients lack a physical interpretation.

27.7 Explaining dependencies in a service system

A route repair changes a delivery's value; electrical restoration can change how that pair sustains refrigeration. The triple isolates this contextual change after smaller contributions are removed. An EBU explanation can therefore show how one service changes an already understood relationship, instead of presenting an unexplained group bonus.

A zero is informative too. With affine response and quadratic valuation, adding a third action leaves the pair relationship unchanged in the full Boolean difference. Oscillatory delivery preserves that result when the response remains affine.

The recursion focuses coordination on combinations whose dependence matters. The next chapter gathers this hierarchy into one exact polynomial.

An exact Taylor language for finite choices

Every finite coalition table has one exact multilinear representation. The coefficients are precisely its Möbius interactions. This gives EBU a compact mathematical language for the full hierarchy and makes clear when a reduced description is exact, rather than an approximation that discards part of the account.

28.1 Why use the word Taylor?

A continuous Taylor expansion builds a function from its value, slopes, curvature and higher derivatives near a point. A coalition table has a related hierarchy: its baseline, single-action changes, pair interactions, and higher interactions. The resemblance becomes an exact mathematical statement when we describe absent/present choices by indicator variables.

There is also a crucial difference. An ordinary truncated continuous Taylor series generally leaves a remainder. The complete multilinear representation of a finite Boolean table leaves none. It reproduces every row exactly because it has exactly the coefficients needed to represent that finite collection of choices.

28.2 From a subset to its indicators

Let z_i be one when action i is included and zero otherwise. Write χ_S for the vector representing S . Define

$$\tilde{E}(z) = \sum_{T \subseteq N} m_E(T) \prod_{i \in T} z_i, \quad z \in \mathbb{R}^n, \quad (28.1)$$

with the empty product equal to one. Each variable appears with exponent at most one, which is what multilinear means here.

Theorem 28.1 (Exact finite discrete Taylor representation). Equation 28.1 is the unique multilinear polynomial satisfying $\tilde{E}(\chi_S) = E(S)$ for all $S \subseteq N$. It has no remainder on the Boolean cube, and

$$\partial_T \tilde{E}(0) = m_E(T) \quad (28.2)$$

for each set of distinct variables T .

Proof. At χ_S , a product is one precisely when $T \subseteq S$ and otherwise zero. Evaluation therefore gives $\sum_{T \subseteq S} m_E(T) = E(S)$. Any multilinear polynomial has one coefficient per such product; evaluating its subsets in increasing size determines these coefficients uniquely by the same inversion. Differentiating once in each variable of T and evaluating all variables at zero selects exactly its coefficient. $\square$

The theorem allows the whole landscape to be reconstructed from its interactions. It does not require small physical actions, differentiable endpoints or an equilibrium probability law. It is a representation theorem for the finite table.

28.3 A complete quadratic example

For the three equal one-unit couriers of the previous chapter,

$$\tilde{E}(z) = \frac{15}{4}(z_1 + z_2 + z_3) - \frac{1}{2}(z_1 z_2 + z_1 z_3 + z_2 z_3). \quad (28.3)$$

There is no $z_1 z_2 z_3$ term. Setting all indicators to one gives $45/4 - 3/2 = 39/4$, exactly the triple value. Setting only z_1, z_2 to one gives $15/2 - 1/2 = 7$, exactly the pair value.

Now compare this with the physical polynomial for fractional control intensities,

$$E_{\text{phys}}(z) = 4(z_1 + z_2 + z_3) - \frac{1}{4}(z_1 + z_2 + z_3)^2. \quad (28.4)$$

At Boolean vertices $z_i^2 = z_i$, so it reduces to the multilinear formula above. Between vertices the two functions generally differ. For one half-active action, the physical polynomial gives $31/16$, whereas the multilinear extension gives $15/8$. Both agree when the action is entirely absent or present.

This small difference carries a large conceptual lesson: the unique multilinear extension is not automatically the physical law for fractional actions. Binary observations determine one interpolating polynomial in the multilinear class. They do not determine every smooth function through the same vertices.

28.4 A probabilistic interpolation with a declared meaning

There is a useful alternative interpretation. Suppose actions are included independently, with inclusion probabilities $z_i \in [0, 1]$, and then the selected coalition receives its table value. The expected table value is

$$\sum_{S \subseteq N} E(S) \prod_{i \in S} z_i \prod_{j \notin S} (1 - z_j) = \tilde{E}(z). \quad (28.5)$$

The equality follows by taking expectations of the product indicators in Equation 28.1. This is a mathematical randomized selection model with independent inclusions. It does not claim that actual actors choose independently or that an infeasible group becomes admissible because it has a probability weight.

A fractional physical dose and a probability of a full dose are different interventions. Confusing them can turn an exact interpolation into a false physical prediction. In an EBU system, the record should say whether a slider changes quantity, participation probability, or merely the displayed interpolation.

28.5 Constants, units and reparametrization

The empty coefficient records the baseline table value. In the normalized action table it is zero. With a natural-drift comparator it may not be, and we must retain it or explicitly center the table. Nonempty differences annihilate a constant, but reconstruction of the full value still needs that constant.

The degree also refers to declared action coordinates. Splitting an action into new controls, changing a resolver or choosing nonlinear intensity coordinates can change the apparent polynomial structure of the physical composite. A representation theorem does not make every coordinate choice equally meaningful. Physical interpretation requires the action labels and response map that supplied the table.

28.6 Why this helps an EBU system

The complete polynomial is a second representation of the same table. It can reconstruct every coalition value from the baseline and interaction coefficients. This supports explanation and reconciliation: a reader can see which single actions, pairs and larger groups contribute to a proposed total.

If a theorem limits the degree, the representation becomes genuinely smaller. A quadratic potential with affine endpoint response needs only the baseline, singleton and pair coefficients. A common linear feedback system preserves that ceiling even if it contains many temporal modes. The simplification follows from the composed function's degree, not from the number of differential equations or the visual smoothness of a trajectory.

Without such a structural result, deleting higher orders changes the table. The omitted terms have an exact remainder on each coalition: their sum over subsets of that coalition. This lets an EBU design state the consequence of a reduced account explicitly instead of hiding it in a convenient approximation.

The efficiency benefit is precise. Use the smallest representation that retains the required information, and preserve the full dependencies where they matter. This reduces unnecessary computation while helping a community understand why a coordinated service has its reported value.

28.7 The boundary with ordinary calculus

The word exact belongs to the finite Boolean representation. The next chapter connects a fixed additive physical response to mixed derivatives of the state potential. That connection explains why small interactions have familiar Taylor orders and why quadratic fields give pair-only structure under affine action maps.

The two constructions will meet, but they will not become interchangeable. The discrete polynomial is determined by all vertices. The local continuous derivative is determined near a point. One is a finite table identity; the other is local geometric information with explicit regularity and remainder conditions.

From finite interaction to continuous curvature

A fixed additive action family connects the finite table to ordinary calculus. Repeated integration of a mixed derivative recovers the finite interaction exactly; small-action expansion then identifies its leading local term. The distinction between these two statements tells us when curvature is a complete explanation and when a remainder still matters.

29.1 Walk across the table through physical space

The Boolean table records endpoints, but some action families give us more: a continuous path between every combination. Suppose each action has a fixed state increment h_i and the increments add. Turning action i from absent to present then moves continuously along h_i . The pair table forms the corners of a parallelogram in state space; a triple forms the corners of a parallelotope. Curvature describes how a slope changes across that shape. Repeated integration of mixed derivatives recovers the alternating endpoint difference exactly. This is the bridge between the finite interaction table and the familiar language of continuous calculus.

29.2 Fixed additive increments are a substantive assumption

Let

$$x_S = x_0 + \sum_{i \in S} h_i, \quad G(t) = V \left(x_0 + \sum_{i \in T} t_i h_i \right). \quad (29.1)$$

The h_i do not change when a different subset is chosen. A shared-capacity resolver that reduces one action's quantity when another is present does not satisfy this declaration merely because it returns a vector. It supplies a more general composite map, treated later.

For the physical calculus interpretation, the entire parallelotope must lie in a domain on which the required derivatives are valid. Admissible vertices alone need not supply all admissible intermediate physical states in a nonconvex domain. If an extension of V is used for mathematical interpolation, label that extension and limit the physical claim accordingly.

29.3 Integrating the mixed derivative

Theorem 29.1 (Repeated fundamental theorem of calculus). For $T = \{i_1, \dots, i_k\}$ with fixed additive increments, suppose $V \in C^k$ on a neighbourhood of the parallelotope in Equation 29.1. Then

$$m_E(T) = - \int_{[0,1]^k} D^k V \left(x_0 + \sum_{i \in T} t_i h_i \right) [h_{i_1}, \dots, h_{i_k}] dt. \quad (29.2)$$

Here $D^k V(x)[h_{i_1}, \dots, h_{i_k}]$ is the k -linear derivative of V applied to the specified directions.

Proof. For one direction the difference $G(1) - G(0)$ is the integral of its derivative. Apply the same identity in a second coordinate to the difference of those integrals, then continue through all k coordinates. Continuity on the compact cube permits the repeated integrals and their rearrangement. The alternating corner sum is $\Delta_T G(0)$, while the chain rule for the affine map gives the displayed directional derivative. Finally $m_E(T) = -\Delta_T G(0)$. $\square$

For a pair this becomes

$$m_E(ij) = - \int_0^1 \int_0^1 h_i^\top \nabla^2 V(x_0 + sh_i + th_j) h_j ds dt. \quad (29.3)$$

A pair coefficient is therefore an integrated cross-curvature over a finite physical region. It need not equal the Hessian at the initial point times the increments unless the curvature is constant or a controlled approximation is used.

29.4 Why the pair sign is not fixed by convexity

A positive-semidefinite Hessian guarantees $h^\top H h \geq 0$ for the same direction twice. It does not guarantee $h_i^\top H h_j \geq 0$ for different directions. If the directions oppose one another, the cross product can be negative, giving a positive EBU pair correction.

This is the continuous explanation of the cancelling-transfer example. Convexity penalizes a nonzero departure from the reference, but opposite departures can cancel in the joint endpoint. The positive correction repairs the overstatement of separate burdens. It is not evidence of a mysterious extra output.

For EBU, the theorem lets a measured interaction be compared with the field's declared geometry. A mismatch can expose an omitted response dependence or a wrong field. Such a comparison is useful for model diagnosis before any institutional reward is attached to the number.

29.5 Small actions and a controlled local approximation

Set $h_i = \varepsilon d_i$, holding directions d_i fixed while shrinking the common amplitude. Continuity of $D^k V$ at x_0 gives

$$m_E(T) = -\varepsilon^k D^k V(x_0)[d_{i_1}, \dots, d_{i_k}] + o(|\varepsilon|^k). \quad (29.4)$$

The notation $o(|\varepsilon|^k)$ means that the remainder divided by $|\varepsilon|^k$ tends to zero. It does not promise a full extra power of ε under only C^k smoothness.

If $D^k V$ is Lipschitz with constant L on the relevant neighbourhood, the stronger estimate is

$$|R_T(\varepsilon)| \leq \frac{L}{2} |\varepsilon|^{k+1} \left(\sum_{i \in T} \|d_i\| \right) \prod_{i \in T} \|d_i\|. \quad (29.5)$$

Indeed, compare the derivative at $x_0 + \varepsilon \sum t_i d_i$ with that at x_0 , use the Lipschitz bound and multilinearity, and integrate t_i over $[0, 1]$, whose mean is $1/2$.

The assumptions determine the error claim. Replacing little- o by $O(\varepsilon^{k+1})$ without stronger regularity would claim precision the theorem has not supplied. For physical measurements this difference can matter when extrapolating interactions from small interventions to larger ones.

29.6 An exact finite table and a local approximation can both be right

The multilinear expansion in Chapter 28 reconstructs the measured finite table without a remainder. Equation 29.4 approximates a coefficient as action amplitudes shrink. Its remainder concerns the replacement of an integrated derivative by one local derivative, not an incompleteness of Boolean inversion.

A practical comparison is the difference between a complete map of several specified trips and a local estimate of road slope. The map can give exact distances between its listed places while a tangent estimate between nearby points remains approximate. Neither kind of knowledge invalidates the other; their domains differ.

For a system designer, the finite table is appropriate when actions are indivisible or large. A derivative model can help identify how changes scale with size and where a small-action approximation is defensible. The record should say which object was used and carry its error conditions.

29.7 Weaker regularity needs a real theorem

Some physical models are piecewise smooth. Repeated FTC can still hold under suitable absolute-continuity and integrability conditions on the pullback $G(t)$, but “derivatives exist almost everywhere” is not a sufficient replacement by itself. Appendix A states a useful formulation and explains a singular-function counterexample.

The finite Möbius coefficient remains defined from the vertices even when no such derivative theorem applies. This gives a robust order of reasoning: retain the exact finite comparison, then add a continuous interpretation only when its mathematical hypotheses are met. A failed smooth approximation need not invalidate the endpoint account.

29.8 A useful question for the EBU system

The integral representation explains an interaction as accumulated mixed curvature over the declared action region. A local derivative describes only the leading contribution near the baseline. A uniform bound on the next derivative controls the difference between that local approximation and the finite interaction.

This gives EBU two complementary calculation methods. A finite endpoint table supplies exact differences directly. A smooth model can explain those differences geometrically and, when bounds are available, support controlled approximations for small actions. The method can be chosen to match the scale of the intervention without changing what the account means.

The practical value is economy with an explicit error boundary. Small adjustments may be described efficiently by local terms, while large transfers retain the full finite calculation. The polynomial case is stronger still: when the potential has finite degree and response is affine, sufficiently high differences vanish exactly at any admissible action size. That result is the subject of the next chapter.

What a quadratic can and cannot explain

A polynomial has a finite hierarchy of derivatives. Under affine response, that hierarchy becomes an exact ceiling on coalition interaction. For the Gaussian EBU potential, only singleton and pair terms survive. We state the complete assumptions because this powerful simplification continues through linear feedback but can fail under nonlinear response.

30.1 A ceiling on interaction order

A quadratic curve has slope and curvature but no third derivative. That familiar fact becomes a statement about action combinations only after the action-to-state map is specified. If the endpoint depends affinely on the action controls, a quadratic potential produces at most pair interactions. If the endpoint depends nonlinearly on the controls, the conclusion can fail.

The distinction is one of the book's most useful structural results. A quadratic potential is a field declaration. Pair-only interaction is a theorem about that field *together with* an affine response. It is not a universal property of everything called Gaussian.

Theorem 30.1 (Polynomial degree ceiling). Let V be a polynomial of total degree at most d on the relevant state coordinates, and let $x(z) = x_0 + \sum_i z_i h_i$. For the complete coalition table $E(S) = V(x_0) - V(x(\chi_S))$,

$$m_E(T) = 0 \quad \text{whenever } |T| > d. \quad (30.1)$$

Proof. The composite $V(x(z))$ is a polynomial in the controls of degree at most d . On Boolean vertices, powers z_i^r with $r \geq 1$ reduce to z_i . This reduction cannot increase total degree. The unique multilinear representation consequently has no monomial involving more than d distinct variables. Its coefficients are the Möbius coefficients, so all higher ones vanish. $\square$

This proof is algebraic. It does not assume infinitesimal increments and carries no Taylor remainder. Restricted physical feasibility still matters: a physical claim about the complete cube requires every row used in the coefficient to be admissible under the stated comparison.

30.2 The quadratic formula in full

Let

$$V(x) = c + g^\top(x - x_0) + \frac{1}{2}(x - x_0)^\top H(x - x_0), \quad H = H^\top. \quad (30.2)$$

Positive definiteness is useful for a homeostatic minimum, but is not needed for the degree argument. Expanding at $x_0 + \sum z_i h_i$ and reducing Boolean squares gives

$$\begin{aligned} m_E(i) &= -g^\top h_i - \frac{1}{2}h_i^\top H h_i, \\ m_E(ij) &= -h_i^\top H h_j, \\ m_E(T) &= 0 \quad (|T| \geq 3). \end{aligned} \quad (30.3)$$

For the Gaussian field, $g = H(x_0 - x^*)$. The pair term is independent of the baseline as long as H and the increments stay fixed. Singleton values change with the baseline because their tangent contributions change.

A diagonal H does not make every pair term zero. Two actions that write the same physical coordinate can have a nonzero weighted overlap. Conversely, with a nondiagonal H , disjoint coordinate writes can interact through the field. The theorem keeps response support and potential coupling distinct.

30.3 Why a null prediction is valuable

An exact zero removes an entire class of terms from the account. Under the theorem's assumptions, third- and higher-order interactions need not be estimated or stored: they are absent from the mathematical model. This saves work and makes a joint value easier to explain.

The conclusion is stronger than a small observed coefficient and more specific than calling a system simple. A linear feedback system can contain pending stocks and decaying oscillations while keeping an affine response to additive inputs. Combined with a quadratic potential, it still has this exact pair-only structure at every admissible horizon.

Conversely, a nonzero higher coefficient in a correctly specified table establishes that the conjunction of these premises does not describe that table. It does not, by itself, identify which premise changed. A nonlinear plant, a coalition-dependent controller, a shared constraint or a nonquadratic potential can each alter the composed map.

For EBU this is a useful design boundary. It separates a class of services that admit an exact economical account from services that require a richer interaction description. The saved complexity can support more complete physical representation and clearer decisions about shared infrastructure.

30.4 Cubic curvature supplies a genuine triple

To see an exact higher-order term without a nonlinear response, take the scalar teaching potential

$$V(x) = \frac{\lambda}{6}x^3, \quad x_S = x_0 + \sum_{i \in S} h_i. \quad (30.4)$$

For three increments a, b, c , its third derivative is the constant λ . Repeated FTC gives

$$m_E(123) = -\lambda abc. \quad (30.5)$$

The result is independent of x_0 because the third derivative is constant. It can also be checked by expanding all eight cubes: only the product involving all three distinct increments survives the alternating sum.

With $x_0 = 0$, $a = b = c = 1$ and $\lambda = 1$, singleton values are $-1/6$, pair values $-8/6$, and the triple value is $-27/6$. Therefore $-27/6 + 3(8/6) - 3(1/6) = -1$. These are exact constructed values. The cubic is used on a declared finite domain; it is not claimed to be a globally confining physical equilibrium energy.

30.5 Quartic curvature continues the pattern

For

$$V(x) = \frac{\lambda}{24}x^4, \quad (30.6)$$

the fourth derivative is λ , so four fixed increments give

$$m_E(1234) = -\lambda h_1 h_2 h_3 h_4. \quad (30.7)$$

With unit increments and zero baseline, the alternating coefficient is $[-256 + 4(81) - 6(16) + 4(1)]/24 = -1$ for $\lambda = 1$. All coefficients of order five or higher vanish under the same additive declaration.

The factorials in these examples normalize the relevant derivative to λ . They do not convert stock into EBU by themselves; the units and calibration of λ must match the chosen coordinate. The examples show a structural possibility, not a measured coefficient for society.

30.6 Degree is a property of a representation and its response

A nonlinear coordinate change can turn a simple expression into a complicated one. If it represents the same physical quantity faithfully, the complete endpoint EBU is unchanged, but the split between potential and response

derivatives can change. We should therefore resist treating the printed degree of V as a coordinate-free explanation of all observed interactions.

For fixed action coordinates, the relevant object is the composite $G(z) = V(x(z))$. A finite table can even have no high-order coefficients while a chosen smooth extension has nonzero high derivatives between vertices whose integrated effects cancel. Absence of a coefficient is a fact about the declared finite contrast, not necessarily the absence of every local nonlinearity.

The pair-only statement with its full subject

A fixed quadratic potential evaluated on an affine action-to-state map has no Möbius coefficients above order two. Quadratic potential alone is insufficient. Nonlinear response alone does not guarantee a nonzero high-order coefficient.

We now know what an affine endpoint would imply. To determine whether an actual response has that form, we need its law of change. The next chapters introduce the motion law and generator, derive a feedback example, and show exactly when superposition survives through time. We then return to the general composite response in Chapter 34.

Why a field is not a motion law

An account can reveal a useful direction without choosing the action that follows. We now specify the additional motion law needed to turn local information into a trajectory. Exact descent conditions, conservation and finite-step limits show how EBU can inform a controller while preserving the distinction between valuation, physical response and institutional choice.

31.1 A map does not move the traveler

The potential maps physical states to numbers. Its gradient describes how that number changes near the present state. Nothing in this definition makes a pump run, a household choose a repair, or a government open a water route. Conflating a measurement with a controller would hide the most important design question in EBU: how could the information guide physically possible conduct?

The distinction already appears in mechanics. A height map identifies downhill directions, but a stone's motion depends on inertia, friction and constraints. In thermodynamics, a driving force and a constitutive response are also different objects. The EBU field can describe a directional tendency; a separate account of mobility and choice is required before a trajectory is predicted.

31.2 The Onsager analogy and its boundary

Lars Onsager's 1931 work on reciprocal relations studied near-equilibrium irreversible processes. In that setting, suitably chosen thermodynamic forces and fluxes are connected by linear response coefficients, subject to assumptions including the relevant reversibility conditions [19, 20]. This historical idea is useful here because it shows why naming a "force" does not yet specify a rate: the response coefficients and their domain matter.

As a proposed EBU controller, one might write

$$\dot{x} = -M(x)\nabla V(x), \quad (31.1)$$

where M maps gradients into physically admissible state velocities. If M is symmetric positive semidefinite, the field is fixed and the motion remains in

the declared domain, then

$$\frac{dV}{dt} = \nabla V^\top \dot{x} = -\nabla V^\top M \nabla V \leq 0. \quad (31.2)$$

That inequality follows from the proposed dynamics and its assumptions. It does not say that the actual economy follows Equation 31.1, nor that every real action should move downhill. Essential service, finite reserves and irreversible transformations can create obligations that a one-line gradient flow does not express.

Onsager's point was not that a gradient alone creates a flow. The classical setting identifies physical fluxes, such as heat or matter transport, and forces related to departures from equilibrium. Close enough to the specified equilibrium, the fluxes can be approximated by a matrix acting on those forces. The matrix's entries describe the system's response. Reciprocity is a further statement about paired coefficients under the relevant microscopic assumptions; it is not a universal property of every social decision process. If an EBU implementation calls a numerical gradient a "force," it still owes a measured or designed map from that force to a permissible physical rate.

For EBU, the useful structure is the explicit map from a directional quantity to a permissible rate. Mobility carries units and response assumptions. Specifying them makes it possible to change delivery speed without silently changing the potential, or to change the physical constraints without pretending the field itself supplied permission.

31.3 The directed threshold response has its own identity

For the earlier response $J_e = M_e[f_e - \theta_e]_+$ with $M_e > 0$ and $\theta_e \geq 0$, suppose the complete motion is $\dot{x} = u + SJ$. Since $f = -S^\top \nabla V$,

$$\dot{V} = \nabla V^\top u - \sum_e f_e J_e = \nabla V^\top u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (31.3)$$

For an active edge, $f_e = J_e/M_e + \theta_e$; for an inactive edge, its product is zero. This proves the equality for that declared response. With no external drive the potential cannot increase along an admissible continuous solution. Thresholds, disconnected routes and constraints can still prevent arrival at the reference. A later cap or projection changes the vector field and requires checking the actual accepted flux.

For a frozen Euler direction w , a quadratic gives the exact change $\Delta V = h \nabla V^\top w + \frac{1}{2} h^2 w^\top H w$. When $D = -\nabla V^\top w > 0$ and $w^\top H w > 0$, nonincrease along this frozen direction holds precisely for $0 \leq h \leq 2D/(w^\top H w)$. Physical feasibility remains a separate condition. On a network, $w = SJ$ brings in $J^\top S^\top H S J$, including shared-node cross terms.

31.4 Conservation restricts motion

Suppose $Ax = b$ expresses conserved homogeneous totals. A permissible internal flow must satisfy $A\dot{x} = 0$. An arbitrary $-\nabla V$ usually will not. One can construct a mobility whose image lies in the conservation tangent space, or write flows through an incidence matrix, $\dot{x} = Sj$, with $AS = 0$. Either way the conservation law is part of the physical permission layer, not a favourable term that can be traded against EBU.

Additional constraints may bound sources, routes, reserve stocks and simultaneous demand. The menu of feasible actions can therefore be empty even when a mathematical descent direction exists. Conversely, a permitted service action can have negative EBU under the current field. A controller has to say how it treats that case; the potential cannot silently answer it.

31.5 Finite steps require a finite check

Even if a continuous trajectory satisfies $dV/dt \leq 0$, a large discrete step based on its initial slope can overshoot. For the fixed quadratic field, Equation 21.6 shows exactly why: the curvature term grows quadratically with the step. If a policy promises non-increasing V on accepted actions, it must check the finite successor, not merely the sign of the initial gradient.

For example, freeze $g = \nabla V(x)$ and propose one Euler step $\delta = -hMg$ with $h > 0$. The exact quadratic calculation is

$$E(h) = hg^T Mg - \frac{h^2}{2}(Mg)^T H(Mg). \quad (31.4)$$

The first term is the infinitesimal improvement suggested by mobility; the second is the finite curvature correction. When both terms are positive and finite, sufficiently small h improves the field, while an arbitrarily large h need not. This statement still presupposes that the proposed increment is physically feasible. It is not a license to step outside a stock or route constraint.

In one dimension, set $V(x) = x^2/2$ and let mobility be the positive number m . The continuous proposal is $\dot{x} = -mx$, so a solution beginning at x_0 is $x(t) = x_0 e^{-mt}$ and its potential is $V(t) = V(0)e^{-2mt}$. The simple finite Euler instruction $x' = x - hmx$ is different. Direct calculation gives

$$E = \frac{1}{2}x^2[1 - (1 - hm)^2] = \frac{1}{2}x^2(2hm - h^2m^2). \quad (31.5)$$

It improves the potential only for $0 < hm < 2$, apart from equality at the endpoints. A larger discrete step can reverse the promised direction even though the continuous law is restoring. This is why a real quote should use

the actual finite successor rather than attach an unlimited entitlement to a favourable local slope.

The same lesson appears in the first transfer example of this book. Moving some stock toward the reference improves balance; moving too much crosses the reference and can worsen it. Exact EBU is therefore a more useful action account than a marginal slogan. Whether a decision-maker chooses the best feasible quantity, a safe bounded quantity, or a quantity required by demand remains a separate rule.

31.6 A usable architecture

A complete design joins four objects: the represented physical state, the feasible transformations, the field that values their consequences and the rule that selects or generates them. Each has a clear mathematical task. Keeping them distinct makes their connections easier to use.

For a pure unforced gradient law with symmetric positive semidefinite mobility, V is nonincreasing. Such a law cannot sustain a nonconstant periodic orbit: a closed orbit would have zero total decrease, forcing the dissipation and hence the velocity to vanish along it. At a nondegenerate minimum, its linearized rates are real and nonpositive under the same mobility assumptions. Appendix C gives the precise argument.

A pipeline controller can instead have complex decaying modes because its pending state and consumption change the response architecture. It may use a receiver-shortage signal to request dispatch while water arrives under another law. That system must be analysed through its full generator; its stability need not be descent of the receiver potential.

This separation supports an efficient EBU design process. Use the field for a consistent comparison, the response law for timing and recovery, and the feasible set for physical safety. The resulting account can explain why a necessary service consumes resources, where correction is delayed and which improvements reduce avoidable activity. The next chapter gives the common generator interface through which all these response laws supply endpoints.

A generator supplies the movement

A generator supplies the endpoint produced by a declared law of change. Once that endpoint exists on the physical domain, it enters the same finite EBU and coalition formulas as an immediate action. This interface connects accounting to duration, background movement and ordered execution without requiring a new definition of value.

32.1 From a map of value to a rule of change

The previous chapter separated the gradient from the rule that operates a pump. We now give a general interface for a declared motion law. Once a law and a duration determine an endpoint for each control setting, that endpoint can enter the same coalition table. The potential definition and Möbius transform need not be reinvented.

A generator therefore sits beside the valuation chain as a source of physical response. It is not the foundation from which EBU must always be derived. Measured finite transitions, fixed increments and jumps can all have endpoint EBU without being solutions of the particular differential equation used here.

32.2 A well-posed flow is needed first

Let a fixed control vector z specify a vector field b_z . Consider

$$\dot{x} = b_z(x), \quad x(0) = x_0, \quad x_z(T_f) = \Phi_z^{T_f}(x_0). \quad (32.1)$$

For each control used in the coalition table, require a unique solution through the entire declared duration T_f , within the admissible state domain. Continuity and local Lipschitz dependence on state are familiar sufficient local conditions; existence up to the chosen horizon and domain invariance still need to be established.

These requirements prevent the endpoint from being ambiguous. The equation $\dot{x} = \sqrt{|x|}$ starting at zero has nonunique solutions: the state can remain zero for an arbitrary waiting time before departing. The equation $\dot{x} = x^2$ starting at one has solution $1/(1 - t)$ and no finite endpoint at or beyond $t = 1$. Writing a generator symbol does not repair either problem.

32.3 The generator-to-coalition theorem

Theorem 32.1 (Flow, value and interaction). Suppose every required control $z = \chi_S$ has a unique admissible endpoint $\Phi_z^{Tf}(x_0)$, and V is finite there. Set

$$E_{Tf}(S) = V(x_0) - V(\Phi_{\chi_S}^{Tf}(x_0)). \quad (32.2)$$

Then for every nonempty T ,

$$m_{E_{Tf}}(T) = -\Delta_T[V(\Phi_z^{Tf}(x_0))]_{z=0}. \quad (32.3)$$

If in addition V and the trajectories satisfy the path theorem,

$$E_{Tf}(S) = -\int_0^{Tf} \nabla V(x_S(t))^T b_{\chi_S}(x_S(t)) dt. \quad (32.4)$$

Proof. The flow supplies a complete endpoint map. Apply the coalition definition and the cancellation of the common constant $V(x_0)$ in every nonempty difference. Along each regular solution, the chain rule gives $dV/dt = \nabla V^T b_z$; its integral gives the path expression. $\square$

The first statement requires endpoints, not a smooth interpolation across all control values. Control derivatives used later need stronger parameter regularity. Keeping those levels separate makes the theorem usable for a broad range of physically declared response models.

32.4 No control can still mean natural movement

An empty coalition need not leave a dynamical system at rest. If $b_0 \neq 0$, the no-action endpoint $x_\emptyset(T_f)$ can differ from x_0 . Then the raw table has

$$E_{Tf}(\emptyset) = V(x_0) - V(x_\emptyset(T_f)) \neq 0. \quad (32.5)$$

It measures the whole interval, including natural drift. To compare controls against that same no-action evolution, define the centered table

$$E_{Tf}^{\text{rel}}(S) = V(x_\emptyset(T_f)) - V(x_S(T_f)). \quad (32.6)$$

Now $E_{Tf}^{\text{rel}}(\emptyset) = 0$. The raw and relative tables differ by a constant, so all their nonempty Möbius coefficients agree. Their total values and interpretations differ by the baseline drift.

The relative table isolates the action comparison within the declared dynamics. The raw table retains the complete interval change. EBU can use both without crediting passive recovery to an idle actor: the background line and the action line remain distinct, and together explain the physical history.

32.5 Historical force, flux and opposition

The historical EBU force–response chain can be read through this interface. A potential supplies directional information, mobility and thresholds determine requested flux, and complete physical balances determine the state velocity. A scalar quantity called opposition in a particular proposal is one input to that construction; it is not by itself the full vector field b_z .

The complete generator includes source withdrawal, pending stock, arrival, consumption and every represented boundary flow. The arrival direction must also include a valued sink when losses are present, as derived in Chapter 13. A controller signal based on eventual completed delivery need not equal the derivative of immediate dispatch.

This explicit construction makes response models interchangeable in a useful sense. Different route or controller laws can supply different endpoints while the EBU definition remains the same. Their assumptions stay visible, so a difference in outcome can be traced to the response instead of being hidden in a redefinition of value.

32.6 Ordered actions ask a different question

A flow under simultaneous controls produces one endpoint. Performing A and then B can produce another, and reversing their order can produce a third. These are ordered words, not interchangeable rows of an unordered subset table.

For a concrete smooth example in the plane, let $X = \partial_x$ and $Y = x\partial_y$, starting from $(0, 0)$. Applying X for time T_f then Y for time T_f reaches (T_f, T_f^2) . Reversing the order reaches $(T_f, 0)$. Applying $X+Y$ simultaneously for time T_f reaches $(T_f, T_f^2/2)$. All statements follow by direct integration of these declared equations. Each sequential protocol lasts $2T_f$, while the simultaneous protocol here lasts T_f ; the example isolates the stated order and timing conventions, rather than claiming a comparison at equal total duration.

The order difference is related at small times to the Lie bracket $[X, Y] = DYX - DXY$. For observables, $[L_X, L_Y] = L_{[X, Y]}$, where $L_X f = Df[X]$. Appendix C gives the convention and expansion. The difference between two orders is not itself an unordered Möbius coefficient; one must first define a coalition protocol for every subset.

32.7 Why the interface matters for EBU

The generator interface gives the account a disciplined time dimension. A repair can alter what a delivery achieves after an hour; pending material can

change the endpoint after a day. Fixing the horizon lets these consequences be compared consistently. A failure of endpoint uniqueness belongs to the response model, while an incorrect potential difference belongs to valuation.

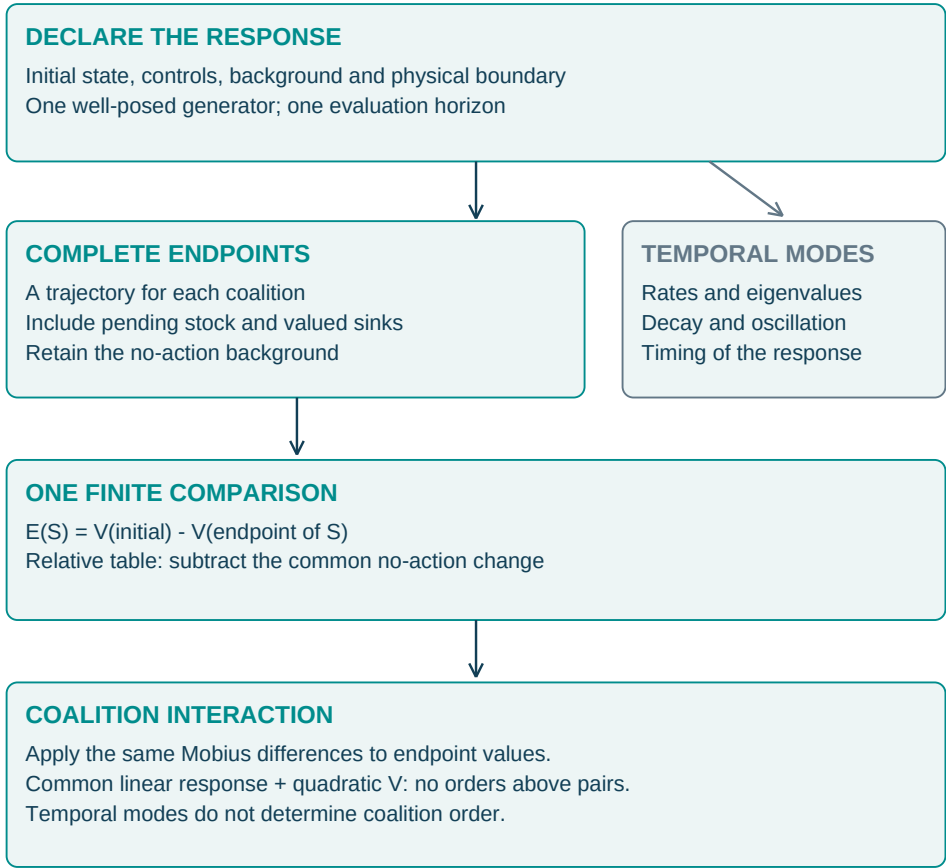


Figure 32.1: The generator supplies a physical endpoint. Finite EBU values that endpoint, and Möbius differences compare the resulting coalition table. Temporal modes describe the dynamics on a separate branch of the same model.

The practical feature is a common account across different physical mechanisms. A community can compare a delivery, a repair and a scheduling change through their complete outcomes while retaining their different timescales and resource boundaries. This supports coordinated maintenance and reduces the temptation to treat an immediate stock reading as the whole future.

The next chapter connects pending stock, memory and oscillation to the affine endpoint theorem, then identifies where nonlinear response creates higher orders.

Feedback, memory and interaction through time

A delivery changes the future as well as the present. This chapter derives how pending stock creates memory, when correction oscillates, and which joint values remain exactly pairwise despite that temporal behaviour. It then identifies the nonlinear response that can create higher-order interaction, connecting physical dynamics to the finite accounting laws.

33.1 Water in the pipe is part of the present

A reservoir can be low even while a large delivery is already on its way. A controller that sees only the reservoir may order more water, then receive both deliveries after the shortage has passed. The response can overshoot and settle through diminishing corrections. Pending stock makes that response a property of the present state, rather than an unexplained surprise.

We now give the generator a concrete physical form. One stock-and-pipeline model supplies exact statements about its operating point, memory and temporal modes. Its coalition table then supplies another exact result: damped oscillation can coexist with zero three-way interaction. A monotone response can, conversely, have a nonzero triple. The two classifications answer different questions.

The model and its historical native units are reconstructed in the feedback reconciliation [47]. We retain its explicit response assumptions and use a separate dimensionless example to connect it to finite EBU. This lets the reader follow the complete theoretical bridge in Book 1; later volumes develop applications, designs and results.

33.2 Declare the stocks, the rule and the unit

Let X be usable receiver stock, Q pending stock, and L a declared receiver target. Constant consumption is $h > 0$. A maintained source dispatches material into the pipeline at rate D , and arrival transfers pending material into the receiver at rate A . On one lossless, unconstrained deficit branch,

declare

$$\begin{aligned} V_B(X) &= w(L - X)^2, & D &= \nu(L - X), \\ \nu &= 2wM, & A &= Q/\tau, & w, M, \tau &> 0. \end{aligned} \quad (33.1)$$

The resulting balances are

$$\dot{X} = Q/\tau - h, \quad \dot{Q} = \nu(L - X) - Q/\tau. \quad (33.2)$$

The first says that arrival replenishes the receiver while consumption empties it. The second says that dispatch fills the pipeline while arrival empties it. Together they show why the pending stock carries information about the future.

These balances do not derive the dispatch policy. That policy uses the shortage signal of a completed delivery, $-V'_B(X) = 2w(L - X)$, and a separately declared response coefficient M . Nor do they derive the residence law $A = Q/\tau$. Both are additional modelling choices. A maintained source means that its supply conditions are assumed; it does not mean that water, replenishment or pumping is free.

The subscript on V_B matters. The historical potential has a native burden unit B. With resource unit U and time unit T, w has units B/U², M has U²/(BT), and ν has 1/T. The residence parameter τ has units of time. We use T_f for the observation horizon, keeping it distinct from τ .

Native burden and normalized EBU retain distinct units. As Chapter 11 explained, a conversion $V = (V_B - c_B)/B_0$ requires a declared fixed burden scale B_0 . The roots of this controller identify response timescales, not that denomination. They depend on the product $\nu = 2wM$, so they also cannot separate burden weight from mobility.

The physical branch also matters: $0 < X < L$, nonnegative pending stock, a maintained source and no active capacity switch. The affine equations exist mathematically outside that branch, but a physical interpretation must establish that the states used remain admissible. These are the assumptions under which the following statements have their physical meaning.

33.3 Steady stocks can retain a shortage

At an operating point, arrival and dispatch must both match consumption. Setting the two derivatives in Equation 33.2 to zero gives

$$Q_* = \tau h, \quad X_* = L - h/\nu. \quad (33.3)$$

The point lies on the interior branch when $L > h/\nu$. Its receiver burden is

$$V_B(X_*) = w(h/\nu)^2 > 0. \quad (33.4)$$

A continuing shortage is what keeps this particular proportional controller dispatching enough to replace consumption. The stock is steady because flows balance, not because the receiver has reached the minimum of V_B .

This is an operating equilibrium of the declared deterministic equations. It is not automatically thermodynamic equilibrium, zero burden or an optimum. For an EBU service design, reliability and adequacy therefore have separate, useful indicators. The operating point describes what this controller sustains; the reference describes the condition against which shortage is valued. Improving a reliable but inadequate service means changing its physical capacity or response, not merely relabelling stability as zero burden.

33.4 Two stocks become two temporal modes

Define deviations $x = X - X_*$ and $q = Q - Q_*$. Subtracting the operating balances yields exactly

$$\dot{x} = q/\tau, \quad \dot{q} = -\nu x - q/\tau. \quad (33.5)$$

There is no linearization error on the stated affine branch. A negative q means less pending stock than usual; it does not mean negative physical inventory.

Theorem 33.1 (Modes of the declared pipeline model). For $\nu, \tau > 0$, the deviation matrix and its characteristic equation are

$$F = \begin{pmatrix} 0 & 1/\tau \\ -\nu & -1/\tau \end{pmatrix}, \quad \tau\lambda^2 + \lambda + \nu = 0. \quad (33.6)$$

The roots are

$$\lambda_{\pm} = \frac{-1 \pm \sqrt{1 - 4\nu\tau}}{2\tau}. \quad (33.7)$$

All modes decay. They are complex precisely when $4\nu\tau > 1$.

Proof. The determinant of $\lambda I - F$ is $\lambda^2 + \lambda/\tau + \nu/\tau$. Completing the square gives $(2\tau\lambda + 1)^2 = 1 - 4\nu\tau$. If the discriminant is nonnegative, its square root is less than one, so both roots are negative. Otherwise their common real part is $-1/(2\tau) < 0$. $\square$

For $4\nu\tau < 1$ there are two distinct real decaying modes. Equality gives the repeated critical rate and solutions of the form $(C_1 + C_2 t)e^{-t/(2\tau)}$. Above one, the free response has an exponentially shrinking sinusoidal component. An observed oscillation still requires excitation of the mode and a quantity in which it is visible; the unperturbed operating point does not start moving merely because its matrix has complex roots.

The threshold is exact for this model. Changing how dispatch uses pending stock, imposing a capacity constraint or introducing a true transport delay changes the model and requires a new analysis. Damped-oscillator roots are standard dynamics, not a new EBU law [49].

33.5 A period is not a price

In the complex-mode regime, write

$$\lambda_{\pm} = -\alpha \pm i\omega, \quad \alpha = \frac{1}{2\tau}, \quad \omega = \frac{\sqrt{4\nu\tau - 1}}{2\tau}. \quad (33.8)$$

The amplitude envelope decays as $e^{-\alpha t}$. The angular frequency is ω , the ordinary frequency is $\omega/(2\pi)$, and the period is $2\pi/\omega$. These quantities describe how the declared model responds over time. None is an EBU price.

For infrastructure coordination, these timescales distinguish an outstanding correction from a new shortage. A response-aware scheduler can use pending deliveries and settling time to avoid repeated orders against the same deficit. EBU then values the complete accepted consequence, including the source and process boundary. The feature supports more reliable service with less avoidable corrective throughput.

33.6 Removing a stock does not remove its memory

Suppose an observer wants an equation for the receiver alone. Multiply $\dot{q} + q/\tau = -\nu x$ by the integrating factor $e^{t/\tau}$ and integrate. The hidden stock becomes

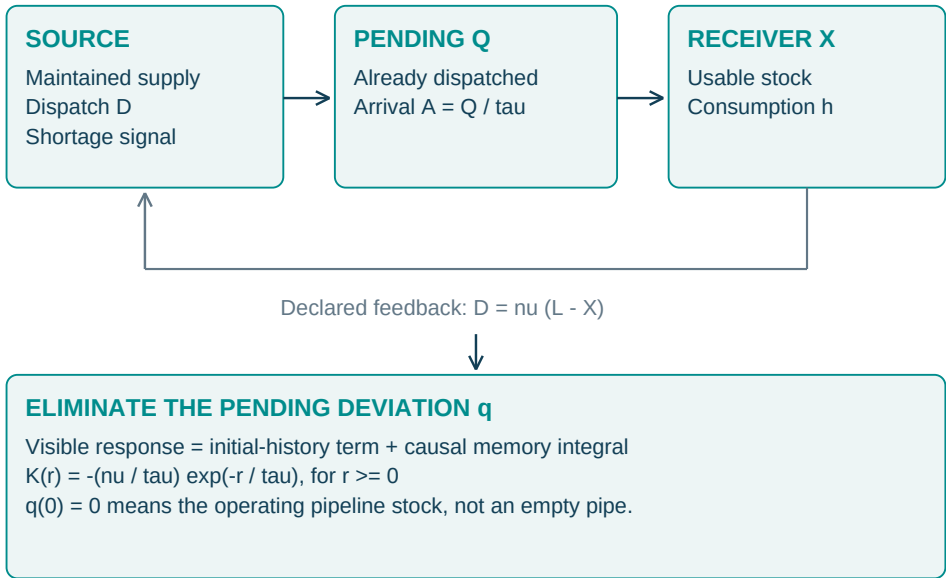
$$q(t) = e^{-t/\tau} q_0 - \nu \int_0^t e^{-(t-s)/\tau} x(s) ds. \quad (33.9)$$

Substituting into $\dot{x} = q/\tau$ gives

$$\begin{aligned} \dot{x}(t) &= \frac{q_0}{\tau} e^{-t/\tau} + \int_0^t \mathcal{K}(t-s)x(s) ds, \\ \mathcal{K}(r) &= -\frac{\nu}{\tau} e^{-r/\tau} \chi_{[0,\infty)}(r). \end{aligned} \quad (33.10)$$

The present rate remembers previous receiver deviations because they changed dispatch and therefore the pending stock. The kernel is causal and restoring. It has units $1/T^2$; integration against stock and time produces a stock rate. Its integral is $-\nu$, so the full kernel is not a probability density.

The initial term cannot be discarded. Here $q_0 = Q(0) - Q_*$; choosing $q_0 = 0$ means the pipeline initially has its operating stock, not that the pipeline is empty. An empty pipeline instead gives $q_0 = -\tau h$. Two receivers with the



Explicit pending state and its memory representation are equivalent.

Deleting both would lose commitments already made.

Figure 33.1: The pending stock is an explicit state in the two-variable model. Eliminating it replaces that state by a memory integral and its initial-condition term. The diagram describes a declared response model, not measured flow data.

same current stock but different pending stocks can have different immediate rates and different futures.

This is a first-order lag or residence model. A pure travel time would instead require dispatch history and a different delay equation. An added dispatch input also contributes its own convolution; Appendix C derives that term. One exponential compartment gives no universal memory kernel for every supply network.

The representation rule gives EBU a concrete feature: retain pending commitments either as explicit state or as equivalent memory with its initial condition. Both descriptions preserve the same response information. Omitting both would invite duplicate interventions, while retaining them supports reliable scheduling and a clearer account of what remains to be delivered.

33.7 A stable controller can move burden upward

The historical stability argument introduces another quantity,

$$\mathcal{E}_B(x, q) = wx^2 + \frac{q^2}{2M\tau}. \quad (33.11)$$

It is positive definite in the deviations. Using $\nu = 2wM$ gives

$$\begin{aligned} \dot{\mathcal{E}}_B &= 2wx\frac{q}{\tau} + \frac{q}{M\tau} \left(-2wMx - \frac{q}{\tau} \right) \\ &= -\frac{q^2}{M\tau^2} \leq 0. \end{aligned} \quad (33.12)$$

The chosen pending term cancels a dynamical cross term. It has not thereby been established as the economic or physical value of pending inventory. Nor is a momentary zero derivative proof of rest: at $q = 0$ and $x \neq 0$, the second equation immediately changes q .

Keep three objects separate. V_B measures the native receiver burden. $\mathcal{E}_B$ measures deviation for this stability proof. Canonical E is a finite difference of an independently declared normalized potential. The shared appearance of a quadratic does not identify them.

For an exact single-state example, use compatible units with $w = 1$, $M = 1/2$, $\tau = 2$, $\nu = 1$, $L = 10$ and $h = 2$. The operating point is $(X_*, Q_*) = (8, 4)$. At $(X, Q) = (9, 3)$, direct substitution gives

$$\dot{V}_B = +1 \text{ B/T}, \quad \dot{\mathcal{E}}_B = -\frac{1}{2} \text{ B/T}. \quad (33.13)$$

Receiver burden is increasing while the stability measure decreases. The system is moving toward an operating point that retains a larger shortage. The difference follows directly from the vector field at the stated point.

The stability function is a mathematical instrument for this response law. Its receiver term is the quadratic remainder of V_B about the operating point: $V_B(X) - V_B(X_*) - V'_B(X_*)(X - X_*) = wx^2$. Its pending term was chosen to cancel the cross term in the derivative. It therefore serves a different purpose from valuing pending inventory. An EBU service record can report both reliability and burden without replacing one by the other, making persistent shortage visible even in an orderly controller.

33.8 Linear feedback preserves an affine coalition response

Return now to the coalition question. A feedback system may have many hidden modes, including oscillatory ones. If every action enters through a

fixed additive input to the same linear system, the endpoint still has the affine form required by the degree theorem.

Theorem 33.2 (Common linear feedback and affine endpoints). Let every required coalition share an initial state y_0 , horizon T_f , field, comparison protocol and well-posed linear dynamics

$$\dot{y} = F(t)y + g_0(t) + \sum_{i \in S} g_i(t). \quad (33.14)$$

Assume integrable coefficients and admissibility throughout the common horizon. Then

$$y_S(T_f) = y_\emptyset(T_f) + \sum_{i \in S} h_i(T_f). \quad (33.15)$$

Proof. For the common state-transition matrix $U(t, s)$, variation of constants gives the empty-input endpoint and each action response as

$$\begin{aligned} y_\emptyset(T_f) &= U(T_f, 0)y_0 + \int_0^{T_f} U(T_f, s)g_0(s) ds, \\ h_i(T_f) &= \int_0^{T_f} U(T_f, s)g_i(s) ds. \end{aligned} \quad (33.16)$$

Linearity of integration and uniqueness prove the decomposition. $\square$

This is standard linear-systems mathematics [42, 48]. A fixed feedback gain can be part of F . A coalition-dependent gain generally cannot. Stability is not needed for the finite-horizon identity if the required endpoints exist and remain admissible. Saturation or switching must not silently replace the common affine model for some rows of the table.

Corollary 33.3 (Oscillatory modes do not raise the quadratic ceiling).

Under Theorem 33.2, let the fixed potential be $V(y) = c + \ell^\top y + \frac{1}{2}y^\top H y$, with $H = H^\top$. For the empty-evolution-relative table,

$$m_E(i) = -(\ell + H\bar{y})^\top h_i - \frac{1}{2}h_i^\top H h_i, \quad (33.17)$$

where $\bar{y} = y_\emptyset(T_f)$, while

$$m_E(ij) = -h_i^\top H h_j, \quad m_E(T) = 0 \quad \text{for } |T| \geq 3. \quad (33.18)$$

Proof. Substitute the affine endpoint into the quadratic. On Boolean indicators the result has degree at most two; Theorem 30.1 applies. A constant raw drift offset changes only the empty coefficient. $\square$

The result gives EBU an exact economical representation even when deliveries oscillate. A common linear response and quadratic valuation require only singleton and pair terms; higher temporal complexity does not add higher coalition orders. A fixed gain belongs inside the common operator. If a coalition changes that gain, a different endpoint map must be analysed.

33.9 An oscillatory response with an exact zero triple

To make the corollary tangible, use a separately declared dimensionless teaching example. This is not a calibration of the historical unit B. Take $\nu = \tau = 1$, $L = 3$, $h = 1$, common initial state $(X, Q) = (2, 1)$ and potential $V(X, Q) = (3 - X)^2/2$. Each of three action labels adds constant dispatch $\epsilon = 1/100$ over the same horizon.

For a coalition of size m , the deviations obey $\dot{x} = q$ and $\dot{q} = -x - q + m\epsilon$. Their exact solution can be written $x = m\epsilon g(t)$ and $q = m\epsilon \dot{g}(t)$, where

$$g(t) = 1 - e^{-t/2} \left[\cos \frac{\sqrt{3}t}{2} + \frac{1}{\sqrt{3}} \sin \frac{\sqrt{3}t}{2} \right]. \quad (33.19)$$

Direct differentiation verifies the equation and zero initial deviations. The nonempty responses oscillate about their input-shifted operating points. Appendix C checks that all eight coalitions stay on the declared physical branch by elementary inequalities, without producing a time series.

At any fixed T_f , put $r = \epsilon g(T_f)$. The complete value table is summarized by its four distinct rows:

Coalition size	Endpoint EBU
0	0
1	$r - r^2/2$
2	$2r - 2r^2$
3	$3r - 9r^2/2$

Every singleton and every pair has its corresponding row. Subtraction gives

$$m_E(AB) = -r^2, \quad m_E(ABC) = E_3 - 3E_2 + 3E_1 - E_0 = 0. \quad (33.20)$$

Thus a nonzero pair may coexist with a zero triple despite damped oscillation. If the same dynamics is valued with the linear supplied potential $V = X$, the table is $E_m = -mr$ and even the pair vanishes. The zero triple is an exact identity at every admissible horizon. A nonzero pair additionally requires $re0$.

33.10 Temporal modes and interaction order ask different questions

Dynamics, value and interaction

A **generator or feedback law** supplies a trajectory and its endpoint. A **potential** values the endpoint. **Möbius differences** compare coalition values. **Eigenvalues or poles** describe temporal modes of the declared dynamics. The number or kind of those modes does not determine the order of a coalition interaction.

The converse example is just as simple. With fixed Boolean controls a_i , let $\dot{y} = a_1 a_2 a_3$, $y(0) = 0$ and $V(y) = y$. Each response is constant or monotone, yet $m_E(ABC) = -T_f \neq 0$ for $T_f > 0$. The action inputs combine nonadditively. No oscillation is needed.

The two remaining combinations are equally explicit. A linear input law $\dot{y} = \sum_i a_i$ with linear $V = y$ has only singleton values and no oscillation. In the pending tank, replacing the additive input by $u = \epsilon a_1 a_2 a_3$ leaves every proper coalition at the original operating point, while the full coalition has the same oscillatory response $\epsilon g(t)$. With $V = X$, its triple is $-\epsilon g(T_f)$, nonzero whenever $g(T_f) \neq 0$. The earlier admissibility bound still applies because this input is no larger than the three-action additive case.

This separation supports two distinct EBU design features. A response display can identify timing and reliability, while an interaction display explains the dependence of joint intervention values. A service can need less oscillation without needing a higher-order valuation model, or need joint coordination without oscillating at all. The distinction directs effort toward the actual problem.

33.11 The first few time terms

For one fixed z , write at x_0

$$v = b_z(x_0), \quad A = Db_z(x_0), \quad B = D^2b_z(x_0).$$

TEMPORAL BEHAVIOUR AND COALITION ORDER

MONOTONE / SINGLETONS

$dy/dt = \text{sum of action inputs}$
 Linear $V(y)$
 Only singleton terms

OSCILLATING / PAIRS

Common linear pending tank
 Quadratic V
 Pairs allowed; triples exactly zero

MONOTONE / TRIPLE

$dy/dt = a_1 a_2 a_3$
 Linear $V(y)$
 Triple = $-T_f$

OSCILLATING / TRIPLE

Pending tank; input = $\text{eps } a_1 a_2 a_3$
 Linear receiver $V = X$
 Triple = $-\text{eps } g(T_f)$, when nonzero

Four declared examples; no sampled trajectory or performance ranking.

Figure 33.2: Temporal behavior and coalition order are independent classifications. These four algebraic examples compare different declared models; they are not simulated outcomes or a performance ranking.

Under the smoothness and bounded-neighbourhood assumptions detailed in Appendix C,

$$\begin{aligned}
 E_z(T_f) = & -T_f DV[v] \\
 & - \frac{T_f^2}{2} \{D^2V[v, v] + DV[Av]\} \\
 & - \frac{T_f^3}{6} \{D^3V[v, v, v] + 3D^2V[v, Av] \\
 & \quad + DV[B[v, v] + A^2v]\} + O(T_f^4). \quad (33.21)
 \end{aligned}$$

The first term evaluates the starting velocity. The next terms account for curvature of V and change in the velocity as the state moves. The expression concerns one declared flow; taking its coalition differences then reveals interaction orders.

The remainder requires more than the existence of the printed derivatives at one point. A convenient sufficient condition is a C^3 vector field and C^4 potential on a neighbourhood containing the short trajectories, with the needed

bounds. This is an asymptotic statement as duration shrinks, not a promise that three terms describe a long interval accurately.

33.12 Control-affine does give a short-time order rule

Suppose

$$b_z = f_0 + \sum_i z_i f_i, \quad (33.22)$$

with adequate smoothness and common admissible short-time solutions. For $|T| = k$, the leading result is

$$m_{E_{T_f}}(T) = -\frac{T_f^k}{k!} \sum_{\pi \in \text{Perm}(T)} L_{f_{\pi_1}} \cdots L_{f_{\pi_k}} V(x_0) + O(T_f^{k+1}). \quad (33.23)$$

A sufficient regularity choice is C^k fields and $V \in C^{k+1}$ on a suitable common neighbourhood. Appendix C proves the result by expanding iterated generators.

Why can no lower time order contain all k actions? A product of fewer than k control-affine generator factors cannot contain every one of the k distinct control labels. The k -fold Boolean difference removes those lower-degree terms. At order k , every surviving word contains each action field once, in some order. Drift words enter at later orders because including drift would use a place needed for one of those action labels.

This is a local order statement. Higher coefficients can become nonzero over a finite interval, and the displayed leading coefficient can itself vanish through cancellation. Its value for EBU is to identify which time orders can first contain a specified combination of controls, with a controlled remainder.

33.13 An exact counterexample at every order

Take the scalar controlled equation

$$\dot{x} = ux, \quad u = \sum_{i \in S} 1 = |S|, \quad x_0 = 1, \quad V(x) = \frac{1}{2}x^2. \quad (33.24)$$

The generator is affine in the controls and the potential is quadratic. Nevertheless the endpoint is $x_S(T_f) = e^{|S|T_f}$, so

$$E_{T_f}(S) = \frac{1}{2}(1 - e^{2|S|T_f}). \quad (33.25)$$

For every nonempty T of size k , binomial expansion gives

$$m_{E_{T_f}}(T) = -\frac{1}{2}(e^{2T_f} - 1)^k. \quad (33.26)$$

At every positive duration this is nonzero at all available orders. As $T_f \rightarrow 0$, it is $O(T_f^k)$, exactly consistent with the short-time theorem.

The coefficient does not contradict the quadratic pair-only result: the endpoint map is exponential, not affine in the controls. The example is an exact analytic response illustration, with no claim that uncontrolled exponential growth describes a viable EBU economy.

33.14 Where nonlinear response changes the account

The bilinear example keeps a quadratic potential but creates all interaction orders because the common-operator assumption fails. Saturation, switching, thresholds, shared constraints and coalition-dependent gains can likewise make endpoint increments depend on the group. Higher-order terms then become possible, although cancellation can still remove them.

For example, $\dot{y} = u/(2y)$ from $y(0) = 1$ gives $y(T_f) = \sqrt{1 + uT_f}$ for $u \geq 0$. With $u = |S|$ and $V = y^2/2$, the nonlinear response is cancelled by the valuation and $E(S) = -T_f|S|/2$ has only singleton terms. The relevant object is always the composite endpoint function.

The complete theory now offers a useful design sequence. Include the pending state needed for a well-posed response. Determine its temporal behaviour and physical admissibility. Evaluate its complete endpoints under one field and horizon. Then use the interaction coefficients to decide how much of the coalition hierarchy is required. This can reduce duplicate dispatch, distinguish persistent shortage from transient correction, and reserve complex coordination for the dependencies that actually require it.

The generator has supplied dynamics, not a thermodynamic probability law. Canonical density and entropy enter under their own assumptions in Chapters 37–39. We first finish the general response argument: the next chapter studies $V(x(z))$ when action endpoints no longer superpose.

When actions change one another's physical effects

Feedback has shown both sides of the interaction boundary: linear response preserves the quadratic ceiling, while nonlinear accumulation can break it. We now state the general object behind both results. The coalition account reads the composition of the endpoint response and the potential, so interaction must be understood through that whole composition.

34.1 The endpoint rule belongs in the mathematics

Two pumps can compete for pressure. One repair can enable a delivery that otherwise stops halfway. A resolver can reduce accepted quantities when a shared source is exhausted. In each case the physical endpoint of a group need not equal the starting state plus a fixed vector for each member.

The appropriate general object is a response map $x(z)$. The controls z describe the actions; the response describes what physical state they produce. The potential then evaluates that state. The composite

$$G(z) = V(x(z)) \quad (34.1)$$

is the landscape seen by the coalition calculation. Keeping both stages visible prevents a nonlinearity in physical response from being misdiagnosed as a new feature of the potential.

34.2 The exact condition for a high-order coefficient

For any declared endpoint map on the Boolean vertices and any nonempty T ,

$$m_E(T) = -\Delta_T G(0). \quad (34.2)$$

Thus a nonzero order- k coefficient occurs exactly when the corresponding k -fold finite difference of the composite is nonzero. This condition requires no smoothness. It is necessary and sufficient for that table and baseline.

It would be too strong to replace it with “nonlinearity creates synergy.” Nonlinearity can create a high-order term, but it can also cancel under composition or vanish on the relevant vertices. A correct result must inspect the actual composite contrast.

34.3 A quadratic potential with a triple interaction

Use a family extending the response of Chapter 27:

$$x(z) = z_1 + z_2 + z_3 + \alpha z_1 z_2, \quad V(x) = \frac{1}{2}x^2, \quad x_0 = 0. \quad (34.3)$$

The square contains $2\alpha z_1 z_2 z_3$ before multiplication by $1/2$. All other terms either involve fewer than three distinct controls or reduce to lower-order Boolean monomials. Therefore

$$m_E(123) = -\alpha. \quad (34.4)$$

At $\alpha = 1$ the table already calculated gives -1 . The field is still quadratic. The third action changes the value of the pair response through the composition.

This is a particularly helpful counterexample for EBU. A finite capacity resolver, a completion rule or an enabling action can change accepted physical effects with membership. A quadratic evaluator downstream of that mechanism does not guarantee a pair-only coalition table. Certifying the evaluator's formula is only half of the structural check.

34.4 Even a linear potential can have high-order response effects

Take a scalar field $V(x) = x$ and the constructed response $x(z) = z_1 z_2 z_3$, again from baseline zero. Every proper subset produces zero, while the full group produces one. Hence $E(123) = -1$ and all proper nonempty values are zero, so $m_E(123) = -1$.

There is no curvature in this potential. The entire three-way dependence is in the response: all three controls are required for the endpoint to change. This example resembles an enabling chain in its logic, but its unit response is a mathematical declaration, not a measured law for any particular machine or institution.

The operational implication is constructive. When a high-order effect is found, examine the response before complicating the field. It may be more faithful to represent the actual enabling mechanism than to force the effect into a fitted potential term.

34.5 A nonlinear response with no interaction at all

Now let

$$x(z) = \sqrt{1 + z_1 + z_2 + z_3}, \quad V(x) = \frac{1}{2}x^2. \quad (34.5)$$

The baseline is $x_0 = 1$. Although $x(z)$ is nonlinear,

$$G(z) = \frac{1}{2}(1 + z_1 + z_2 + z_3), \quad E(S) = -\frac{1}{2}|S|. \quad (34.6)$$

All interactions above singletons vanish. The nonlinear response and the nonlinear field cancel in the composite. This exact example is why a list of nonlinear ingredients is not enough to establish a nonzero interaction.

Nor does a zero coefficient prove that the ingredients are simple. In an application, one may need additional measurements to distinguish additive response from cancellation. The final group values alone identify the finite landscape, not every internal mechanism that could produce it.

34.6 How a resolver changes the response

Suppose A and B each request four units from a source holding six. A proportional resolver accepts three each when both are present, while accepting four when either acts alone. The table is well defined, but each action's accepted increment now depends on the coalition.

The same issue arises when feedback gains, capacity switches or completion thresholds change with the group. They change $x(z)$ before the potential evaluates it. The common-linear-operator theorem no longer supplies an affine endpoint automatically. A quadratic V can remain perfectly correct while the composite needs higher orders.

For EBU, retain the request, the resolver's acceptance and the resulting endpoint. Those three objects explain why a joint service differed from isolated proposals. This is particularly valuable around scarce infrastructure: the account can show whether the limiting cause was shared water, pump capacity or an explicit allocation rule. A useful design improvement then targets that cause rather than adding an unexplained adjustment to the final score.

34.7 What can be learned from a finite table

A complete finite table determines its coefficients exactly. Several internal mechanisms can nevertheless produce the same table. The nonlinear cancellation example and an additive response can agree at every coalition endpoint while differing between them.

The invariant object is the composite finite contrast. A decomposition into physical causes requires the corresponding response description. This gives EBU a clear hierarchy of explanation: report the exact group values first, then explain them through the declared mechanism at the level of detail that mechanism supports.

A shared-capacity problem belongs to the resolver; a coupled consequence belongs to the field. The next chapter organises their contributions through the composite chain rule.

Two sources of interaction, one measurable total

The potential can be nonlinear, and the response can be nonlinear. Their derivatives enter one chain rule whose integral gives the finite interaction. This chapter explains that structure at every order and distinguishes the stable coalition total from a representation-dependent division into named origins.

35.1 The field bends, and the response can bend too

A curved potential changes how successive increments are valued. A nonlinear response changes the increments themselves as controls are combined. Both effects can contribute to the same interaction coefficient. To separate them algebraically, we need a smooth response model between the Boolean endpoints.

For this chapter assume the pullback $G = V \circ x$ is sufficiently smooth on a neighbourhood of the relevant control cube. Write $x_i = \partial_i x$, $x_{ij} = \partial_i \partial_j x$, and similarly for higher derivatives. These symbols denote derivatives of the response, not the endpoint of action i alone. The context and subscripts distinguish them from the earlier physical stock coordinates.

35.2 The pair chain rule

Differentiate once: $\partial_i G = DV(x)[x_i]$. Differentiate again. The derivative can act on DV , because the state changes, or on x_i , because the response direction changes. Thus

$$\partial_{ij} G = D^2 V(x)[x_i, x_j] + DV(x)[x_{ij}]. \quad (35.1)$$

The first term pairs the two first-order responses through field curvature. The second evaluates the mixed response through the field slope. Integrating their sum over the pair face, with a minus sign, gives $m_E(ij)$.

This is useful when diagnosing a measured pair effect. The same total can arise from different combinations of field curvature and response dependence. If quantities are certified fixed and additive, $x_{ij} = 0$ and the second term vanishes. Without that certification, attributing the whole effect to curvature would be unjustified.

35.3 The triple chain rule

Differentiate Equation 35.1 in a third distinct control. The result is

$$\begin{aligned}\partial_{ijk}G &= D^3V[x_i, x_j, x_k] \\ &\quad + D^2V[x_{ij}, x_k] + D^2V[x_{ik}, x_j] + D^2V[x_{jk}, x_i] \\ &\quad + DV[x_{ijk}].\end{aligned}\tag{35.2}$$

All derivatives of V are evaluated at $x(z)$. There is one pure third-derivative term, three terms mixing field curvature with second response derivatives, and one slope acting on the third response derivative.

For a quadratic V , the first term vanishes but the others need not. For an affine response, every response derivative of order two or more vanishes, leaving only the pure derivative of V . The formula therefore unifies the two counterexamples of the preceding chapter without changing the meaning of EBU.

35.4 A mixed example with both contributions

Take $s = z_1 + z_2 + z_3$, response $x = s + \alpha z_1 z_2$, and potential $V = x^3/6$. Boolean expansion gives

$$m_E(123) = -1 - \frac{5}{2}\alpha - \frac{1}{2}\alpha^2.\tag{35.3}$$

At $\alpha = 1$ this is -4 . Both cubic field structure and response dependence contribute.

We can see the split directly from the smooth chain rule. At $\alpha = 1$, $x_1 = 1 + z_2$, $x_2 = 1 + z_1$, $x_3 = 1$, $x_{12} = 1$, and the other relevant higher response derivatives vanish. Since $V''' = 1$ and $V'' = x$,

$$\partial_{123}G = (1 + z_1)(1 + z_2) + x.\tag{35.4}$$

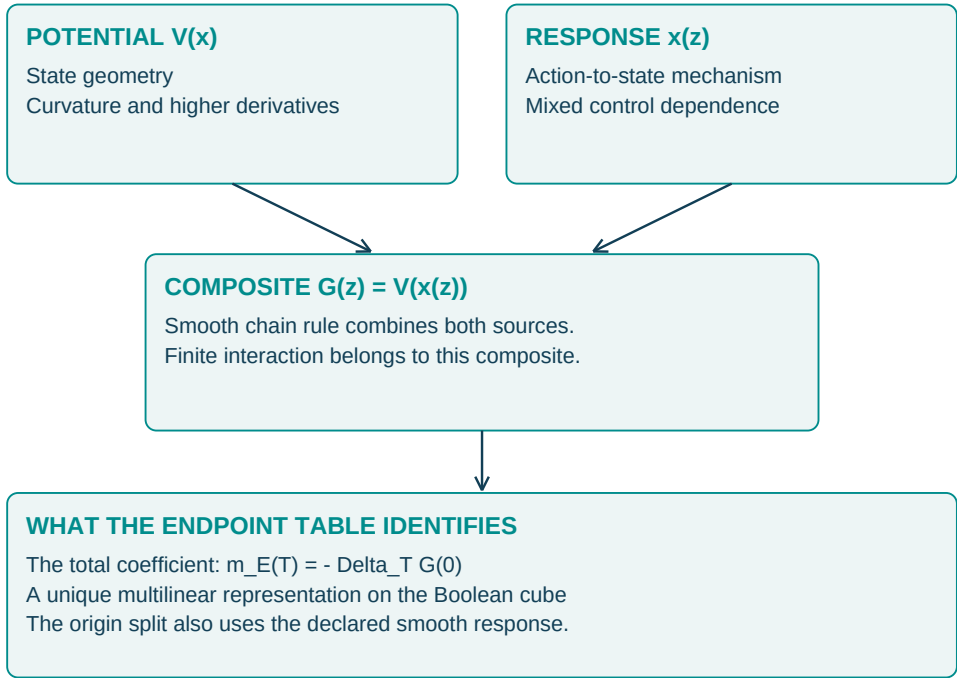
The first term integrates over the unit cube to $9/4$, and the second to $7/4$. Their sum is four, so the EBU coefficient is minus four. This checks the finite table by a different route.

The split is informative within this declared model. It would be misleading to describe $9/4$ and $7/4$ as separately observed physical substances. They are terms in the derivative decomposition of a chosen representation.

35.5 Why the origin split is not unique

Consider two smooth maps on the same pair cube:

$$x^{(a)} = z_1 + z_2, \quad x^{(b)} = z_1 + z_2 + z_1(1 - z_1)z_2,\tag{35.5}$$



Quadratic V + nonlinear response can have higher-order interaction.

Figure 35.1: The coalition coefficient belongs to the composite $G = V \circ x$. Potential curvature and response dependence can both contribute. Their smooth derivative decomposition depends on the chosen coordinates and interpolation; the finite table coefficient does not depend on that interpolation.

with $V = x^2/2$. The extra term vanishes at every Boolean vertex. Both maps therefore produce the same four endpoints and the same pair coefficient $m_E(12) = -1$.

For the first interpolation, the integrated curvature and response terms in Equation 35.1 are 1 and 0. For the second, they are $7/6$ and $-1/6$. They still sum to one. Appendix A gives the elementary integrations.

No observation of the four endpoints alone can choose between these interpolations. Thus the finite interaction is invariant under this change, while the named origin split is not. A nonlinear change of state coordinates can similarly move complexity between V and x . The total composite contrast is the stable object; a causal or physical interpretation of its components needs additional evidence.

35.6 The all-order rule

Let $\Pi(T)$ be the set of partitions of the distinct action indices in T . For a block B , write $x_B = \partial_B x$. The general chain rule is

$$\partial_T(V \circ x) = \sum_{\pi \in \Pi(T)} D^{|\pi|} V(x) [x_B : B \in \pi]. \quad (35.6)$$

Each partition groups together the differentiations that act on one response derivative. The partition into singletons gives the highest field derivative; the one-block partition gives the highest response derivative. All mixed cases lie between them.

For distinct labeled derivatives, each set partition appears once. There is no extra factorial to insert into this form. Appendix B proves the rule by induction and relates it to established multivariable chain-rule work [40].

35.7 What an explanation of an interaction contains

An explanation of a coefficient should identify the endpoint table, the response model and the smooth coordinates used for its derivative decomposition. Within those declarations, the chain rule establishes exactly which terms contribute. Another interpolation with the same vertices can redistribute the terms while preserving their total.

The feedback example gives a useful reference case. Under a common linear operator, higher response derivatives vanish, so a quadratic potential gives only pairs. A coalition-dependent gain or nonlinear plant can introduce response derivatives; a higher-degree field can introduce higher potential derivatives. The partition formula gathers both kinds of term without assigning a universal causal ownership to either.

For EBU, this makes improvement more specific. A shared pressure limit calls attention to the response mechanism. A field that omits water quality calls attention to the valuation boundary. A changed starting state calls attention to the comparator. Locating the relevant part of the model helps direct resources toward a physical or institutional change that can improve service, rather than treating every large coefficient as the same problem.

We have now connected finite differences, continuous curvature and generated response. The next chapter examines how their dependency structures determine which combinations require joint treatment and which can be kept local.

Which connections matter for interaction?

Not every connection is a pipe, and not every interaction follows geographical proximity. We distinguish the physical network, potential factors, feasible groups and response dependencies. Exact structural ceilings then show when EBU can simplify a coalition account and when the full combinatorial burden remains.

36.1 A pipe drawing is only one map

A network diagram can show where water may flow without showing which measurements a potential reads, which actions share a machine, or which combinations are feasible. These are different structures. Confusing them can make an account appear local while it omits a remote dependency, or make two actions appear independent while they compete for one capacity.

The physical graph lists transformations. The factor graph lists dependencies of the valuation. The feasible-action poset records which subsets are permitted. The value-interaction hypergraph records which sets have nonzero coefficients. A hyperedge can join three or more actions; it is not necessarily a pipe, a causal link or a contractual group.

36.2 Four structures in one water account

It helps to follow the same delivery through four mathematical structures. Here “topology” includes the connectivity and composition structures used by the account; it does not mean that a combinatorial subset lattice is the same object as a topological state space.

State topology concerns the admissible states and the transitions that connect them. A tank with nonnegative stock and a capacity ceiling has a restricted domain. A pipe makes certain changes possible. A supplied single-valued potential assigns a value throughout the compared part of that domain; the path theorem does not additionally require that the domain be simply connected.

Path composition concerns a history through that domain. Replenish the tank, serve the clinic, then replenish again: the endpoint of each transition

is the start of the next. Joined receipts telescope. A complete loop under one field has zero net value. The order can alter intermediate conditions and service even when the final potential difference is unchanged.

Coalition topology concerns alternatives from a common baseline. With actions A and B, the four corners are neither, A, B and AB. Inclusion relates the action sets. An arrow from A to AB means adding B to that alternative, not necessarily executing B after A in physical time. A complete Boolean lattice permits the familiar inclusion–exclusion formulas; a restricted feasible family has its own order structure.

Interaction topology is extracted from that table. A nonzero pair coefficient joins A and B in the interaction representation. A nonzero triple joins three actions through a dependence that singleton and pair terms do not exhaust. The coefficients carry sign and magnitude; a drawing of the hyperedges alone loses that information.

These four structures answer successive questions: which states are possible, how a history joins them, which action alternatives are compared, and where their joint value fails to decompose. The physical network is another structure: its vertices may be places or machines, while a state-space point describes the whole represented system at once.

For the town’s water service, this distinction prevents a common mistake. Two pumps can be far apart in the pipe network yet interact because both deplete one reserve. Conversely, a complicated delivery history can have an affine endpoint response and therefore no interaction above pairs under a quadratic potential. Distance, sequence length and coalition order are not interchangeable measures of complexity.

The gain is a more useful map of cooperation. The account can identify the shared reserve that needs coordination without treating every physical connection as a financial relationship or every statistical coefficient as proof of responsibility.

36.3 Factor support can impose an exact ceiling

Suppose the composite landscape has a declared decomposition

$$G(z) = \sum_{a \in \mathcal{A}} G_a(z_{K_a}), \quad (36.1)$$

where K_a contains every action control on which factor a depends. The dependence must be checked after composing the state response with the potential. A local state factor can acquire broad control dependence through a coupled response.

Theorem 36.1 (Interaction support cancellation). For nonempty T , if no K_a contains T , then $m_E(T) = 0$.

Proof. For each factor there is an index in T on which it does not depend. Taking the difference in that index gives zero. Therefore its full T -difference vanishes. Linearity of differences gives $\Delta_T G = 0$, and the EBU sign relation gives the result. $\square$

This is a sufficient structural condition, not a converse. Even when a factor depends on all of T , its finite difference can cancel. A drawn connection means an interaction is possible under that factor structure; it does not guarantee a nonzero coefficient.

36.4 Shared sources and distant factors

Under the diagonal quadratic, two deliveries from the same source interact through the source coordinate, even if their destinations are far apart. Their pair term contains the product of the two source withdrawals weighted by the source curvature. The relationship comes from shared state support.

A coupled potential can instead link actions with disjoint write supports. For example, a factor representing jointly required water and energy can change when either service changes. A nonlinear response can add a third layer: one action may alter how another affects a remote coordinate. The correct support is obtained by tracing the actual response and factor dependencies, not by measuring distance on the physical drawing.

For an EBU system this matters both computationally and socially. Complete affected-factor evaluation can reduce unnecessary data collection. But declaring a narrow support merely to avoid inconvenient measurements can conceal an external effect. Mathematical locality should be demonstrated from the model and then challenged against the physical claim.

36.5 Small motifs can be studied without claiming universal compression

A motif is a small pattern worth examining repeatedly: two actions sharing a source, a relay, an enabling triple, or a small coupled factor. If two motifs have the same complete coalition table up to a relabeling, their coefficients match under that relabeling. This follows immediately by substituting the bijection into the alternating sum.

The premise is stronger than graph isomorphism. Two identical pipe drawings can have different stocks, capacities, field scales, boundary conditions, histories and timing rules. Any of those differences can change the table. Reusing

a cached coefficient requires a certificate that all result-affecting inputs and semantics agree, together with invalidation when a dependency changes.

The earlier EBU motif programme requires its A1–A8 conditions jointly for the stronger runtime and history reuse claims. These include complete boundary and composition information, the necessary subset interactions, cache identity and provenance, dependency invalidation, and history-wide boundary equivalence. This book does not replace that full certificate with a drawing or implement a compressed runtime. The algebraic relabeling result proves only what its premise states.

36.6 Aggregation can hide the relationship one wants to measure

Combining two tanks into a single total preserves mass but loses their internal distribution. A transfer between them then leaves the coarse state unchanged even though the fine potential changes. To preserve the fine value, a coarse representation must retain enough information about internal deviation or include a justified summary of it.

The same issue appears in interaction tables. A region summarized only by its total may hide which local actions share a constrained source. A group may then appear additive at the coarse level because the variables carrying the interaction were discarded. Coarse additivity is not proof of fine independence.

There is a useful exact criterion. If a summary map $r(x)$ and coarse potential $\bar{V}$ satisfy $V(x) = \bar{V}(r(x)) + c$ on all compared states, then every endpoint difference is preserved. If not, a residual must be retained or the claim narrowed. This statement does not require a particular compression algorithm; it gives a test that any proposed one must meet.

36.7 The exponential burden is real

A general table of n Boolean actions contains 2^n rows. Möbius inversion reorganises those rows without losing information; it does not make arbitrary missing rows known. A short-looking formula can still require a large family of alternatives.

Structure supplies real reductions. If the composite value decomposes into factors involving at most r action labels, all coefficients above order r vanish. If the response is affine and the potential quadratic, the ceiling is two. Both are mathematical reasons for a smaller representation. Sparsity guessed from a diagram is not the same reason.

Dynamics must be included when identifying support. An action can propagate through a common linear network to many endpoint coordinates while preserving a low degree. A nonlinear bottleneck can create higher-order dependence in a small physical region. Reach, temporal complexity and coalition order are therefore three different measures of complexity.

An EBU system can use these distinctions to spend its effort well. It can separate independent components, retain shared constraints and compute only the interaction orders justified by the composed model. For a community, the benefit is an account that explains consequential dependencies without requiring every local service decision to enumerate the whole economy.

36.8 From topology to a physical probability interpretation

Everything so far has concerned states, responses and finite values. None of it has required an equilibrium distribution. That is deliberate: a coalition table can be useful in a deterministic or driven setting without claiming canonical physics.

We can now add a probability layer where independently justified. The next chapter shows exactly what the canonical relation contributes and why it does not, by itself, prove reversibility, a motion law, or a universal economic denomination.

When a potential is also a probability landscape

The finite theory does not need a probability law. When a physical canonical model supplies one, however, the potential and relative surprisal become two expressions for the same landscape. We derive that bridge, including its units and measure, and distinguish it from the separate dynamics that may preserve a density.

37.1 Two ways of learning about the same states

A physical model can assign energy to a state. Repeated observations can estimate how often nearby states occur. In a properly specified thermal equilibrium benchmark, these two descriptions are related. This supplies a second mathematical reading of the same potential and, in applications, a way to compare physical and statistical descriptions.

The relationship is conditional. A useful inventory potential does not become a thermal energy by resemblance to a Gaussian curve. A stationary histogram does not establish detailed balance. We first state the density relation, then identify the additional assumptions required for its dynamical and thermodynamic interpretations.

37.2 Canonical normalization

Suppose a system has a specified energy $U(x)$, one fixed temperature $T > 0$, an appropriate canonical ensemble, and a declared reference measure ν . Assume its density relative to that measure is

$$p(x) = Z_U^{-1} \exp[-\beta_{\text{thermo}} U(x)], \quad \beta_{\text{thermo}} = \frac{1}{k_B T}, \quad (37.1)$$

with finite nonzero normalizer. Define the dimensionless potential

$$V(x) = \frac{U(x) - U(x^*)}{k_B T}. \quad (37.2)$$

The constant reference energy can be absorbed into the normalizer, giving $p = Z^{-1} e^{-V}$.

Here β_{thermo} has units of inverse energy. It is not the dimensionless coefficient β_{bridge} used to compare an independently defined EBU potential with relative surprisal. Keeping those symbols separate prevents a normalization convention from looking like a fitted universal constant.

Corollary 37.1 (The normalized bridge in its canonical scope). Under the canonical assumptions and Equation 37.2, define $J_{\text{prob}}(x) = -\ln p(x)$ and relative surprisal $J_{\text{rel}}(x) = J_{\text{prob}}(x) - J_{\text{prob}}(x^*)$. Then

$$J_{\text{rel}}(x) = V(x), \quad \beta_{\text{bridge}} = 1. \quad (37.3)$$

Proof. Taking logarithms gives $J_{\text{prob}} = V + \ln Z$. Subtracting its value at x^* removes the constant and uses $V(x^*) = 0$. $\square$

The equality is exact in this canonical normalization. It identifies the dimensionless bridge coefficient as one because the energy scale $k_B T$ has already been included in V . Comparing independently defined fields uses a different question: whether they admit a common physical normalization. Chapter 40 retains that distinction.

37.3 Why an independent physical ruler matters

An internally consistent score could still be multiplied by an arbitrary positive constant. Paths would still integrate, histories would still telescope and the signs of all coalition coefficients would remain unchanged. Those laws alone therefore cannot choose the size of the unit. Calibration enters because a denomination needs more than internal consistency.

The canonical benchmark supplies an independent physical scale: energy measured relative to the thermal energy $k_B T$. Under its assumptions, a difference of one in V corresponds to a factor e in the appropriate endpoint density ratio. This gives the number a meaning beyond an arbitrary label attached to a landscape.

For an independently proposed EBU field, the ambitious question is whether its physical construction agrees with the independently observed probability landscape at the predicted scale. Defining the potential from that same histogram and then recovering the histogram would only check an identity. Agreement between independent descriptions is the physically informative comparison.

This explains the role of the later E1a benchmark. A controlled optical system offers a clean setting in which field changes and normalization can be examined before heterogeneous environmental or economic actions are compared. Its

importance lies in the denomination question, not in any claim that optical traps model an economy. The benchmark has not been executed by this integration.

A failure of the proposed denomination in its declared benchmark would challenge the broader claim. Success would support that limited bridge and motivate further work; it would not on its own establish a universal resource conversion. The theory here states exactly what the comparison means. Apparatus, test design and results belong to the later volumes.

37.4 A density is relative to a measure

For a continuous variable, the probability of an exact point is normally zero. A density tells us probability per unit of a declared measure. Saying a state has density twice another state's is meaningful only within that coordinate and measure convention.

If $y = f(x)$ is an invertible smooth change of coordinates and densities are reported relative to the new Lebesgue measure, then

$$p_y(f(x)) = \frac{p_x(x)}{|\det Df(x)|}. \quad (37.4)$$

A nonconstant Jacobian contributes a nonconstant log term. It need not cancel in an endpoint ratio or an interaction contrast. Transforming the reference measure consistently with the state description preserves the original scalar relation; silently changing the measure does not.

A density-of-states factor creates the same kind of issue. If an observed macrostate density is proportional to $g(x)e^{-V(x)}$, its negative log is $V(x) - \ln g(x)$ plus a constant. One must account for g rather than announce a failure or confirmation of $J = V$ from a comparison using incompatible definitions.

37.5 Curvature agrees more generally than covariance does

Where the density relation holds smoothly in the declared coordinates,

$$\nabla^2 J_{\text{prob}}(x) = \nabla^2 V(x) = H(x). \quad (37.5)$$

This follows by differentiation of the log relation. It does not require a Gaussian density, and H may vary with state.

A stronger result applies when $V = \frac{1}{2}(x - x^*)^\top H(x - x^*)$ on all of $\mathbb{R}^d$, with constant symmetric positive-definite H , Lebesgue reference measure and no truncation. Then the normalized density is a global Gaussian with

$$\text{Cov}(x) = \Sigma = H^{-1}. \quad (37.6)$$

Completing the quadratic, or transforming to unit-normal coordinates, proves this covariance statement. It is not a general consequence of matching Hessians at one point.

Truncate the Gaussian to nonnegative stocks, impose a finite-mass boundary, or change the reference measure, and the global covariance need not equal the ambient inverse Hessian. A local quadratic approximation can still be useful, but it must be labeled as an approximation. In particular, the two-cell closed stock world of the opening chapters is not an unconstrained full-support Gaussian distribution.

37.6 Conservation changes which directions can fluctuate

If physical states satisfy $Ax = b$, an interior accessible displacement lies in the null space of A . Choose an orthonormal basis matrix Q for that tangent space and write $x = x_c + Qy$. The restricted potential has Hessian $Q^T H Q$ in the linear coordinates y .

Only in a full untruncated Gaussian on that affine subspace, with its induced measure and positive-definite restricted Hessian, does

$$\text{Cov}(y) = (Q^T H Q)^{-1}, \quad \text{Cov}(x) = Q(Q^T H Q)^{-1}Q^T \quad (37.7)$$

follow. The ambient covariance is singular along conserved directions. For curved manifolds, coordinate curvature and the chosen intrinsic measure need explicit treatment; one should not casually apply this affine formula.

This is a practical protection against interpreting a conserved total as a failed fluctuation measurement. The inaccessible direction is not supposed to have the variance of an unconstrained system.

37.7 The same density can support different dynamics

A probability density tells us how mass is distributed across states, not how it circulates among them. For a diffusion with constant positive diffusion coefficient D , a drift of the form

$$b = -D\nabla V + c, \quad \nabla \cdot (cp) = 0, \quad (37.8)$$

can preserve $p \propto e^{-V}$ under appropriate boundary conditions while carrying stationary current cp . A nonzero current generally violates the reversible equilibrium interpretation for ordinary even state variables.

Appendix D gives a rotating Gaussian example and a driven jump ring. They show why density agreement, stationarity, detailed balance, local detailed balance and entropy accounting must be treated as separate claims. A histogram alone cannot establish all of them.

37.8 What this bridge offers EBU

Within its canonical scope, the bridge gives EBU a physical and a probabilistic reading of the same state comparison. Energy reduction in units of $k_B T$ equals the logarithm of an endpoint density ratio. The unknown normalization cancels, and the curvature of relative surprisal equals the curvature of the potential in the stated coordinates.

The feedback tank illustrates why this is an additional branch of the theory. Its deterministic operating point, decaying modes and storage inequality specify no reservoir temperature, noise law or invariant density. They therefore supply no canonical normalization by themselves. A physical stochastic model may add such a structure, but it must be part of that model.

This precision makes the connection more useful to EBU. A system can use finite accounting for deterministic delivery and use the canonical bridge where a thermal model actually supplies it, without forcing all services into the same physical interpretation. The next chapter carries the density relation through the exact coalition transform.

Interaction as a contrast of probabilities

A canonical density gives the coalition table a second language: endpoint values become log density ratios, and interactions become alternating log contrasts. The same Möbius operation carries the account across this bridge. We also distinguish these endpoint contrasts from probabilities of choosing actions, so that physical dependence is not confused with a behavioural rule.

38.1 The same table in a second language

Under the canonical density relation, a decrease in potential corresponds to an increase in equilibrium density. Each coalition value can therefore be read as a log density ratio. Applying the same alternating signs to the log densities gives the same interaction coefficient.

This is a continuity result: the object being transformed has changed language, while the finite coalition operation has stayed the same. It is not a new source of value to add to the physical account. When its assumptions hold, the two expressions describe one comparison.

Theorem 38.1 (Canonical-density Möbius identity). Assume a complete coalition table, one fixed reference measure and support, $0 < Z < \infty$, and positive finite densities $p(x_S) = Z^{-1}e^{-V(x_S)}$ at all required endpoints. Then

$$E(S) = \ln \frac{p(x_S)}{p(x_\emptyset)}, \quad (38.1)$$

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} \ln p(x_S) \quad (T \neq \emptyset). \quad (38.2)$$

Equivalently, the coefficient is the logarithm of the product of densities raised to the corresponding alternating powers.

Proof. The density relation gives $\ln p(x) = -V(x) - \ln Z$. Subtracting the two endpoint logs gives Equation 38.1. A nonempty Boolean difference annihilates both constant normalization terms. The remaining signed log sum is the EBU coefficient. Exponentiating the sum gives the product form. $\square$

The algebra uses the density relation at the named points. To call it a physical canonical result, the ensemble and calibration conditions of Chapter 37 must be justified independently. Neither stationarity nor reversibility is proved by this finite algebra.

38.2 The pair product

Writing $p_S = p(x_S)$ and $p_0 = p(x_0)$, a normalized two-action table gives

$$m_E(AB) = \ln \frac{p_{AB}p_0}{p_A p_B}. \quad (38.3)$$

The denominator combines the two standalone density readings. The numerator combines the joint endpoint with the shared baseline. The ratio equals one precisely when this pair contrast is zero.

For the courier table, potential values are 16, 9, 9, 4. Their densities have a common unknown normalization. Substitution gives $\ln[e^{-4}e^{-16}/(e^{-9}e^{-9})] = -2$, with the normalizer canceling. The normalizer cancels exactly. The same pair correction has been expressed in density language, providing two compatible readings of one declared landscape.

38.3 The triple product

The eight-term contrast becomes

$$m_E(ABC) = \ln \frac{p_{ABC}p_A p_B p_C}{p_{AB}p_{AC}p_{BC}p_0}. \quad (38.4)$$

The same recursive interpretation survives: it measures how a pair log contrast changes when the third action is present. The powers and signs are fixed by Boolean inversion. They must not be guessed from a verbal claim that synergy should be positive.

When densities are very small, direct products can be numerically unstable. A future implementation would normally add log densities with their signed coefficients. That is a numerical representation choice, not a change in the physical definition. Zeros require separate treatment: a zero-density endpoint produces an undefined or infinite logarithm, not a finite coefficient to be smoothed away without a model.

38.4 Endpoint densities are not action probabilities

The entries $p(x_S)$ describe density at physical endpoints. They are not generally probabilities of the binary action pattern S . Different coalitions can

reach the same endpoint, and a physical density need not sum to one when sampled at a handful of endpoints.

One can define artificial normalized weights $q(S) = p(x_S) / \sum_R p(x_R)$ on the action patterns. Their nonempty log contrasts match the endpoint contrasts because this new normalizer also cancels. But that construction does not show that actors actually choose according to q , or that the resulting binary variables are causally independent.

If a positive binary joint distribution q is independently specified, its pair contrast on a fixed face is an ordinary log odds ratio. A zero pair contrast in the face with all other variables absent does not establish unconditional independence. For example,

$$\ln q(z) = \text{constant} + az_1z_2z_3 \quad (38.5)$$

has zero pair contrast for actions 1 and 2 when $z_3 = 0$, but pair contrast a when $z_3 = 1$. The context changes the relationship exactly as the recursion predicts.

38.5 Several quantities called information remain different

A pointwise log-density contrast is not an alternating sum of marginal entropies. Interaction information uses marginal entropy combinations; multi-information compares a joint distribution with the product of its marginals. Connected information compares successive maximum-entropy models. Cumulants use a log moment-generating function and a different combinatorial structure.

These constructions can all describe dependence, but they answer different questions and need not give the same number or sign. A claim that EBU interaction is one of them requires a precise derivation under the relevant distribution and coding convention. The common word “information” does not establish the equality.

For EBU, precise names keep each capability useful. A pointwise contrast explains the endpoints of a particular intervention family; an entropy statistic summarises a distribution. Choosing the quantity that matches the service question avoids a score whose meaning changes as it moves between reports.

38.6 The prior-art connection

Gibbs and log-linear representations, Möbius inversion, cooperative-game dividends and interaction decompositions have extensive antecedents. Jansma explicitly connects higher-order energy interactions, probability contrasts and

game-theoretic decompositions in this general mathematical family [38, 39]. The EBU formula here should be read in that context, not presented as a newly invented inversion principle.

The derivation above fixes its own sign from $E = V(x_0) - V(x_S)$ and $\ln p = -V - \ln Z$. This provides a direct check without relying on an isolated intermediate equation in another text. The versioned equation-level source comparison remains available in the project's prior-art appendix [30].

38.7 Two expressions for one physical comparison

The physical-potential expression and the log-density expression agree under the same canonical assumptions. Their equality is a theorem, not an extra value to add to the account. If $E(\emptyset)$ is nonzero because the background evolves, every nonempty contrast still agrees: the raw-to-relative shift is constant across the table.

This gives EBU a consistent statistical representation of shared effects. Pair and triple contributions can be explained through physical endpoint values or through canonical density contrasts, with the same sign convention and reference measure. The representation can reveal a missing normalization factor or an incompatible state description before an institutional rule uses the number.

For actual measurements, the estimates of the two expressions have their own errors and covariance; Chapter 41 gives the theoretical propagation rule. Detailed measurement designs belong to the later books. The next theoretical step is the narrower thermodynamic connection to medium entropy under the accepted P4 assumptions.

The entropy connection and its boundary

The canonical bridge can be extended to entropy in a precisely specified conservative thermal model. This chapter states that scope once, derives the signs and shows why Möbius contrasts inherit the same scaling. It also explains why a driven feedback service retains its finite EBU account without automatically acquiring this thermodynamic interpretation.

39.1 A thermal statement needs a thermal model

The word entropy is often used loosely for disorder, waste or difficulty. Here it has a precise role in a specified physical benchmark. A conservative overdamped system coupled to one reservoir at fixed temperature can exchange energy as heat while its stochastic system entropy changes. Under the accepted equilibrium conditions, both changes are related to the same endpoint potential difference, with opposite signs.

This is a narrow and useful connection. It does not say that every EBU improvement is universal entropy production or that an economy's total entropy vanishes. The earlier accounting identities remain valid wherever their own assumptions hold, even when this thermodynamic interpretation is unavailable.

39.2 State the complete benchmark before the formula

For each coalition entry, assume the same fixed conservative potential energy U , the same fixed temperature T , a conservative overdamped single-reservoir model, and the stationary canonical density at both endpoint times. Require compatible paths and boundary conventions, with no omitted external work, hidden reservoir, or additional entropy channel. Use $V = (U - U(x^*)) / (k_B T)$ in one declared reference measure.

These assumptions must hold for every row of the coalition comparison. If applying a control changes the trap or injects work, the energy balance needs that work term. Merely knowing both endpoint potentials does not establish that all released energy went into the same medium in the way the benchmark assumes.

Theorem 39.1 (Accepted P4 entropy relation). Under the preceding conditions, for each compared process,

$$\begin{aligned}\Delta s_{\text{med}}(S) &= k_B E(S), \\ \Delta s_{\text{sys}}(S) &= -k_B E(S), \\ \Delta s_{\text{tot}}(S) &= 0.\end{aligned}\tag{39.1}$$

Consequently, applying the same Möbius transform to the entropy tables gives

$$\begin{aligned}m_{\Delta s_{\text{med}}}(T) &= k_B m_E(T), \\ m_{\Delta s_{\text{sys}}}(T) &= -k_B m_E(T), \\ m_{\Delta s_{\text{tot}}}(T) &= 0.\end{aligned}\tag{39.2}$$

Proof. With no additional work channel, conservative heat delivered to the medium is $U(x_0) - U(x_S)$. Division by T gives $k_B E(S)$. Stochastic system entropy is $s_{\text{sys}}(x) = -k_B \ln p(x)$; its endpoint change is $-k_B \ln[p(x_S)/p(x_0)] = -k_B E(S)$. The two changes cancel. Möbius inversion is a linear signed sum, so applying it to each entrywise equality gives the coefficient relations. $\square$

The second part is not a new thermodynamic law. It is linearity applied to an already conditional physical identity. Stochastic thermodynamics supplies the heat, work and entropy distinctions; the EBU notation organizes their relationship to the finite account [43, 44].

39.3 Why the signs make sense

Suppose the endpoint energy is lower. Under this benchmark, energy leaves the system as heat to the medium, so medium entropy increases and EBU is positive. The lower-energy endpoint has greater canonical density, so its stochastic surprisal and stochastic system entropy are lower. Their changes have opposite signs.

The cancellation does not mean that nothing happened, nor that every thermal trajectory in every model is reversible. It uses the stationary equilibrium density at both endpoint times and the specified conservative one-reservoir balance. A nonequilibrium initial distribution changes the stochastic system entropy calculation. Driving, multiple reservoirs or unrepresented internal variables can change the total balance.

A stochastic entropy change belongs to a specified realization; its ensemble average additionally uses the law of the process. The endpoint identity and its average therefore retain the physical model and probability convention that define them.

39.4 An interaction changes units, not its logical meaning

If an admissible pair in this benchmark has $m_E(AB) = -2$, its medium-entropy contrast is $-2k_B$. That is a contrast between the entropy changes of four declared processes. It is not necessarily the entropy change of one newly executed process called “the interaction.” The coefficient retains its counterfactual table meaning.

The same is true for a triple. Multiplying by k_B gives the thermal units under the benchmark; it does not turn an eight-corner comparison into a causal allocation among three actors. The joint outcome must still be distinguished from an attribution or institutional payment.

39.5 A current-carrying system exposes the limit

A stationary density can agree with e^{-V} while probability circulates. A driven ring can have uniform stationary probabilities and constant state potential while its clockwise and counterclockwise rate ratios carry nonzero cycle affinity. Its density-based EBU differences vanish, but its dynamics can produce entropy through the drive.

This is not a defect in the endpoint definition. It shows that a state-potential difference and a general nonequilibrium path functional are different objects. Appendix D gives the ring explicitly. A loop returns the state potential to its start, while dissipative currents need not cancel their thermodynamic cost.

If a broader account includes nonconservative work, it must declare that work and its boundary. One cannot add an unspecified process charge to the fixed physical EBU definition and claim the original theorem proved the new expression. A separately defined net functional can be analyzed, but its additional terms require their own physical justification.

39.6 Macrostate entropy is another declaration

A constrained macrostate model may use a relation such as $S = S_{\text{eq}} - \kappa V$. Its entropy can concern a coarse-grained ensemble with internal multiplicity. That object is not automatically the stochastic endpoint entropy $-k_B \ln p(x)$ used above. The ensemble, state resolution, reference measure and coefficient must be reconciled before equating them.

This distinction is important for EBU’s broader ambition. A physically meaningful account of living systems may need several scales and internal states. The thermal benchmark offers a precise testbed; it does not license erasing those additional structures when moving to ecology or institutions.

39.7 What the conditional connection is good for

Within the stated P4 model, medium entropy changes and EBU values differ only by the factor k_B , while stochastic system entropy changes have the opposite sign. Linearity carries these relations through every coalition order. This is a complete theoretical consequence of the benchmark assumptions.

The distinction helps EBU keep physical claims precise. A thermal account can connect the finite unit to heat and temperature. A stock-and-pipeline controller can instead use its endpoint potential and stability function, each in its stated units. The tank's decaying oscillations supply no work–heat balance and no stationary canonical density, so they do not invoke P4.

The practical value is a modular account whose quantities retain their meanings. Heat, state improvement, accumulated exposure and controller storage can be displayed together where relevant without being added as though they were one substance. This supports decisions that recognise physical burdens at the appropriate level and makes the reason for each claimed connection clear.

We have completed the fixed-field chain. Two further theoretical questions remain before we assemble it: how to preserve the account when its yardstick changes, and how uncertainty in its inputs propagates through the exact formulas.

When the yardstick changes

A useful account must be able to learn without rewriting what people physically did. When references or scales change, the potential changes even at a fixed state. We derive the extra field-change term, distinguish it from a change of controller and show how a versioned history can remain coherent through both.

40.1 Learning about the world is not the same as changing it

A community can improve a water reserve by moving water. It can also learn that its previous reserve target was unsuitable. The first changes the physical state; the second changes the field used to interpret that state. Both can be important, but only one is the physical action described by the original receipt.

Let $V(x, \theta)$ be a family of potentials, with θ identifying references, scales and other declared field parameters. A fixed-field action still has $E_\theta = V(x_0, \theta) - V(x_1, \theta)$. Once θ changes, an account spanning both versions needs to show that change explicitly.

40.2 The full differential has two components

Assume V is C^1 along a regular path $(x(t), \theta(t))$. The chain rule gives

$$dV = \nabla_x V \cdot dx + \partial_\theta V \cdot d\theta. \quad (40.1)$$

Consequently,

$$V(x_1, \theta_1) - V(x_0, \theta_0) = \int \nabla_x V \cdot dx + \int \partial_\theta V \cdot d\theta. \quad (40.2)$$

The total is an exact state-function difference on the enlarged state-and-parameter space. The two component integrals are generally path dependent separately. One cannot discard the parameter component and retain a universal endpoint theorem for the physical-action component across changing fields.

For example, with $V(x, \theta) = \theta x$, the physical component is θdx and the parameter component is $x d\theta$. Moving x first and changing θ second allocates

different amounts to the two components than doing them in reverse order. Both sums give the same final minus initial product.

40.3 A finite version makes the ordering explicit

For a transition from (x, θ) to (y, θ') , insert the intermediate value $V(y, \theta)$:

$$\begin{aligned} V(x, \theta) - V(y, \theta') &= [V(x, \theta) - V(y, \theta)] \\ &\quad + [V(y, \theta) - V(y, \theta')]. \end{aligned} \quad (40.3)$$

The first bracket values the physical change under the original field. The second records revaluation at the final physical state. An alternative ordering evaluates the physical change under the new field and revalues at the initial state. Both are valid decompositions with different meanings. A protocol must choose and record its ordering.

At (14, 6), changing both Gaussian scales from two to four changes potential from four to one without moving water. That three-unit difference is a field-change entry. An actor who was idle during the revision has not performed a three-unit restoration.

40.4 A closed physical loop can exploit an untracked scale change

Let $W(0) = 0$, $W(1) = 1$, and define fields $V_A = 2W$, $V_B = W$. An action $1 \rightarrow 0$ under A gives +2. A later reverse action $0 \rightarrow 1$ under B gives -1. Returning the field to A completes a loop in both physical state and field version, but the two action entries alone sum to +1.

The missing field-change entries reconcile the loop. No material was created and the fixed-field theorem was not violated: its premise of one field did not hold for the action-only sum. Treating those heterogeneous entries as a persistent spendable claim without a bridge would create an accounting opportunity from changes in the ruler.

This example identifies a concrete institutional risk without deciding a universal remedy. A canonical capacity reference, an explicit conversion rule or separated accounts may each be considered in a future design. Each needs a justified meaning and its own analysis. The book does not adopt one by naming the problem.

40.5 Historical receipts retain their original meaning

An original receipt retains its physical endpoints, field version and signed value. A later field can evaluate the same event retrospectively, but that

new comparison has a different ruler. Preserving both makes learning visible instead of turning revised interpretation into a claim that the past event changed.

A controller revision is another kind of change. Altering mobility, residence time or feedback gain changes the response map and therefore future end-points. It does not directly revalue the same fixed state when V is unchanged. Changing the potential weight changes valuation; if the controller uses that weight, it may also change response. Both effects must then remain explicit.

The native tank makes the distinction concrete. Its roots depend on $\nu = 2wM$ and τ . Keeping ν fixed while changing w and compensating M can preserve the dynamics while changing the native burden scale. Observing unchanged oscillations therefore does not establish an unchanged accounting denomination.

For EBU this is an important trust feature. The system can improve control and revise its representation while participants can still reconstruct why a historical contribution received its value. A service operator gains a clearer basis for comparing equipment or scheduling changes, and a community avoids windfalls created solely by an unrecorded change in scale.

40.6 A persistent entry and a changing action price

Suppose a restoration is registered in 2030. Its record identifies the endpoints, the 2030 field and the value computed under that field. By 2040 the reserve target, physical surroundings or state description may have changed. A new restoration is valued under those current conditions. The historical 2030 entry remains the entry that was made; a retrospective calculation under the 2040 field is a separate comparison.

This persistence has a concrete purpose. Participants can reconstruct what an earlier rule recognised without having every change in scientific understanding rewrite their entire history. An error correction can still be recorded explicitly, linked to the affected event. Non-repricing is therefore compatible with correction and learning.

Nor does a stable denomination imply a constant action price. Even with one fixed quadratic field, a litre transferred from a full tank into a shortage has a different value from the same litre transferred after that shortage has disappeared. The unit can remain unchanged while the state-dependent difference changes. Across fields, retaining one denomination additionally requires a justified calibration or conversion; a common label alone is insufficient.

The analogy with monetary prices is limited but useful: knowing the unit in which a price is expressed does not fix the price of every activity. EBU's

proposed physical denomination addresses the ruler. State, boundary, field and response determine a new action's consequence. This distinction does not establish monetary price stability or eliminate inflation; those belong to a different institutional analysis.

40.7 A common coefficient across fields remains a hypothesis

The canonical benchmark gave $J_{\text{rel}} = V$ under a particular energy normalization. For more general fields, one might investigate

$$J_{\text{rel},\theta}(x) \stackrel{?}{=} \beta_{\text{bridge}} V_{\theta}(x) \quad (40.4)$$

with one coefficient across the specified fields. The question is stronger than agreement within one fitted field. It requires independently defined potentials, compatible density measures and a justified common denomination.

A curvature comparison $K_{\theta} \stackrel{?}{=} \beta_{\text{bridge}} H_{\theta}$ can be part of that programme, with accessible tangent directions treated correctly. Local curvature agreement alone does not establish a global identity or a universal conversion among resources. Nor may the potential branch be defined from the probability branch and then presented as independent evidence of agreement.

40.8 Why a fixed yardstick is not the final social answer

Knowledge and physical conditions change. Drought can alter reserve requirements; technology can alter the relation between energy use and service; a fuller boundary can make a previously omitted consequence visible. A permanent freeze would keep arithmetic simple at the cost of a less useful representation.

The differential and finite split provide the alternative. Preserve physical-action entries under their declared field, record revaluation separately, and make any conversion of persistent institutional claims explicit. This allows the account to remain both adaptable and inspectable.

The benefit is a form of continuity that people can use: maintenance history remains legible while the system learns. Better definitions can guide future choices without erasing prior work or disguising changed judgement as fresh restoration. The next chapter keeps the field fixed and asks how imperfect input knowledge affects an exact interaction calculation.

How much confidence belongs beside an interaction?

Exact laws can be evaluated with uncertain inputs. This does not weaken the law; it determines the uncertainty of the resulting number. We derive the propagation rules for a coalition contrast, including shared baselines and common calibration, so that the mathematical account remains precise when its physical inputs are estimated.

41.1 Exact subtraction is not exact observation

A Möbius coefficient is an exact signed sum of its input table. If the inputs are estimated, the coefficient is estimated too. Alternating signs do not automatically cancel measurement errors, and a shared baseline can create correlations that must be retained.

This matters especially at high order. A genuine signal may shrink as the k th power of a small action amplitude while the contrast still combines 2^k noisy readings. A theorem can be exact and its empirical target difficult to resolve. The resulting distinction is precise: algebra determines the target and its error propagation, while a measurement procedure determines the distribution of the inputs.

41.2 A linear contrast has a complete covariance formula

Collect the required estimates into a vector $\hat{e}$ and let $c_S = (-1)^{|T|-|S|}$. Then $\hat{m} = c^\top \hat{e}$. If the estimates have finite covariance matrix C , elementary bilinearity gives

$$\text{Var}(\hat{m}) = c^\top C c. \quad (41.1)$$

This is exact for the specified linear contrast, whether or not the errors are Gaussian. It includes variances on the diagonal and pairwise error covariances off the diagonal.

Proof. Subtract the mean from each estimate and expand the square of $\sum_S c_S(\hat{e}_S - \mathbb{E}\hat{e}_S)$. Taking expectations gives $\sum_{S,R} c_S c_R \text{Cov}(\hat{e}_S, \hat{e}_R)$, which is the matrix expression. $\square$

For independent equal-variance readings of all 2^k potentials, each with variance σ^2 , the corresponding alternating potential contrast has variance $2^k \sigma^2$. Independence is an assumption about errors, not a property supplied by the transform.

41.3 Why the shared baseline does not simply disappear

For a normalized triple table, $E(\emptyset) = 0$ is exact. The seven nonempty values may be calculated as $\hat{E}(S) = \hat{V}_0 - \hat{V}_S$, all using the same estimated baseline potential. If one treats these seven estimates as independent, one ignores their shared error.

The signed coefficients of the seven nonempty rows sum to one. Therefore the baseline estimate enters the triple contrast once. If the eight potential estimates are mutually independent with equal variance σ^2 , the result has variance $8\sigma^2$: one baseline contribution and seven endpoint contributions.

Equivalently, each nonempty $\hat{E}(S)$ has variance $2\sigma^2$, and two different rows have covariance σ^2 through the baseline. Substituting this full covariance into Equation 41.1 again gives $8\sigma^2$. It is not $14\sigma^2$, obtained by incorrectly treating those differences as independent, and not $7\sigma^2$, which would describe seven independently measured E-values of variance σ^2 under a different design.

What cancels, and what does not

A deterministic constant added to every table entry cancels in a nonempty difference. An uncertain baseline reused inside normalized differences can leave a correlated error contribution. These statements concern different data-generating constructions.

41.4 Shared calibration creates further dependence

All endpoints can use the same estimated reference, temperature, Hessian or response parameter. A common parameter error can therefore move several table entries together. In the feedback example, uncertainty in ν or τ changes the generated endpoints across the whole coalition table; it is not a new independent error for each row.

If $m = f(\eta)$ is differentiable in a parameter vector and a first-order approximation is appropriate, the delta method gives

$$\text{Var}(\hat{m}) \approx \nabla f(\eta)^\top \text{Cov}(\hat{\eta}) \nabla f(\eta). \quad (41.2)$$

This approximation has a different status from the exact linear-contrast identity $c^\top C c$. The latter is exact once the covariance of the table estimates is

specified; the delta method approximates how parameter uncertainty produces that covariance or the coefficient's variance.

For log-density inputs, small densities can magnify relative error. A finite-bin probability is an integral over a region, whereas a point density is relative to a measure at a point. Their uncertainty and bias refer to different targets. Keeping the target explicit preserves the canonical identity's meaning without turning a numerical estimate into a different theorem.

41.5 What a nonzero triple can establish

Within an exactly specified model, affine response and quadratic valuation imply a zero triple. A mathematically nonzero triple in its endpoint table disproves the conjunction of those premises for that table. It does not select a unique alternative mechanism.

With estimated inputs, the contrast additionally carries the error distribution just derived. A departure from zero must be interpreted relative to that uncertainty. Potential nonlinearity, response nonlinearity, changing baselines and a mismatched field are distinct explanations of the same apparent discrepancy.

A zero can likewise arise from the structural degree ceiling or from cancellation in a nonlinear composite. The square-root response of Chapter 34 demonstrates exact cancellation. Thus the theorem's logical subject remains the complete set of assumptions, not a claim that every zero implies simple physics or every triple implies oscillation.

41.6 A deterministic error bound

Error bounds can also be stated without a probability model. If each input satisfies $|\hat{e}_S - e_S| \leq \epsilon_S$, the triangle inequality gives

$$|\hat{m} - m| \leq \sum_{S \subseteq T} |c_S| \epsilon_S = \sum_{S \subseteq T} \epsilon_S. \quad (41.3)$$

The bound is conservative because it allows all errors to align with the contrast signs. Covariance information can give a more informative probabilistic description, but it is not needed for this deterministic guarantee.

A structural zero has a further advantage: it removes terms by proof rather than by an uncertain estimate. Factor support and degree ceilings therefore reduce both computational and informational demands. For EBU, this encourages the simplest account that is exact under the declared model, with explicit error bounds wherever approximations enter.

41.7 Communicating a result people can challenge

An interaction record identifies the group, baseline, field, endpoint table and coefficient. When inputs are estimated, it also identifies their uncertainty and dependence. This connects the mathematical result to the information used to evaluate it.

The resulting account is more useful to communities because disagreements can be located. A challenge to a sensor reading differs from a challenge to the reference, and both differ from a challenge to an allocation rule. Error propagation preserves those distinctions instead of hiding them inside a single authoritative-looking score.

We now have the tools to join the theory in one place: exact endpoint laws, generated response, finite interaction, canonical probability and uncertainty each connect to the same declared comparison.

What is the glue?

The book's continuity is the preservation of one comparison through several mathematical languages. This chapter gathers the results into a usable hierarchy and explains the capability each adds to EBU: complete history, local computation, joint valuation, time-aware response and, where applicable, a physical probability and entropy interpretation.

42.1 One comparison carried through several languages

We began with an ordinary concern: how can an account of action say something about the physical conditions that keep life and service possible? The answer developed here is deliberately specific. Declare a state, a fixed potential and a complete physical transition. The before-minus-after difference is its EBU. Every later mathematical layer either represents that same difference or supplies a declared way to construct its endpoints.

The glue is therefore not a metaphor about balance. It is the preservation of one mathematical object across changes of representation. The chain rule connects endpoints to a path integral. Telescoping connects successive transitions. Möbius inversion connects group values to irreducible finite interactions. Smooth response and potential derivatives explain those interactions when their hypotheses hold. A canonical density can express the same values as log ratios, and the accepted thermal benchmark can express them as medium entropy changes.

Each arrow has conditions. Keeping those conditions visible is what makes the continuity rigorous rather than merely persuasive.

42.2 The central continuity statement

Theorem 42.1 (Fixed-field EBU continuity hierarchy). For a complete represented transition $x_0 \rightarrow x_1$ and one finite, fixed potential V , the defining account is $E = V(x_0) - V(x_1)$. Under the additional assumptions stated beside each expression below, this same number has the following representations.

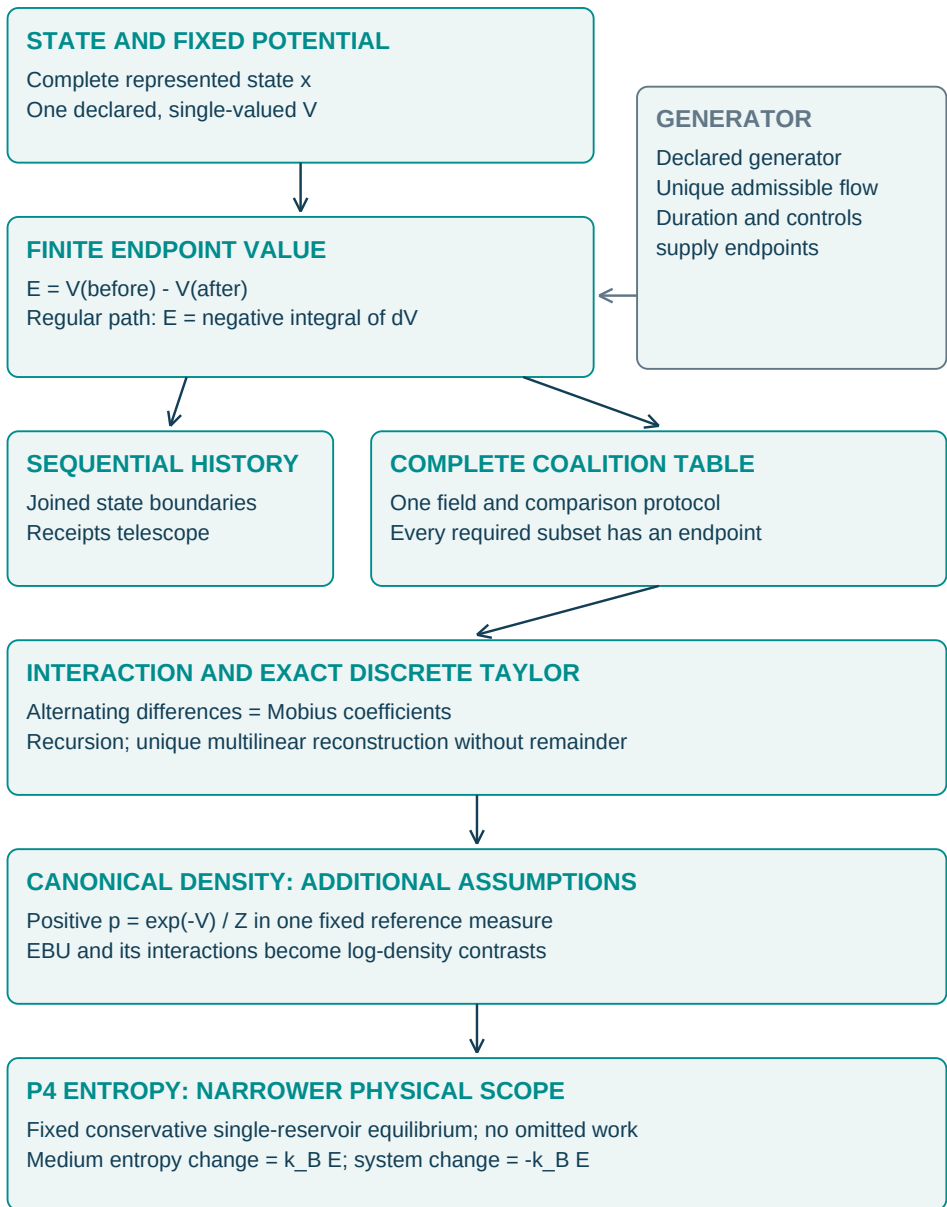


Figure 42.1: The continuity hierarchy. The generator is an optional interface supplying endpoints. The main finite theory needs neither equilibrium nor an actor account. Canonical probability and P4 entropy are additional conditional branches of the same declared comparison.

Central continuity: preserve the assumptions

D: definition. One fixed field and complete endpoints:

$$E = V(x_0) - V(x_1).$$

P: regular path. Theorem 14.1:

$$E = - \int_{\gamma} dV = - \int_{\gamma} \nabla V \cdot dx.$$

Q: canonical density. Fixed measure, positive endpoints, $p = Z^{-1}e^{-V}$:

$$E = \ln[p(x_1)/p(x_0)].$$

T: accepted P4 scope. Fixed conservative overdamped single-reservoir equilibrium, same temperature and no omitted work or entropy channels:

$$E = \Delta s_{\text{med}}/k_B, \quad \Delta s_{\text{sys}} = -k_B E.$$

The proof consists of the already established chain rule, log-density identity and conditional heat balance. It does not remove any premise when the results are placed together. In particular, D does not imply Q or T. A stock account can satisfy D and P without being a thermal canonical system.

42.3 The interaction form of the same continuity

For a nonempty action set T , use one complete coalition table and write $G(z) = V(x(z))$. The exact finite statement is

Central interaction continuity

B: complete Boolean table.

$$m_E(T) = \Delta_T E(\emptyset) = -\Delta_T G(0).$$

A: additive response and $C^{|T|}$ potential.

$$m_E(T) = - \int_{[0,1]^{|T|}} D^{|T|} V(x(t)) [h_i : i \in T] dt.$$

Q: canonical density at every corner.

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} \ln p(x_S).$$

T: accepted P4 scope for every entry.

$$m_{\Delta s_{\text{med}}}(T) = k_B m_E(T).$$

The derivative formula is one specialization of the general composite difference. With nonlinear smooth response, integrate $\partial_T(V \circ x)$ instead and use its partition chain rule. With no smooth extension, the finite table identity still stands. With a restricted feasible poset, use the applicable incidence inversion rather than assume a missing Boolean cube.

42.4 What the generator contributes

A generator supplies $x_S = \Phi_{x_S}^{Tf}(x_0)$ under a declared initial state, background and horizon. The finite definition values those endpoints. A measured jump or fixed increment can supply an endpoint directly; a continuous generator is one response interface rather than a prerequisite for every receipt.

Dynamics and interaction in the same account

Response. A well-posed generator supplies the complete endpoint for each coalition.

Temporal behaviour. Pending state can produce memory; eigenvalues describe decay and oscillation of the declared dynamics.

Affine ceiling. A common linear operator with additive action inputs gives $x_S = x_\emptyset + \sum_{i \in S} h_i$. With fixed quadratic V , all $m_E(T)$ for $|T| \geq 3$ vanish at every admissible horizon.

Nonlinear response. In general the exact object is $-\Delta_T(V \circ x)$. Non-linearity can create higher orders or cancel under composition.

These results give a service system two coordinated descriptions. Dynamics explains whether a delivery arrives, overshoots or settles. The interaction table explains how combined interventions compare with their parts. A period is not an EBU denomination, and the order of a temporal mode is not a coalition order.

42.5 What is conserved, and what merely closes

Carrier conservation concerns physical quantities within a boundary. EBU closure concerns differences of a state function. Probability normalization concerns total probability. Institutional balance concerns a declared update rule. Each identity is exact in its own setting.

For sequential receipts, the connection is

$$\sum_{r=1}^m [V(x_{r-1}) - V(x_r)] = V(x_0) - V(x_m). \quad (42.1)$$

Intermediate values cancel because the states join. A cycle has zero total; subdividing an event preserves its total. With external changes or field revisions, their explicit lines complete the same reconciliation.

The practical consequence is reliable composition. A local receipt can join a route, a route can join a service history, and that history can be reconciled to its complete endpoint change. Joint events use their group total once. Autonomous relaxation remains in the physical history without becoming a stream of fresh actor claims.

Each discrepancy then has a place: a missing litre belongs to the carrier account, a stale baseline to the comparison, a scale revision to the field history, and an allocation dispute to the institution. This makes a large system easier to inspect and improve.

42.6 Where each connection applies

The finite comparison and Möbius reconstruction require complete declared values. The path and derivative representations add regularity. Degree ceilings add structure in both potential and response. The canonical branch adds a probability density in a fixed measure, and P4 adds the specified conservative thermal balance.

The distinction is particularly clear in the feedback tank. Its equations establish its operating point, stability, memory and affine endpoint response. They do not supply a temperature or invariant density, so its finite EBU and interaction theory stand without invoking the canonical branch. Likewise, a canonical density can be preserved by a current-carrying process, so density and reversibility remain separate properties.

This organisation lets EBU use every result at full strength within its domain. A theorem is not weakened by stating its hypotheses; the hypotheses identify the systems for which its conclusion is guaranteed.

42.7 What the continuity does offer

The completed theory gives EBU a common language for physical action at several scales. Exact cancellation makes local computation possible. Telescoping preserves the signed history. Möbius inversion explains shared dependence without adding value twice. Feedback theory includes pending commitments and timing. Degree bounds simplify the account where structure permits it. Canonical normalization connects appropriate physical models to probability and entropy.

These capabilities address concrete sources of inefficiency: duplicate claims on the same improvement, repeated dispatch against a pending delivery, omitted shared constraints, unnecessary high-order calculation and hidden changes in the yardstick. Removing those errors leaves more room for useful maintenance and more legible decisions about collective infrastructure.

The remaining chapters use this theoretical foundation to describe attribution, routes and institutional interfaces. They explain how a physical account can support cooperation, access and continuing service while keeping the choices about rights and allocation explicit. Detailed test designs, simulations and results are developed in the later books.

Dividing a result does not define an entitlement

A joint physical result has one total before anyone divides it. Attribution adds a declared rule for assigning parts of that total to contributors. We derive two exact constructions, explain when they agree and show how making the rule visible can support shared work without treating a mathematical decomposition as an automatic entitlement.

43.1 The joint account is already known

We have now defined a complete coalition landscape and its interaction hierarchy. None of that analysis required an actor wallet. A joint transition has one value before anyone decides how to divide it. We can therefore examine an attribution rule without mistaking its closure for a new physical law.

A path attribution can be useful for reporting contributions under a declared convention. Its assumptions, path and child decomposition must be retained. It is not automatically an observed causal contribution, a fair share or a payment.

43.2 One declared path through the joint change

Consider the straight path

$$\gamma(s) = x^b + s\delta_G, \quad 0 \leq s \leq 1. \quad (43.1)$$

The parameter s says what fraction of the joint increment has been included in this mathematical path. If the physical model explicitly stipulates constant-rate simultaneous execution, that path can also be its realized state path. If only endpoints are specified, it is a declared interpolation convention.

For each child, define the simultaneous-path field attribution

$$R_a = - \int_0^1 \nabla V(x^b + s\delta_G)^T \delta_a ds. \quad (43.2)$$

The minus sign follows the book's convention: reducing potential gives positive field value. The integral sums the child's directional contribution along

the common path. Every child uses that same path; choosing a different path for each would require a different argument for closure.

This construction has Aumann–Shapley-type common-path mathematics. Naming that relationship is an attribution of mathematical structure, not an adoption of a fairness theorem or a claim that physical matter follows a cooperative-game allocation rule.

43.3 Why the path contributions close

Add equation (43.2) over children. Finite summation and integration commute, and the child increments sum to $\delta_{\mathcal{G}}$:

$$\sum_a R_a = - \int_0^1 \nabla V(x^b + s\delta_{\mathcal{G}})^T \delta_{\mathcal{G}} ds \quad (43.3)$$

$$= - \int_0^1 \frac{d}{ds} V(x^b + s\delta_{\mathcal{G}}) ds \quad (43.4)$$

$$= V(x^b) - V(x^b + \delta_{\mathcal{G}}). \quad (43.5)$$

The last step is the fundamental theorem of calculus, under the differentiability and path-domain assumptions. Our fixed quadratic satisfies the required smoothness.

Common-path closure

For fixed additive child increments and one declared differentiable common path of the stated form, the signed derivative integrals sum to the exact group field reduction. The theorem establishes the sum. It does not establish ownership, causal necessity, fair shares or permission to spend an attribution.

The closure result concerns the complete state and potential that were declared. If a group's physical operation changes additional energy, loss or equipment coordinates, those coordinates belong in its complete increment and in any potential that claims to value them. The field integral does not authorize a second, unspecified charge. Nor does its exact sum determine a unique owner allocation.

43.4 The Gaussian integral is explicit

Because the Gaussian gradient is affine,

$$\nabla V(x^b + s\delta_{\mathcal{G}}) = \mu(x^b) + sH\delta_{\mathcal{G}}.$$

Substitute into equation (43.2) and integrate 1 and s :

$$R_a = -\mu(x^b)^T \delta_a - \frac{1}{2} \delta_a^T H \delta_g. \quad (43.6)$$

No numerical quadrature is needed for this quadratic. The expression includes the way each child overlaps the full group increment. It does not merely divide the total by the number of names on the record.

For the symmetric two-courier example, each attribution is 6 and their sum is 12. The same-baseline singleton value was 7, while the first and second sequential values were 7 and 5. These are three different questions with compatible arithmetic.

43.5 Unequal quantities on the same edge

Let A send one unit and B send three from the same baseline (18, 2). The group still sends four units and has field value 12. The singleton values are

$$E_A^{(0)} = 4 - 1/4 = 3.75, \quad E_B^{(0)} = 12 - 9/4 = 9.75,$$

summing to 13.5. The cross-term correction is -1.5 .

Along the common path, A's increment is one quarter of the total and B's is three quarters, so their field attributions are 3 and 9. This proportionality follows here because the increments are collinear and use the same path. It is not a universal justification for dividing all group values by sent quantity.

If the actions use different destinations, different scales or different physical maps, equal sent quantities need not have equal directional field effects. A proportional policy can still be defined, but it answers a policy question distinct from the common-path calculation.

43.6 A fan-out example with distinct destinations

Use three locations with $x^b = (18, 2, 10)$, references (10, 10, 10) and scales all equal to 2. Let A send two units from location 1 to 2 and B send two from location 1 to 3. The increments are $(-2, 2, 0)$ and $(-2, 0, 2)$.

Initial potential is 16. The joint successor is (14, 4, 12), whose terms are 2, 4.5 and 0.5, totaling 7. Group value is 9. Acting alone, A gives 7 and B gives 3, totaling 10. Their one-unit discrepancy comes from the shared source coordinate.

The common-path attributions are 6.5 to A and 2.5 to B. They sum to 9. Equal sent quantities did not yield equal attributions because one destination began below reference while the other began at reference. The arithmetic describes the declared path; whether these are appropriate economic shares remains a separate question.

43.7 Why an allocation remains an additional choice

Several allocations can sum to the same total. Equal shares, quantity shares and marginal constructions can agree in a symmetric example while diverging elsewhere. An institution may also consider responsibility, effort, access, risk and the financing of common equipment.

The common-path rule has precise structural properties. Relabelling fixed children does not change their corresponding values. Proportionally splitting a fixed child preserves its summed attribution. These properties apply to the declared decomposition; they do not imply that changing a request leaves physical acceptance unchanged.

A feedback-generated endpoint requires the same care. Under common linear dynamics, $h_i(T_f)$ can be used as the additive endpoint contribution of action i . A straight interpolation through these endpoint increments then supplies a valid common-path allocation under the usual domain assumptions. That interpolation need not be the actual oscillating trajectory. If controls change the operator or activate a shared constraint, the endpoint may no longer have this additive form.

Making the convention explicit is useful for EBU cooperation. Participants can agree on how a known joint result will be divided while separately inspecting the physical quantities and the response that produced it. This keeps disagreement over shares from becoming an unexplained alteration of the total.

43.8 Dividends can inform an allocation without choosing it

For a normalized cooperative table, one familiar allocation divides every dividend $m_E(T)$ equally among its members and sums the assigned pieces. This gives the Shapley allocation

$$\phi_i = \sum_{T \ni i} \frac{m_E(T)}{|T|}, \quad \sum_i \phi_i = E(N). \quad (43.7)$$

The sum follows because each dividend is distributed exactly once. This is established cooperative-game mathematics, not a rule adopted for EBU ownership [32, 33]. Other rules can close on the same total under different axioms or priorities.

For the symmetric two-courier table, equal dividend splitting gives six each, as does the common straight path. Coincidence in a symmetric example does not make all allocation methods equivalent. Nonlinear response, asymmetric actions and different baselines can distinguish them.

43.9 A transparent division of shared work

An explicit allocation makes shared work reconstructible. A participant can see how much of each dividend was assigned to them, or how their direction contributed along a common path. The institution can choose a rule suited to its purpose and preserve that rule with the record.

This supports cooperation through clarity. Shared maintenance can receive a transparent division of its result; essential service can receive support through a separate access rule; responsibility for a harmful event can use its own evidence and procedure. None requires changing the physical endpoint value to force a preferred distribution.

The key efficiency is that the group result is computed once and distributed once. Interaction is already inside that result. It is not an additional reward pool, and the continuing ring-down of a controller is not an additional actor contribution. We now carry the same discipline along routes and into persistent institutions.

Routes, resource accounts and learning

A service links actions across space and time. Routes, energy conversions and pending deliveries must compose without losing their individual boundaries. We apply the finite and dynamic laws to these chains, then explain why resource accounts, service requirements and access belong together in a useful EBU design.

44.1 A local equation in a connected world

The antidote a clinic needs may be made far away. A pump can depend on a spare part, an electrical supply and a skilled repairer. A water transfer can depend on a chain of transformations that began with fuel. Such chains prevent the local equation from becoming an isolated tank puzzle.

Local accounting does not mean that the world consists of disconnected actions. It means that each calculation uses a declared physical boundary and the complete factors affected within it. Links between actions retain sequence, dependence, quantities and provenance. They do not automatically make the whole chain one indivisible action.

The same structure that values one transfer can connect a whole service: complete physical stages supply the states, telescoping joins their receipts, and the response model describes what remains pending. This lets the account recognise shared infrastructure without hiding fuel, time or losses inside an all-purpose delivery label.

44.2 A route is an ordered physical record

Let a stock travel from A to a hub and then to B. The path has two directed edges. The first action decreases A and increases the hub; the second decreases the hub and increases B. Each action has a finite quantity, time interval and predecessor state.

If the actions occur successively under one fixed potential and no intervening external change, their field differences telescope. The hub may temporarily carry extra stock, but its intermediate potential cancels when the route total

is added. This cancellation requires using the actual intermediate state, not evaluating both edges independently against the original one.

Suppose the hub already contains stock. The later outgoing action may use stock that was present earlier rather than a physically distinguishable parcel just received. A quantity ledger can still track flows. A provenance claim about which shipment served which patient needs an explicit tracing convention or evidence. Algebraic endpoint equality does not identify the history of individual molecules.

44.3 A complete two-edge Gaussian route

Use three cells with references $(10, 10, 10)X$, scales $(2, 2, 2)X$ and initial state $(18, 10, 2)X$. The order is A, hub, B. Potential is $8 + 0 + 8 = 16$.

Send four units from A to the hub. The state becomes $(14, 14, 2)X$, with potential $2 + 2 + 8 = 12$. The first action's value is 4. Now send four from the hub to B. The state becomes $(14, 10, 6)X$, with potential $2 + 0 + 2 = 4$. The second action's value is 8. Their route total is 12.

A direct lossless transfer of four from A to B has the same endpoint and the same field reduction. Under this ideal field alone, adding an intermediate hub has not created a geographical fee. If the route has additional energy use, time effects or material losses, those are additional physical facts that must be represented and compared.

Stage	State	V	Field value
Before	$(18, 10, 2)$	16	–
A to hub	$(14, 14, 2)$	12	4
Hub to B	$(14, 10, 6)$	4	8

The example provides a calculation, not a recommendation that every delivery should pass through a hub. It omits time, handling and equipment. The scope is stated precisely so that a later extension can identify what changed.

44.4 Distance enters through physical mechanisms

A longer route may require more energy, storage or exposure to delay. Distance matters through those mechanisms. Two routes of equal length can have different grades, loads and efficiencies; two routes of different lengths can supply the same service with different complete effects.

Pending stock gives delay a physical place. A sequence of residence compartments can represent material in several stages, with each stage's output

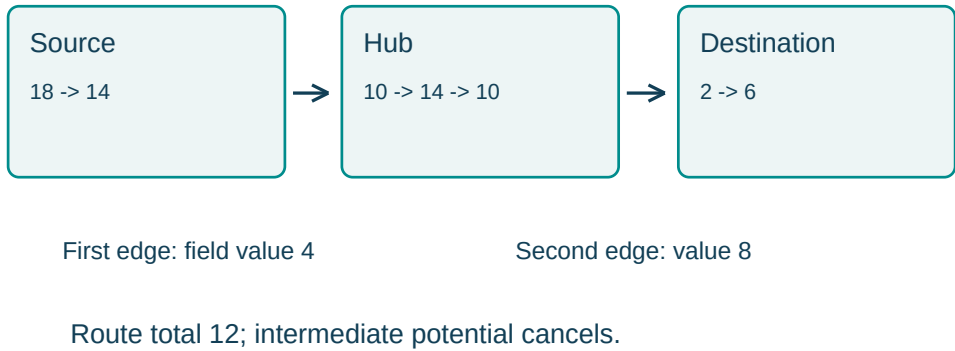


Figure 44.1: The relay route has two real transformations and two consecutive baselines. Their signed values add to the endpoint difference.

entering the next. Eliminating those stocks generally produces a memory kernel rather than an instantaneous delivery rule. A fixed travel time requires an explicit history and a different delay equation. Appendix C explains both representations.

This matters to the clinic because water already dispatched is a commitment, while medicine still travelling is not yet usable medicine. A response-aware account can distinguish those states, avoid duplicate orders and retain a deadline or temperature condition alongside endpoint EBU. If delay changes quality, the complete state changes; if it changes only the duration of shortage, the separate exposure measure records it.

A useful route comparison therefore considers the same service requirement and the complete physical consequences. Lower energy use and reliable arrival are both intelligible features, with their own quantities, rather than competing labels attached to one unexplained score.

44.5 A loss ledger is a physical extension

Suppose ten units traverse stages with retention fractions 0.9, 0.8 and 0.95. The useful amounts are 9, 7.2 and 6.84. The successive losses are 1, 1.8 and 0.36, giving the complete quantity identity $10 = 6.84 + 3.16$.

Each lost amount enters a named compartment or crosses a declared boundary. Water entering another basin is not annihilated; medicine becoming unusable has changed its condition. If the potential values a receiving sink, its derivative

contributes to the relevant directional field. For a pending arrival, the full term is $\mu_Q - \eta\mu_X - (1 - \eta)\mu_W$.

The product of retention fractions is a quantity ratio. EBU values instead compose by consecutive potential differences under one field. Distinguishing these operations prevents an apparently efficient route from hiding a burden in the sink or multiplying values as though they were delivered fractions.

For EBU, complete stage records make physical improvement locatable. A repair that raises one stage's retention changes a specific carrier transformation. The account can then connect reduced waste to the service actually delivered and to the resources no longer required upstream.

44.6 Fuel, electricity and water stay linked and distinct

A generator converts fuel energy into electricity and other outputs. A delivery system conveys electricity. A pump uses electricity to move water and generates heat. These are linked transformations, but the fuel quantity, electrical energy and water quantity do not share one stock unit.

An adequate physical description names each input and output, including boundary exchange and losses of useful form. It does not call the entire chain one water action while treating the upstream transformation as invisible. Nor does it subtract an additional burden at every arrow merely because the arrows exist.

The accounting boundary determines which effects enter the physical state and the valuation. Double-counting can arise if the same electrical consumption lowers a represented energy account and is also subtracted unchanged as a generic water-process cost. The finite EBU of the declared complete transition is the potential difference, not that difference followed by a second hidden deduction.

The original book's practical lesson survives: decompose enough to measure physical transformations, and preserve enough links to understand the service. Neither a disconnected list nor an all-purpose mega-action provides both.

44.7 Several resource accounts form a vector

Suppose one project reports +8 in a declared water field and -2 in a declared energy field. The honest record can display

$$\mathbf{E}_a = \begin{pmatrix} +8 \text{ EBU}_{\text{water}} \\ -2 \text{ EBU}_{\text{energy}} \end{pmatrix}. \quad (44.1)$$

Even if both numbers are mathematically dimensionless after standardization, their scales and interpretations can differ. Adding them to obtain 6 chooses

a conversion or weighting rule. Dimensionless notation alone does not justify that choice.

A scalar combination may be useful for a specified decision, but its weights need an explicit meaning and sensitivity analysis. If one resource is non-substitutable for an essential function, an easy gain in another account cannot automatically compensate for its loss. Physical feasibility and minimum service conditions may remain separate constraints.

Within each resource account, the relevant endpoint and process terms can close. The vector preserves those separate reconciliations. It allows comparison without pretending that one universal token has already solved the problem of valuing heterogeneous systems.

44.8 Planning is a different use of the records

A planner can compare feasible routes, examine repair strategies or forecast future needs. Such planning may use information that a local action quote does not: deadlines, inventories elsewhere, expected failures and future scenarios. This is acceptable when the layer and its assumptions are declared.

The planner should not be smuggled into the EBU formula. The formula evaluates an action given its state and field. A route optimizer chooses among possible sequences using an objective. A random actor chooses by a different rule. Both can use the same value calculation while producing different decisions.

If a planner's prediction later differs from reality, the forecast and actual record should be compared. Replacing the actual physical result with the predicted favourable one would destroy the account's purpose. A forecast can justify a decision under uncertainty while remaining a forecast.

44.9 Receipts as evidence for maintenance

A series of well-specified records may reveal that a pump repeatedly uses more electricity for the same delivered water, that a route often misses a deadline or that emergency parts come from one fragile source. These patterns can motivate investigation.

They do not automatically identify a cause. A higher apparent burden may reflect changing loads, altered references, better measurement or a genuine equipment problem. Comparisons need consistent conditions and versioned definitions. A report should retain these explanatory variables instead of ranking devices by one unqualified average.

Maintenance itself has physical consequences. It can consume parts now and improve later performance. Its present construction or repair record and the later operating records are connected, but the later improvement cannot be issued in advance as an observed current field reduction unless a justified present capacity coordinate already represents it.

44.10 Infrastructure changes the menu of possible actions

Building a bridge, laying a pipe or adding a cold store can change which actions are feasible. This is a deeper change than selecting a different quantity on an existing edge. The action menu, physical map and possibly the represented state must be updated with explicit versions.

The field can describe a consequence only through its declared variables. If a new bridge changes travel time but the model contains neither time nor service constraints, the benefit is not captured by merely adding an edge label. The limitation should be reported and the representation revised where justified.

Once the new infrastructure operates, future receipts can supply evidence about actual use and consequences. That evidence may differ from the investment forecast. A good archive lets readers study the difference, including unused capacity, maintenance demands and unexpected access effects.

44.11 Cooperation through shared physical functions

Two clinics can share a cold store, transport capacity or specialist maintenance. Their joint response determines whether that combination meets the required service with a lower represented burden. The coalition table separates the contributions of individual actions from the effect of combining them.

The recursion is particularly useful here. A route repair can change how water delivery and electrical service work together. A third-order coefficient describes that contextual change. If the response remains affine under a quadratic field, the exact ceiling says that pairs are enough, even when the common delivery system oscillates.

Shared provision can also concentrate a bottleneck. The physical boundary should retain the common capacity and the service constraints of both clinics. Joint valuation then gives a coherent total, while attribution and access rules determine how the shared service is supported and made available.

This is a concrete role for EBU in cooperation: reveal where a combination changes the physical account, preserve the dependencies needed for continuity of service, and make the distributional choice explicit. Better coordination

can reduce redundant equipment and corrective activity while maintaining access for the places the service exists to support.

44.12 Poverty can constrain the choice set

A household may use an inefficient device because it cannot obtain a replacement. Another household may have capital, secure housing and access to shared infrastructure. Their physical records can differ because their feasible opportunities differ, not because one values the environment more.

An institution that attaches consequences to EBU records must confront that asymmetry. It could provide access, collective investment, support for essential services or other arrangements, but the equation does not choose among them. A physical burden should remain visible while responsibility and assistance are considered separately.

This is why socioeconomic questions were placed at the beginning of the revised book. They are not a decorative appendix to an otherwise finished currency. An account intended to serve life must be assessed for how people actually encounter its rules, including people with disability, limited resources or unavoidable high-burden needs.

44.13 Observation does not require unlimited surveillance

A record can identify a transformation without publicly exposing every personal detail of its beneficiaries. The information needed to verify a physical event depends on its quantities, timing, boundaries and evidence. Some identity links may be held by authorized auditors; other information may be aggregated or withheld.

The choices need explicit governance: purpose, access, retention, correction and appeal. Mathematical locality offers a way to describe necessary physical inputs. It does not itself design secure storage or justify collecting personal data unrelated to the calculation.

More observations can improve an estimate, but more data can also multiply inconsistent definitions and opportunities for misuse. The quality of a physical account depends on what its measurements mean and how they can be challenged, not simply on how many records exist.

44.14 Money and physical accounts can coexist

A payment can finance a repair while a physical account describes its effects. The two ledgers may inform the same decision without being interchangeable.

Money can redistribute claims or fund access; it cannot retroactively change the litres moved or the field used in an observed account.

The physical ledger can serve disclosure, maintenance planning or a more ambitious capacity institution. Those uses share the finite account but add different decision rules. Keeping the role explicit makes it possible to introduce useful information without making every physical receipt a spendable token.

An institutional capacity rule is not identical to the physical account. If a future system uses balances to decide which offers are affordable, that rule must state whose balances, which obligations, and how an essential service is protected. Property rights, income support, appeals, labour agreements and democratic authority do not follow from the physical identity; they have to be designed and assessed openly.

44.15 Guided problem: compare complete routes

Two routes move the same amount of water between the same endpoints under one fixed field. The first uses less electrical energy than the second. If energy is represented in the complete state and the potential, their final states differ, so their EBU values may differ too. Without the energy coordinates and their valuation, the difference cannot honestly be assigned a number. If both routes reach exactly the same complete represented endpoint, their EBU values are equal; a hidden route charge cannot change that identity.

Now suppose the lower-energy route misses a required deadline. It no longer satisfies the same service problem. A report should say that it fails the deadline, not rank it as the better solution by ignoring time. If time-dependent degradation changes the physical endpoint, even equality of the water quantities is insufficient.

44.16 Practice and reflection

Recalculate the two-edge Gaussian route with quantity two. The first state becomes (16, 12, 2) with potential 13; the second becomes (16, 10, 4) with potential 9. The values are 3 and 4, totaling 7, equal to the direct endpoint reduction.

Then choose a familiar service, such as drinking water or a refrigerated meal. Draw three linked physical transformations and name the carrier at each. Identify one quantity ledger, one possible service constraint and one institutional question that cannot be read from the physical ledger.

Finally describe an infrastructure improvement and separate its present construction effects, measured capability, future forecast, authorization and com-

pensation. State which claim can be observed now and which must wait for later evidence.

What to carry forward

Routes compose physical actions under explicit states and times. Resource accounts retain their meanings as a vector unless a conversion is justified. Records connect maintenance and cooperation to visible physical consequences; access and governance remain explicit institutional choices.

The route links physical records across space and time. The next chapter asks a further institutional question: when could a signed history alter anyone's future capacity to act, and which parts of that choice remain outside the physical law?

From a signed history to a capacity institution

A signed physical history can inform future action, but a capacity institution adds rules that the potential does not choose. We derive its aggregate invariant and examine ownership, netting and access. The aim is an institution that can use precise physical information while supporting necessary service and preserving a transparent history.

45.1 A record is not yet a permission to spend

The physical account gives a signed value to a complete transition. A declared attribution can place parts of a joint value beside actors. A further institutional rule might use those records to alter future decision capacity. These are successive choices, not synonyms.

A signed historical account can record harmful and beneficial contributions without requiring its balance to stay nonnegative. A spendable capacity balance can instead impose a nonnegativity rule, a lending rule or another constraint. The potential theorem selects none of them. This distinction is essential if an EBU institution is to support people whose necessary actions sometimes have negative physical value under the current field.

45.2 One conditional balance identity

Suppose a separately declared policy maintains accounts B_i and updates them so that $\sum_i \Delta B_i = E_t$ for the complete actor transition at event t . Let $A_{\text{ext},t}$ accumulate external potential changes, using $D_{\text{ext},t} = V(x_t^b) - V(x_t)$ before the actor's transition $x_t^b \rightarrow x_{t+1}$. Under one fixed field,

$$V(x_t) + \sum_i B_{i,t} - A_{\text{ext},t} = K \quad (45.1)$$

is invariant when the updates and initial value K are defined consistently.

Proof. The change in the left side is $V(x_{t+1}) - V(x_t) + E_t - D_{\text{ext},t}$. Substitute $E_t = V(x_t^b) - V(x_{t+1})$ and the definition of $D_{\text{ext},t}$. Every term cancels. $\square$

This is a conditional accounting theorem, not adoption of a wallet. External audit is not a pool that the next actor may spend. Different ownership and update rules can satisfy the same aggregate identity while giving people very different opportunities.

45.3 Positive group value does not settle owner permission

Imagine an attribution rule gives two owners changes $(+3, -1)$, totaling a positive group value of two. Under a proposed rule requiring each post-event capacity to remain nonnegative, two initially empty accounts would refuse the group because the second owner cannot cover the debit. If one owner held both lines and same-owner netting were permitted, the net $+2$ could pass.

The physical endpoint and group EBU have not changed. Ownership and netting changed the institutional decision. A policy that pools all owners, lends capacity or protects an essential action would produce another decision. Those alternatives deserve explicit comparison rather than being hidden inside a routine called settlement.

The possible social benefit of separating these layers is practical: one can retain the physical cost of a necessary service while designing access rules that do not punish its recipient. Hiding the negative physical line would make the system less informative; treating it as an automatic prohibition would make an unproved moral choice.

45.4 Zero genesis is a narrow observation

At a strict quadratic reference, every nonzero net displacement has negative EBU. Suppose an institution begins with zero balances and requires every account change to be nonnegative. Their sum cannot equal a negative group value, so that institution refuses such an action. This is a consequence of the specified policy, not a new physical prohibition.

A different policy can protect essential service, lend capacity or provide a public initial endowment. Those choices change opportunity without changing the sign of the physical transition. After disturbance, positive restoration can also alter the available institutional balances.

The aggregate invariant constrains all such histories once an update rule is chosen. It does not determine their distribution or dynamics. A controller's stability likewise does not establish the stability of the augmented state containing physical stocks and institutional balances. The complete response of that enlarged system is the relevant object.

This distinction gives EBU an important design freedom: preserve an honest physical account while choosing access rules appropriate to the service. A clinic need not be denied water merely because its users cannot first generate a positive historical balance.

45.5 Repeated damage and repair need complete provenance

Under a fixed field, a complete closed physical cycle has zero aggregate EBU. This can block one simple accounting exploit when every signed transition is recorded. It does not by itself identify who caused damage, prevent an actor from concealing the negative event, or guarantee that gains and losses are assigned to the same owner.

A person who damages a shared asset and another who repairs it may have opposite historical records. Whether compensation or responsibility follows needs evidence and governance. The aggregate theorem cannot infer intent from endpoints. Essential maintenance can also recur because of genuine external wear or demand, so repeated positive restoration is not automatically suspicious.

A useful institution would therefore preserve event identities, causal evidence where available, correction procedures and appeals. These are social and engineering requirements supporting the account, not conclusions secretly contained in the equation.

45.6 Atomic commitment protects a joint account

An institution that authorizes a joint group should connect its commitment to that exact physical event and apply its update once. Crediting one child while losing another's debit would change the group total. Reissuing a receipt after a communication failure must not duplicate the contribution.

Here a coherent commitment preserves the group identity, accepted transformation, field and attribution rule together. The underlying physical stages remain visible: dispatch, pending stock, arrival, loss and energy use do not disappear because the institutional update is grouped.

A controller's continuation is also distinguished from a new action. Repeated observations during ring-down update knowledge of the response; they do not independently authorize new actor credit. A registered finite-horizon event settles on its stated comparison, and a later action receives a new identity.

This supplies a concrete institutional feature: one authorized contribution is recognised once, with its complete signed consequences. It reduces acci-

dental duplication, supports reliable shared maintenance and makes disputes traceable to an event rather than to an unexplained balance change.

45.7 Access changes whether information can help

Physical information becomes useful when people have feasible opportunities to act on it. A household with an inefficient heater may lack the means to replace it. A remote clinic may rely on an energy-intensive route because no alternative exists. Their account can reveal where investment would reduce future burden without treating current need as a personal fault.

An EBU institution can use that information to support shared repairs, public reserves or access to better infrastructure. The relevant feature is a change in the feasible set: people gain an option that meets the same need with lower resource consequences. A favourable score alone would not create that option.

The theoretical architecture supports this separation. Physical value describes the consequence; attribution describes a contribution; capacity policy describes permission; rights protect access. An appeal can challenge the observation, the field or the institutional rule at the layer where the disagreement belongs.

Such a design can make efficiency socially useful. Savings in energy, water or corrective work can support more reliable provision, while essential services remain available during the transition to better equipment and routes.

45.8 A place for learning without retrospective rewriting

If a field turns out to omit an important consequence, revise it through a visible process. If an allocation produces unfair outcomes, revise that rule while retaining its historical version. The separation of physical value, attribution and capacity makes such changes more intelligible: one can identify which layer failed and what evidence motivated the revision.

That separation is a strength of the proposed architecture. It gives a future society a way to learn without pretending that either a mathematical theorem or a political rule was infallible. The next chapter translates the same discipline into the responsibilities of an implementation.

The interfaces of a reproducible account

A theorem names a mathematical object. An EBU system must preserve that object as observations become proposals, proposals become accepted actions and actions become records. This chapter specifies the theoretical interfaces that keep those meanings intact. It explains why a reproducible account can remain local, composable and open to correction without becoming a single opaque score.

46.1 Every quantity has a role

A physical stock, a reference, a rate, a finite quantity and an institutional balance can all be stored as numbers. Their meanings are nevertheless different. A stock is available material; a reference is a comparison condition; a rate acts over time; a finite quantity is the resulting amount; a balance belongs to a policy. Substituting one for another can produce plausible arithmetic while changing the question.

The account therefore preserves the type, unit and identity of each object. Stocks and their units travel with the cell map. References, scales and factor definitions travel with the potential version. Action directions and accepted quantities travel with the complete transformation. Time horizons, pending state and controller parameters travel with the response declaration.

This is more than careful naming. It is the mathematical condition for combining records. Two sites can add carrier flows only in compatible units and boundaries. Two consecutive EBU receipts telescope only when their states join and their field is held fixed, or when changes are separately recorded. A capacity rule can use a receipt without pretending that EBU is another stock of water.

46.2 Permission, response and valuation

Permission identifies the admissible transformations. The response map supplies the complete endpoint of an accepted transformation. Valuation compares that endpoint with the declared baseline. An action selector chooses among alternatives using a separately specified policy.

For an immediate lossless transfer, the response may simply be $x' = x + S_e q$. For a maintained service it may be the endpoint of a differential equation at T_f . For a shared source it may include a resolver that reduces competing requests. These are different response interfaces to the same finite definition.

The interfaces can be assembled operationally, but their responsibilities remain distinct. An evaluator that silently changes a requested quantity has also become a resolver. A permission rule that chooses the largest EBU has also become a selector. A recorder that changes the baseline to make a total close has changed the event it claims to describe.

A clear separation supports efficient improvement. A better route model can replace the old response while preserving the valuation definition. A fuller potential can be introduced with an explicit field version. A different access rule can use the same physical receipts. Each revision then has an identifiable purpose and effect.

46.3 The event that the receipt describes

A complete event record connects its identity, predecessor, accepted transformation, successor and field. It retains the service condition and the physical evidence relevant to those inputs. A joint event retains its child actions and the convention used to attribute the total.

For a dynamic service, the record also identifies the horizon and background comparison. The endpoint includes whatever remains pending at that time. The raw interval change and the action-relative change can both be recorded, but they are not interchangeable. Their difference is the no-action background term.

This lets a participant reconstruct the number from the same operands. It also makes disagreement specific. A mistaken meter reading changes an observed endpoint; a changed reference changes the potential; a disputed owner changes attribution. The original record remains visible when a correction is appended.

46.4 Local evaluation follows exact support

If $V = \sum_{\alpha} \phi_{\alpha}$, factors with unchanged inputs cancel from the finite difference. The evaluator therefore needs only the affected factors and their complete arguments. The efficiency follows from the theorem, not from an assumption that distant places never matter.

Response can enlarge the affected endpoint support. A dispatched action can reach a downstream compartment before the horizon, or alter a shared reserve

through feedback. Its local calculation must include the factors changed by that complete response. Pending state cannot be dropped merely because it is outside the receiver's visible tank.

Where a composed factor depends on at most r action labels, its higher differences vanish above order r . Where a common linear response feeds a quadratic field, the ceiling is two. These results let an implementation retain an exact smaller representation when its declarations satisfy the premises.

The practical benefit is less redundant work. A small action need not trigger an entire planetary recalculation. At the same time, the relevant shared resource is included when it matters. This combination is central to an account that can serve many small participants rather than only organisations able to maintain a large central model.

46.5 Three equivalent calculations, one value

The four-unit transfer from $(18, 2)$ to $(14, 6)$ has EBU twelve under the running field. Endpoint subtraction gives $16 - 4$. Quadratic expansion gives $4(4) - (1/4)4^2$. Integrating the changing field gives $\int_0^4 (4 - q/2) dq$. Their equality follows from the earlier laws.

These representations explain different parts of the same event. Endpoints establish the total; curvature explains why the initial slope overstates it; the path integral explains how the marginal changes. They are not three contributions to add.

The same discipline applies to interaction. A group total reconstructed from its Möbius coefficients equals the original table entry. The coefficients explain the total and must not be issued again as additional value. A common-path allocation or dividend allocation then divides the known total under its declared rule.

46.6 History has an exact reconciliation

For consecutive complete states under one field, the receipt total is $V(x_0) - V(x_m)$. External movement has its own line; a field revision has its own line; an attribution has its own sum. Each identity checks a particular connection between records.

This makes missing or duplicate claims visible. A positive restoration cannot be detached from a preceding negative action when the purpose is to report the complete net history. A split transfer cannot reuse the original baseline for every piece. An autonomous controller cannot create a new registered actor event with every oscillation.

The benefit is a history that can support both accountability and maintenance. An institution can recognise useful work once, preserve necessary burdens honestly and see where recurring resource use suggests a physical improvement.

46.7 Approximation remains an explicit object

A local Taylor expression may approximate a finite result; an estimated end-point may approximate a physical state; a reduced coalition model may omit higher terms. Each approximation has a target and a remainder or error description.

For a truncated multilinear account, the remainder at a coalition is exactly the sum of omitted coefficients contained in that coalition. For smooth small actions, the derivative assumptions give the Taylor order. For uncertain table entries, covariance or deterministic bounds propagate through the signed contrast. The word approximation therefore need not mean an unexplained departure from the theory.

This gives a system a useful flexibility. It can use economical calculations where a controlled error is acceptable and exact finite formulas where the event is consequential or the simplification is available. The reason for the choice remains visible to the people relying on the result.

46.8 A reproducible account remains an understandable account

A reader should be able to move from the service being supplied to the physical change, from that change to its EBU, and from the EBU to any institutional consequence. These are the same connections developed throughout the book. Their implementation should preserve the explanation rather than replace it with an unexplained authority.

The next chapter assembles the declarations into one coherent theoretical model. Detailed software verification, measurement designs, simulation protocols and results belong to the later books; the present task is to specify exactly what those implementations and studies will be about.

From laws to a coherent model

The preceding chapters establish a family of connected results. To use them together, we must specify one model whose states, actions, response and potential are mutually compatible. This chapter assembles that theoretical specification and follows a maintained water service through it. It provides the foundation that the later books use for their designs and results.

47.1 Begin with the function to be maintained

The clinic needs usable water when care requires it. That service statement guides the choice of stocks, routes, capacities and time description. It does not yet assign a potential or select a controller. A model begins by identifying which physical conditions determine whether the function is available.

For the simplest instantaneous lesson, two usable stocks and a lossless edge were enough. For a maintained delivery, the source, pending stock, receiver, consumption and relevant losses belong in the physical account. Energy and quality enter when the claim concerns their consequences. The level of representation follows the function being described.

This is the first source of efficiency in the theory: represent what the service requires, and use exact cancellation to avoid recomputing what an action leaves unchanged. Simplification comes after the dependency is understood, rather than by omitting the dependency at the start.

47.2 State the domain and the transformations

Let $\mathcal{X}$ be the admissible state space. It includes nonnegative stocks, conservation relations and any relevant capacity or service bounds. Let $\mathcal{A}(x)$ describe the transformations available at state x . An action has a complete carrier direction or a more general response rule, with units that match the state.

For a pending route, dispatch and arrival are distinct transformations. The directions $s_D = (-1, 1, 0, 0)^\top$ and $s_A = (0, -1, \eta, 1 - \eta)^\top$ in (S, Q, X, W) preserve the complete carrier. Consumption or external replenishment has its own boundary term.

A group can share source stock or equipment even if its members have individually valid requests. The model therefore states how joint permission is resolved. If a resolver changes accepted quantities, that dependence belongs in the response map used by every coalition row.

The result is a consistent physical language. Each proposal can be interpreted as a possible transformation before its numerical value is considered.

47.3 Declare one field for one comparison

Choose a finite potential V on the states to be compared, with its reference, scale, factorization and units. Under the standardized Gaussian construction, V is dimensionless and its differences are expressed in the declared EBU convention. A native burden V_B retains its own unit until an explicit normalization is supplied.

The field is fixed across one finite receipt and across one coalition table. A later revision is a separate field-change event. This condition is what makes the path identity, telescoping and interaction contrast refer to a single object.

A quadratic is useful because it gives exact finite curvature and a low degree under affine response. A more general field may be appropriate for a declared function, but its increased complexity should represent a physical or functional dependence. Adding a term merely to force a desired score changes the account without explaining the service.

47.4 Choose how the endpoint is supplied

The response may be immediate, $x' = x + \delta$, or generated through time. In the latter case the model states the initial state, background inputs, control family, boundary rules and horizon T_f . It supplies a unique admissible endpoint for each required action setting.

The stock-and-pipeline model illustrates a complete choice. The receiver gains Q/τ and loses consumption h ; pending stock gains shortage-driven dispatch and loses arrival. Its operating point, memory kernel and roots follow from those equations. Changing dispatch to use pending stock is a new controller, with a new operating point and characteristic equation.

This theoretical completeness is useful even before an institutional rule is attached. It explains what is pending, which timescale governs correction and whether a chosen interval remains inside the physical branch. The endpoint can then enter the same finite account used for an immediate transfer.

47.5 Fix the alternatives and the background

For an action set N , each required subset S has an endpoint under the same initial state, horizon, field and background. If the no-action system moves, retain both its raw interval change and the relative action table:

$$E^{\text{rel}}(S) = V(x_{\emptyset}(T_f)) - V(x_S(T_f)). \quad (47.1)$$

The relative table starts at zero. Its nonempty coefficients agree with those of the raw table because the background difference is constant across rows.

This declaration makes a coalition comparison meaningful. A pair of repairs is compared with the same repairs alone under the same background, not with unrelated starting conditions. Pending stock is part of that background, as are fixed gains and source assumptions.

An infeasible row is a missing physical alternative, not a zero-valued action. One can restrict the motif, declare an extension or use the actual feasible poset. The appropriate inversion follows the structure of the declared alternatives.

47.6 Use the strongest exact simplification available

Once the composite endpoint value is specified, the interaction hierarchy is determined. Möbius inversion reconstructs the table exactly. Additive response and smoothness permit the repeated-FTC representation. Polynomial degree and affine response can impose a finite ceiling.

The maintained linear service is an instructive case. Suppose each action adds an input g_i to one common linear operator $F(t)$. Variation of constants gives $y_S = \bar{y} + \sum_{i \in S} h_i$ at the horizon. A fixed quadratic potential then requires only singleton and pair terms, regardless of whether F has oscillatory modes.

If actions change F , activate saturation or alter a shared resolver, this simplification must be reconsidered. The exact table still exists where its endpoints do, and its coefficients remain $-\Delta_T(V \circ x)$. The general theory therefore survives when the simple ceiling does not.

For an EBU system, this is a practical rule for allocating effort: use proved structure to simplify, and retain the full composite dependence where structure requires it.

47.7 Keep timing and value connected but distinct

A horizon fixes the endpoint comparison. Peak shortage, settling time and time-integrated burden describe the path leading to it. Their units and definitions remain separate. The same endpoint EBU can accompany different interruptions of service.

A controller's storage function provides another description. It can establish decay of deviations while the valued receiver burden rises toward a driven operating point. The two functions answer different questions, so their agreement is not assumed.

This lets a service design pursue several compatible improvements: lower resource burden, less overshoot, fewer duplicate orders and reliable access at the required time. The finite account keeps its exact meaning while the broader physical model describes the function people need.

47.8 Add the probability branch only with its structure

When a canonical physical model supplies $p = Z^{-1}e^{-V}$ in a fixed reference measure, finite EBU becomes a log density ratio and each interaction becomes the corresponding log contrast. Full-support Gaussian assumptions additionally give inverse-Hessian covariance. The specified P4 model adds the medium-entropy scaling.

These are complete theoretical bridges with named hypotheses. The deterministic delivery model does not require them to have a valid endpoint account. Conversely, a stationary density alone does not supply reversibility or a heat balance.

The advantage is a foundation broad enough to describe different physical services while precise about each interpretation. A thermal benchmark and a maintained delivery can share finite accounting without being forced into identical dynamics.

47.9 Attach an institutional rule explicitly

The physical model supplies a group value. An attribution rule divides it. A capacity policy may use the attributed history to affect future choices. Rights, essential access and public investment determine how that policy serves people.

The aggregate balance identity constrains a declared capacity update, but several distributions can satisfy it. A household's limited opportunity to make a repair is a property of its feasible set and resources, not a defect in the finite difference. An institution can use the account to identify where collective support creates a better option.

This separation gives the theory a constructive social role. It can make restoration legible, expose shared burdens and guide investment while allowing the institution to protect service during the transition to more efficient arrangements.

47.10 A complete foundation for the later books

The model now has a physical domain, feasible transformations, a fixed valuation, a response and horizon, a coalition comparator, exact interaction laws and explicit institutional interfaces. Its optional canonical and thermodynamic branches have their own conditions. Its uncertain inputs have a defined propagation rule.

These declarations are the objects to which subsequent designs, simulations and results refer. Book 1 establishes the theory and its meaning. The next chapter returns to the economy as a whole and explains how the pieces can form a useful system around the functions it is meant to keep alive.

From a calculation to an EBU system

A useful EBU system connects precise physical information to the decisions that maintain life and service. Its purpose is not to maximise the number of receipts. It is to make restoration, shared dependence and resource consequences visible enough to support better choices. We now assemble the theory around that purpose.

48.1 What the account adds to an economy

Money coordinates exchange, finances equipment and redistributes claims. A payment alone does not describe the physical conditions left behind by the activity it finances. EBU adds a signed account of those conditions under a declared field and boundary.

A leaking water line makes the difference concrete. Its repair uses labour, materials and energy now, then changes the physical capability of the route. A commercial invoice describes payment for the work. The physical account describes the represented change of condition, the resource consequences of the repair and the later delivery records. Where the potential contains a justified capacity or condition coordinate, its present improvement can be valued directly; otherwise future benefits remain attached to future outcomes or to an explicitly named forecast.

This makes maintenance legible alongside production. A community can see why preventing loss matters even when it creates less immediate sales activity. It can also see the burden of constructing the improvement, rather than treating a desirable project as physically free.

48.2 Why not simply use monetary prices?

Money provides a common unit for prices, a means of exchange and a way to carry purchasing power through time. Those functions make specialization and exchange possible without requiring every pair of participants to want one another's goods at the same moment.[2] Contracts and financing can also make a long-lived water repair possible before its service benefits have accumulated. An EBU account should acknowledge these strengths.

The question it adds is different: what happened to the represented physical state when the action occurred? A purchase price can carry useful information about scarcity, opportunity, bargaining and institutional rules. It is not, by definition, the before-minus-after difference of a specified physical potential. The same activity can therefore have a monetary price and an EBU consequence without the two numbers being interchangeable.

The economic motivation includes externalities: effects on others that are not included in the relevant transaction price can separate private cost from broader social cost.[4] This is a conditional economic problem, not a claim that every price omits every environmental consequence. Taxes, regulations, liability, contracts and public provision can change what private decisions take into account.

Existing environmental accounting also goes well beyond a list of payments. The United Nations System of Environmental-Economic Accounting brings physical and monetary information about environmental flows and assets into an integrated statistical framework.[50] EBU should be compared with that work as well as with a simple cash ledger. Its contribution proposed here is the particular potential-difference architecture, explicit coalition decomposition and separation of response, registration and history; the book does not establish superiority over existing systems.

Question	Monetary account	EBU architecture
Recorded object	Payments, claims, prices and valuations under the applicable convention.	Registered physical transitions valued under a declared potential.
Unit	Defined by the monetary and institutional system.	Declared EBU convention, with physical calibration in the appropriate domain.
Changing prices	Prices can vary while remaining expressed in the same currency.	New action values can vary with state, response and field.
Physical consequences	Can enter prices, contracts, regulation and linked physical accounts.	Enter only when represented in the boundary, state and potential.
Joint effects	Depend on the economic model, contract and accounting purpose.	The group endpoint is valued before its interaction is decomposed.
Pending effects	Orders, commitments and inventories can be recorded; physical detail depends on the system.	Future-relevant pending state belongs in the response description.

Question	Monetary account	EBU architecture
History	Treatment of transactions and asset revaluation depends on the convention.	Original event entries persist; revaluation and corrections have explicit records.
Rights and fairness	Supplied through legal, social and institutional arrangements.	Also require explicit institutional choices; no derivative assigns them.

A plausible future arrangement could use money for exchange and financing, an EBU-style account for declared physical consequences, and ordinary legal and public processes for rights and allocation. A public authority might use the physical account to explain a maintenance decision without making EBU transferable or spendable at all. None of the finite theorems requires replacing money.

48.3 One purchase can have several physical consequences

Consider a workshop buying an inexpensive cleaning input for the water system. Its financial record includes the purchase and payment. The physical account may need to include extraction of the input, useful cleaning, energy use, residual contamination, disposal capacity and material still in transit. The example assigns no numerical EBU: those values would require a defined field and observed or modelled endpoints.

The useful question is whether the declared boundary follows the relevant consequences. If a cleaner tank is obtained by moving the contamination into an unrepresented sink, the local improvement is incomplete as a claim about the wider system. Once that sink is part of the state and its burden part of the potential, its change enters the same finite difference whether or not a separate invoice exists for it.

EBU can miss the consequence too. Omitting the sink, ignoring a future-relevant pending state or assigning it no justified sensitivity leaves the account incomplete. The architecture makes those choices inspectable; it cannot make an omitted coordinate reappear through algebra.

The societal opportunity is more precise than promising to eliminate externalities. A fuller physical description can help people see where a low private cost has displaced a burden, compare ways to meet the same service requirement, and decide whether a contract or public rule should change. Monetary cost remains relevant to whether the alternative can be financed.

48.4 From a physical event to a historical entry

The architecture can now be stated as a sequence of responsibilities. The physical state says what is present. The current field says which potential and units govern the comparison. A registered action identifies the actor event and its declared boundary. Its response supplies the complete endpoint, or the group endpoints needed for a coalition comparison. Finite valuation produces the EBU amount. A chosen attribution and institutional rule determine any actor entry, which retains its original event and field identity.

The account has a physical layer and an event layer

Physical state and current field determine a potential comparison. Registration identifies which actor event is being accounted for. Joint physical value is established before its allocation. The resulting historical entry retains the comparison under which it was made.

Field evolution alone is not a registered action. No registered action means no actor transaction. Current conditions price new actions; they do not automatically reprice historical entries.

The distinction matters during a delivery. Water dispatched yesterday may arrive today, changing the current state. A registered dispatch-and-delivery event can encompass that interval under its specified outcome rule. If the model instead registers distinct physical stages, each stage has its own complete boundary and chronology. Either way, repeated observation of the same arrival does not create repeated credit.

Nor should a more favourable reserve target manufacture an actor contribution. The new target changes the field; it does not show that someone moved water. The physical history may contain the field revision and natural evolution even when the actor ledger has no new entry. This lets the system remain informative between human decisions without turning every fluctuation into a payment.

48.5 The physical state includes commitments already made

A system that records only usable stock can miss water in transit, repair work in progress or a reserve already committed to another service. Pending coordinates make these commitments part of the present description.

The feedback theory explains why this matters. Eliminating a pending stock produces memory and an initial-condition term. Ignoring both can make two identical tank readings appear to require the same response even though one tank has a large delivery approaching and the other does not.

An EBU service interface can therefore show current availability, pending delivery and the horizon of its quote together. This can reduce repeated orders, unnecessary pumping and avoidable overshoot. The resulting efficiency supports continuity of service because corrective effort is directed to an actual remaining need.

48.6 Joint projects receive one physical total

Water delivery, electrical repair and refrigeration can depend on one another. Separate invoices can describe the work, but isolated EBU quotes need not sum to the joint value. The coalition landscape and Möbius hierarchy explain the difference.

The total is evaluated once from the complete group endpoint. Its coefficients show which dependencies are already explained by single actions, which belong to pairs and which require a larger context. The recursion makes the explanation readable: a third action changes a pair relationship, and the third-order term isolates that change.

Where common linear response and quadratic valuation apply, the hierarchy stops at pairs exactly. That simplification can make a large service account less expensive to maintain. Where a nonlinear resolver or plant creates higher-order dependence, the richer model has a specific job: explain a physical effect that the simpler account cannot represent.

This supports coordination without creating an extra reward merely for grouping names together. Shared effects are inside the total, and a separate attribution rule determines how the institution recognises contributors.

48.7 History supports trust and learning

A complete signed history telescopes. Splitting an action does not remove its curvature; reversing a complete state change cancels its fixed-field value; external consumption or disturbance has its own line. These laws make the account coherent over time.

The same history can reveal recurring burdens. Repeated electricity use for a fixed delivered service can direct attention to equipment condition, route choice or operating conditions. The record retains those circumstances so that a change in the field or load is distinguishable from a change in the machine.

Model revisions remain visible too. A new reserve target is a field change. A new feedback gain is a response change. A corrected observation is a correction to knowledge of an event. Their separation allows the system to improve while preserving the meaning of earlier contributions.

For participants, this is a practical form of trust: the record explains how a number was obtained, what changed later and which rule governed any institutional consequence.

48.8 A useful incentive preserves essential service

An institution can use physical information to support maintenance, finance shared repairs or alter future action capacity. It can reward represented restoration while retaining negative entries for activities that consume reserves. The physical sign stays honest; the access rule states how necessary service is protected.

A remote clinic may have no low-burden route today. Denying its users care would not improve the physical opportunity set. An institution can instead identify that route as a priority for public investment, while supporting current service through an explicit rule. The account then helps create a better feasible option rather than blaming people for the options they lack.

Similarly, an efficient household device may be unaffordable to the household that would benefit most. A collective repair or financing programme changes the feasible set. EBU can describe the represented physical improvement and preserve its provenance; the institution supplies the means to act.

The constructive relationship is therefore between information and opportunity. Better physical accounts become socially useful when they help people maintain function with fewer avoidable burdens and when access to those improvements is part of the design.

48.9 Efficiency has a physical explanation

Each major law has supplied a particular efficiency feature. Local cancellation avoids recalculating unchanged factors. Telescoping reconciles many events without inventing new value at each subdivision. Möbius inversion removes double counting of shared effects. Degree ceilings eliminate unnecessary higher-order terms. Feedback memory keeps pending commitments visible. Complete boundaries prevent a local improvement from hiding a shifted physical burden.

These are specific consequences of the theory. Their combination offers a system that can spend less effort on redundant calculation and correction, while giving more attention to maintenance, shared bottlenecks and useful service.

Efficiency does not mean maximizing one scalar without constraints. A reserve has value because it supports future function; a deadline matters because care

is time-sensitive; redundancy can be useful because a single route can fail. The state, service constraints and response model give those requirements their explicit place.

48.10 A common account does not erase different resources

Water, energy, habitat and institutional capacity have different physical and social meanings. Several EBU fields can be displayed as a vector, preserving each one's scale and boundary. A scalar combination adds a conversion or weighting rule whose meaning must be stated.

The canonical bridge provides a precise normalization in its physical scope. It does not automatically turn every dimensionless field into an interchangeable currency. A useful institution can still coordinate several accounts, treating an essential resource constraint as a constraint rather than allowing an easy gain elsewhere to cancel it numerically.

This preserves the strengths of a common mathematical language without erasing the distinctions that make life-support functions non-substitutable.

48.11 What these capabilities could mean for society

The following uses are prospective interpretations of the architecture. They describe why its structure may be useful if the chosen physical models and institutional arrangements work in their intended domains. They are neither measured outcomes nor a claim that one potential has already been calibrated for every resource.

48.11.1 Restoration and depletion become different questions from expenditure

Return to the watershed supplying the town. Degradation lowers its ability to support the declared function; restoration aims to recover that condition. A monetary account can describe what the restoration costs. The physical account asks how the represented state actually changed. Expenditure on an unsuccessful intervention and expenditure on successful restoration can be similar while their physical consequences differ.

A complete signed history also keeps earlier degradation visible. Exact restoration under one fixed field reverses the earlier potential change. This helps distinguish maintenance of a functioning system from a story that counts only the repair and forgets the damage. Genuine external wear can justify recurring restoration, so the account must preserve where each change came from instead of treating recurrence itself as wrongdoing.

For public decisions, this offers a way to discuss both affordability and consequence. A community may need a repair even when its financial return is low, and may reject an expensive intervention that fails to restore the relevant function. The account supplies the physical comparison; public priorities and finance determine which feasible repair is undertaken.

48.11.2 Infrastructure can be evaluated as a working combination

Storage, delivery and reliable energy are useful together. Improving an empty store without a supply route may achieve little; increasing delivery without safe storage may merely shift the bottleneck. The coalition table asks about each alternative and the joint endpoint under one baseline. Möbius terms explain which part of the result requires the combination.

The same conceptual question can be asked of transport, housing and a grid connection, provided a suitable physical state and boundary are declared. The joint effect may be favourable, unfavourable or additive. No sign is guaranteed by the fact that the project sounds cooperative.

This helps a public discussion distinguish “three good projects” from “a good combined system.” Higher-order recursion can show why a third component changes the relationship between the first two. The benefit is more informed coordination and a clearer explanation of dependencies. Who pays and who receives any shared gain remains a separate institutional question.

48.11.3 Energy and water decisions can include pending consequences

A storage device changes when energy becomes available; it does not create energy by shifting the time of use. Likewise, a water order changes pending stock before it changes water available at the clinic. The response model describes those stages, and the carrier account retains conversion and loss.

The feedback lesson is useful in both settings: a controller can repeatedly correct a shortage while previous correction is still on its way. Explicit pending state, or its correctly derived memory representation, distinguishes an uncovered need from one already being addressed. This can guide a choice of controller and reduce unnecessary repeated intervention.

Stability alone does not decide whether the service is satisfactory. A system can settle at a persistent shortage; a well-valued final state can also have been reached after an unacceptable interruption. A useful service description therefore retains endpoint value, timing and access as distinct requirements. The theory explains how to compare them without disguising one as another.

48.11.4 Shared sinks make displaced burdens visible

A town can improve its own immediate environment while exporting waste elsewhere. A boundary that follows the relevant material into its sink can represent that relocation. The valued-sink edge formula makes the arithmetic explicit: the receiver's benefit and the source, pending and sink consequences enter the complete transition.

This can help discussions about pollution, resource depletion and supply chains begin from the same physical description. It does not imply that every effect is measurable or that every harm is substitutable. Where a constraint protects an essential ecological function, a gain in another coordinate need not authorize violating it. Permission and valuation remain separate.

The efficiency sought here is maintaining the required function with fewer avoidable burdens across the represented system. It is not simply improving one visible score by moving its cost out of sight.

48.11.5 Long-lived assets connect present action to future options

A reserve, a functioning pipe or a restored soil condition can matter because it changes which future services remain possible. If that capability is part of a justified present state and potential, the current improvement can be represented there. If it is only a forecast of later outcomes, the forecast remains explicitly conditional on its response model and horizon.

This distinction prevents counting a promised future restoration as though it had already occurred. It also prevents ignoring every long-lived benefit merely because it is not consumed today. The theoretical requirement is to say which present condition has changed, how it affects future possibilities and what valuation is being assigned to that condition.

An intergenerational decision additionally needs views about rights, acceptable risks and obligations to people who cannot participate now. EBU does not derive those views. It can help preserve a legible record of depletion, maintenance, pending commitments and changed capacity on which those decisions can draw.

48.12 The institutional questions the mathematics leaves open

A physical scalar does not choose fairness, political legitimacy, ownership or distributional justice. An interaction coefficient does not by itself identify causal responsibility or credit entitlement. An aggregate capacity invariant does not supply a market design, a tax rule, a monetary policy or a guarantee

of financial stability. Human preferences, social welfare aggregation and legal enforcement require their own concepts and institutions.

These are substantial responsibilities, not details to be filled in by an unexplained weight in V . An institution must decide who may revise the field, challenge observations, inspect a group comparison and appeal an allocation. It must also decide which rights and essential services constrain the use of the account.

The theory remains useful precisely because those decisions need an intelligible physical starting point. It prevents later choices from beginning with a total that double-counts a joint outcome, omits an in-transit commitment, confuses a changed ruler with restoration or treats a stable shortage as success. People can then disagree about policy while knowing which physical comparison is actually at issue.

This is the constructive role of the accounting layer: a common description of represented consequences that can support better institutional judgement. It supplies neither a constitution nor a command to maximize one number.

48.13 The system remains explainable to its participants

A person should be able to understand the route from a service to its record: what changed, which boundary was used, why the sign has its value and what the institution does with it. The explanations developed in this book are part of that design. A formula hidden behind an unexplained score would lose much of their value.

The physical account, the response model, attribution and access rules can be reviewed at their own level. A dispute about a sensor does not require rejecting the theorem. A dispute about fairness does not require altering the endpoint arithmetic. This makes correction possible without making the whole system opaque.

The final chapter gathers what the theory has established and returns to the opening question: which conditions must an economy continue to maintain so that its success today does not consume the basis of tomorrow's life?

What the theory establishes

Book 1 has developed a complete theoretical foundation for the declared EBU framework. Its central object is a finite physical comparison, and its later laws explain how that comparison composes through paths, histories, coalitions, dynamics and canonical probability. We can now state the result positively and see why the connections matter.

49.1 One definition holds the argument together

For a fixed potential and complete endpoints,

$$E = V(x^b) - V(x^a). \quad (49.1)$$

This finite definition does not depend on an initial-slope approximation or on one chosen motion law. A regular path integrates to it. A sequence of joined transitions telescopes to its net value. A complete cycle under one field has zero total.

These results make an account stable under changes of representation. Dividing a transfer into smaller records does not create extra value. Choosing a convenient evaluation path does not alter the total. Keeping the negative half of a damage–repair history prevents the positive half from being mistaken for net creation.

The practical consequence is a coherent history of physical change. Local work can be recorded, combined and inspected without losing the relation to the whole represented outcome.

49.2 Cooperation has an exact mathematical description

A coalition table specifies the alternatives under a common comparison. Möbius inversion extracts the contribution associated with each subset and reconstructs every group value exactly. Its recursion explains each higher order as a change in a lower-order relationship.

The exact multilinear representation gives the finite analogue of a Taylor hierarchy. Repeated integration connects additive physical increments to continuous mixed derivatives. Polynomial degree imposes a ceiling when the endpoint is affine. A quadratic then needs no terms above pairs.

These statements give shared projects a clear account. They distinguish complementarity from repeated claims on the same improvement, and they show when a compact explanation is complete. A larger group need not receive an unexplained bonus merely because its members act together; its extra contribution has a precise comparator and a precise place in the total.

49.3 Time enriches the response without changing the unit

A generator supplies endpoints at a stated horizon. Pending stock can create memory when it is eliminated from the visible state. A shortage-driven pipeline can have a stable operating point with nonzero burden and can approach it through damped oscillations.

The derived roots describe temporal behaviour. The storage function describes stability. The finite potential difference describes EBU. Their roles remain distinct. Under one common linear operator with additive inputs, endpoint responses superpose and the quadratic pair ceiling survives at every admissible horizon.

Nonlinear response can create higher orders even with a quadratic or linear potential; it can also cancel under composition. The exact object is always the finite difference of $V(x(z))$. This tells a designer where complexity enters instead of equating every oscillation with synergy or every smooth response with independence.

For a service system, the benefit is concrete: pending commitments remain visible, duplicate correction can be reduced, and timing can be evaluated alongside the physical endpoint consequence.

49.4 The physical probability bridge has a defined scope

In the specified canonical normalization, relative surprisal equals the dimensionless potential. Finite EBU is then a log density ratio, and interaction coefficients are the corresponding alternating log contrasts. Under full-support Gaussian assumptions, covariance is the inverse Hessian in the accessible coordinates. Under the accepted P4 conditions, medium entropy contrasts scale by k_B .

The assumptions identify exactly where each equality applies. A driven deterministic controller supplies no canonical density by itself, and a stationary density can coexist with probability current. Keeping these distinctions explicit allows the theory to connect to established physics without forcing unlike systems into one interpretation.

This gives EBU both breadth and precision: a common finite language for declared physical changes, with stronger interpretations where the physical model supplies their conditions.

49.5 A trustworthy institution can build on the separation

The theory distinguishes a physical result from its attribution and from a permission to spend. A group total can be exact while several allocation rules remain possible. A capacity invariant can hold while access rules differ. The institution therefore states its choices openly rather than presenting them as consequences of a derivative.

That separation is useful to society. It permits an honest account of resource burdens while protecting essential care, supports shared maintenance without double-counting its result and allows public investment to improve the options available to people with limited resources. The physical account can expose a problem without turning its recipient into the problem.

Versioned fields, complete boundaries and explicit corrections make learning compatible with a durable history. A community can improve its understanding of function while retaining the meaning of work already performed.

49.6 Three levels of the completed argument

The first level is established mathematics. Potential differences, telescoping, finite-difference inversion, linear feedback and canonical logarithms give exact conclusions under their stated hypotheses. An experiment is not needed to make the cancellation in a telescoping sum valid.

The second level is the EBU synthesis: these results are connected into one architecture of complete transitions, response, interaction, registration and persistent history. The book explains the definitions and choices that make that connection coherent. It does not claim that the classical mathematics originated with EBU.

The third level is the empirical and institutional programme. Whether a chosen state is adequate, whether a proposed physical denomination transfers across its intended fields and how people should use the resulting account remain substantive questions. Their study begins with the theory established here. Keeping the three levels visible lets a proved theorem be used confidently without turning it into an unsupported conclusion about an economy.

49.7 What belongs to the later books

The mathematical results established here stand with their stated assumptions. The later books specify physical and institutional applications, measurement and test designs, simulations, implementation checks and results. They use this theoretical foundation rather than asking experiments to prove algebraic identities anew.

The distinction also keeps the present book readable. Its task is to explain what each object means, how each law follows, where the laws connect and which system features they enable. Detailed protocols and outcome records have their own place in the series.

49.8 What an economy must keep alive

An economy succeeds through the conditions it continues to make possible: usable water, energy, repair, food, health, ecological function and access to the arrangements that sustain them. Transactions help coordinate those activities, but their number or price alone does not describe the physical future they leave behind.

EBU makes a precise contribution to that problem. It brings complete physical change into the account of action, connects local work to shared outcomes and preserves the information needed to distinguish restoration from displacement of burden. Its laws can support less redundant calculation, less duplicate correction and more visible maintenance.

Homeostasis is not motionless perfection. A clinic consumes water while it serves patients. Equipment requires repair while it continues to operate. A community adapts its references and infrastructure as its conditions change. The theory developed here is built to describe that active continuity: states change, actions interact, responses take time, and the account remains connected to what the system must keep functioning.

That is the purpose of the complete foundation. It gives the subsequent work a coherent language for designing institutions and physical services whose efficiency helps people meet present needs while preserving the conditions on which future life depends.

Finite differences, regularity and exact checks

This appendix completes the finite-difference proofs and verifies the examples that connect endpoint tables to smooth curvature. It also states the regularity and constraint conditions needed to use those connections exactly.

A.1 The alternating-sum identity by induction

Let $T = \{1, \dots, k\}$ and let F be defined on the corresponding Boolean cube. For $k = 1$, $\Delta_1 F(0) = F(1) - F(0)$. Suppose the k -fold formula is known. Taking one more difference subtracts the same k -fold sum with the new indicator absent from the sum with it present. The first sum gives all subsets containing the new index with sign $(-1)^{k+1-|S|}$, and the second gives all subsets lacking it with that same rule. Thus

$$\Delta_T F(0) = \sum_{S \subseteq T} (-1)^{|T|-|S|} F(\chi_S) \quad (\text{A.1})$$

at every finite order. No assumption about the values other than their existence is used.

This proof also establishes commutation of distinct difference operators: the resulting signed sum depends only on the set of indices, not the order of expansion. Ordered physical flows can still fail to commute because their endpoint protocol is a different object.

A.2 Repeated FTC with a useful weaker formulation

The main theorem assumes $G \in C^k$ on a neighbourhood of the cube. One may weaken this if the repeated one-dimensional FTC steps remain valid. A sufficient formulation is that, in a chosen coordinate order, the successive mixed partials have absolutely continuous representatives on the relevant coordinate sections, their endpoint traces agree with the lower-order derivatives used in the iteration, and the final mixed derivative is integrable with the hypotheses needed for Fubini's theorem. The conditions must cover the boundary sections appearing in the repeated differences, not merely an unspecified almost-everywhere interior derivative.

Under these conditions, apply the one-dimensional absolutely continuous FTC successively. Each step gives an endpoint difference equal to an integral, and the integrability conditions justify the repeated integration. This is a valid weaker route, but it is not a license to replace the hypotheses with “smooth enough” without checking them in a piecewise or singular model.

The Cantor function illustrates the risk. It is continuous, increases from zero to one, and has derivative zero almost everywhere, yet the integral of that derivative is zero. It is not absolutely continuous. A statement based only on almost-everywhere differentiability would miss its finite change. For a nonsmooth EBU model, the finite endpoint definition remains available even when a proposed differential representation needs repair.

A.3 The remainder estimate in detail

For $h_i = \varepsilon d_i$, factor ε^k out of the multilinear derivative in the repeated-FTC formula. Subtract the derivative at x_0 . If $D^k V$ is continuous, uniform continuity on shrinking compact neighbourhoods gives a remainder $o(|\varepsilon|^k)$.

If $D^k V$ is Lipschitz in operator norm with constant L , the integrand difference is bounded by

$$L|\varepsilon| \left\| \sum_i t_i d_i \right\| \prod_i \|d_i\| \leq L|\varepsilon| \sum_i t_i \|d_i\| \prod_i \|d_i\|. \quad (\text{A.2})$$

Multiplying by $|\varepsilon|^k$ and integrating each t_i gives the $L/2$ bound in Equation 29.5. This argument does not assume the leading derivative is nonzero. If it vanishes, the bound still controls the remainder but does not identify the next nonzero order without more information.

A.4 Checking the mixed cubic example from vertices

For $x = z_1 + z_2 + z_3 + \alpha z_1 z_2$ and $V = x^3/6$, the singleton states are one, the 12 state is $2 + \alpha$, the other pair states are two, and the triple state is $3 + \alpha$. Baseline is zero. Therefore

$$\begin{aligned} m_E(123) &= \frac{-(3 + \alpha)^3 + (2 + \alpha)^3 + 2(2^3) - 3}{6} \\ &= -1 - \frac{5}{2}\alpha - \frac{1}{2}\alpha^2. \end{aligned} \quad (\text{A.3})$$

At $\alpha = 0$ the additive cubic value is -1 ; at $\alpha = 1$ it is -4 . The continuous split in Chapter 35 must add to this vertex result, providing a useful independent sign check.

A.5 Checking interpolation dependence explicitly

For the second pair interpolation in Chapter 35, write the actual map

$$x(s, t) = s + t + s(1 - s)t = s + (1 + s - s^2)t. \quad (\text{A.4})$$

With $V = x^2/2$, the curvature term is $x_s x_t$ and the response term is $x x_{st}$. Here

$$x_s = 1 + (1 - 2s)t, \quad x_t = 1 + s - s^2, \quad x_{st} = 1 - 2s.$$

Integrate in t first. The curvature contribution is

$$\int_0^1 (1 + s - s^2)(3/2 - s) ds = \frac{7}{6}. \quad (\text{A.5})$$

The response contribution is

$$\int_0^1 (1 - 2s) \{s + \frac{1}{2}(1 + s - s^2)\} ds = -\frac{1}{6}. \quad (\text{A.6})$$

Their sum is one. For $x = s + t$, the corresponding pair is $(1, 0)$. Both maps agree at all four vertices, so both EBU coefficients equal minus one. The contrast belongs to the vertices; the named derivative split belongs to the smooth interpolation as well.

A.6 Affine constraints and quadratic geometry

For a constant symmetric positive-definite H , minimizing $\frac{1}{2}(x - x^*)^\top H(x - x^*)$ on an affine plane $Ax = b$ is a constrained problem. A stationary interior candidate satisfies

$$H(x - x^*) = A^\top \lambda. \quad (\text{A.7})$$

The sign convention for λ is arbitrary and fixed here by that equation. Solving the constraint gives the appropriate projected point when $AH^{-1}A^\top$ is invertible. Nonnegativity or other active boundaries require their own constrained conditions.

For one conserved total and diagonal $H_{ii} = \sigma_i^{-2}$, the candidate is

$$x_i = x_i^* + \lambda \sigma_i^2, \quad \lambda = \frac{M - \sum_i x_i^*}{\sum_i \sigma_i^2}. \quad (\text{A.8})$$

It solves the nonnegative problem only when every coordinate is nonnegative. Equal nonzero marginals can therefore characterize a constrained stationary point even when the ambient gradient is not zero. This explains why a zero edge field is not universally the same as reaching the declared reference.

The all-order composite chain rule

The composite chain rule organises every possible placement of a control derivative. Its partition form explains the pair and triple formulas and provides the complete higher-order theory without inventing a new rule at each order.

B.1 Partitions organize every derivative placement

Let T be a finite set of distinct control labels. A set partition π divides T into nonempty disjoint blocks whose union is T . For example, the partitions of $\{1, 2, 3\}$ are

$$1|2|3, \quad 12|3, \quad 13|2, \quad 23|1, \quad 123.$$

The notation records which differentiations act together on one derivative of the response. It is not a partition of physical actors into institutions or a grouping decision for execution.

Assume x and V are C^k on suitable neighbourhoods, with $k = |T|$, and define $x_B = \partial_B x$. The chain rule is

$$\partial_T(V \circ x) = \sum_{\pi \in \Pi(T)} D^{|\pi|} V(x)[x_B : B \in \pi]. \quad (\text{B.1})$$

Since mixed derivatives commute under these hypotheses, the order in which the blocks are listed does not change the term.

B.2 An induction proof

For one label, the only partition has one block and the formula is the ordinary chain rule $\partial_i(V \circ x) = DV[x_i]$.

Assume the formula holds for T and differentiate in a new label j . In any term, the derivative can act on $D^{|\pi|} V(x)$ through x . This creates a new argument x_j and corresponds to adding the singleton block $\{j\}$ to π .

Alternatively, it can act on one argument x_B , replacing it by $x_{B \cup \{j\}}$. This corresponds to inserting j into that block. Every partition of $T \cup \{j\}$ has

exactly one of these forms: either j is a singleton, or it belongs to one unique larger block. Removing j recovers a unique predecessor term. Thus every new partition appears once and only once, proving the formula.

This labeled form explains why no factorial coefficient accompanies a partition. Multiplicities arise in alternative versions when several differentiation labels are identified or terms with equal form are collected. Those versions must be translated carefully before numerical coefficients are borrowed.

B.3 The fourth order displayed

For labels 1, 2, 3, 4, the fifteen partitions fall into five patterns:

$$\begin{aligned}
 \partial_{1234}(V \circ x) = & D^4V[x_1, x_2, x_3, x_4] \\
 & + \sum_{\{i,j\} \subset \{1,2,3,4\}} D^3V[x_{ij}, x_k, x_l] \\
 & + D^2V[x_{12}, x_{34}] + D^2V[x_{13}, x_{24}] + D^2V[x_{14}, x_{23}] \\
 & + D^2V[x_{123}, x_4] + D^2V[x_{124}, x_3] \\
 & + D^2V[x_{134}, x_2] + D^2V[x_{234}, x_1] \\
 & + DV[x_{1234}].
 \end{aligned} \tag{B.2}$$

In the six-term sum, $\{k, l\}$ is the complementary pair of labels. The five patterns have counts 1, 6, 3, 4, 1, adding to fifteen.

A quadratic potential leaves the D^2V and DV terms. An affine response leaves only the D^4V term. If both conditions hold, the fourth derivative vanishes, in agreement with the finite degree ceiling. If neither holds, several algebraic origins can contribute to one total.

B.4 From the derivative back to the coefficient

Suppose the complete control cube and the regularity needed for repeated FTC are available. Then

$$m_E(T) = - \int_{[0,1]^{|T|}} \partial_T(V \circ x)(z_T, 0_{N \setminus T}) dz_T. \tag{B.3}$$

Substituting the partition formula gives a finite sum of integrated derivative terms. This can support a model-based diagnostic, but it still depends on the chosen smooth response across the cube. The vertices alone determine the sum, not its decomposition into those terms.

A nonzero local mixed derivative also need not produce a nonzero finite coefficient: positive and negative contributions can cancel in the integral. Conversely, a nonzero coefficient proves that the integrated mixed derivative is

nonzero when the representation applies, not that its sign is uniform throughout the cube.

B.5 How to use the formula without overclaiming it

The formula can identify which additional response measurements a proposed explanation needs. A claim that a triple is entirely due to D^3V should justify why the mixed response derivatives vanish. A claim of direct three-way response should identify x_{123} rather than infer it solely from the total coefficient.

The theorem is established multivariable calculus; Hardy's treatment explains the combinatorics of such partial derivatives [40]. In EBU it serves as an interpretation tool inside a declared model. Under common affine feedback, every response derivative above first order vanishes, leaving only the pure potential term at each order. A nonlinear response retains additional partitions. This gives EBU a precise way to identify which mathematical dependency must be represented, while the invariant finite total remains the comparison shared across representations.

Feedback, memory and generator proofs

This appendix completes the dynamic theory used in the main text: local time expansions, ordered response, exact hidden-state elimination, alternative delay laws and the admissibility of the oscillatory example. The proofs identify which simplifications preserve the endpoint account and which change the physical response.

C.1 Differentiating one solution

For an autonomous vector field b with solution $x(t)$, differentiate the equation at $t = 0$. With $v = b(x_0)$, $A = Db(x_0)$ and $B = D^2b(x_0)$,

$$x'(0) = v, \tag{C.1}$$

$$x''(0) = Av, \tag{C.2}$$

$$x'''(0) = B[v, v] + A^2v. \tag{C.3}$$

For $F(t) = V(x(t))$, the ordinary chain rule then gives

$$F'(0) = DV[v], \tag{C.4}$$

$$F''(0) = D^2V[v, v] + DV[Av], \tag{C.5}$$

$$F'''(0) = D^3V[v, v, v] + 3D^2V[v, Av] + DV[B[v, v] + A^2v]. \tag{C.6}$$

Subtracting the Taylor expansion of F from $F(0)$ proves Equation 33.21. A C^3 field and C^4 potential on a common neighbourhood containing the short trajectories give a bounded fourth time derivative and hence the stated $O(T_f^4)$ remainder. With less regularity, the remainder must be weakened rather than retained by notation alone.

These terms describe the local time dependence of the declared endpoint observable. Their controlled remainder makes the distinction between short-time structure and an exact finite-horizon identity explicit.

C.2 The observable generator

For a smooth observable f , define $L_b f = Df[b]$. Along the flow, $df(x(t))/dt = L_b f(x(t))$. Iteration gives the formal short-time expansion

$$V(\Phi_z^{Tf}(x_0)) = \sum_{r=0}^k \frac{T_f^r}{r!} L_{b_z}^r V(x_0) + O(T_f^{k+1}) \quad (\text{C.7})$$

under the corresponding smoothness, existence and boundedness assumptions. This is a finite expansion with a controlled remainder, not an unconditional claim that an infinite series converges.

If $b_z = f_0 + \sum_i z_i f_i$, then $L_{b_z} = L_{f_0} + \sum_i z_i L_{f_i}$. The operator product of r such factors expands into ordered words. Each word carries a control monomial of degree at most r . The operators need not commute, so their order must be retained.

C.3 Proof of the control-affine order result

Take a nonempty set T of k labels. The T -difference annihilates every term of degree less than k . Therefore all time orders $r < k$ vanish from the coefficient.

At order $r = k$, a surviving monomial must contain each label in T at least once. There are exactly k factors, so it contains each once, no label outside T , and no drift factor. The surviving operator words are all permutations of those k action fields. The endpoint EBU sign gives

$$m_{E_{T_f}}(T) = -\frac{T_f^k}{k!} \sum_{\pi \in \text{Perm}(T)} L_{f_{\pi_1}} \cdots L_{f_{\pi_k}} V(x_0) + O(T_f^{k+1}). \quad (\text{C.8})$$

A finite number of Boolean differences preserves the remainder order. The coefficient can vanish through cancellation; then this expression gives an upper-order bound rather than identifying the first nonzero term.

The generator-series and word language has established antecedents in nonlinear systems and control theory, including Fliess and Sontag [41, 42]. The EBU use here is to apply the finite coalition contrast to the endpoint observable, with the comparison assumptions kept explicit.

C.4 The all-time affine endpoint proof

For $\dot{x} = A(t)x + b_0(t) + \sum_i z_i b_i(t)$, let $\Phi(t, s)$ be the fundamental matrix of the common homogeneous equation. Variation of constants gives

$$\begin{aligned} x_z(T_f) &= \Phi(T_f, 0)x_0 + \int_0^{T_f} \Phi(T_f, s)b_0(s) ds \\ &\quad + \sum_i z_i \int_0^{T_f} \Phi(T_f, s)b_i(s) ds. \end{aligned} \quad (\text{C.9})$$

The first line is $\bar{x}(T_f)$ and the integral multiplying each z_i is $h_i(T_f)$. The endpoint is affine in controls. A quadratic potential therefore yields a Boolean polynomial of degree at most two for every horizon for which the equations and physical domain assumptions hold.

If A depends on z , the fundamental matrix does too. It cannot be pulled into a common affine expression in this way. The bilinear example $\dot{x} = (\sum_i z_i)x$ makes that failure explicit and produces all interaction orders at finite time.

C.5 Ordered noncommutativity and the sign convention

Let X and Y be smooth vector fields. Applying X for time T_f and then Y gives

$$\Phi_Y^{T_f} \Phi_X^{T_f}(x) = x + T_f(X + Y) + T_f^2 \left\{ \frac{1}{2} DX X + DY X + \frac{1}{2} DY Y \right\} + O(T_f^3). \quad (\text{C.10})$$

Reversing the order replaces the cross term $DY X$ by $DX Y$. Hence

$$\Phi_Y^{T_f} \Phi_X^{T_f}(x) - \Phi_X^{T_f} \Phi_Y^{T_f}(x) = T_f^2 [X, Y](x) + O(T_f^3), \quad (\text{C.11})$$

with $[X, Y] = DY X - DX Y$. For observables, $L_X L_Y f - L_Y L_X f = Df[DY X - DX Y]$, so the convention is consistent.

The planar example $X = \partial_x$, $Y = x\partial_y$ has $[X, Y] = \partial_y$, agreeing with the positive T_f^2 vertical difference when Y follows X . Keeping the convention explicit avoids an otherwise easy sign reversal.

This is a statement about ordered execution. An unordered coalition table must state whether its group row means simultaneous flow, a designated order, an average over orders or another protocol. The table transform cannot choose that semantics afterward.

C.6 Eliminating a linear hidden state

Let a constant-coefficient system be partitioned into visible state x and hidden state c :

$$\dot{x} = Ax + Bc, \quad \dot{c} = Cx + Dc. \quad (\text{C.12})$$

Variation of constants in the second equation gives

$$c(t) = e^{Dt}c_0 + \int_0^t e^{D(t-s)}Cx(s)ds. \quad (\text{C.13})$$

Substitution produces the exact visible equation

$$\dot{x}(t) = Ax(t) + Be^{Dt}c_0 + \int_0^t Be^{D(t-s)}Cx(s)ds. \quad (\text{C.14})$$

The matrix kernel is $K(r) = Be^{Dr}C$ for $r \geq 0$. Its form follows from the hidden dynamics; it is not selected independently after deleting the state. The initial term records information that the visible initial condition alone does not contain. Matrix exponentials and input convolutions are standard linear-systems constructions [42, 48].

For the pending tank, take visible x , hidden q , $A = 0$, $B = 1/\tau$, $C = -\nu$, $D = -1/\tau$. This gives the restoring kernel $-(\nu/\tau)e^{-r/\tau}$. Its negative integral $-\nu$ confirms that the kernel is a response coefficient, not a probability density.

An added dispatch input $u(t)$ enters the hidden equation as $\dot{q} = -\nu x - q/\tau + u$. The visible equation becomes

$$\begin{aligned} \dot{x}(t) &= \frac{q_0}{\tau}e^{-t/\tau} - \frac{\nu}{\tau} \int_0^t e^{-(t-s)/\tau} x(s)ds \\ &\quad + \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s)ds. \end{aligned} \quad (\text{C.15})$$

A coalition of additive inputs enters through the last convolution. This is another direct route to affine endpoint response. For EBU, it shows why a smaller state description can preserve the same account only when it retains the corresponding memory and initial conditions.

C.7 A residence compartment and a fixed travel time

The law $A = Q/\tau$ describes a first-order residence compartment. A pure fixed transport delay $d > 0$ instead has $A(t) = D(t-d)$, with the necessary dispatch history prescribed on the preceding interval. With shortage-driven dispatch and constant consumption, deviations obey

$$\dot{x}(t) = -\nu x(t-d). \quad (\text{C.16})$$

An exponential mode $x(t) = e^{\lambda t}$ therefore satisfies

$$\lambda + \nu e^{-\lambda d} = 0. \quad (\text{C.17})$$

This is a delay characteristic equation, not the quadratic $\tau\lambda^2 + \lambda + \nu = 0$ of the residence model. The required initial data and spectrum differ. Replacing a

pipe's travel history by one pending compartment is a modelling choice, even if both are described informally as delay.

The distinction is useful for EBU service design because both descriptions can supply endpoints, but their timing and admissibility claims come from different equations. The finite valuation need not change when a more appropriate route response replaces the old one.

C.8 Changing the dispatch signal changes the modes

Consider the alternative pending-aware policy $D = \nu(L - X - Q)$, still with $A = Q/\tau$ and consumption h . On its positive-dispatch branch, the operating point is

$$Q'_* = \tau h, \quad X'_* = L - h/\nu - \tau h. \quad (\text{C.18})$$

For deviations about that point,

$$\dot{x} = q/\tau, \quad \dot{q} = -\nu x - (\nu + 1/\tau)q. \quad (\text{C.19})$$

Its characteristic polynomial factors:

$$\lambda^2 + (\nu + 1/\tau)\lambda + \nu/\tau = (\lambda + \nu)(\lambda + 1/\tau). \quad (\text{C.20})$$

Both rates are real and negative for positive parameters; when they coincide, the repeated-root solution can contain a polynomial prefactor. This contrasts with the shortage-only policy's possible complex pair.

The operating point also changed. Fewer oscillatory modes do not by themselves imply a better matched service: the reference, feasible branch and sustained stock must be compared as well. The theoretical benefit is specificity. A different treatment of pending stock changes the response through an explicit term, so a design can identify which property it seeks to improve.

C.9 Storage, forcing and the receiver potential

With $\nu = 2wM$, the positive storage $\mathcal{E}_B = wx^2 + q^2/(2M\tau)$ satisfies

$$\dot{\mathcal{E}}_B = -\frac{q^2}{M\tau^2} + \frac{qu}{M\tau} \quad (\text{C.21})$$

under added dispatch input u . The second term is the supply term in this particular storage identity. It is not an independently established thermodynamic work or institutional credit.

Without forcing, the derivative vanishes on $q = 0$. A solution that remains on that set must also have $x = 0$, because $\dot{q} = -\nu x$. Thus the only invariant

zero-dissipation state is the operating point. The negative-real-part roots already prove asymptotic stability of the affine system; this storage calculation explains the cancellation that supports it.

The receiver term is exactly

$$V_B(X) - V_B(X_*) - V'_B(X_*)(X - X_*) = wx^2. \quad (\text{C.22})$$

Consequently storage differs from raw receiver burden by its linear operating-point contribution and the pending term. At $w = 1$, $M = 1/2$, $\tau = 2$, $L = 10$, $h = 2$, and $(X, Q) = (9, 3)$, one has $\dot{X} = -1/2$, $V'_B = -2$, hence $\dot{V}_B = 1$. At the same point $q = -1$ gives $\dot{\mathcal{E}}_B = -1/2$. The opposite signs are therefore exact.

C.10 Admissibility of the oscillatory coalition example

In Chapter 33, each of three actions adds $\epsilon = 1/100$ to dispatch, and the common initial state is the unforced operating point $(X, Q) = (2, 1)$. For a coalition of size $m \leq 3$,

$$x(t) = m\epsilon g(t), \quad q(t) = m\epsilon g'(t), \quad (\text{C.23})$$

where

$$g(t) = 1 - e^{-t/2} \left(\cos \frac{\sqrt{3}t}{2} + \frac{1}{\sqrt{3}} \sin \frac{\sqrt{3}t}{2} \right), \quad (\text{C.24})$$

$$g'(t) = \frac{2}{\sqrt{3}} e^{-t/2} \sin \frac{\sqrt{3}t}{2}. \quad (\text{C.25})$$

Differentiation gives $g'' + g' + g = 1$, $g(0) = g'(0) = 0$. Thus the expression solves the declared initial-value problem exactly.

For all $t \geq 0$, the sine-cosine amplitude bound gives

$$|g(t)| \leq 1 + 2/\sqrt{3} < 3, \quad |g'(t)| \leq 2/\sqrt{3} < 2. \quad (\text{C.26})$$

Therefore $|x| < 0.09$ and $|q| < 0.06$ for every coalition. It follows that $1.91 < X < 2.09 < 3$, $Q > 0.94$, and total dispatch $D = 1 - x + m\epsilon > 0.91$. The receiver and pending states remain on the stated physical branch for every nonnegative time under the maintained-source assumptions. These are analytic bounds, with no sampled trajectory required.

At horizon T_f , let $r = \epsilon g(T_f)$. Since $V(X, Q) = (3 - X)^2/2$, direct substitution gives $E_m = mr - m^2 r^2/2$. Its pair contrast is $-r^2$ and its triple contrast is

$$(3r - 9r^2/2) - 3(2r - 2r^2) + 3(r - r^2/2) = 0. \quad (\text{C.27})$$

The example demonstrates that oscillatory service response and pair-only valuation are compatible with an admissible complete coalition cube.

C.11 Why pure gradient descent has different local modes

Let $\dot{x} = -M(x)\nabla V(x)$ on an admissible domain, with $M(x)$ symmetric positive semidefinite. Along a regular solution, $\dot{V} = -\nabla V^\top M \nabla V \leq 0$. A periodic solution would return to the same V , so the nonnegative dissipation must vanish throughout the orbit. For a symmetric positive semidefinite matrix, $g^\top M g = 0$ implies $M g = 0$. The velocity therefore vanishes, excluding a nonconstant periodic orbit of this pure law.

At a critical point with positive-definite Hessian H , the linearization is $-M_* H$; derivatives of M multiply the zero gradient and vanish. This matrix is similar to $-H^{1/2} M_* H^{1/2}$, which is symmetric negative semidefinite. Its eigenvalues are real and nonpositive. Zero rates require further analysis for attraction; positive-definite M_* makes all rates strictly negative.

This result applies to the specified unforced gradient architecture. The pending tank has a different state coupling and a driven operating point with nonzero receiver gradient, so its complex modes create no contradiction. For EBU, the distinction prevents a descent theorem from being attached to a controller whose actual response does not satisfy its premises.

C.12 A stochastic generator values expected observables

For a Markov process X_t^z with a well-defined transition semigroup P_t^z , an expected-value table can be declared as

$$\bar{E}_t(S) = V(x_0) - \mathbb{E}_{x_0}[V(X_t^{XS})] = V(x_0) - P_t^{XS}V(x_0). \quad (\text{C.28})$$

Require the expectation to be finite. Its nonempty coefficients are the corresponding finite differences of $-P_t^z V(x_0)$. This is an expected observable, not a particular realized endpoint receipt.

Writing a time expansion in a Markov generator additionally requires that V belong to the relevant generator domains and that the iterated-generator and remainder conditions hold. Differential symbols alone do not establish those functional-analytic facts. Boundary conditions and admissibility can matter as much as the interior formula.

This extension shows how the same table logic can organize stochastic response. It does not establish a stationary law, detailed balance or the thermodynamic interpretation of every Markov process. Those questions belong to the next appendix's physical distinctions.

Density, current and thermodynamic scope

These examples complete the probability branch by separating a state density from the dynamics that preserve it. They explain why canonical endpoint identities can coexist with current, and why the thermodynamic interpretation requires its own physical balance.

D.1 Stationarity and reversibility are separate properties

For a diffusion with constant scalar diffusion coefficient $D > 0$ and density p , the probability current has the form $j = bp - D\nabla p$. A stationary density satisfies $\nabla \cdot j = 0$, with appropriate boundary conditions. Zero divergence permits circulating current; it does not require $j = 0$.

If $p = Z^{-1}e^{-V}$, then $\nabla p = -p\nabla V$. A drift $b = -D\nabla V + c$ therefore gives $j = cp$. Any sufficiently regular c satisfying $\nabla \cdot (cp) = 0$ and the boundary requirements preserves the same density. Reversible gradient diffusion is the special case with no such current in the usual setting of even state variables.

This mathematical distinction is one reason nonreversible sampling dynamics can share a target density with reversible ones [45]. An observed canonical-shaped density cannot determine which of those mechanisms produced it.

D.2 A rotating Gaussian with the same density

On $\mathbb{R}^2$, take $V(x) = \frac{1}{2}x^\top x$, $D = 1$, and a nonzero constant skew-symmetric matrix

$$Q = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}. \quad (\text{D.1})$$

Consider drift $b(x) = -x + Qx$. The standard Gaussian density satisfies $\nabla p = -xp$. Its current is $j = Qxp$.

To check stationarity, use $\text{tr } Q = 0$ and $x^\top Qx = 0$:

$$\nabla \cdot (Qxp) = \text{tr}(Q)p + (Qx)^\top \nabla p = 0. \quad (\text{D.2})$$

The Gaussian decay supplies the usual integrability and boundary behavior. Yet the current is nonzero away from the origin when $\omega \neq 0$. Density agreement with e^{-V} therefore coexists with circulation.

An EBU endpoint log-density identity still holds for that declared density and potential. The accepted P4 interpretation cannot be inferred from the density alone. In a physical model, the nonconservative rotational drive and its work or entropy accounting need their own treatment.

D.3 Detailed balance for a jump process

For a jump process with rates k_{xy} and a positive stationary mass function p , detailed balance means

$$p(x)k_{xy} = p(y)k_{yx} \quad (\text{D.3})$$

for the paired transitions. When $p \propto e^{-V}$, it gives

$$\ln \frac{k_{xy}}{k_{yx}} = \ln \frac{p(y)}{p(x)} = V(x) - V(y). \quad (\text{D.4})$$

The rate-ratio conclusion requires detailed balance, not merely stationarity. A physical local-detailed-balance relation, in turn, connects rate ratios to specified reservoir entropy changes under a physical model. It should not be used as a synonym for global detailed balance in a driven network [44, 43].

D.4 A driven ring with constant state potential

Take a three-state ring, clockwise rate two and counterclockwise rate one on every edge. Uniform probability $p_i = 1/3$ is stationary: each state has total inflow and outflow three times the same mass. Pairwise detailed balance fails because $p_i k_{i,i+1} = 2/3$ and $p_{i+1} k_{i+1,i} = 1/3$.

The uniform density corresponds to a constant V up to normalization, so every endpoint EBU is zero. Nevertheless each clockwise edge has log rate ratio $\ln 2$, and the clockwise cycle affinity is $3 \ln 2$. If a compatible physical local-detailed-balance model identifies those ratios with reservoir entropy flow, the drive has a nonzero thermodynamic cost. The state potential does not account for it as an endpoint difference.

The example isolates the logical boundary without inventing an EBU charge. A broader physical account must explicitly represent the drive, work and reservoirs. It cannot recover their path dependence by claiming that a constant state potential already contained them.

D.5 Stationary curvature and global Gaussian covariance

If a positive density satisfies $J_{\text{prob}} = V + \ln Z$ in fixed coordinates, Hessian equality follows wherever the derivatives exist. No statement about global covariance follows without additional structure.

For a full-support Gaussian with positive-definite constant H , diagonalize $H = R^\top \Lambda R$ with orthogonal R . The variable $u = \Lambda^{1/2} R(x - x^*)$ has density proportional to $e^{-u^\top u/2}$, so $\text{Cov}(u) = I$. Transforming back gives $\Sigma = H^{-1}$.

If a boundary truncates the density, the transformed support is not all of $\mathbb{R}^d$. Its moments need not be those of the untruncated standard normal. A locally fitted Hessian can describe local log curvature while the tails and boundaries determine different global moments. This distinction is particularly relevant to nonnegative conserved stocks.

D.6 Different statistical interactions use different operations

A log-linear coefficient on a binary cube is an anchored alternating log-probability contrast. Interaction information instead applies inclusion–exclusion to marginal entropies. Marginalization changes the objects before the alternating sum. One should therefore not expect the two operations to commute or produce equal coefficients.

As a simple structural example, a joint log density with only $az_1z_2z_3$ has no anchored pair term at the all-zero background, but conditioning on the third variable being one produces a pair log odds ratio a . Marginalizing over the third variable produces yet another pair expression involving a sum of exponentials. Background, conditioning and marginalization are distinct procedures.

Connected-information and maximum-entropy decompositions offer another perspective on dependence [46]. They can complement an EBU study when the relevant joint distribution is defined, but their terms should retain their own names and estimands. Pointwise endpoint densities alone do not supply the full statistical objects those methods require.

D.7 The boundary is scientifically productive

These examples do not weaken the canonical-density theorem. They identify which additional observations are needed to advance from density to dynamics and entropy. Currents, rate ratios, work and reservoir conditions can distinguish processes that endpoint histograms cannot.

For EBU, the hierarchy permits the strongest interpretation actually supplied by the model. The deterministic feedback tank supplies a response and storage law without a density. A canonical model supplies the log-ratio identity. A conservative single-reservoir model under the P4 premises supplies the entropy relation. These levels connect through their explicit assumptions, keeping every quantity useful within its physical meaning.

Following one claim all the way through

One complete example now follows the argument from endpoints through interactions, calculus and conditional probability. A dynamic realization then connects it to the feedback distinction. The purpose is to show how the theoretical pieces can be used together in a single reproducible calculation.

E.1 A declaration small enough to inspect

Use three dimensionless binary controls and the scalar response

$$x(z) = z_1 + z_2 + z_3 + z_1 z_2, \quad V(x) = \frac{1}{2}x^2, \quad x_0 = 0. \quad (\text{E.1})$$

Assume all eight endpoints are admissible for this constructed example. The declaration specifies a response and a potential, not a real apparatus, a selection policy or an economic institution.

For each subset S , define $E(S) = -V(x_S)$. The table, including every interaction coefficient, is

S	x_S	$E(S)$	$m_E(S)$
$\emptyset$	0	0	0
1	1	$-1/2$	$-1/2$
2	1	$-1/2$	$-1/2$
3	1	$-1/2$	$-1/2$
12	3	$-9/2$	$-7/2$
13	2	-2	-1
23	2	-2	-1
123	4	-8	-1

The coefficients sum on the full group to $-3/2 - 7/2 - 1 - 1 - 1 = -8$. The three-way value is not the whole difference between the group and its singletons. The pair terms must be removed first.

E.2 The exact finite polynomial

The unique multilinear extension is

$$\tilde{E}(z) = -\frac{1}{2}(z_1 + z_2 + z_3) - \frac{7}{2}z_1 z_2 - z_1 z_3 - z_2 z_3 - z_1 z_2 z_3. \quad (\text{E.2})$$

Evaluating this polynomial at any Boolean vertex recovers the table. It has no remainder there. Its values at nonbinary controls need not match the physical smooth composite $-x(z)^2/2$, because the multilinear reduction replaced repeated powers using a Boolean identity.

E.3 The smooth derivative check

For the physical smooth response, the only nonzero mixed second derivative is $x_{12} = 1$, and $x_3 = 1$. Since $V'' = 1$ and $V''' = 0$, the triple chain rule gives $\partial_{123}(V \circ x) = 1$. Its integral over the unit cube is one, so the EBU triple is minus one.

This matches the table and demonstrates response-generated interaction under a quadratic field. It does not prove a unique physical cause in another system whose response has not been measured. The response is known here because it was explicitly declared.

E.4 The probability check is conditional

If, separately, these scalar states are evaluated under the normalized Gaussian density $p(x) = (2\pi)^{-1/2}e^{-x^2/2}$ on $\mathbb{R}$, the density relation holds. The triple log product is

$$\ln \frac{p(4)p(1)^3}{p(3)p(2)^2p(0)} = -1. \quad (\text{E.3})$$

The normalizers cancel. The density readings are at response endpoints; they are not probabilities that each coalition will be selected.

The stipulated density makes this equality exact. It supplies a probability interpretation of the same coefficient, while a law of motion would separately determine stationarity and detailed balance.

E.5 The entropy check needs still more

If every entry were realized by a process satisfying the full fixed conservative overdamped single-reservoir P4 conditions, the medium-entropy coefficient would be $-k_B$. That conclusion does not follow merely because a Gaussian density was written down.

The response law above does not specify heat, work, a reservoir, or how controls are applied. Therefore the example as declared does not claim that any actual entropy was produced or measured. It illustrates exactly where the thermal premise would enter and which additional physical assumptions complete the connection.

E.6 A joint outcome still leaves attribution open

The full group has value minus eight. Many triples of signed numbers sum to minus eight. None is selected by the Möbius decomposition as a unique owner payment. The dividends describe subset interactions; an allocation rule may use them, but must state how and why.

A common-path decomposition would need a declared path and a decomposition of its velocity into child contributions. A Shapley-type allocation would average conditional increments across orders under its own axioms. A service institution might use another rule entirely. All can be discussed without changing the already determined group value.

E.7 An uncertainty statement needs the measurement design

If the eight potential values were independently estimated with common variance σ^2 , the triple contrast would have variance $8\sigma^2$. If the table instead came from shared fitted parameters, its covariance would differ. The exact table cannot tell us which observation model was used.

Thus a complete empirical report would append the calibration and covariance model, not simply say that the coefficient is exactly minus one. Exactness describes the constructed mathematics here. Measurements would need their own uncertainty and possible bias assessment.

E.8 A monotone dynamic realization

The response table also arises at horizon $T_f = 1$ from

$$\dot{x} = z_1 + z_2 + z_3 + z_1 z_2, \quad x(0) = 0. \quad (\text{E.4})$$

Each Boolean control produces a constant or increasing trajectory. The endpoint is exactly the table's $x(z)$, so the quadratic potential still gives a triple of minus one. No oscillation is present.

Compare this with the admissible pending-tank example: its nonempty response has damped oscillatory modes, yet its quadratic coalition table has an exact zero triple. The two examples establish both directions of the distinction between temporal behaviour and interaction order. They also show why the composite endpoint, rather than a visual impression of complexity, determines the required EBU hierarchy.

E.9 The claim ledger

Claim	What supports it
Complete endpoint table	The declared response and admissible vertices.
Exact coefficients	Boolean finite differences and reconstruction.
Triple under quadratic V	Nonlinear response and the composite derivative.
Log-density equality	An additionally stipulated canonical density and measure.
Thermal entropy interpretation	Not established by this response declaration; full P4 premises would be needed.
Owner entitlement	Not defined by the table.
System capability	Exact reconstruction, explicit response dependence and a transparent basis for a separately chosen allocation.

This sequence is the book’s method in miniature. Start from a clear physical comparison, derive what follows, and name every additional premise when moving to a stronger interpretation. The result is a coherent theoretical specification: each equality is established under its premises, and each additional interpretation names the structure that supplies it.

Plain-language dictionary and symbol guide

Action. A declared physical transformation with a specified boundary, quantity or other input, predecessor state and successor. It can be a transfer or another represented change. Naming a useful activity does not by itself define its state effect.

Actor. The person, organization or modeled decision-maker associated with an action. Its identity is distinct from the action's field value. The field equation does not assign moral worth to the actor or recipient.

Affected factors. Terms in a potential that depend on at least one changed coordinate. A complete local calculation includes every such term. In the separable Gaussian model, these are the changed endpoint terms.

Baseline, x^b or x_0 . The common state from which an action and its no-action alternative are compared. In the prospective forcing-first order it is the post-forcing, pre-action state. A baseline must identify both data and model version.

Capacity account, B_i . A proposed persistent account controlling later modeled affordability. It is not the physical stock, storage capacity, household income or an automatically legitimate entitlement. Its update and use require a separately declared rule.

Cell or node. An indexed location or subsystem holding a represented state. A cell is a modeling boundary rather than necessarily a household, person or geographic square.

Common path, $\gamma(s)$. A single declared state path used to calculate the group integral and its child contributions. The path parameter is not automatically clock time. A physical-path reading needs a physical model that realizes it.

Conservation. A complete physical quantity balance within a named boundary. Lossless internal transfers preserve the sum of homogeneous stocks. Transformations, sinks and boundary crossings must be named when they are present.

Curvature. The way a slope changes as the state changes. It is represented locally by the Hessian $H(x)$, constant for a quadratic. Curvature explains why an initial slope times a finite quantity differs from the exact finite value.

EBU, E . The signed difference $V_{\text{fixed}}(x_{\text{pre}}) - V_{\text{fixed}}(x_{\text{post}})$ across a declared complete physical transition. In the Gaussian example it is dimensionless

standardized-deviation reduction. The name does not make it physical energy or money.

Edge. A declared possible directed physical process. For a lossless transfer, its incidence column has -1 at the source and $+1$ at the destination. A line drawn between places is not sufficient evidence that a real process is available.

Equilibrium. A property of a specified dynamical system, typically a state unchanged by its evolution rule. It is distinct from a declared reference or a minimum of a chosen potential unless the required relationship is established.

External audit, A_{ext} . The proposed cumulative signed potential change from external forcing. It explains externally introduced or removed deviation. It is not a spendable “nature wallet.”

Factor. One term in a decomposed potential, with a declared set of input coordinates. A factor can be local without depending on only one coordinate. Its dependencies determine the required evaluation neighbourhood.

Feasibility. Satisfaction of the declared physical conditions on an action, such as nonnegative stock, permitted edges and shared capacity. A favourable EBU does not establish feasibility.

Field. The declared state, reference geometry and relevant action relationships used for valuation. Its gradient supplies local directional information. The word does not assert a new fundamental physical force.

Field attribution, R_a . A child’s signed derivative integral along a declared group path. It sums to the group field reduction under the stated conditions. It is not automatically a uniquely observed causal contribution or an ownership share.

Forcing. A declared external physical change preceding the action baseline in the prospective study. Raw random proposals and physically applied forcing can differ if feasibility rules reject some proposals.

Functional balance. A description of conditions under which relevant functions can continue. A single numerical reference is one representation of part of that idea, not a complete definition for every living system.

Gaussian geometry. The quadratic relative negative-log shape used here to measure standardized deviation. It does not establish an observed normal distribution of actual states, nor an autonomous restoring process.

Gradient, $\mu = \nabla V$. The vector of derivatives of the potential. Each entry describes local change in potential per small change of its coordinate. The gradient is not an actor, a physical stock or a selection algorithm.

Group, $\mathcal{G}$ or S . Accepted actions evaluated through a common physical transition because of their declared dependencies. The child actions remain identifiable. A group total does not by itself determine every child allocation.

Hessian, H . The matrix of second derivatives. For the separable Gaussian model, $H_{ii} = 1/\sigma_i^2$ and off-diagonal entries vanish. Shared action coordinates can still produce group cross terms.

Homeostasis. Maintaining relevant function through continuing change and regulation. The term does not mean that every component is static, that resources are unlimited or that any proposed accounting rule has already achieved regulation.

Incidence column, S_e . A signed template turning an edge quantity into its state increment. Multiplying S_e by a finite quantity gives the source and destination changes in stock units.

Locality. A statement about necessary support or information. Mathematical locality, distributed computation and decentralized governance are distinct properties. None should be inferred from the others without a specific argument.

Marginal field, f_e . For a lossless directed transfer, $\mu_i - \mu_j$. It gives the initial rate of field reduction per added action quantity. Exact finite value also includes curvature and all represented physical effects.

No action. A zero control can still permit natural drift. A zero physical increment has zero fixed-field EBU; a drifting no-control comparator must be retained or explicitly centered as in Chapter 32. Missing evidence is not verified inactivity.

Potential, V . The declared scalar function of state. In this model, $V = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2$. It measures represented standardized discrepancy, not total welfare or conserved matter.

Physical carrier or sink. A represented place where energy, matter or useful capacity goes during an action. It must be accounted for in the state or boundary when the corresponding physical effect is in scope; it is not a second subtraction from the finite EBU identity.

Quantity, q . The finite source-side amount in this edition. A rate j has stock-per-time units and becomes quantity after integration over time. For a constant rate, $q = \Delta t j$.

Receipt. A provenance-linked record of a physical event, its accepted comparison, evidence and signed calculation. A record can exist before an institution has decided whether its number changes a transferable balance.

Reference, x_i^* . A declared local comparison value. It may be informed by functional evidence, but it is not automatically an observed mean, ethical optimum or dynamic equilibrium.

Scale, $\sigma_i > 0$. A fixed positive stock-unit width determining how much a given deviation contributes to the potential. It is separate from measurement uncertainty unless an explicit model establishes a relationship.

Service. A separately specified outcome such as delivery of a required quantity before a deadline. A positive field difference can coexist with unmet service; a negative one can coexist with necessary service.

State, x . The ordered collection of represented physical coordinates. Complete state identity includes units, meanings and ordering. Equal potential does not imply equal state.

Support, W . Coordinates directly changed by the action. With coupled factors, the set of coordinates required to evaluate the effect can be larger than this write support.

Telescoping. Cancellation of intermediate values in a sum of consecutive differences. It requires consistent definitions and correctly linked boundaries; it does not reconstruct missing physical history.

Vector account. Separate resource-specific results retained together without an undeclared conversion. Even dimensionless scores can have different calibrations and meanings.

Coalition. A subset of declared actions whose endpoint is specified under one comparison protocol. An unordered subset does not automatically specify an execution order.

Möbius coefficient, $m_E(T)$. The alternating finite contrast that removes all proper-subset contributions before naming the interaction of T .

Composite, $G(z) = V(x(z))$. The potential evaluated on the action-to-state response. Its finite differences determine interaction, including when the response is nonlinear.

Discrete Taylor representation. The unique complete multilinear polynomial reproducing a Boolean table. It has no remainder at the vertices; it need not be the physical response at fractional controls.

Generator, b_z or L_z . A declared state-velocity law or its action on observables. It supplies response, not the definition of EBU. Historical J_e denotes a flux; a scalar opposition Ψ is not automatically a vector generator.

Canonical density. A density of the form $p = Z^{-1}e^{-V}$ relative to a specified measure, with finite normalizer. A physical canonical interpretation needs the appropriate ensemble and calibration.

Relative surprisal, J_{rel} . $-\ln[p(x)/p(x^*)]$ in a fixed density convention. Absolute $J_{\text{prob}} = -\ln p$ differs by a reference constant.

Stationarity and reversibility. Stationarity preserves a probability law under a declared process. Reversibility is a stronger symmetry of the process; stationary currents can distinguish them.

Thermal and bridge coefficients. $\beta_{\text{thermo}} = 1/(k_B T)$ has inverse-energy units. β_{bridge} is the dimensionless comparison coefficient between a declared EBU potential and relative surprisal. The canonical normalized benchmark gives the latter as one.

Entropy tables. Medium, stochastic-system and total entropy changes have separate meanings. Their EBU scaling holds only under the stated P4 physical conditions for every compared process.

Pending stock, Q . Material dispatched but not yet available at the receiver. Its deviation $q = Q - Q_*$ can be negative while physical pending stock remains nonnegative. In earlier transfer chapters, q denotes a finite action quantity; the context identifies the use.

Residence time, τ , **and horizon**, T_f . The first is a parameter of the pending-stock arrival law. The second is the duration at which a response is evaluated. Changing either can change an endpoint, but they are different declarations.

Native burden, V_B , **and storage**, $\mathcal{E}_B$. V_B carries the historical burden unit B . The feedback storage $\mathcal{E}_B$ proves a stability property. Neither is automatically the normalized potential V or its finite EBU difference.

Memory kernel, K . The causal weighting of earlier visible states obtained by eliminating hidden dynamics. The accompanying initial-condition term preserves the hidden initial state. The restoring tank kernel is not a probability density.

Temporal mode. A component of the response described by a root or eigenvalue of the declared dynamics. Decay and oscillation classify time behaviour, independently of coalition interaction order.

Affine feedback response. A common linear operator with additive action inputs produces an endpoint affine in coalition indicators. With a fixed quadratic potential, this implies an exact zero at every interaction order above two, for every admitted horizon.

The compact symbol card

Symbol	Object	Units or interpretation
x_i, x_i^b	Stock at a location	Stock, X
x_i^*	Declared reference	Stock, X
σ_i	Positive reference scale	Stock, X
q	Finite action quantity	Stock, X
j	Action rate	Stock per time
δ	State increment	Stock in each coordinate
V	Declared state potential	Dimensionless account value
μ_i	Marginal potential	Inverse stock
H	Curvature matrix	Inverse squared stock
E	Signed finite value	Compatible potential units
R_a	Common-path attribution	Compatible potential units
B_i	Proposed action balance	Declared capacity-account units
A_{ext}	External potential audit	Compatible potential units
z_i	Action control or indicator	Dimensionless; distinct from stock baseline
$G(z)$	Composite potential	Potential units
$m_E(T)$	Interaction coefficient	Same units as E
p, Z	Density and normalizer	Relative to declared measure
J_{prob}	Negative log density	Declared density convention
β_{bridge}	Bridge coefficient	Dimensionless
k_B	Boltzmann constant	Entropy units
$Q, q =$	Pending stock and deviation	Stock units in the feedback model
$Q - Q_*$		
τ, T_f	Residence parameter and horizon	Time
ν	Tank response coefficient $2wM$	Inverse time
$V_B, \mathcal{E}_B$	Native burden and stability storage	Native burden units B
F, λ	Linear response matrix and mode rate	Rates; distinct from interaction order

Notation alone establishes neither physical calibration nor spending rights.

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How to follow a mathematical claim

The social and environmental references support the bounded claims attached to them in the opening chapters, not a claim that EBU succeeds. The project sources identify definitions, conditional mathematics and historical records. Source versions, PDF hashes and the full revision disposition accompany the editable manuscript.

The local-to-global argument in this edition is the cancellation of unchanged factors. Its Gaussian examples use a newly declared quadratic; they do not reinterpret historical threshold fixtures. The exact finite expansion is derived directly from that quadratic. The path sum uses linearity, the chain rule and the fundamental theorem of calculus. The sequence and closed-cycle identities use consecutive endpoint cancellation.

The field definition, the path convention and any proposed capacity rule have different roles. A proof about the sum of path contributions cannot supply a missing ownership map. Likewise, a physical conservation equation cannot by itself choose which consequences belong in a potential. These boundaries are part of the claim, including when a source file is committed in Git.

The continuity chapters rewrite the cleared R and S-MG derivations into a teaching sequence [28, 29, 30]. The proofs in the text are explicit; the references locate antecedents and physical scope. No theorem is presented as newly discovered on the basis of this integration.

Edition and scientific-status note

This volume rewrites the author’s 160-page *What an Economy Must Keep Alive* manuscript from “Book writer.” The repository report identifies its exact sources. Earlier editions and the original worktree remain preserved.

The central integration uses the repaired R-stage source at commit `6c1d082` and the S-MG theorem and prior-art package at commit `a7655f3`, and the feedback reconciliation at commit `f3ef451`, with the independent-clearance disposition supplied in the author’s integration request. Their original reports are unchanged. Clearance of those inputs does not constitute independent audit of this newly edited book.

The governing definition remains $E = V(x_{\text{pre}}) - V(x_{\text{post}})$ for a complete declared transition under one fixed field. The frozen physical foundation and working theory baseline retain precedence over exploratory or historical material. The manuscript changes no scientific authority, runtime, preregistration, seed, execution seal or result artifact.

The finite and interaction theorems are established under their stated assumptions. The canonical density and accepted P4 entropy relations carry their additional physical assumptions. Cross-field denomination, general nonequilibrium interpretation, actor mechanisms, economic effectiveness and institutional fairness remain separate research questions. No completed experiment or successful economy is inferred from this book’s rendering.

All new numerical examples are constructed teaching calculations. No scientific random stream, model trajectory, synthetic campaign or optical-trap experiment was executed for this integration. The current E1a execution pause is unchanged. A future study requires its own scientific and operational authorization.

Book 1 supplies the complete cleared theoretical foundation within 49 chapters and five appendices. Later books develop implementation, test designs, simulations, applications and results. The eight-part plan retains its numbering; Part VII revisits coordination and system dynamics through advanced applications. Length follows the argument, with no page ceiling.

Combined Book 1 theoretical integration complete — independent book audit required.